This report is intended solely for cbass@marketstreetpartners.com

EQUITY RESEARCH

Capital Markets

RBC Technology, Media & Telecom Equity Team
Click here for contributing analysts' contact information

April 28, 2022

# RBC Software Primers: ARR, Billings, or RPO — What's the Best Leading Indicator?

RBC Capital Markets, LLC
Rishi Jaluria (Analyst)
Matthew Hedberg (Analyst)
Richard Poland (Senior Associate)
Philip Rigby (Senior Associate)
Dan Bergstrom (Analyst)
Matthew Swanson (Analyst)
Anushtha Mittal (Associate)
Simran Biswal (Associate)

RBC Europe Limited
Sherri Malek (Analyst)

**Our view:** In our fifth software primer, we focus on ARR, billings, and RPO, the primary leading indicators of growth. Based on our analysis, we highlight CRWD, GWRE, HUBS, NOW, and ZI as showcasing the most encouraging leading trends and cautious on COUP, DCT, DOCU, and ZM. We also believe ESTC, FSLY, MDB, NICE, TWLO, and ZM could benefit from disclosing ARR as a leading metric.

- **In our fifth software primer, we continue our practice of providing a deep dive on a key topic in software,** informed by our conversations with investors and companies. Previous primers include Net Retention Rates, Gross Margins, Sales Efficiency, and Software in a Post-Pandemic World.
- **We also highlight ongoing key themes,** including performance, valuations, the "Rule of 40", IPOs, and M&A; highlight our top picks; and discuss our outlook. The RBC Software Index is down 24% YTD in 2022 (vs. NASDAQ down 22%).
- **This quarter, we focus on leading indicators in software** and determining which metrics investors should focus on to predict future growth. Leading indicators are especially important in SaaS companies because revenue is a backward-looking metric, as 80% of revenue in a typical SaaS company rolls off deferred revenue in any given quarter.
- **Traditionally, calculated billings** (revenue plus the change in deferred revenue) **has been the "gold standard"** for SaaS investors, but billings has numerous puts and takes that have made the metric more "noisy" than predictive.
- **We see investors increasingly turn to RPO** (a newer metric disclosed under ASC 606) and RPO-derived metrics (e.g. bookings), and we see companies also increasingly disclosing non-GAAP metrics like ARR or ACV — metrics that had traditionally only been used by companies undergoing a subscription transition. Having said that, we see RPO and ARR bringing their own puts and takes, including inconsistency in how companies calculate ARR.
- **In this note, we discuss the "death of billings"**, outline our framework for working with RPO, discuss ARR disclosures and the differing definitions (as well as its importance for on-premise companies), and how we deal with consumption companies.
- **In addition, we provide company-by-company analysis for 73 companies in our extended coverage universe and determine the best leading indicator for each company (beginning on page 14).** We also analyze the varying definitions of ARR and rationale for disclosure across software.
- The companies with the **best leading trends**, in our view, are **CRWD, GWRE, HUBS, NOW, and ZI.**
- On the other hand, **we would be cautious on COUP, DCT, DOCU, and ZM** given the forward-looking indicators.
- **We believe ESTC, FSLY, MDB, NICE, TWLO, and ZM could benefit from disclosing ARR** as a potential leading metric. In addition, we believe CWAN and DBX, which disclose ARR, could benefit from guiding to the metric.
- **Finally, we give credit to PEGA, GWRE, PD and VEEV for solid disclosures** and how they deal with leading indicators.

Priced as of prior trading day's market close, EST (unless otherwise noted). All values in USD unless otherwise noted
**Disseminated: Apr 28, 2022 04:46EDT; Produced: Apr 28, 2022 04:46EDT**
**For Required Non-U.S. Analyst and Conflicts Disclosures, see page 117**

This report is intended solely for cbass@marketstreetpartners.com



# Table of Contents

**ARR, Billings, and RPO — What's the Best Leading Indicator? ..... 4**
Why Leading Indicators Matter ..... 4
The Death of Billings ..... 5
Remaining Performance Obligations ..... 6
Annual Recurring Revenue (ARR) ..... 9
What Do We Do With Consumption Companies? ..... 13

**Company-Specific Analysis ..... 14**
Akamai Technologies, Inc. (AKAM) – covered by Rishi Jaluria ..... 14
ANSYS (ANSS) – covered by Matt Hedberg ..... 15
Altair Engineering (ALTR) – covered by Matt Hedberg ..... 17
Atlassian Corp. Plc (TEAM) – Not Covered ..... 18
Alteryx Inc. (AYX) – Not Covered ..... 19
Autodesk (ADSK) – covered by Matt Hedberg ..... 20
Asana, Inc. (ASAN) – covered by Rishi Jaluria ..... 21
Box, Inc. (BOX) – covered by Rishi Jaluria ..... 22
Bentley Systems (BSY) – covered by Matt Hedberg ..... 23
Check Point Software (CHKP) – covered by Matt Hedberg ..... 24
Cloudflare (NET) – covered by Matt Hedberg ..... 25
Coupa Software, Inc. (COUP) – covered by Rishi Jaluria ..... 26
Couchbase (BASE) – covered by Matt Hedberg ..... 27
Coursera Inc. (COUR) – covered by Rishi Jaluria ..... 29
Crowdstrike (CRWD) – covered by Matt Hedberg ..... 30
Clearwater Analytics Holdings Inc. (CWAN) – covered by Rishi Jaluria ..... 32
Datadog (DDOG) – covered by Matt Hedberg ..... 33
DocuSign, Inc. (DOCU) – covered by Rishi Jaluria ..... 34
DoubleVerify (DV) – covered by Matt Hedberg ..... 35
Dropbox, Inc. (DBX) – covered by Rishi Jaluria ..... 36
Dynatrace (DT) – covered by Matt Hedberg ..... 37
Duck Creek Technologies, Inc. (DCT) – covered by Rishi Jaluria ..... 38
Elastic (ESTC) – covered by Matt Hedberg ..... 39
EverCommerce (EVCM) – covered by Matt Hedberg ..... 40
F5 (FFIV) – covered by Matt Hedberg ..... 41
Fastly, Inc. (FSLY) – covered by Rishi Jaluria ..... 42
Fortinet (FTNT) – covered by Dan Bergstrom ..... 43
Gitlab (GTLB) – covered by Matt Hedberg ..... 45
Guidewire Software, Inc. (GWRE) – covered by Rishi Jaluria ..... 46
HubSpot Inc. (HUBS) – covered by Rishi Jaluria ..... 48
Informatica (INFA) – covered by Matt Hedberg ..... 49
Jamf (JAMF) – covered by Matt Hedberg ..... 50
MongoDB, Inc. (MDB) – covered by Rishi Jaluria ..... 51
Microsoft Corporation (MSFT) – covered by Rishi Jaluria ..... 52
N-Able (NABL) – covered by Matt Hedberg ..... 53
NetScout (NTCT) – covered by Matt Hedberg ..... 54
New Relic, Inc. (NEWR) – covered by Rishi Jaluria ..... 55
NICE Ltd. (NICE) – covered by Rishi Jaluria ..... 56

This report is intended solely for cbass@marketstreetpartners.com



NortonLifeLock (NLOK) – covered by Matt Hedberg ..... 57
Nutanix (NTNX) – covered by Matt Hedberg ..... 58
Okta (OKTA) – covered by Matt Hedberg ..... 59
Olo (OLO) – covered by Matt Hedberg ..... 60
PagerDuty (PD) – covered by Matt Hedberg ..... 61
Palantir Technologies Inc. (PLTR) – covered by Rishi Jaluria ..... 62
Palo Alto Networks (PANW) – covered by Matt Hedberg ..... 63
Pegasystems Inc. (PEGA) – covered by Rishi Jaluria ..... 65
Ping Identity (PING) – covered by Matt Hedberg ..... 68
PowerSchool (PWSC) – covered by Matt Hedberg ..... 69
PTC (PTC) – covered by Matt Hedberg ..... 70
Qualys (QLYS) – covered by Matt Hedberg ..... 71
Rapid7 (RPD) – covered by Matt Hedberg ..... 72
Smartsheet, Inc. (SMAR) – covered by Rishi Jaluria ..... 73
Salesforce, Inc. (CRM) – covered by Rishi Jaluria ..... 74
Samsara (IOT) – covered by Matt Hedberg ..... 75
ServiceNow (NOW) – covered by Matt Hedberg ..... 76
Snowflake Inc. (SNOW) – Not Covered ..... 77
SolarWinds (SWI) – covered by Matt Hedberg ..... 78
Splunk (SPLK) – covered by Matt Hedberg ..... 79
Sumo Logic (SUMO) – covered by Matt Hedberg ..... 80
TeamViewer AG (TMV) – covered by Sherri Malek ..... 81
Teradata (TDC) – covered by Matt Hedberg ..... 82
Twilio, Inc. (TWLO) – covered by Rishi Jaluria ..... 83
UiPath (PATH) – covered by Matt Hedberg ..... 84
Varonis (VRNS) – covered by Matt Hedberg ..... 85
Veeva Systems Inc. (VEEV) – covered by Rishi Jaluria ..... 86
VMware (VMW) – covered by Matt Hedberg ..... 87
Workday, Inc. (WDAY) – Not Covered ..... 88
Xometry (XMTR) – covered by Matt Hedberg ..... 90
Ziff Davis, Inc. (ZD) – covered by Rishi Jaluria ..... 91
Zendesk, Inc. (ZEN) – Not Covered ..... 92
ZoomInfo Technologies, Inc. (ZI) – covered by Rishi Jaluria ..... 93
Zoom Video Communications, Inc. (ZM) – covered by Rishi Jaluria ..... 94
Zscaler (ZS) – covered by Matt Hedberg ..... 95

**Corporate Access Recap ..... 96**

**Returns by Style ..... 97**

**Valuation ..... 98**

**Rule of 40 ..... 101**

**IPOs ..... 103**

**SPACs ..... 106**

**M&A ..... 107**

**Top Picks ..... 111**

**LinkedIn/Glassdoor Trackers ..... 113**

This report is intended solely for cbass@marketstreetpartners.com



# ARR, Billings, and RPO — What's the Best Leading Indicator?

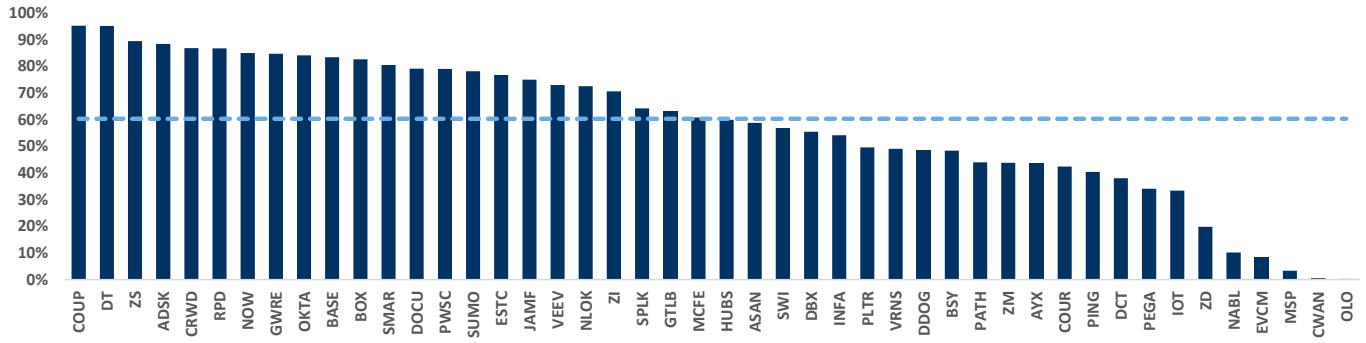
## Why Leading Indicators Matter

The primary reason leading indicators are crucial in a SaaS model is that revenue is ratable and, as such, quarterly revenue is largely a backward-looking metric, measuring business that has already been booked, as opposed to new business generated.

In order to quantify this, we look at two different metrics: 1) the percentage of revenue recognized from pre-existing deferred revenue; and 2) RPO coverage. Interestingly, both of these are new disclosures under ASC 606 and, arguably, the only positives coming out of ASC 606. In our view, since ASC 606 became effective, it has caused more confusion and made analyzing companies more difficult than before, but we suppose it's justified in securing jobs for accountants that should be done by automation software instead.

Exhibit 1 shows the percentage of subscription revenue (or total revenue) recognized in the most recent Q1 that was already in deferred revenue on the balance sheet at the beginning of the quarter. To us, this is the most significant indicator that revenue is a backward-looking metric and that deferred revenue is a meaningful metric. Across our extended SaaS coverage universe (non-consumption), the median is 63%. For companies where this percentage is significantly lower, billings becomes a less-relevant metric.

Exhibit 1 - Percent of Revenue Recognized from Prior Period Deferred Revenue (Most Recent Q1)



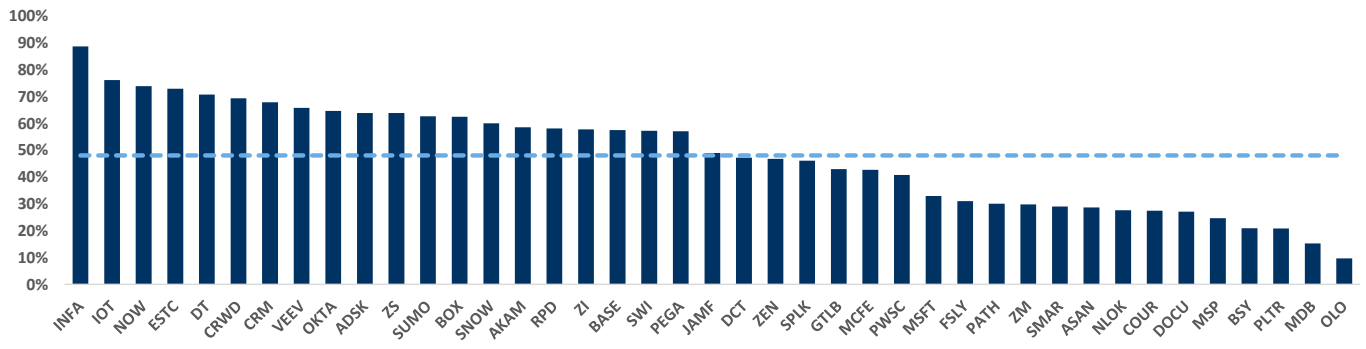

Source: Company reports, RBC Capital Markets

Exhibit 2 shows RPO coverage, a metric we follow closely, defined as current RPO, or cRPO, (NTM) divided by the following four quarters. We use historicals for this metric, since SaaS companies tend to guide conservatively. Interestingly, this coverage ratio can provide a glimpse into how conservative a company is being with its guidance and perhaps give us a sense as to the upside case to future numbers (as we discuss in the RPO section later in this note). We note we only include those companies that provide a NTM cRPO figure for the sake of comparability and those that don't exclude any contracts with initial duration of one years or less under the ASC 606 "practical expedient" (even when ASC 606 gets it right, it messes things up).

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 2 - RPO Coverage




Source: Company reports, RBC Capital Markets

The company-by-company analysis for 73 names within our extended coverage universe (AKAM, ANSS, ALTR, TEAM, AYX, ADSK, ASAN, BOX, BSY, CHKP, NET, COUP, BASE, COUR, CRWD, CWAN, DDOG, DOCU, DV, DBX, DT, DCT, ESTC, EVCM, FFIV, FSLY, FTNT, GTLB, GWRE, HUBS, INFA, JAMF, MDB, MSFT, NABL, NTCT, NEWR, NICE, NLOK, NTNX, OKTA, OLO, PD, PLTR, PANW, PEGA, PING, PWSC, PTC, QLYS, RPD, SMAR, CRM, IOT, NOW, SNOW, SWI, SPLK, SUMO, TMV, TDC, TWLO, PATH, VRNS, VEEV, VMW, WDAY, XMTR, ZD, ZEN, ZI, ZM, ZS) begins on page 14.

## The Death of Billings

Before the days of ASC 606, investors turned to calculated billings (revenue plus the change in deferred revenue) as the best leading indicator. We're not sure who the first clever investor, analyst, or CFO was to look at billings, but the hegemony of billings has effectively stood unchallenged for years.

We note there are a number of reasons to like billings:

- **It is a forward-looking indicator and generally attempts to measure new business generated in the quarter.** As we have seen historically, on a longer-term basis, a slowdown in billings foreshadows a slowdown in revenue growth.
- **As the name implies, calculated billings can be derived from GAAP** metrics (revenue from the income statement, deferred revenue from the balance sheet) and doesn't require any financial trickery or judgement calls from management.
- **It is a flexible metric.** In addition to standard calculated billings, investors can look at the change in deferred revenue from the cash flow statement (ostensibly to control for FX fluctuations) or look only at subscription revenue (higher-quality revenue) and current deferred revenue to calculate subscription or current billings (and attempt to control for duration). In addition, investors can look at TTM billings to control for volatility (we have seen companies like DOCU, PD, and Slack emphasize this as a metric).

However, there are a number of issues with billings that have cropped up over the years and, arguably, been increasingly exposed during COVID:

- **Billings is rather sensitive to duration.** While investors can try to control for this by looking at current billings, shortening duration is hard to control for. We note this came up a lot during the pandemic, as many customers (especially SMB customers) were opting for monthly or quarterly billings instead of traditional annual billings.

This report is intended solely for cbass@marketstreetpartners.com



- **Billings is effectively useless for consumption-based companies (e.g. TWLO, FSLY**). As we discuss later in this note, consumption-based companies likely need to figure out a better leading indicator.
- **Billings can force unnatural behavior.** We have seen examples of companies discounting to close deals early or discounting for longer-duration deals, so that a company can "make" its billings numbers. We have also seen companies sign longer-duration deals to "flatter" FCF margins (the simple fix for this is to calculate OCF using only the change in current deferred revenue, and exclude LT deferred revenue).
- **It can be a rather volatile metric.** Billings exhibits meaningful seasonality and changes in seasonality make the number more noisy. In addition, billings can be meaningfully impacted by early renewals or changes to co-terminus billings.

Ultimately, while we do pay attention to and continue to model billings as a metric, we believe the introduction of RPO under ASC 606 (along with investors now having ample history of the metric) and more companies introducing ARR have made billings less relevant over time.

## Remaining Performance Obligations

RPO or remaining performance obligations capture future obligations as a result of contractual relationships. This is effectively a measure of backlog and includes future obligations that have been billed (typically as deferred revenue) and the rest will be unbilled backlog, which isn't on the balance sheet (since there is nothing to "balance" it out).

Companies will typically disclose current RPO as well (often expressed as a percentage of total RPO). Most companies define performance obligations remaining over the next 12 months (NTM) as current, but others use measures like next 24 months (e.g. WDAY). We prefer companies that use NTM, as it's the most commonly accepted standard and allows for comparability between companies, and NTM seems like a fair benchmark to us.

We note that some companies had historically disclosed off-balance-sheet backlog on an annual basis, including BOX and NOW. If we take historical disclosures for off-balance-sheet backlog and add the total deferred revenue on the balance sheet that typically approximates what is now referred to as RPO. This is helpful knowledge since it allows us to analyze RPO trends for a longer time, as opposed to just the past four years.

### How we use RPO as a metric

The two great things about RPO are: 1) it is a GAAP metric; and 2) it is very flexible.

We can use RPO to gauge the health of the business in a number of different ways, including: 1) total RPO growth; 2) current RPO (cRPO) growth; 3) calculated bookings (similar to billings: revenue plus the change in RPO); 4) current calculated bookings (using cRPO instead of total RPO); and 5) RPO coverage (cRPO divided by NTM subscription revenue).

Exhibit 3 shows our case study for BOX, where we look at all the different potential leading indicators and discuss the trends in detail. What is clear is that the leading indicators did show a future slowdown in growth and also predicted the reacceleration we've seen recently (although we do often question why the metrics never seem to all point in the right direction in the past several quarters).

This report is intended solely for cbass@marketstreetpartners.com



## Exhibit 3 - Box Case Study

| | FY16A | FY17A | FY18A | FY19A | FY20A | FY21A | FY22A | FY23E |
|---|---|---|---|---|---|---|---|---|
| Revenue[1] | 302,704 | 398,605 | 506,142 | 608,386 | 696,264 | 770,770 | 874,332 | 993,104 |
| *YoY Growth (%)* | *40%* | *32%* | *27%* | *20%* | *14%* | *11%* | *13%* | *14%* |
| Billings[2] | 369,060 | 454,176 | 585,081 | 672,897 | 745,075 | 812,509 | 939,234 | |
| *YoY Growth (%)* | *50%* | *23%* | *29%* | *15%* | *11%* | *9%* | *16%* | |
| ST Billings[3] | 362,862 | 459,210 | 569,388 | 670,653 | 750,170 | 807,181 | 946,161 | |
| *YoY Growth (%)* | | *27%* | *24%* | *18%* | *12%* | *8%* | *17%* | |
| Ending Deferred Revenue[4] | 186,413 | 241,984 | 320,923 | 375,041 | 423,849 | 465,613 | 534,242 | |
| *YoY Growth (%)* | *55%* | *30%* | *33%* | *17%* | *13%* | *10%* | *15%* | |
| *Total Deferred/ST Deferred*[5] | *1.11* | *1.06* | *1.10* | *1.06* | *1.04* | *1.05* | *1.03* | |
| Ending ST Deferred Revenue[6] | 168,051 | 228,656 | 291,902 | 353,590 | 407,493 | 443,929 | 519,485 | |
| *YoY Growth (%)* | | *36%* | *28%* | *21%* | *15%* | *9%* | *17%* | |
| RPO[7] | 370,813 | 500,084 | 589,109 | 686,300 | 767,800 | 896,900 | 1,070,800 | |
| *YoY Growth (%)* | *46%* | *35%* | *18%* | *16%* | *12%* | *17%* | *19%* | |
| cRPO[8] | | | | 446,095 | 499,070 | 551,000 | 653,188 | |
| *YoY Growth (%)* | | | | | *12%* | *10%* | *19%* | |
| *Coverage (% of NTM Rev)*[9] | | | | *64%* | *65%* | *63%* | *66%* | |
| Bookings[10] | 419,460 | 527,876 | 595,167 | 705,577 | 777,764 | 899,870 | 1,048,232 | |
| *YoY Growth (%)* | | *26%* | *13%* | *19%* | *10%* | *16%* | *16%* | |
| Current Bookings[11] | | | | | 749,239 | 822,700 | 976,520 | |
| *YoY Growth (%)* | | | | | | *10%* | *19%* | |

| **Ratios**[12] | FY16A | FY17A | FY18A | FY19A | FY20A | FY21A | FY22A |
|---|---|---|---|---|---|---|---|
| Billings/FY+1 Revs | 0.93 | 0.90 | 0.96 | 0.97 | 0.97 | 0.93 | 0.95 |
| ST Billings/FY+1 Revs | 0.91 | 0.91 | 0.94 | 0.96 | 0.97 | 0.92 | 0.95 |
| RPO/FY+1 Revs | 0.93 | 0.99 | 0.97 | 0.99 | 1.00 | 1.03 | 1.08 |
| Bookings/FY+1 Revs | 1.05 | 1.04 | 0.98 | 1.01 | 1.01 | 1.03 | 1.06 |
| Current Bookings/FY+1 Revs | | | | | 0.97 | 0.94 | 0.98 |

Source: RBC Capital Markets, Company Reports
Footnotes discussed below

[1] BOX only reports a total revenue line, although it has noted that roughly 4% of revenue is services and the rest is subscription (although that includes some premium services, which we don't agree is software subscription revenue). As seen, BOX has dramatically decelerated from 40% growth in FY16 to 11% growth in FY21 (with 8% growth in F4Q21), before reaccelerating to 13% growth in FY22 and 14% in FY23E (based on guidance).

[2] Billings (revenue plus the change in total deferred revenue) has historically been the most important metric for BOX (management even used to be compensated on billings). While there have been a number of moving parts in the metric, including early renewals and duration, it has generally foreshadowed a slowdown in revenue, while the acceleration in FY22 provides some level of support for the acceleration in revenue growth baked into FY23 guidance.

[3] Short-term or current billings (revenue plus the change in short-term deferred revenue) is likely the better metric, as duration becomes less of a factor. Short-term billings has generally moved in a similar direction as billings and accelerated in FY22.

[4] We also look at growth rates in total deferred revenue as a leading indicator.

[5] Interestingly, the ratio of total deferred revenue to short-term deferred revenue has narrowed over the years, suggesting BOX is signing fewer long-term deals than before (and that effective durations have decreased).

[6] Short-term deferred revenue is also an important leading indicator, and we note the growth rates have differed between total deferred revenue and short-term deferred revenue.

This report is intended solely for cbass@marketstreetpartners.com



[7] RPO is a new metric disclosed under ASC 606, beginning in FY19. Prior to this, however, BOX did disclose off-balance-sheet backlog, and we use the sum of backlog and deferred revenue to estimate RPO for the years prior to ASC 606. RPO has generally been a good leading indicator and, in fact, FY22's acceleration showed up in RPO before billings.

[8] cRPO is also an important metric we track, although there is more limited history with the metric. We do note cRPO did decelerate in FY21, while total RPO accelerated, which is curious.

[9] cRPO coverage is an important metric, as discussed earlier in this whitepaper, and measures cRPO/NTM revenue. That number has been between 63% and 66% since BOX began disclosing the metric, which gives us pretty good visibility into future revenue. It is interesting that cRPO coverage was relatively low in FY21, suggesting FY22's acceleration included more new business than normal. We would also note if we apply that 63% coverage ratio to FY22's cRPO balance, it would imply BOX would do ~$1,036M in revenue in FY23 or 4% above the high end of guidance.

[10] Calculated bookings (revenue plus the change in total RPO) may be a better metric than billings, although this metric has shown to be a little bit more volatile, especially on a quarterly basis. For example, bookings growth was 39% in F2Q22 and 5% in F4Q22 (against a much easier comp).

[11] The same is the story with current bookings (revenue plus the change in cRPO), although the quarterly volatility is less pronounced.

[12] We also look at ratios for forward-looking metrics by looking at the metric divided by NTM revenue. While bookings tends to be a little more volatile, the rest of the metrics seem to show a pretty close correlation with future growth, supporting the use of these as leading indicators. We would make two further points.

## Advice for Company Management

We believe more companies should emphasize RPO and cRPO as leading metrics (assuming they are relevant metrics) and have two additional pieces of advice for management teams.

First, given software companies do tend to be acquisitive, we would like to see companies disclose RPO contribution from acquisitions, particularly large acquisitions (as CRM has done with Slack), since this allows investors to come up with an organic RPO number.

Second, we believe more companies should start guiding to RPO. We give credit to CRM, NOW, and WDAY as companies that guide to the metric. We believe this has helped shift focus away from billings.

## Companies with incomplete RPO disclosures

As much as we appreciate RPO as a disclosure, we note a number of companies offer an RPO figure that excludes certain contracts and, therefore, is significantly less useful. Typically, this move, known as the "practical expedient", is to exclude performance obligations related to contracts with 12 months or less in duration.

Companies with incomplete RPO disclosures include COUP, DOCU, GWRE, MDB, NICE, PD and PLTR. Note DCT/VEEV apply the practical expedient to exclude professional services RPO for contracts less than 12 months. WDAY excludes professional services RPO completely.

This report is intended solely for cbass@marketstreetpartners.com

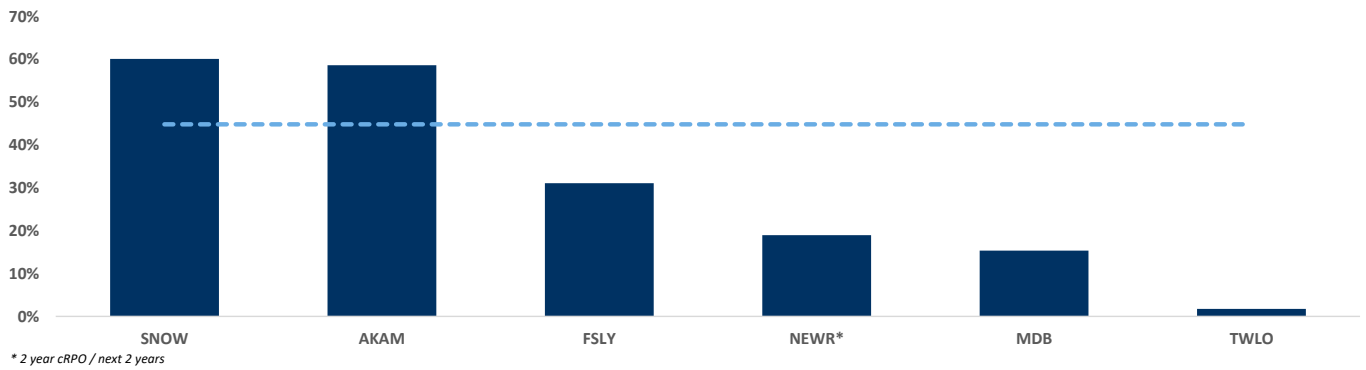


In our view, it would be better for companies to offer a full RPO metric in the interest of providing additional transparency to investors and, again, shifting attention away from billings. We would provide the example of Slack, which, as a public company, saw a notable deceleration in billings during the pandemic, then would spend massive amounts of time and attention explaining why billings was not a great metric and how a slowdown in billings wasn't a good predictor of a slowdown in revenue growth (even though this ultimately was the case). In our view, if Slack offered a full RPO metric, it would be less impacted by changes in payment terms during the pandemic, but the company made and stuck to the choice of adopting the practical expedient.

### Is RPO a useful metric for consumption companies?

While consumption-based companies do get certain levels of minimum commitment from customers, the low levels of coverage in RPO (or cRPO) suggest that RPO isn't a particularly useful metric for consumption companies, as seen in Exhibit 4.

Exhibit 4 - RPO Coverage – Consumption Companies




Source: Company reports, RBC Capital Markets

## Annual Recurring Revenue (ARR)

### Why ARR can be an important metric

Once upon a time, ARR was a metric almost exclusively reserved for companies undergoing a cloud transition (to normalize for the revenue declines typically seen early in a cloud transition) or utilized as a vanity metric by private companies to show scale.

Since then, however, we've seen more companies disclose ARR in three general buckets:

- Cloud transition stories, like PEGA (which discloses ACV), GWRE, or SPLK.
- On-premise, term license companies that use ARR to normalize for ASC 606, which forces upfront revenue recognition of the license component, including PATH and AYX.
- SaaS companies that disclose ARR as a better leading indicator than billings, including DBX.

We note that for SaaS companies that do not disclose ARR, investors generally estimate ARR by taking quarterly revenue times four.

Exhibits 5-7 show all the companies that provide ARR (or similar) disclosures on a regular (or semi-regular) basis, the definition of the metric, and the rationale for disclosure.

This report is intended solely for cbass@marketstreetpartners.com



## Exhibit 5 - ARR / ACV Metric Definitions: Part One

| Name | Metric Disclosed | Frequency | Definition | Primary Delivery Model | Reasons for Disclosures/Notes |
|---|---|---|---|---|---|
| **Ansys** | ACV | Quarterly | From 10-K: *"Annual Contract Value (ACV) is one of our key performance metrics and is useful to investors in assessing the strength and trajectory of our business. Given that revenue is more volatile due to the upfront revenue recognition of perpetual licenses and multi-year lease license sales, we provide ACV as a supplemental metric to help evaluate the annual performance of the business. Summed over the long term, ACV and revenue are equal. However, there will be years in which ACV growth lags revenue growth and other years in which ACV growth leads revenue growth. It is used by management in financial and operational decision-making and in setting sales targets used for compensation. ACV should be viewed independently of revenue and deferred revenue as ACV is a performance metric and is not intended to be combined with any of these items. There is no GAAP measure comparable to ACV. ACV is composed of the following: The annualized value of maintenance and lease contracts with start dates or anniversary dates during the period, plus the value of perpetual license contracts with start dates during the period, plus the annualized value of fixed-term services contracts with start dates or anniversary dates during the period, plus the value of work performed during the period on fixed-deliverable services contract."* | Mixed | Introduced metric as it normalizes for potential volatility in revenue due to the duration and timing of multiple year leases |
| **Bentley Systems** | ARR | Quarterly | From 10-K: *"ARR is defined as the sum of the annualized value of our portfolio of contracts that produce recurring revenues as of the last day of the reporting period, and the annualized value of the last three months of recognized revenues for our contractually recurring consumption-based software subscriptions with consumption measurement durations of less than one year. We believe that the last three months of recognized revenues, on an annualized basis, for our recurring software subscriptions with consumption measurement period durations of less than one year is a reasonable estimate of the annual revenues, given our consistently high retention rate and stability of usage under such subscriptions"* | Mixed | Management views as a significant KPI |
| **Couchbase** | ARR | Quarterly | From 10-K: *"We define ARR as of a given date as the annualized recurring revenue that we would contractually receive from our customers in the month ending 12 months following such date. Based on historical experience with customers, we assume all contracts will be automatically renewed at the same levels unless we receive notification of non-renewal and are no longer in negotiations prior to the measurement date. ARR excludes revenue from on-demand arrangements and services. ARR should be viewed independently of revenue, and does not represent our revenue under U.S. GAAP on an annualized basis, as it is an operating metric that can be impacted by contract start and end dates and renewal dates. ARR is not intended to be a replacement for forecasts of revenue. Although we seek to increase ARR as part of our strategy of targeting large enterprise customers, this metric may fluctuate from period to period based on our ability to acquire new customers and expand within our existing customers. We believe that our ARR is an important indicator of the growth and performance of our business."* | Subscription | Eliminates volatility due to the timing of project initiations |
| **Clearwater Analytics** | ARR | Quarterly | From 10-K: *"Annualized recurring revenue is calculated at the end of a period by dividing the recurring revenue in the last month of such period by the number of days in the month and multiplying by 365… Because a substantial majority of the assets on our platform have lower levels of volatility with respect to their market value, the growth in annualized recurring revenue is generally not attributable to the fluctuating market value of the assets on our platform. Rather, the growth in annualized recurring revenue is due to an increase in the number of clients using our offering as well as from onboarding more assets of our existing clients onto our platform."* | SaaS | Management views as best leading indicator, which makes sense given customers are billed monthly in arrears and pricing is based on a percentage of the average daily AUM on the platform |
| **CrowdStrike** | ARR | Quarterly | From 10-K: *"ARR is calculated as the annualized value of our customer subscription contracts as of the measurement date, assuming any contract that expires during the next 12 months is renewed on its existing terms. To the extent that we are negotiating a renewal with a customer after the expiration of the subscription, we continue to include that revenue in ARR if we are actively in discussion with such an organization for a new subscription or renewal, or until such organization notifies us that it is not renewing its subscription."* | SaaS | Management views as as a key metric |
| **Dropbox** | ARR | Quarterly | From 10-K: *"We primarily focus on total annual recurring revenue ("Total ARR") as the key indicator of the trajectory of our business performance. Total ARR represents the amount of revenue that we expect to recur annually, enables measurement of the progress of our business initiatives, and serves as an indicator of future growth. In addition, Total ARR is less subject to variations in short-term trends that may not appropriately reflect the health of our business, however the changes in ARR throughout the year could be subject to seasonality. Total ARR is a performance metric and should be viewed independently of revenue and deferred revenue, and is not intended to be a substitute for, or combined with, any of these items. Total ARR consists of contributions from all of our revenue streams, including subscriptions and add-ons. We calculate Total ARR as the number of users who have active paid licenses for access to our platform as of the end of the period, multiplied by their annualized subscription price to our platform. We include ARR related to acquired companies in our total ARR in the period of the acquisition. We adjust the exchange rates used to calculate Total ARR on an annual basis at the beginning of each fiscal year."* | SaaS | Management views as best leading indicator, rather than billings |
| **Duck Creek** | SaaS ARR | Quarterly | From 10-K: *SaaS Annual Recurring Revenue ("SaaS ARR"). We calculate SaaS ARR by annualizing the recurring subscription revenue recognized in the last month of the measurement period (the latest twelve-month period). Our SaaS ARR was $135.3 million, $95.6 million, and $51.7 million, for fiscal 2021, 2020, and 2019, respectively. Our calculation excludes one existing contract for a service no longer offered on a standalone basis by the Company. We believe SaaS ARR provides important information about our ability to acquire new subscription SaaS customers and to maintain and expand our relationship with existing subscription SaaS customers. SaaS ARR is included in a set of metrics that we calculate quarterly to review with management as well as periodically with members of our board of directors.* | SaaS (77% of non-services revenue was SaaS) | Transition to SaaS and one of the metrics used to determine executive bonus payouts |
| **Dynatrace** | ARR | Quarterly | From 10K: *"We define annual recurring revenue, or ARR, as the daily revenue of all subscription agreements that are actively generating revenue as of the last day of the reporting period multiplied by 365. We exclude from our calculation of ARR any revenues derived from month-to-month agreements and/or product usage overage billings, where customers are billed in arrears based on product usage."* | Mixed | Management believes ARR is the best metric because it removes the variability associated with billings and contracting |
| **Guidewire** | ARR<br>Cloud ARR<br>IS Cloud ARR<br>Fully-Ramped ARR | Quarterly<br>Annually<br>Annually<br>Annually | From 10-K: *We use ARR to quantify the annualized recurring value outlined in active customer contracts at the end of a reporting period. ARR includes the annualized recurring value of term licenses, subscription agreements, support contracts, and hosting agreements based on customer contracts, which may not be the same as the timing and amount of revenue recognized. All components of the licensing and other arrangements that are not expected to recur (primarily perpetual licenses and professional services) are excluded. In some arrangements with multiple performance obligations, a portion of recurring license and support or subscription contract value is allocated to services revenue for revenue recognition purposes, but does not get allocated for purposes of calculating ARR. This revenue allocation only impacts the initial term of the contract. This means that as we increase arrangements with multiple performance obligations that include services at discounted rates, more of the total contract value will be recognized as services revenue, but our reported ARR amount will not be impacted. In fiscal year 2021, the recurring license and support or subscription contract value recognized as services revenue was $5.5 million.*<br><br>*If a customer contract contains invoicing amounts that increase over the contract term, then ARR reflects the annualized invoicing amount outlined in the contract for the current reporting period. For example, given a contract with annual invoicing of $1.0 million at the beginning of year one, $2.0 million at the beginning of year two, and $3.0 million at the beginning of year three, and the reporting period is subsequent to year two invoicing and prior to year three invoicing, the reported ARR for that contract would be $2.0 million.* | On-premises (30% of non-services revenue was cloud SaaS in FY21 and 40% of ARR was cloud SaaS) | Transition to cloud SaaS; also, management views total ARR as the best overall leading indicator, while InsuranceSuite cloud ARR is seen as the best indicator for cloud transition progress and cloud ARR / fully-ramped cloud ARR provide insight into the overall health of the cloud business (cloud ARR includes InsuranceNow and cloud add-ons, and fully-ramped ARR normalizes for large multi-year cloud contracts where ARR ramps throughout the life of the contract) |

Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Exhibit 6 - ARR / ACV Metric Definitions: Part Two

| Name | Metric Disclosed | Frequency | Definition | Primary Delivery Model | Reasons for Disclosures/Notes |
|---|---|---|---|---|---|
| **HubSpot** | ARR | Bi-anually | Analyst Day slides definition of ARR: *"We define "ARR" as the annual value of our customer subscription contracts as of the specified point in time excluding any commissions owed to our partners. For each Hub, this is the sum of customer ARR for the Starter, Basic, Professional and Enterprise subscriptions, plus applicable Contacts (Marketing Only) or Add-Ons (e.g. Reporting or Ads). For multi-product customers, their ARR would be distributed across based on the value of each SKU/Hub for which they pay. ARR can differ from Revenue due to several factors. ARR is converted into USD at fixed rates that are held consistent over time and may vary from those used for Revenue or Deferred Revenue. ARR would exclude any impact for Bad Debt and Partner Commissions (as noted above) and would also differ from Revenue due to timing of revenue recognition."* | SaaS | One of the metrics used to determine executive bonus payouts |
| **Informatica** | ARR<br><br>Cloud ARR<br><br>Subscription ARR | Quarterly | From 10-K: *"Annual Recurring Revenue (ARR) represents the expected annual billing amounts from all active maintenance and subscription agreements. ARR is calculated based on the contract Monthly Recurring Revenue (MRR) multiplied by 12. MRR is calculated based on the accounting adjusted total contract value divided by the number of months of the agreement based on the start and end dates of each contracted line item. The aggregate ARR calculated at the end of each reported period represents the value of all contracts that are active as of the end of the period, including those contracts that have expired but are still under negotiation for renewal."* | Mixed | Management considers ARR as an important metric since it tracks the recurring cash flow characteristic of the business |
| **Jamf** | ARR | Quarterly | From 10-K: *"Annual Recurring Revenue ("ARR") represents the annualized value of all subscription and support and maintenance contracts as of the end of the period."* | Mixed (86% of CY/21 revenue from SaaS subscription and support and maintenance, 8% from on-premise subscription and 6% from professional services and perpetual licenses) | Maangement notes that ARR is an important metric because it mitigates fluctuations from seasonality, sales mix for subscriptions of SaaS and licenses, and contract terms |
| **Nutanix** | ARR<br><br>Run-rate ACV | Quarterly<br><br>Quarterly | From 10-Q: *"Annual Contract Value, or ACV, is defined as the total annualized value of a contract, excluding amounts related to professional services and hardware. The total annualized value for a contract is calculated by dividing the total value of the contract by the number of years in the term of such contract, using, where applicable, an assumed term of five years for contracts that do not have a specified term.*<br><br>*Annual Recurring Revenue, or ARR, for any given period, is defined as the sum of ACV for all non life-of-device contracts in effect as of the end of a specific period. For the purposes of this calculation, we assume that the contract term begins on the date a contract is booked, unless the terms of such contract prevent us from fulfilling our obligations until a later period, and irrespective of the periods in which we would recognize revenue for such contract.*<br><br>*Run-rate ACV, at the end of any period, is the sum of ACV for all contracts that are in effect as of the end of that period. For the purposes of this calculation, the Company assumes that the contract term begins on the date a contract is booked, irrespective of the periods in which the Company would recognize revenue for such contract."* | Mixed | Management views ARR as the best leading indicator, reflecting pure subscription while ACV includes life-of-device (LOD) transactions in addition to subscription |
| **PagerDuty** | ARR | Annual | From 10-K: *"We define ARR as the annualized recurring value of all active contracts at the end of a reporting period."* | SaaS | Management views ARR as a valuable long-term metric vs. billings which is impacted by co-terming and duration. We believe ARR should be the primary growth metric for the business |
| **Palo Alto Networks** | NGS ARR | Quarterly | From F2Q22 earnings deck: *"Next-Gen Security ARR is annualized allocated revenue of all active contracts as of the final day of the reporting period for Prisma and Cortex offerings inclusive of the VM-Series and related services, and certain cloud-delivered security services."* | Mixed | Management views NGS ARR as the primary metric, which is more indicative of the underlying drivers of the business than billings |
| **Pegasystems** | ACV | Quarterly | From 10-K: *ACV represents the annualized value of our active contracts as of the measurement date. The contract's total value is divided by its duration in years to calculate ACV for term license and Pega Cloud contracts. Maintenance revenue for the quarter then ended is multiplied by four to calculate ACV for maintenance. ACV is a performance measure that we believe provides useful information to our management and investors, particularly during our subscription transition.* | Mixed (36% of ACV was cloud in CY21, 33% was maintenance, and 31% was term) | Transition to SaaS and normalizes for term licenses under ASC 606. |
| **Ping Identity** | ARR | Quarterly | From 10-K: *"ARR represents the annualized value of all subscription contracts as of the end of the period. ARR mitigatesfluctuations due to seasonality, contract term and the sales mix of subscriptions for term-based licenses and SaaS.ARR only includes the annualized value of subscription contracts. ARR does not have any standardized meaning andis therefore unlikely to be comparable to similarly titled measures presented by other companies. ARR should be viewed independently of revenue and deferred revenue and is not intended to be combined with or toreplace either of those items. ARR is not a forecast and the active contracts at the end of a reporting period used in calculating ARR may or may not be extended or renewed by our customers."* | Mixed (SaaS ARR is over 25% of total ARR) | Management views as best leading indicator as the company transitions to SaaS based model |
| **PowerSchool Holdings** | ARR | Quarterly | From 10-K: *"ARR represents the annualized value of all recurring contracts as of the end of the period. ARR mitigates fluctuations due to seasonality, contract term, one-time discounts given to help customers meet their budgetary and cash flow needs and the sales mix for recurring and non-recurring revenue. ARR does not have any standardized meaning and is therefore unlikely to be comparable to similarly titled measures presented by other companies. ARR should be viewed independently of revenue and deferred revenue and is not intended to be combined with or to replace either of those items. ARR is not a forecast, and the active contracts at the end of a reporting period used in calculating ARR may or may not be extended or renewed by our customers."* | Mixed | ARR helps to smooth out the cyclicality of the business given their end market is the education sector |
| **PTC** | ARR | Quarterly | From 10-K: *"ARR (Annual Run Rate) represents the annualized value of our portfolio of active subscription software, cloud, SaaS, and support contracts as of the end ofthe reporting period. ARR includes orders placed under our Strategic Alliance Agreement with Rockwell Automation, including orders placed to satisfy contractualminimum commitments. We believe ARR is a valuable operating metric to measure the health of a subscription business because it captures expected subscription and support cashgeneration from customers. Because this measure represents the annualized value of customer contracts as of a point in time, it does not represent revenue for anyparticular period or remaining revenue that will be recognized in future period."* | Mixed, focused on transitioning to SaaS | Management noted this is the best leading indicator of PTC's top-line performance |
| **Rapid7** | ARR | Quarterly | From 10-K: *"Annualized Recurring Revenue (ARR) is defined as the annual value of all recurring revenue related to contracts in place at the end of the quarter. ARR should be viewed independently of revenue and deferred revenue as ARR is an operating metric and is not intended to be combined with or replace these items. ARR is not a forecast of future revenue, which can be impacted by contract start and end dates and renewal rates and does not include revenue reported as perpetual license or professional services revenue in our consolidated statement of operations. We use ARR and believe it is useful to investors as a measure of the overall success of our business."* | Mixed (recurring revenue was 92% of the total revenue in CY/21) | Management views as best leading indicator, and is what they guide to. |
| **Samsara** | ARR | Quarterly | From 10-K: *"We believe that ARR is a key indicator of the trajectory of our business performance, enables measurement of the progress of our business initiatives, and serves as an indicator of future growth. We define ARR as the annualized value of subscription contracts that have commenced revenue recognition as of the measurement date. ARR highlights trends that may be less visible from the face of our financial statements due to ratable revenue recognition. ARR does not have a standardized meaning and is not necessarily comparable to similarly titled measures presented by other companies. ARR should be viewed independently of revenue and is not intended to be combined with or to replace it. ARR is not a forecast and the active contracts at the date used in calculating ARR may or may not be extended or renewed. Our ARR has grown in each of the past three fiscal years, reflecting growth in new customers as well as expanded sales to existing customers."* | SaaS | Management views as best leading indicator, and is what they guide to |

Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Exhibit 7 - ARR / ACV Metric Definitions: Part Three

| Name | Metric Disclosed | Frequency | Definition | Primary Delivery Model | Reasons for Disclosures/Notes |
|---|---|---|---|---|---|
| **SolarWinds** | ARR<br><br>Subscription ARR | Quarterly | From 10-K: *"We use Subscription Annual Recurring Revenue, or Subscription ARR, and Total Annual Recurring Revenue, or Total ARR, to better understand and assess theperformance of our business, as our mix of revenue generated from recurring revenue has increased in recent years. Subscription ARR and Total ARR each provides anormalized view of customer retention, renewal and expansion, as well as growth from new customers. Subscription ARR and Total ARR should each be viewedindependently of revenue and deferred revenue, and are not intended to be combined with or to replace either of those items."* | Mixed | Management views as best leading indicator as they increasingly move to subscription |
| **Splunk** | ARR<br><br>Cloud ARR | Quarterly | From 10-K: *"We use cloud annual recurring revenue ("Cloud ARR") and total annual recurring revenue ("Total ARR") to identify the annual recurring value of customer contracts at the end of a reporting period and to monitor the growth of our recurring business as we continue to shift to a cloud services delivery model. Cloud ARR represents the annualized revenue run-rate of active cloud services contracts at the end of a reporting period. Total ARR represents the annualized revenue run-rate of active cloud services, term license and maintenance contracts* | Mixed | A significant foreward looking KPI to help investors monitor the pace of the transition towards subscription and SaaS and normalize for the near-term impact on revenue |
| **Sumo Logic** | ARR | Annual | From 10-K: *"We define ARR as the annualized recurring revenue run-rate from all customers that are under contract with us at the end of the period or with which we are negotiating a renewal contract. Given our historical experience of customer renewals, if we are in active discussions for a renewal, we continue to include customers with expired contracts in our ARR until the customer either renews its contract or negotiations terminate without renewal. For certain customers whose revenue may fluctuate from month to month based upon their specific contractual arrangements, we calculate ARR using the annualized monthly recurring revenue, or MRR, run-rate (MRR multiplied by 12). This enables us to calculate our anticipated recurring revenue for all customers based on our packaging and licensing models, which we believe provides a more accurate view of our anticipated recurring revenue."* | SaaS | Management views as a significant KPI to help normalize for contract duration and timing |
| **Teradata** | ARR<br><br>Public Cloud ARR | Quarterly | From 10-K: *"Annual Recurring Revenue ("ARR") - annual value at a point in time of all recurring contracts, including subscription, cloud, software upgrade rights, and maintenance. ARR does not include managed services and third-party software.*<br>*Public Cloud ARR (included within total ARR) - annual value at a point in time of all contracts related to public cloud implementations of Teradata Vantage and does not include ARR related to private or managed cloud implementations."* | Mixed | Help investors to normalize during the transition towards greater subscription and recurring revenue as well as monitor the growing portion of ARR made up of public cloud and subscription as opposed to maintenance |
| **UiPath** | ARR | Quarterly | From 10-K: *"We define ARR as annualized invoiced amounts per solution SKU from subscription licenses and maintenance obligations assuming no increases or reductions in their subscriptions. ARR does not include the costs we may incur to obtain such subscription licenses or provide such maintenance, and does not reflect any actual or anticipated reductions in invoiced value due to contract non-renewals or service cancellations other than for specific bad debt or disputed amount."* | Mixed | Management views ARR as a key metric to show the ability to acquire, maintain and expand relationships with subscription customers and as a leading indicator for topline growth. Additionally, given lumpy revenue recognition under ASC 606 due to different deployment models, management runs and manages the business on ARR |
| **Varonis** | ARR | Quarterly | From 10-K: *"Annual recurring revenues is a key performance indicator defined as the annualized value of active term-based subscription license contracts and maintenance contracts related to perpetual licenses in effect at the end of that period. Subscription license contracts and maintenance for perpetual license contracts are annualized by dividing the total contract value by the number of days in the term and multiplying the result by 365."* | Subscription | Introduce to help investors monitor the transition toward subscription revenue as a primary topline metric for the near-term headwinds towards revenue. Now viewed as a KPI for topline growth. |
| **Vmware** | ARR<br><br>Subscription and SaaS ARR | Quarterly | From 10-K: *"Annual recurring revenue ("ARR") represents the annualized value of our committed customer subscription and SaaS contracts as of the end of the reporting period, assuming any contract that expires during the next 12 months is renewed on its existing terms, except that, for consumption-based subscription and SaaS offerings, ARR represents the annualized quarterly revenue based on revenue recognized for the current reporting period. ARR is an operating measure we use to assess the strength of our subscription and SaaS offerings. ARR is a performance metric and should be viewed independently of, and not as a substitute for or combined with, revenue and unearned revenue."* | Mixed | Key metric for the company's growing subscription and SaaS revenue basis and helping investors with compares to previous years where license made up a greater percentage of revenue. |

Source: Company reports, RBC Capital Markets

## Potential limitations of ARR

As much as we like companies disclosing ARR, we acknowledge a few limitations:

**Differing definitions.** As seen above, every company defines ARR differently and there is no one consistent definition, which makes comparability a little difficult. We encourage investors to look critically at the definition of ARR and push back on companies that use misleading definitions of ARR. We have found a few ways to "test" ARR definitions, including comparing it to TTM or annualized billings or comparing it to calculated ARR (for SaaS companies).

**ARR can be misleading.** If ARR is telling a substantially different story than revenue, then it may signal issues with how the metric is calculated. We would point to the example of DOMO, which disclosed $100MM in ARR as a private company back in early 2015, but took until 2018 to reach $100MM in TTM GAAP revenue. This is because DOMO's definition of ARR was an annualized billings run rate, which is very far off from GAAP revenue.

**It is a non-GAAP metric.** Related to the prior two points, ARR is ultimately a non-GAAP metric and, as such, companies can change their method of calculating ARR or exercise judgment in what the metric captures.

**Many companies don't guide to ARR.** Every public company CFO knows managing expectations is critical, but for some reason, many companies don't guide to ARR. We note there are a few companies that do, such as PATH, AYX, and PEGA, but we would like to see more companies guide to ARR.

This report is intended solely for cbass@marketstreetpartners.com



**We note a number of companies in our coverage universe would benefit from ARR disclosures:**

- **ZM:** Given the ongoing churn Zoom is seeing with prosumer customers, we believe an ARR metric would give investors a better feel for the magnitude of the churn, as well as potentially highlight the strength in direct customers.
- **NICE:** NICE is undergoing a cloud transition and given that CXone bills in arrears (and has a consumption-based element to it), we believe an ARR figure for the business as a whole (that also incorporates maintenance and term license) would be beneficial for investors and serve as an important leading metric.
- **MDB:** With Atlas becoming a bigger portion of the business and Enterprise Advanced having somewhat lumpy revenue recognition due to ASC 606, an ARR disclosure could be helpful as a forward-looking indicator and move attention away from billings.
- **FSLY/TWLO/NEWR/ESTC:** As we discuss in the following section, ARR or ACV is a potential disclosure for consumption-based companies to explore as a leading indicator.

## What Do We Do With Consumption Companies?

The one unanswered question from this analysis is what to do with consumption companies. As we have shown, billings and RPO tend to be meaningfully less useful for consumption companies, given these companies bill in arrears. So what do we do with consumption companies like TWLO, FSLY, SNOW, NICE (the CCaaS portion), NEWR, and MDB (increasingly with Atlas)?

For now, we look at NRR as the best leading indicator for consumption companies. As we outlined in our NRR whitepaper (see here), improvement or deterioration in NRR generally indicates a coming acceleration or slowdown for consumption companies, especially since the metric is more volatile and shows usage patterns. We find this to be the case especially with TWLO, whose NRR metric is a quarterly, not a TTM, metric (which we applaud).

We do wonder if ARR might be a good disclosure for consumption companies, but see a number of puts and takes. On one hand, it can help normalize for volatility in usage trends and, in theory, extrapolating from usage trends at the end of a period may be a good forward-looking indicator. On the other hand, seasonality may make the metric a little tricky, especially for companies with strong Q4s or holiday seasons (e.g. FSLY, TWLO) – having said that, a Y/Y growth rate would normalize for seasonality, and billings itself is prone to seasonality as well.

This report is intended solely for cbass@marketstreetpartners.com



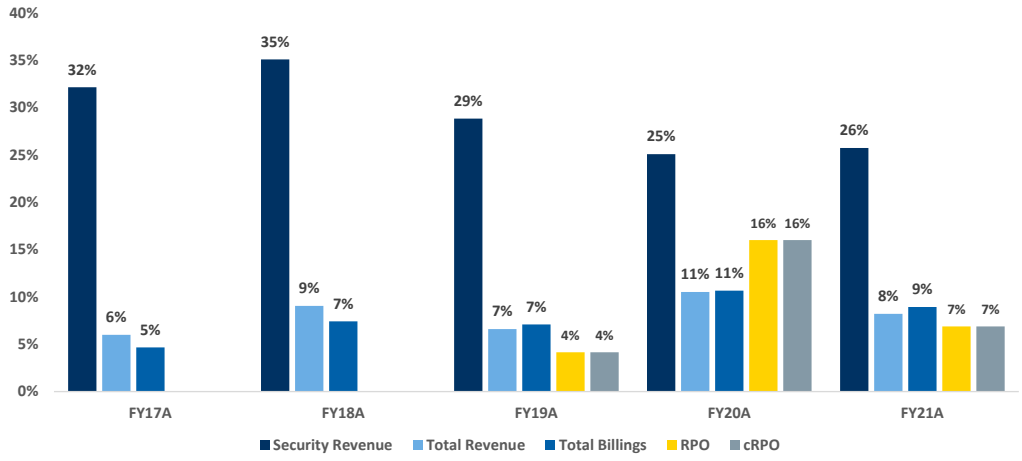
# Company-Specific Analysis

## Akamai Technologies, Inc. (AKAM) – covered by Rishi Jaluria

**Leading metric practice/relevance:** Akamai is a primarily consumption-based business, with 61% of revenue coming from its CDN/Edge segment. We do not view leading metrics as useful for Akamai's business but do expect this to change over time as security (primarily recurring revenue) continues to outpace total revenue growth and grow as a percent of total.

Exhibit 8 shows Y/Y growth of AKAM's security (recurring) revenue, total revenue, billings, RPO, and cRPO.

Exhibit 8 - AKAM Leading Metrics Y/Y Growth Rates, 2017A-2022A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## ANSYS (ANSS) – covered by Matt Hedberg

**Key metric(s):** ACV

**Leading metric practice/relevance:** The company introduced annual contract value (ACV) as its primary growth metric in 2017 as it normalizes for potential volatility in revenue due to the duration and timing of multiple-year leases. ACV from recurring sources accounted for 81% of total ACV in CY/21. We expect ACV to grow faster than revenue as the business model shift to subscription lease license continues. Deferred revenue and cRPO do not contribute a substantial amount to revenue in any given year with cRPO coverage trends at ~32% over the past several years. The company guides to ACV; cRPO and RPO are disclosed in 10-Q/Ks.

**Relevant commentary:** (CFO Nicole Anasenes on the Q4/21 earnings call) *"As a reminder, annual ACV is the best metric to observe the momentum in our business."*

(Former CFO Maria Shields at the 2019 Investor Day) *"The third element of our business model is a new key financial metric that we introduced at Investor Day back in 2017, ACV or annual contract value. And you can see, since that time, we've had strong performance in ACV with 17% constant currency growth last year. While some of our software peers have shifted to 100% subscription licensing model, as you heard Rick speak to earlier, our model will continue to offer our customers flexibility in both licensing and delivery options, so that they are aligned with the business needs of our customers. And this has enabled us to still have strong recurring sources with 77% of our ACV of a recurring nature last year."*

(Former CFO Maria Shields at the 2019 Investor Day) *"Now, you may be asking, why have we pivoted to an ACV target rather than a revenue target? Since our transition to ASC 606, we've recognized the potential volatility in revenue that arises from the timing and duration of multiyear leases. And as such, we thought it would be better to pick a target metric that is more reflective of the real time operating performance or our business."*

ACV 10-K definition and explanation as a key metric: "*Annual Contract Value (ACV) is one of our key performance metrics and is useful to investors in assessing the strength and trajectory of our business. Given that revenue is more volatile due to the upfront revenue recognition of perpetual licenses and multi-year lease license sales, we provide ACV as a supplemental metric to help evaluate the annual performance of the business. Summed over the long term, ACV and revenue are equal. However, there will be years in which ACV growth lags revenue growth and other years in which ACV growth leads revenue growth. It is used by management in financial and operational decision-making and in setting sales targets used for compensation. ACV should be viewed independently of revenue and deferred revenue as ACV is a performance metric and is not intended to be combined with any of these items. There is no GAAP measure comparable to ACV. ACV is composed of the following: The annualized value of maintenance and lease contracts with start dates or anniversary dates during the period, plus the value of perpetual license contracts with start dates during the period, plus the annualized value of fixed-term services contracts with start dates or anniversary dates during the period, plus the value of work performed during the period on fixed-deliverable services contracts.*"

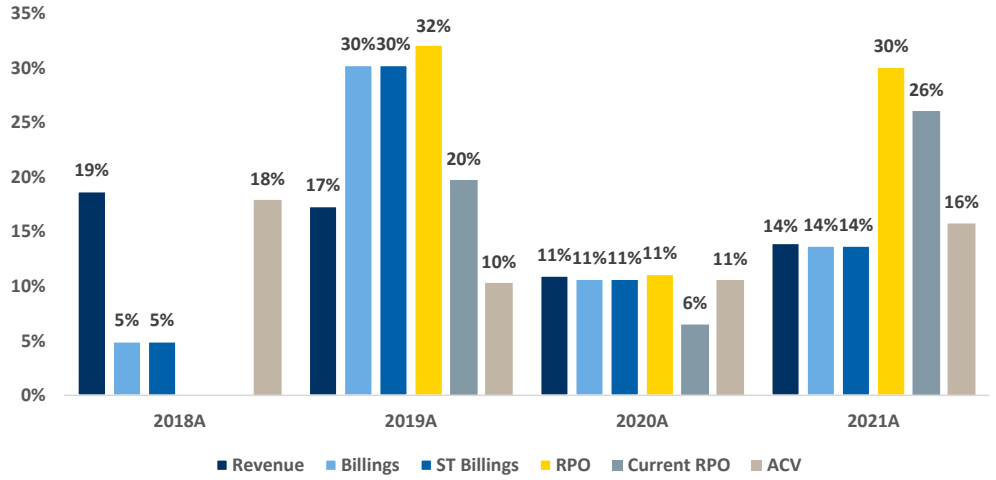
Remaining Performance Obligations 10-K definition and explanation as a key metric: "*Revenue allocated to remaining performance obligations represents contracted revenue that has not yet been recognized, which includes both deferred revenue and backlog. Our backlog represents installment billings for periods beyond the current quarterly billing cycle*."

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 9 shows Y/Y growth of ANSS's key leading metrics since CY/18A.

Exhibit 9 - ANSS Leading Metric Y/Y Growth Rates CY18-21A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Altair Engineering (ALTR) – covered by Matt Hedberg

**Key metric(s):** N/A

**Leading metric practice/relevance:** 90%+ of software revenue is recurring in nature. TTM growth trends can normalize the impacts around timing and seasonality of quarterly trends. Deferred revenue and cRPO do not contribute a substantial amount to revenue in any given year with cRPO coverage trends at ~20% over the last several years. The company does not guide to billings or RPO and does not disclose ARR. cRPO and RPO are disclosed in 10-Q/Ks. With 90%+ of software revenue recurring in nature, we think an additional disclosure around ARR could make sense.

**Relevant commentary:** (CFO Matt Brown at 2021 Investor Day) *"…90%-plus of our Software revenue is recurring in nature."*

(CFO Matt Brown on the Q1/21 earnings call) *"It's worth reminding you of the seasonal nature of our business, where our first and fourth quarters have higher Software billings and revenue, and we expect that pattern to continue."*

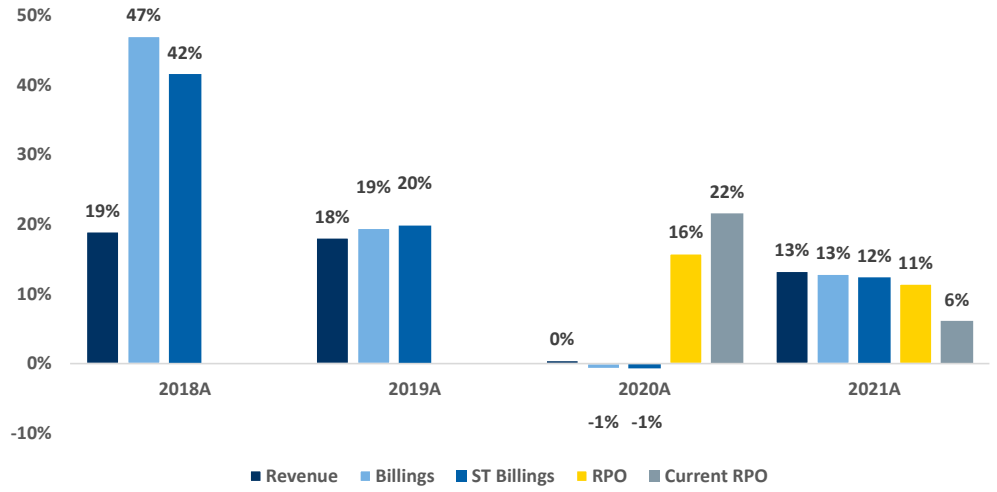
(CFO Matt Brown on the Q4/20 earnings call) *"We tend to view billings over longer time periods due to the impact that variations in timing of renewals, expansions, and new customer arrangements can have quarter to quarter."*

Billings 10-K definition and explanation as a key metric: "*Billings consists of our total revenue plus the change in our deferred revenue, excluding deferred revenue from acquisitions during the period. Given that we generally bill our customers at the time of sale but typically recognize a portion of the related revenue ratably over time, management believes that Billings is a meaningful way to measure and monitor our ability to provide our business with the working capital generated by upfront payments from our customers.*"

Remaining Performance Obligations 10-K definition and explanation as a key metric: "*Revenue allocated to remaining performance obligations represents contracted revenue that has not yet been recognized, which includes deferred revenue and amounts that will be invoiced and recognized as revenue in future periods.*"

Exhibit 10 - ALTR Leading Metric Y/Y Growth Rates CY18-21A

Exhibit 10 shows Y/Y growth of ALTR's key leading metrics since CY/18A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



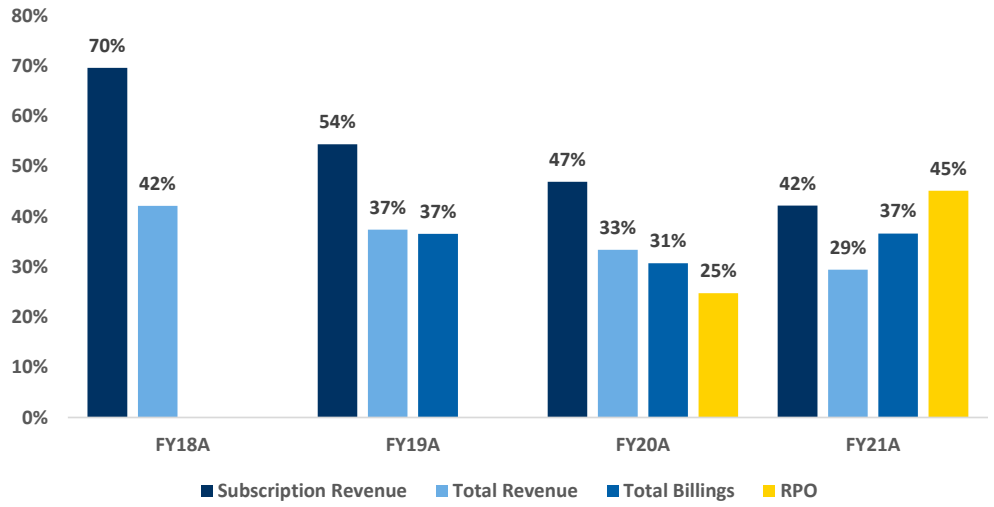
## Atlassian Corp. Plc (TEAM) – Not Covered

**Key metric(s):** Billings

**Leading metric practice/relevance:** Atlassian encourages investors to focus on revenue instead of billings, especially given the number of monthly subscriptions and the ongoing cloud transition. We believe investors do still focus on billings, although the metric has become slightly less useful with the cloud transition.

**Relevant commentary:** (CFO James Beer on F4Q20 earnings call) *"We do continue to believe that revenue growth is the best way for investors to think about our business. When you consider that now around 90% of our total revenue is recurring in nature, we're very comfortable that revenue is a strong reflection of the underlying demand for our products and services. So one thing I'll certainly note is that the cloud business has a blend of subscription terms. So about three quarters of our customers take monthly subscriptions as compared to our data center and server businesses being typically annual-type terms. We note in the shareholder letter to illustrate this point that at the end of the fiscal year, about 25% of the deferred revenue was cloud-based, while a little less than 50% of the revenue was cloud-driven. So that's the only thing I would add. But we still very much focus on revenue growth as a very fair measure of our progress."*

Exhibit 11 - TEAM Leading Metrics Y/Y Growth Rates, FY18A-FY22A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Alteryx Inc. (AYX) – Not Covered

**Key metric(s):** ARR, Billings

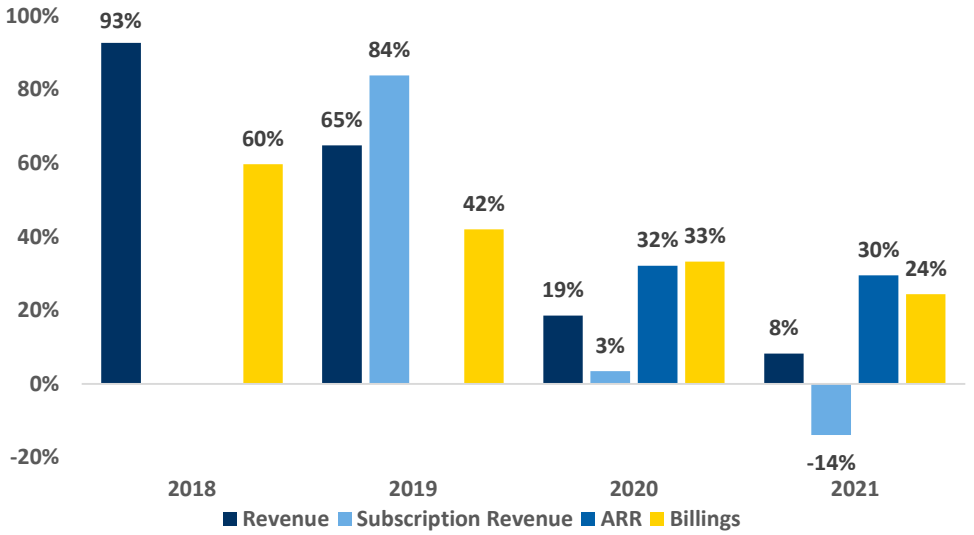
**Leading metric practice/relevance:** Alteryx began fully reporting and guiding to ARR in 3Q20 and views it as the cleanest metric for gauging business performance. Although the company is a subscription model, it's technology is primarily on-premise based, which has interesting dynamics for revenue recognition under ASC 606 (~40% of revenue recognized upfront), meaning revenues are highly variable from deal signings, renewals, and changes in contract duration (all dynamics that played out materially during the pandemic). We view ARR as the best leading metric for Alteryx and like that the company provides quarterly guidance for investors. Billings has become less relevant for Alteryx under the shift to ASC-606 as the license component of subscription deals is no longer recognized ratably (i.e. does not hit deferred revenue). That said, we believe investors do continue to track billings with the caveat that Alteryx's accounting can make the metric particularly noisy on a quarter-to-quarter basis.

**Relevant commentary:** (CFO Kevin Rubin at 2022 Investor Conference) **–** "*Yeah, so both the designer and server products today, which make-up the lion's share of our ARR and revenue, are on-premise deployed technologies. So although it's subscription and it's always been a term license, we give that license to our customers, they download it, they install it in their environments. The server product is often actually installed in cloud services, but it's ultimately managed by our customers. So it's still treated as if it's an on-premise technology, so we will recognize as much as – up until 12/31 of last year we recognized up to 40% of the value of the contracts underlying those products upfront and the remainder ratably over time.*"

(10-K Key Metrics commentary on billings): "*ARR represents the annualized recurring value of all active subscription contracts at the end of a reporting period and excludes the value of non-recurring revenue streams, such as certain professional services. Both multi-year contracts and contracts with terms less than one year are annualized by dividing the total committed contract value by the number of months in the subscription term and then multiplying by twelve.*"

Exhibit 12 - AYX Leading Metrics Y/Y Growth Rates, 2018A-2021A

Exhibit 12 shows Y/Y growth of AYX's key leading metrics since 2018.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Autodesk (ADSK) – covered by Matt Hedberg

**Key metric(s):** cRPO, RPO, Billings

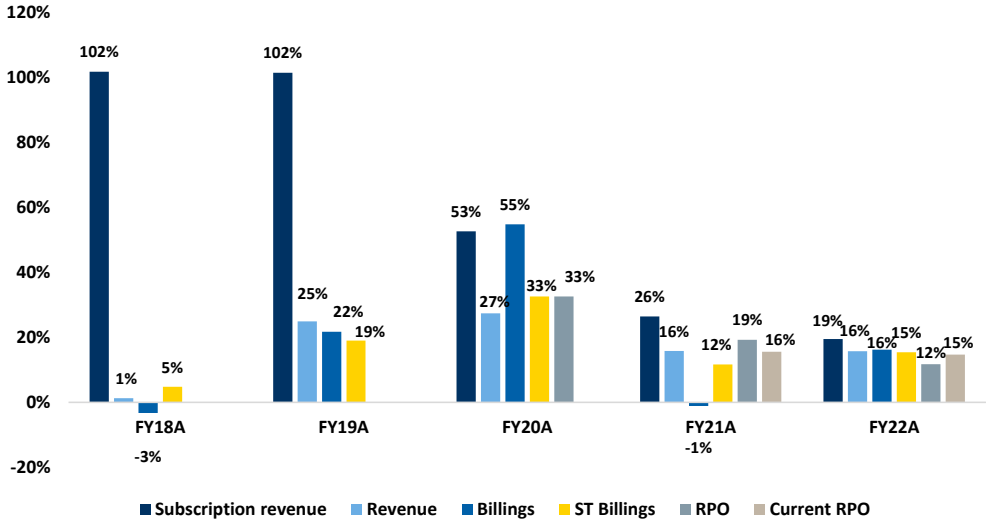
**Leading metric practice/relevance:** With 95% of revenue generated from subscriptions, we believe cRPO and RPO are good leading indicators for the overall health of the business. Looking at cRPO coverage trends, it has remained consistent at ~62% of FY+1 revenue for the past three years, which points to the stability of the metric, in our view. The company moved away from disclosing ARR starting in FY/21 following the subscription transition and shifted the focus to revenue, billings, cRPO and RPO. In the absence of ARR, investors closely watch billings, with management providing annual guidance for the metric. To that point, we note the ongoing transition of multi-year paid-upfront EBA customers to annual billings (a majority are already annual) creates a more ratable revenue and FCF model longer term. It does introduce some level of variability/compression in the near-term FCF. While management does not guide to them, we believe cRPO and RPO are cleaner metrics than billings, though still impacted by timing and volume of EBA contracts. Management expects growth rates for cRPO and revenue to converge over time. We also note less than 60% of revenue in any given year comes from deferred revenue.

**Relevant commentary:** (CFO Debbie Clifford at 2021 Investor Day) *"RPO is the sum of total short-term, long-term and deferred revenue. RPO is a strong leading indicator of our future revenue potential and has grown consistently in double digits over the last five years even during the height of the pandemic."*

(Former CFO Scott Herren at 2020 Investor Day) *"As we evolve from a company in transition to a growth SaaS business, the key metrics we focus on are changing. Our prior focus on metrics like ARR, subs, ARPS, M2S and spend will be deemphasized. These metrics served us well during the transition but have less relevancy today. Instead, we will increase focus on metrics like revenue, net revenue retention rate or NR3, RPO and current RPO, free cash flow and operating margin, which are more relevant indicators of our performance in this new growth phase of our business."*

Exhibit 13 - ADSK Leading Metric Y/Y Growth Rates FY18-22A

Exhibit 13 shows Y/Y growth of ADSK's key leading metrics since FY18.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Asana, Inc. (ASAN) – covered by Rishi Jaluria

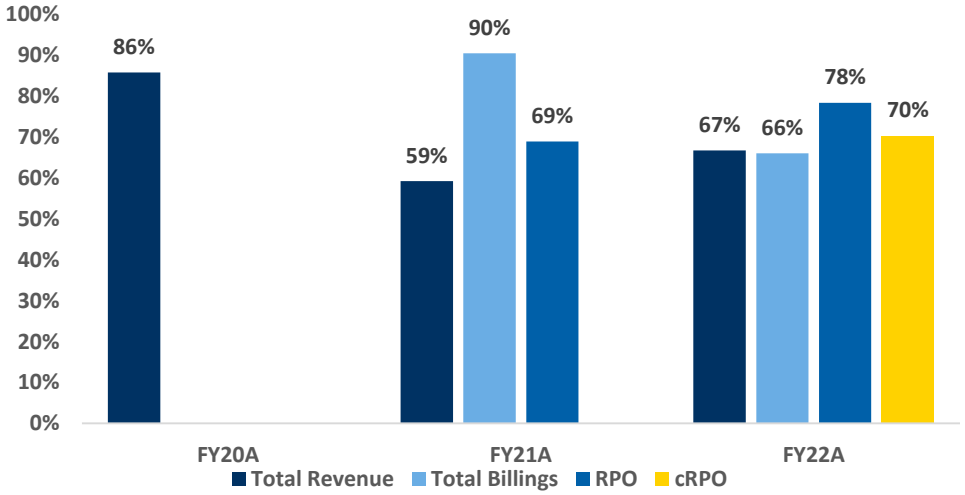
**Key metric(s):** Billings

**Leading metric practice/relevance:** In our view, billings is the most closely watched leading metric by investors, although management has long viewed the metric as unreliable considering the shorter contract durations of Asana's user base (~1/3 of customers are on monthly contracts). Over time, management believes RPO/cRPO and billings will become more relevant metrics as the company moves more upmarket and shifts from a more "land grab" GTM to enterprise expansions. We view billings to be the best-available forward indicator for Asana as the COVID slowdown showed up first in billings, then revenue, and the same happened with the reacceleration coming out of COVID.

**Relevant commentary:** (CFO Tim Wan on F4Q22 earnings call) *"But I would also point out that billings isn't necessarily the one metric we run the business on. We certainly look at the business across a number of different metrics, and I think GAAP revenue is probably a better indicator of how we think about our growth rate. Now, with that said, we are – as you can see, one of the things that we are focused on is building out our go-to-market around the enterprise motion, in that you would expect RPO as well as our CRPO to – as the one metric that you should be looking to over time where we're getting more enterprise traction, especially as we go beyond – as we do more longer term deals."*

Exhibit 14 - ASAN Leading Metrics Y/Y Growth Rates, FY20A-FY22A

Exhibit 14 shows Y/Y growth of ASAN's key leading metrics since FY20A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Box, Inc. (BOX) – covered by Rishi Jaluria

**Key metric(s):** Billings

**Leading metric practice/relevance:** Investors closely watch billings (and deferred revenue) as a leading indicator for Box and management does guide to and comment on the metric. Historically, billings (especially ST billings) has been a helpful leading indicator for Box's performance, indicating future revenue slowdowns and accelerations, although FY21/CY20 was a notable exception (FY21 ST Billings would have indicated further deceleration while FY22 revenue actually accelerated meaningfully). We note that Box does not explicitly disclose professional services revenue but, based on management commentary, we believe that services has typically ranged from 3%-4% of total revenue. We note that Box does refer to its "premium services" as subscription, which we do not agree with.

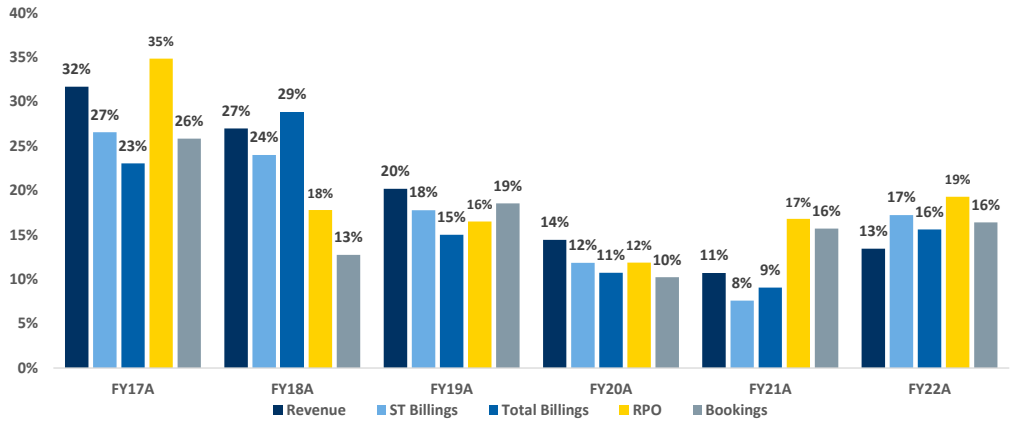
**Relevant commentary:** (CFO Dylan Smith at 2022 Analyst Day) *"Billings and deferred revenue are the leading indicators that are most directly correlated with our forward revenue growth, as our average payment durations have been stable this past year."*

(10-K Key Metrics commentary on billings): "*Billings represent our revenue plus the changes in deferred revenue and contract assets in the period. Billings we record in any particular period primarily reflect subscription renewals and expansion within existing customers plus sales to new customers, and represent amounts invoiced for all of our products and professional services... Billings help investors better understand our sales activity for a particular period, which is not necessarily reflected in our revenue given that we recognize subscription revenue ratably over the contract term. We consider billings a significant performance measure. We monitor billings to manage our business, make planning decisions, evaluate our performance and allocate resources."*

(10-K Key Metrics commentary on RPO): "*RPO consists of deferred revenue and backlog, offset by contract assets. Backlog is defined as non-cancellable contracts deemed certain to be invoiced and recognized as revenue in future periods. Future invoicing is determined to be certain when we have an executed non-cancellable contract or a significant penalty that is due upon cancellation, and invoicing is not dependent on a future event such as the delivery of a specific new product or feature, or the achievement of contractual contingencies.*"

Exhibit 15 - BOX Leading Metrics Y/Y Growth Rates, FY17A-FY22A

Exhibit 15 shows Y/Y growth of BOX's key leading metrics since FY17A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Bentley Systems (BSY) – covered by Matt Hedberg

**Key metric(s):** ARR

**Leading metric practice/relevance:** Subscriptions account for ~85% of revenue with perpetual licenses accounting for ~5-8% of revenue. We view ARR in conjunction with the recurring revenue dollar-based net retention rate as a good leading indicator of revenue growth. Deferred revenue and cRPO do not contribute a substantial amount to revenue in any given year with cRPO coverage trends at ~23% over the last several years. The company guides to ARR; cRPO and RPO are disclosed in 10-Q/Ks.

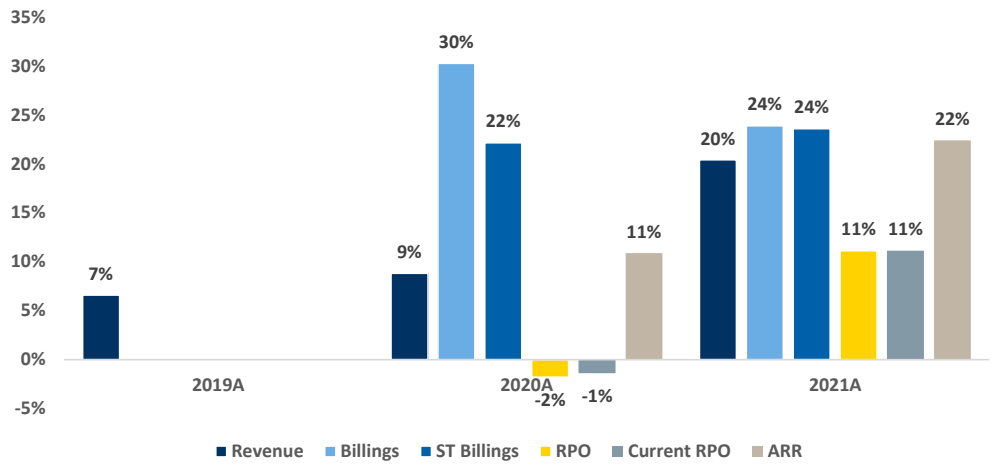
**Relevant commentary:** (CEO Greg Bentley on the Q3/20 earnings call) *"Our SELECT subscriptions are for those with perpetual licenses, but more than just maintenance. Our enterprise subscriptions are either Enterprise – ELS, Enterprise License Subscriptions or E365, which we'll look at, at greater length today. And we offer quarterly and monthly term licenses. So that's the recurring majority of our revenues. Within our ARR, most, by far, renews annually. As I say, we have term licenses that are monthly and quarterly. The E365 program, enterprise program, is based on daily consumption. And that would be subject to the greatest volatility…"*

(CEO Greg Bentley on the Q4/20 earnings call) *"…the bulk of our revenues and ARR are associated with annually reset contracts paid in advance and thus not sensitive to short-term usage fluctuations."*

Annualized Recurring Revenue 10-K definition and explanation as a key metric: "*ARR is defined as the sum of the annualized value of our portfolio of contracts that produce recurring revenues as of the last day of the reporting period, and the annualized value of the last three months of recognized revenues for our contractually recurring consumption-based software subscriptions with consumption measurement durations of less than one year. We believe that the last three months of recognized revenues, on an annualized basis, for our recurring software subscriptions with consumption measurement period durations of less than one year is a reasonable estimate of the annual revenues, given our consistently high retention rate and stability of usage under such subscriptions.*"

Exhibit 16 - BSY Leading Metric Y/Y Growth Rates CY19-21A

Exhibit 16 shows Y/Y growth of BSY's key leading metrics since CY/19A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Check Point Software (CHKP) – covered by Matt Hedberg

**Key metric(s):** Billings, RPO

**Leading metric practice/relevance:** With a transition toward subscription revenue underway, security subscription revenue has been the focus metric over the last several years. Billings has been less of a focus for management, but discussion around the metric has increased over the last several quarters as several consecutive quarters of accelerating trends could point to an acceleration in revenue growth over future periods. Deferred revenue and cRPO contribute a moderate amount to revenue in any given year with cRPO coverage trends at ~54% over the last several years. cRPO and RPO are reported annually in the company's 20-F filings.

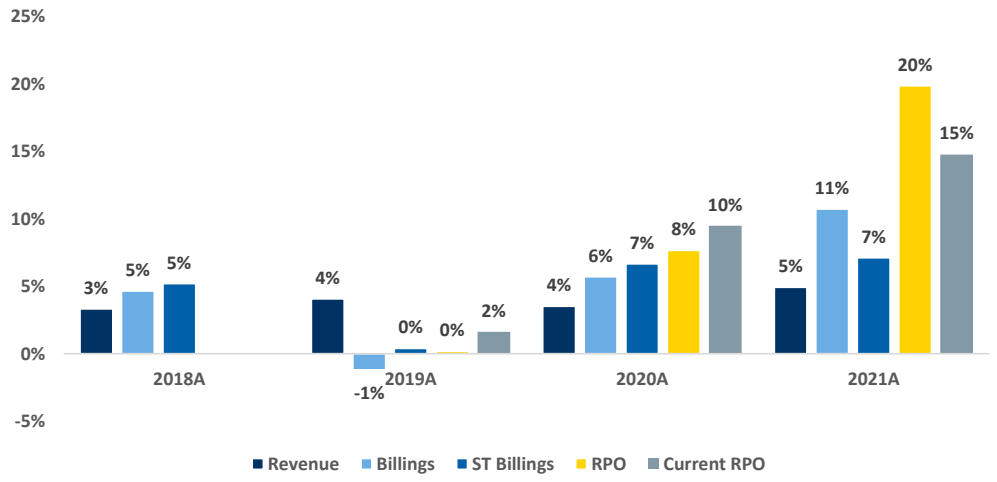
**Relevant commentary:** (CEO Gil Shwed on the Q4/21 earnings call) *"As you know, every quarter, we are trying to play down the billing number, because again, in many cases, it doesn't represent. This quarter, I think we've seen some very good indicators. So we are not just showing these numbers because they are impressive. But because at least last quarter, I don't know if you predict the next quarter, but last quarter, we've seen very good internal metrics."*

(CEO Gil Shwed on the Q4/21 earnings call) *"I think I already stated that some of the assumptions that we have during the projections, we feel that our billing can keep growing, growing in a nice rate. Our internal projections are definitely for – our internal goals are definitely for double-digit growth. Not all of that will translate immediately into revenues, partly because we are growing the subscription part of our business and the deferred revenues, partly because there's risks associated with the supply chain and others that may slow us down on some things…"*

Performance Obligations 20-F definition and explanation as a key metric: "*We recognize revenues under the core principle that transfer of control to our customers should be depicted in an amount reflecting the consideration we expect to receive in revenue. Therefore, we identify a contract with a customer, identify the performance obligations in the contract, determine the transaction price, allocate the transaction price to each performance obligation in the contract and recognize revenues when (or as) we satisfy a performance obligation.*"

Exhibit 17 - CHKP Leading Metric Y/Y Growth Rates CY18-21A

Exhibit 17 shows Y/Y growth of CHKP's key leading metrics since CY/18A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Cloudflare (NET) – covered by Matt Hedberg

**Key metric(s):** cRPO

**Leading metric practice/relevance:** With one revenue line item and nearly all revenue subscription based, we view revenue as a more relevant metric vs. billings. Additionally, the company transitioned key reporting metrics to revenue-based in Q1/20 from billings-based when the company entered the public markets. There are both monthly pay-as-you-go customers and contracted customers on enterprise plans with contracts from one to three years, with revenue recognized ratably over the life of the contract. Deferred revenue and cRPO contribute a moderate amount to revenue in any given year with cRPO coverage trends at ~42% over the last several years. cRPO and RPO are reported with quarterly results.

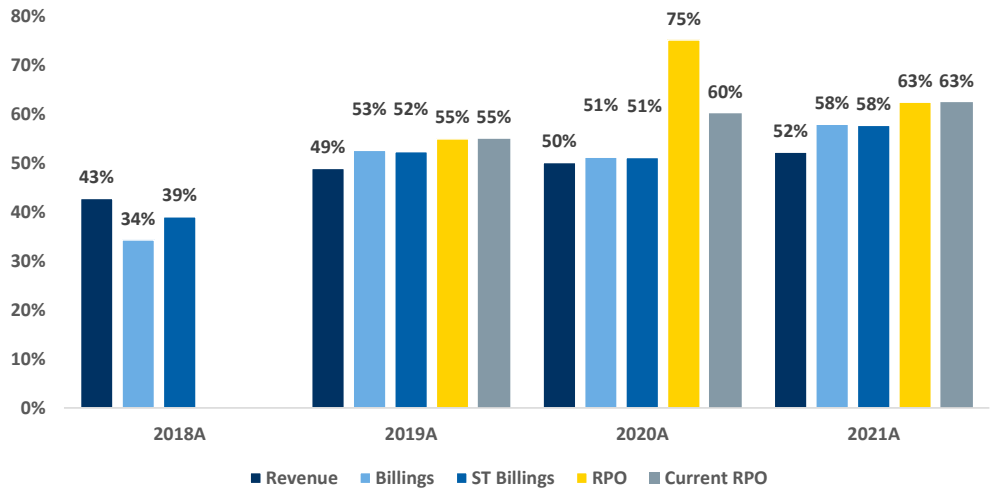
**Relevant commentary:** (Company 10-K) *"Beginning with the quarter ended March 31, 2020, we transitioned the method for calculating our key business metrics from a billings-based methodology to a revenue-based methodology. We believe the change in methodology to GAAP-based metrics provides improved disclosures for our investors by better aligning our key business metrics with GAAP and our financial statements and will provide a better representation of these important components of our operating model and business performance as we continue to grow our business."*

(Company 10-K) *"Our contracted customers, which consist of customers that enter into contracts for our Enterprise subscription plan, have contracts that range from one to three years and are typically billed on a monthly basis. Our contracted customer sales cycle typically lasts less than one quarter. Our agreements with contracted customers are tailored and priced to meet their varying needs and requirements. Enterprise subscription plan agreements for our contracted customers generally include a base subscription and a smaller portion based on usage."*

Performance Obligations 10-K definition and explanation as a key metric: "*Performance obligation primarily consists of subscription and support services, as they are provided over the same service period.*"

Exhibit 18 - NET Leading Metric Y/Y Growth Rates CY18-21A

Exhibit 18 shows Y/Y growth of NET's key leading metrics since CY/18A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Coupa Software, Inc. (COUP) – covered by Rishi Jaluria

**Key metric(s):** Billings, cRPO/RPO

**Leading metric practice/relevance:** Management sees billings as the primary leading indicator but references RPO/cRPO from time to time to complement billings. As highlighted below, recent acquisitions have blurred the organic billings line, and management has pointed to RPO as a way to reduce some of the near-term noise in billings and provide a cleaner look at the underlying health of the core business. That said, COUP excludes from cRPO contracts that are 12 months or less and is based on revenue expected to be recognized in the next 24 months, both which make the metric less useful as a leading indicator, in our view.

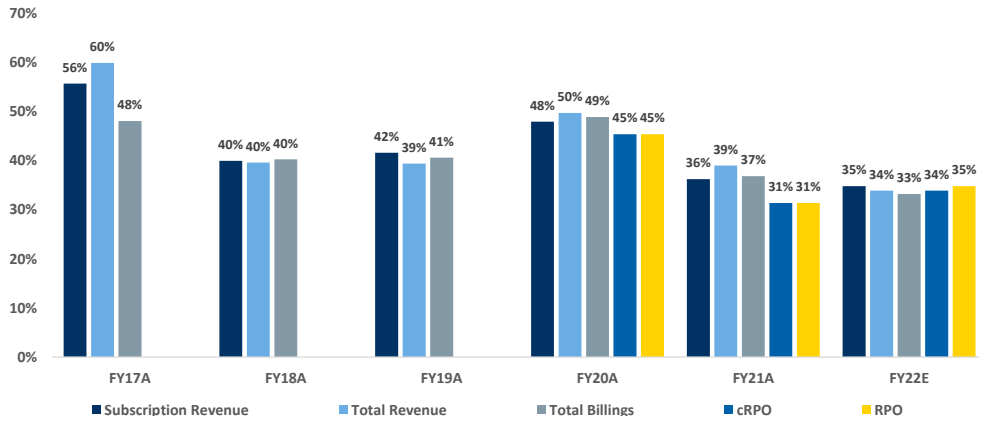
**Relevant commentary:** (10-K Key Metrics commentary on billings): "*The trailing twelve months calculated billings is intended to provide information about our subscription revenue growth over time, and can typically be seen as an early indicator of trends in revenue growth. While trailing twelve months calculated billings can increase as our revenues grow, it may fluctuate up or down from period to period for several reasons, including amounts, timing, and duration of customer contracts, as well as the timing of billing cycles for each order*."

(10-K Key Metrics commentary on RPO): "*In calculating the RPO amount, we elected to apply the two expedients under the revenue standard to exclude RPO related to contracts that are <=12 months and contracts where revenue is recognized under the as-invoiced method. We generally execute multiple year subscription contracts and invoice an initial amount at contract signing followed by subsequent annual invoices. At any point in the contract term, there might be amounts that are not due for billing yet. These amounts are not recorded in our consolidated financial statements, and are considered to be part of the RPO amount.*"

(CFO Anthony Tiscornia on F3Q22 earnings call): "*New business deals are on average 3 years. We often do large deals that are even longer than that. When you look at RPO, if you do a 3-year deal, of course, in billings, you get a 1-year benefit in the numerator for the current period. But in RPO, you get a full 3-years' benefit. When you look at cRPO, there are some near-term… headwinds… that pertain to some of the legacy agreements from LLamasoft and… other acquisitions we've done… the license migration that's going on and our push for professional services to continue to have our partners very strategically perform those... So some of the near-term noise in calculated billings, really, you cut through some of that when you look at the cRPO numbers… We don't typically share those but this quarter we felt like it was particularly important to give a clear picture of the underlying health of the business.*"

Exhibit 19 - COUP Leading Metrics Y/Y Growth Rates, FY17A-FY22A

Exhibit 19 shows Y/Y growth of COUP's key leading metrics since FY17A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Couchbase (BASE) – covered by Matt Hedberg

**Key metric(s):** ARR, cRPO, RPO

**Leading metric practice/relevance:** With ~94% of revenue generated from subscriptions, we believe ARR, cRPO and RPO are good leading indicators for momentum in the business. When deals are booked, they go into ARR, which looks out 12 months but deals won't be billed until start dates in the future. The company guides to ARR on a quarterly and annual basis. Deferred revenue and cRPO contribute a moderate amount to revenue in any given year with cRPO coverage trends at ~59% over the last several years. ARR, cRPO and RPO are reported with quarterly results.

**Relevant commentary:** (CFO Greg Henry on the Q2/22 earnings call) *"In order to provide investors with better transparency into our bookings as well as provide a framework that aligns how we plan, forecast and manage our business internally, we're providing annual recurring revenue or ARR, which represents the annualized recurring revenue at the end of the period that is currently contracted and committed over the forward 12-month period. We believe ARR best represents our business performance by accounting for timing variability among our customer's implementation times."*

(CFO Greg Henry on the Q2/22 earnings call) *"Because our software is often implemented in large scale and often complex enterprise application environments, our revenue experiences a degree of volatility due to the timing of the initiation of certain projects. In fact, we saw this dynamic play out this quarter as several very large deals renewed with significant upsell earlier than expected, contributing upside to our RPO, but did so with project implementation timing later than we had expected and the start dates for revenue and billings later than we had originally forecasted. Later start dates can impact near-term revenue and billings performance as it did this quarter, but are still very healthy for our long term business."*

(CFO Greg Henry on the Q2/22 earnings call) *"Look enterprises buy when they want and need it. We don't always have the exact timing. There's always these types of deals that are sort of floating around and potential. We've landed a couple of large ones today. And if you think about buying this kind of software, when I buy software from a budgeting, I look for the largest most enterprise critical and by those first. And so we're happy to see customers are doing the same, same with us. So the way they would flow though is we booked the deal in the quarter, say a future start dates. So we put it in ARR today and obviously it goes in RPO. But because the dates are starting in the future, we don't bill for those nor do we start recognizing revenue until we have those new subscriptions start date. So that's the way it plays in terms of what the near-term metrics versus some of the longer term metrics are. And again that's why we've really focused on ARR as the best metric to view Couchbase, because we've aligned that to take some of that noise out of the system when we get some of these deals that have future start date. So that's really how it flows."*

Annual Recurring Revenue 10-K definition and explanation as a key metric: "*We define ARR as of a given date as the annualized recurring revenue that we would contractually receive from our customers in the month ending 12 months following such date. Based on historical experience with customers, we assume all contracts will be automatically renewed at the same levels unless we receive notification of non-renewal and are no longer in negotiations prior to the measurement date. ARR excludes revenue from on-demand arrangements and services. ARR should be viewed independently of revenue, and does not represent our revenue under U.S. GAAP on an annualized basis, as it is an operating metric that can be impacted by contract start and end dates and renewal dates. ARR is not intended to be a replacement for forecasts of revenue. Although we seek to increase ARR as part of our strategy of targeting large*

This report is intended solely for cbass@marketstreetpartners.com

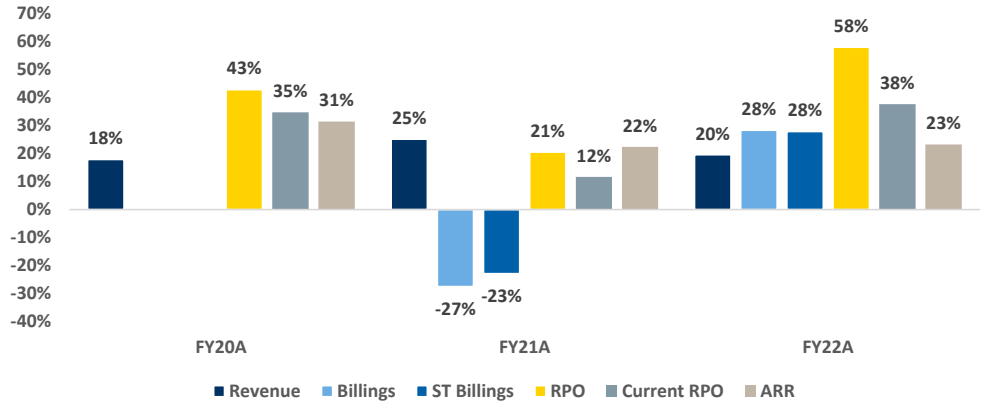


*enterprise customers, this metric may fluctuate from period to period based on our ability to acquire new customers and expand within our existing customers. We believe that our ARR is an important indicator of the growth and performance of our business."*

Remaining Performance Obligations 10-K definition and explanation as a key metric: "*Remaining performance obligations ("RPOs") represent the amount of contracted future revenue that has not yet been recognized, including both deferred revenue and non-cancelable contracted amounts that will be invoiced and recognized as revenue in future periods."*

Exhibit 20 shows Y/Y growth of BASE's key leading metrics since FY/20A.

Exhibit 20 - BASE Leading Metric Y/Y Growth Rates FY20-22A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Coursera Inc. (COUR) – covered by Rishi Jaluria

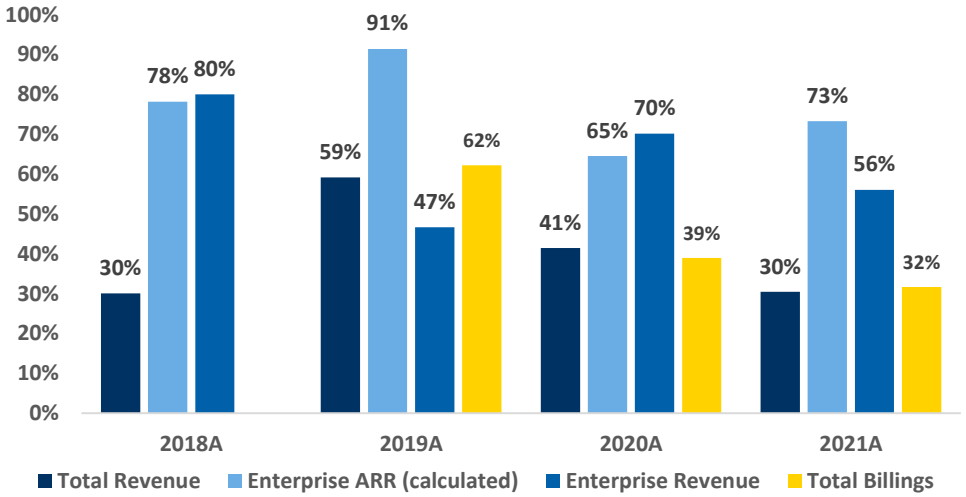
**Key metric(s):** Enterprise ARR (calculated)

**Leading metric practice/relevance:** With the exception of Coursera's Enterprise segment (29% of 2021 revenue), very little revenue is truly recurring meaning that typical leading metrics are not useful barometers of future business. Management does occasionally comment on Enterprise ARR but does not explicitly disclose performance of the metric. Additionally, we note that RPO and billings/deferred revenue are not useful in forecasting Enterprise segment performance as they include Consumer subscription plans (primarily Coursera, ~25% of Consumer revenue).

**Relevant commentary:** (S-1 Key Metrics commentary on enterprise ARR) "*We calculate annual recurring revenue ("ARR") by annualizing each customer's monthly recurring revenue ("MRR") for the most recent month at period end. We define an Enterprise cohort as Enterprise customers that purchase Coursera via our direct sales force and who purchased their first subscription with us in a given year*."

Exhibit 21 - COUR Leading Metrics Y/Y Growth Rates, 2018A-2021A

Exhibit 21 shows Y/Y growth of COUR's key leading metrics since 2018.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Crowdstrike (CRWD) – covered by Matt Hedberg

**Key metric(s):** ARR, Backlog, cRPO, RPO

**Leading metric practice/relevance:** With subscriptions accounting for ~94% of revenue, we believe ARR, backlog cRPO, and RPO are good leading indicators. The company views ARR as the key focal point for its business, though it does not guide to the metric while investors have been focused on net-new ARR. Deferred revenue and cRPO contribute a moderate amount to revenue in any given year with cRPO coverage trends at ~64% over the last several years. ARR is reported with quarterly results. Backlog, cRPO and RPO are disclosed in 10-Qs/Ks.

**Relevant commentary:** (CFO Burt Podbere on the Q1/20 earnings call) *"We view annual recurring revenue, or ARR, as a key metric to measure our business, given it is driven by our ability to acquire new subscription customers and to maintain and expand our relationships with existing subscription customers. We define ARR as the annualized value of our customer subscription contracts at the end of the quarter, assuming any contract that expires during the next 12 months is renewed on its existing terms."*

(CFO Burt Podbere on the Q2/20 earnings call) *"…when we think about ARR, of course, we don't guide to it but we do provide it, we think about ARR and the way we calculate it very simply. We take the ending ARR and we divide that by four and then we pick out some churn and contraction, and we add in some positive dollars for revenue in the quarter. So I think that when we look at our ARR and we look at our projections, we really do look at what we know in-house, not what we don't know. We've had some success in the past in running the tables. And we clearly don't want to guide based on the fact that we're going to run the table on all future quarters. So, we've been happy with the results to date and that's how we think about our ARR."*

(CFO Burt Podbere on the Q4/22 earnings call) *"…when you think about net new ARR it could come from anywhere, it could come from new logos, it could come from our existing base. Our focus, as you know, has been and always will be net new ARR wherever it comes from. So, when we think about looking out into this year, we're doing the same thing from a compensation perspective with our sales team as we've done last year. Meaning, we're going to pay our sales team the same whether they bring in net new logo or whether they bring it from an existing customer. So, we pay on net new ARR, we don't care where it comes from. The great news is, of course, that we have tremendous amount of headroom in both, right, our expanding base really helps in terms of cross-sell for sure, but we still have a tremendous amount of headroom in terms of being able to grab the new logos. And that's kind of a great position to be in and it really talks to the fact that we think about CrowdStrike in the early innings of our journey with a lot of headroom to go. So, hopefully that makes sense to you."*

Annual Recurring Revenue 10-K definition and explanation as a key metric: "*ARR is calculated as the annualized value of our customer subscription contracts as of the measurement date, assuming any contract that expires during the next 12 months is renewed on its existing terms. To the extent that we are negotiating a renewal with a customer after the expiration of the subscription, we continue to include that revenue in ARR if we are actively in discussion with such an organization for a new subscription or renewal, or until such organization notifies us that it is not renewing its subscription*."

Performance obligation 10-K definition and explanation as a key metric: "*Performance obligations promised in a contract are identified based on the services that will be transferred to the customer that are both capable of being distinct, whereby the customer can benefit from the service either on its own or together with other resources that are readily available from the Company or from third parties, and are distinct in the context of the contract, whereby the*

This report is intended solely for cbass@marketstreetpartners.com

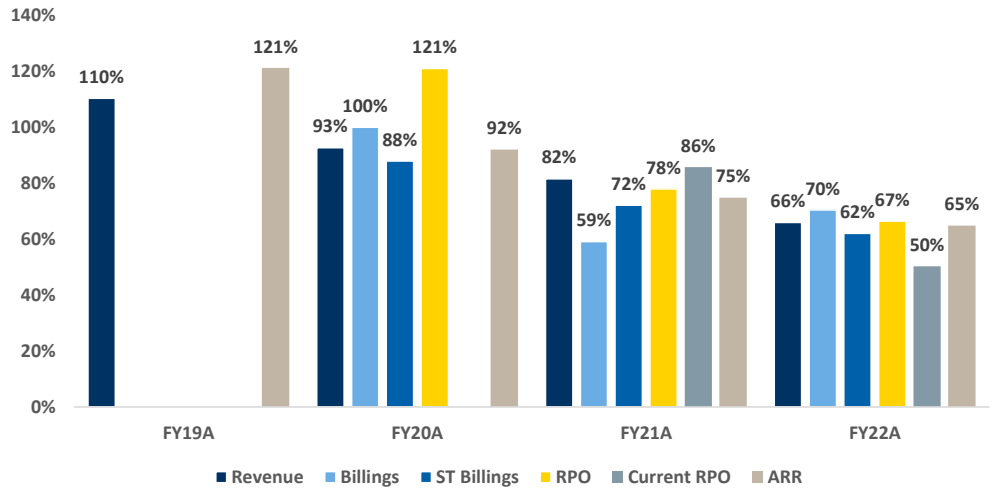


*transfer of the services is separately identifiable from other promises in the contract. The Company's performance obligations consist of (i) subscriptions and (ii) professional services*."

Backlog 10-K definition and explanation as a key metric: "*We enter into both single and multi-year subscription contracts for our solutions. We generally invoice our customers at contract signing prior to commencement of subscription period. Until such time as these amounts are invoiced, they are not recorded in deferred revenue or elsewhere in our consolidated financial statements, and are considered by us to be backlog. As of January 31, 2022, we had backlog of approximately $735.8 million. We expect backlog will change from period to period for several reasons, including the timing and duration of customer agreements, varying billing cycles of subscription agreements, and the timing and duration of customer renewals. Because revenue for any period is a function of revenue recognized from deferred revenue under contracts in existence at the beginning of the period, as well as contract renewals and new customer contracts during the period, backlog at the beginning of any period is not necessarily indicative of future revenue performance. We do not utilize backlog as a key management metric internally*."

Exhibit 22 - CRWD Leading Metric Y/Y Growth Rates FY19A-FY22A

Exhibit 22 shows Y/Y growth of CRWD's key leading metrics since FY/19A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Clearwater Analytics Holdings Inc. (CWAN) – covered by Rishi Jaluria

**Key metric(s):** ARR

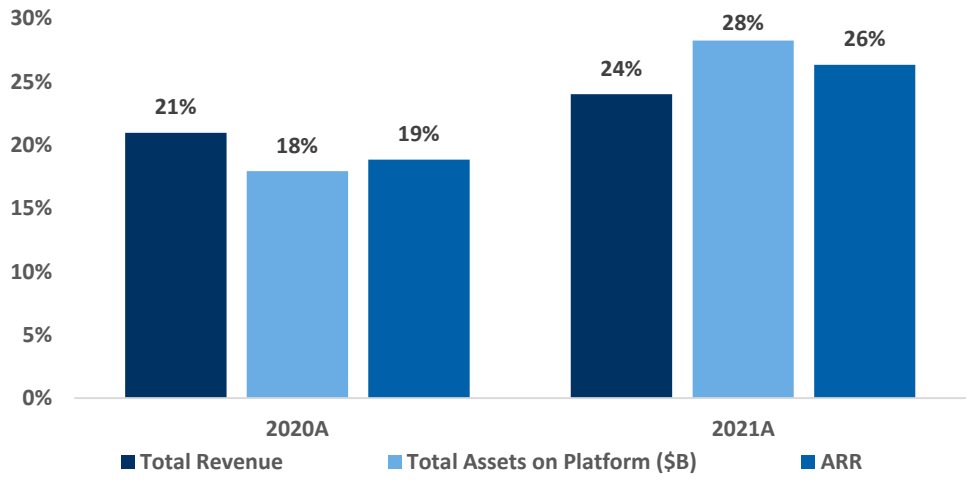
**Leading metric practice/relevance:** Clearwater reports Annualized Recurring Revenue, but does not guide to the metric. Clearwater's ARR figure is based on recurring revenue in the last month of the quarter, which aligns to how the company bills customers (monthly in arrears), though might result in a rounding error for new ARR (i.e. when contracts start halfway into the month). Note visibility into Clearwater's financial model and forward-looking metrics are a bit challenging, considering the shortage of FY historical data, the large land nature of the platform (and thus more limited expand motion), and billing in arrears. Based on our conversations with investors, we believe assets under management (AUM) are a closely watched metric as Clearwater pricing is based on a take rate of average daily AUM. However, Clearwater only discloses AUM on an annual basis, leaving ARR as lone primary performance metric. Neither deferred revenue nor RPO are relevant for Clearwater given customers are usually billed in arrears on a monthly basis and, therefore, deferred revenue only includes implementation services and RPO is not material.

**Relevant commentary:** (CFO Jim Cox at 2021 Investor Conference) *"So, some of the important non-P&L metrics that we talk about is we talk about ARR, which is our annualized recurring revenue, so it's kind of the run rate of our revenue in that last month of the period that we reported."*

(10-K Key Metrics commentary on ARR): "*Annualized recurring revenue is calculated at the end of a period by dividing the recurring revenue in the last month of such period by the number of days in the month and multiplying by 365… Because a substantial majority of the assets on our platform have lower levels of volatility with respect to their market value, the growth in annualized recurring revenue is generally not attributable to the fluctuating market value of the assets on our platform. Rather, the growth in annualized recurring revenue is due to an increase in the number of clients using our offering as well as from onboarding more assets of our existing clients onto our platform."*

Exhibit 23 shows Y/Y growth of CWAN's key leading metrics since 2020.

Exhibit 23 - CWAN Leading Metrics Y/Y Growth Rates, 2020A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Datadog (DDOG) – covered by Matt Hedberg

**Key metric(s)**: N/A

**Leading metric practice/relevance:** Datadog generates revenue primarily on monthly or annual non-cancellable subscription agreements with the majority of revenue from annual subscriptions based on the number of hosts monitored. The company views revenue as the best metric for the business as billings and RPO can vary based on duration of customer contracts and timing around invoices. RPO is disclosed with quarterly results. ARR and cRPO are not disclosed, though cRPO has consistently been quantified as "substantially all" of RPO in 10-Qs/Ks.

**Relevant commentary:** (CFO David Obstler on the Q4/21 earnings call) *"We continue to believe revenue is a better indicator of our business trends than billings and RPO, as those can fluctuate relative to revenue based on the timing of invoicing and the duration of customer contracts."*

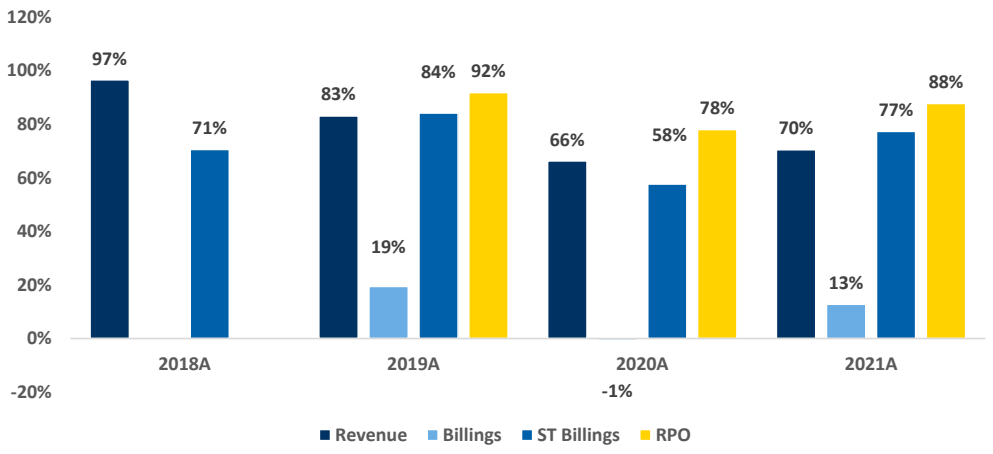
(CFO David Obstler on the Q2/21 earnings call) *"I should note that we continue to believe revenue is a better indicator of our business trends than billings and RPO as those can fluctuate relative to revenue based on the timing of invoicing and the duration of customer contracts."*

(CFO David Obstler on the Q3/19 earnings call) *"Given the mix of annual and monthly invoicing and the variability in billing, we do not believe that calculated billing is the most useful metric for investors to evaluate our business performance. As in any one period, billings growth can vary substantially from revenue growth."*

Annual Recurring Revenue 10-K definition and explanation as a key metric: "*We define ARR as the annual run-rate revenue of subscription agreements from all customers at a point in time. We calculate ARR by taking the monthly run-rate revenue, or MRR, and multiplying it by 12. MRR for each month is calculated by aggregating, for all customers during that month, monthly revenue from committed contractual amounts, additional usage, usage from subscriptions for a committed contractual amount of usage that is delivered as used and monthly subscriptions. We updated the definition of MRR as of the quarter ended September 30, 2021 to capture usage from subscriptions with committed contractual amounts and applied this change retrospectively.*"

Exhibit 24 - DDOG Leading Metric Y/Y Growth Rates 2018A-2021A

Exhibit 24 shows Y/Y growth of DDOG's key leading metrics since 2018.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## DocuSign, Inc. (DOCU) – covered by Rishi Jaluria

**Key metric(s):** Billings

**Leading metric practice/relevance:** The company reports on billings and guides to both quarterly and annual billings, which makes sense to us given DocuSign bills the majority of customers annually in advance. As highlighted below, management specifically prefers TTM billings as quarterly billings can be volatile driven by duration, early renewals, co-terming, discounts, and FX rates. We, however, believe investors continue to focus on quarterly billings as trailing calculations are naturally less powerful as a leading indicator. While DOCU's RPO excludes contracts that are 12 months or less, the company does provide dollar-weighted average contract length (~18 months) and the % of contracts that are 12 months or less (65%) and >12 months (35%), which we can use to back into an adjusted RPO/cRPO figure.

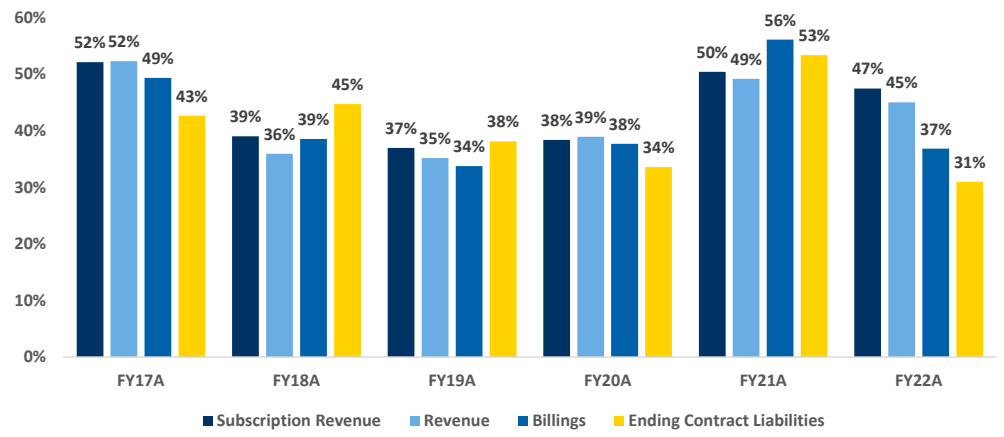
**Relevant commentary:** (CEO Daniel Springer during an investor conference) "*Billings is imperfect and billings is much more volatile. And we always say, please look at the four-quarter moving average for billings or you're going to get yourself either super hyped up or super concerned in a way that I think you will find out you are mistaken, if you go either direction there. And it is better to look at the four-quarter moving average but, that's more forward-looking so people are excited about that metric.*"

(10-K Key Metric commentary): "*We define billings as total revenues plus the change in our contract liabilities and refund liability less contract assets and unbilled accounts receivable in a given period. Billings reflects sales to new customers plus subscription renewals and additional sales to existing customers. Only amounts invoiced to a customer in a given period are included in billings. We believe billings is a key metric to measure our periodic performance. Given that most of our customers pay in annual installments one year in advance, but we typically recognize a majority of the related revenue ratably over time, we use billings to measure and monitor our ability to provide our business with the working capital generated by upfront payments from our customers*."

(CFO Cynthia Gaylor on F2Q22 earnings call) "*We don't use RPO just given the duration of our contracts and the way that's calculated. We don't think that is an indicative number, but it's – when you see it, it will be quite strong and the growth rate year-on-year on that metric is also quite strong.*"

Exhibit 25 - DOCU Leading Metric Y/Y Growth Rates, FY17A-FY22A

Exhibit 25 shows Y/Y growth of DOCU's key leading metrics since FY17.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



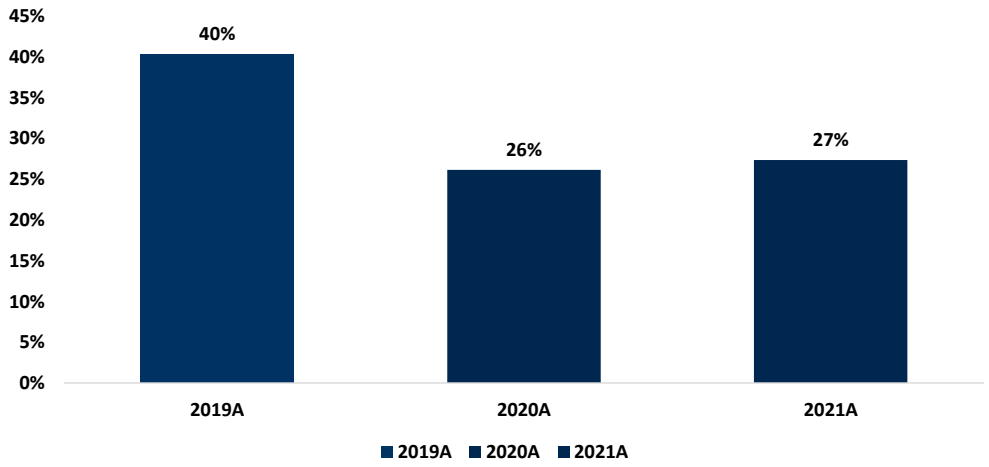
## DoubleVerify (DV) – covered by Matt Hedberg

**Key metric(s):** N/A

**Leading metric practice/relevance:** We think that the key growth metric for DV is revenue given the company doesn't have much in the form of deferred revenue. Management guides to revenue but doesn't note focus on ARR, cRPO, RPO, billings or deferred revenue due to its business model.

Exhibit 26 shows Y/Y growth of DV's key leading metrics since CY19.

Exhibit 26 - DV Leading Metrics Y/Y Growth Rates 2018A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Dropbox, Inc. (DBX) – covered by Rishi Jaluria

**Key metric(s):** ARR

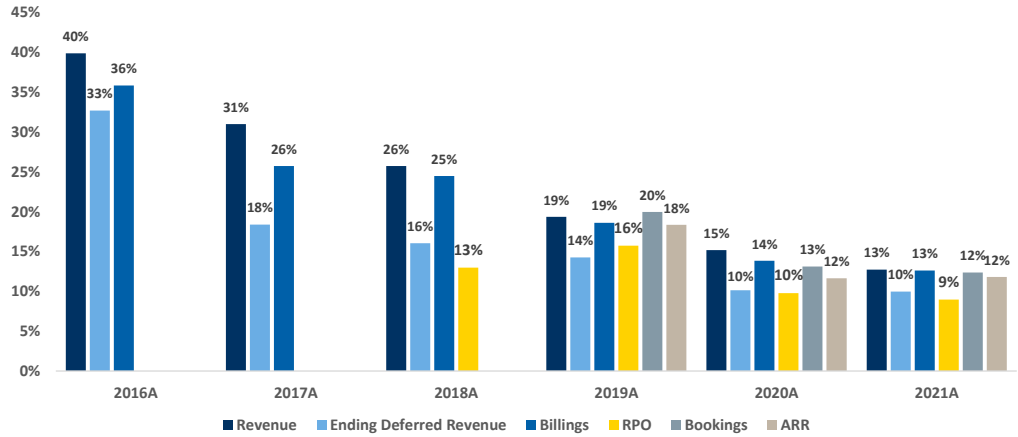
**Leading metric practice/relevance:** Management views ARR as the primary leading metric for investors and notes that Dropbox manages the company for driving ARR growth. This makes sense to us, although we would like to see Dropbox guide to the metric and do think that basing executive bonuses on ARR (currently Dropbox bases bonuses on revenue and EBIT) would send a clearer message to the Street. Although investors do closely watch billings, we acknowledge that the metric is less relevant for Dropbox than other SaaS companies considering the high mix of customers on monthly plans (less than 60% of revenue in any given quarter comes from deferred revenue vs. the 70%-90% range found in companies with longer duration).

**Relevant commentary:** (10-K Key Metrics commentary) "*We primarily focus on total annual recurring revenue ("Total ARR") as the key indicator of the trajectory of our business performance. Total ARR represents the amount of revenue that we expect to recur annually, enables measurement of the progress of our business initiatives, and serves as an indicator of future growth. In addition, Total ARR is less subject to variations in short-term trends that may not appropriately reflect the health of our business, however the changes in ARR throughout the year could be subject to seasonality. Total ARR is a performance metric and should be viewed independently of revenue and deferred revenue, and is not intended to be a substitute for, or combined with, any of these items. Total ARR consists of contributions from all of our revenue streams, including subscriptions and add-ons. We calculate Total ARR as the number of users who have active paid licenses for access to our platform as of the end of the period, multiplied by their annualized subscription price to our platform. We include ARR related to acquired companies in our total ARR in the period of the acquisition. We adjust the exchange rates used to calculate Total ARR on an annual basis at the beginning of each fiscal year.*"

(CFO Tim Regan at 2021 Investor Conference): "*ARR is our primary metric, as focusing on driving ARR is how we actually operate our business, and it's less impacted by short-term trends. And I think that what we're seeing this year in paying users and ARPU really illustrates why we focus on ARR as opposed to these price and quantity metrics.*"

Exhibit 27 - DBX Leading Metric Y/Y Growth Rates 2016A-2021A

Exhibit 27 shows Y/Y growth of DBX's key leading metrics since 2016.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Dynatrace (DT) – covered by Matt Hedberg

**Key metric(s):** ARR, cRPO, RPO

**Leading metric practice/relevance:** Management believes that ARR is a meaningful metric that helps investors better understand performance because it normalizes out fluctuations in contracting and billings. We like ARR as its leading metric and look to ARR drivers such as new logo expansion, net retention rates, growth from partners, and macro trends. Additionally, management noted that while cRPO/RPO are not focal points now, over time they could become more relevant metrics. Looking at cRPO coverage trend, it has remained consistent at low ~70% of FY+1 revenue for the past two years, which we would like to continue to monitor as the company begins to focus more on cRPO. The company expects to recognize 57% of this amount as revenue in the year ended March 31, 2022 and the remainder thereafter.

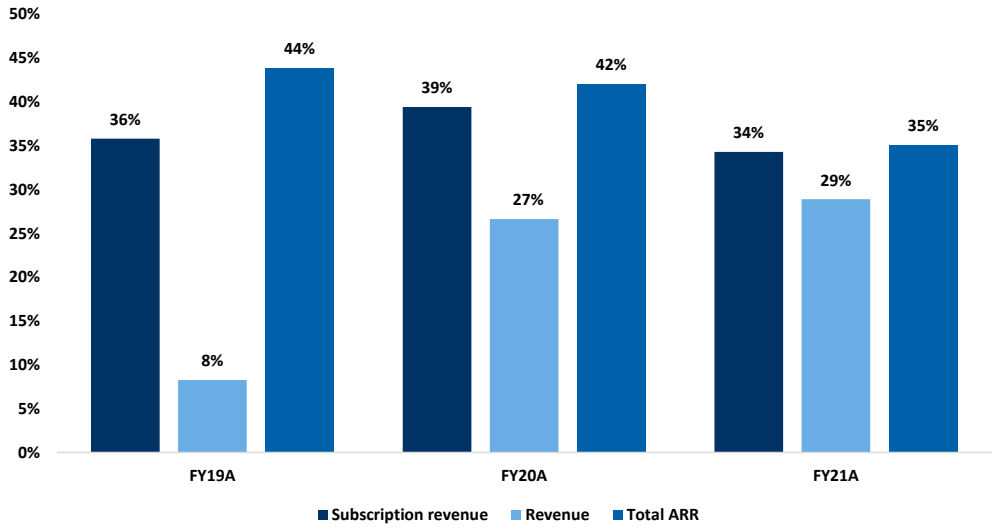
**Relevant commentary:** (CFO Kevin C. Burns) "*We continue to believe ARR is the best metric to understand the business' performance as it removes variability associated with billings and contracting.*"

(CFO Kevin C. Burns Q4 2021 earnings call) *"As I had mentioned in the past, we believe ARR is the best metric to remove billings and contracting noise, but we do provide RPO as we believe, over time, it will become a more meaningful metric."*

(ARR 10-K definition and explanation as a key metric*) "We define annual recurring revenue, or ARR, as the daily revenue of all subscription agreements that are actively generating revenue as of the last day of the reporting period multiplied by 365. We exclude from our calculation of ARR any revenues derived from month-to-month agreements and/or product usage overage."*

Exhibit 28 - DT's Leading Metrics Y/Y Growth Rates FY19A-FY21A

Exhibit 28 shows Y/Y growth of DT's key leading metrics since FY19.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Duck Creek Technologies, Inc. (DCT) – covered by Rishi Jaluria

**Key metric(s):** SaaS ARR, Total ARR (calculated)

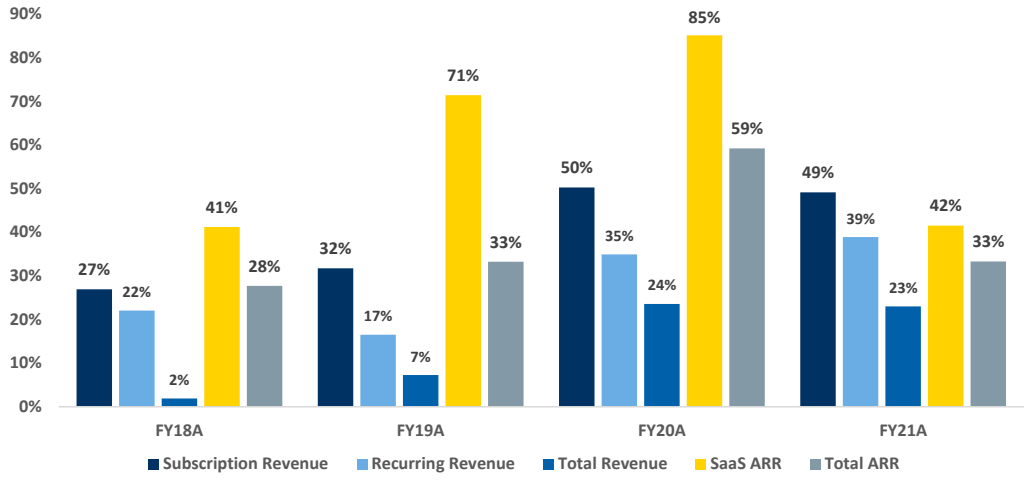
**Leading metric practice/relevance:** Management discloses and comments on SaaS ARR, based on subscription revenue (48% of total FY21 revenue) but does caution investors on the quarter-to-quarter variability of the metric due to DCT's larger lands. SaaS ARR is a focus metric for management and reflects the company's ability to land, maintain, and expand SaaS deployments (we note SaaS ARR is a component of EOY executive bonuses). In our view, investors do pay close attention to SaaS ARR, although the metric has historically been less relevant as a forward-looking indicator as it excluded revenue from a large legacy deployment. Since the legacy deployment finally churned off in June 2021 (during F4Q21), SaaS ARR has more closely tracked subscription revenue growth, and we believe the metric will be a more relevant indicator of future performance as DCT fully laps the comp. Additionally, we note that DCT applies the practical expedient in reporting RPO and excludes certain professional services revenues. We also calculate total ARR, which includes recurring maintenance revenues.

**Relevant commentary:** (CFO Vinny Chippari on F1Q21 earnings call) *"When we sign a deal or contract a deal, that booking, that begins recognizing revenue once the deal is provisioned or the service is available to the customer. That generally takes about 30 days. So, if we sign a deal late in the quarter it is more typical than not that it does not make it into the quarter. So, the quarter that really was a little bit of an outlier, actually, it was Q4 where we had really strong ARR growth because every deal we sold in Q4 was provisioned and generating revenue by the end of Q4… Because we are low-volume, high-dollar deals, it is difficult to get a really consistent trend on a quarter-to-quarter basis. So, we do tend to think about ARR growth in longer time periods in a quarter."*

(10-K Key Metrics commentary on SaaS ARR): "*We calculate SaaS ARR by annualizing the recurring subscription revs recognized in the last mos. of the measurement period (the latest twelve-month period)… excludes one existing contract for a service no longer offered on a standalone basis… We believe SaaS ARR provides important information about our ability to acquire new subscription SaaS customers and to maintain and expand our relationship with existing subscription SaaS customers. SaaS ARR is included in a set of metrics that we calc. quarterly to review with mgmt. as well as periodically with members of our board of directors."*

Exhibit 29 - DCT Leading Metric Y/Y Growth Rates FY18A-FY21A

Exhibit 29 shows Y/Y growth of DCT's key leading metrics since FY18A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Elastic (ESTC) – covered by Matt Hedberg

**Key metric(s):** ST Billings, Billings

**Leading metric practice/relevance:** Management believes that revenue growth is the most meaningful indicator of its performance since the company leverages a consumption model. While it reports billings and RPO, management noted that it does not consider it as meaningful as revenue due to the variability in a customer's consumption patterns. While we don't like billings as a growth metric, TTM billing and ST billings can be helpful indicators. Looking at the billings coverage trend, it has grown steadily now to the mid 80% of FY+1 revenue as of FY20. We believe this provides some more insight into billings trends but would like to see the company disclose ARR in the future as we think ARR can help capture monthly cloud contracts. We'd also noted that the company expects to recognize about 85% of its RPO, which includes deferred revenue, over the next 2 years and the remainder thereafter.

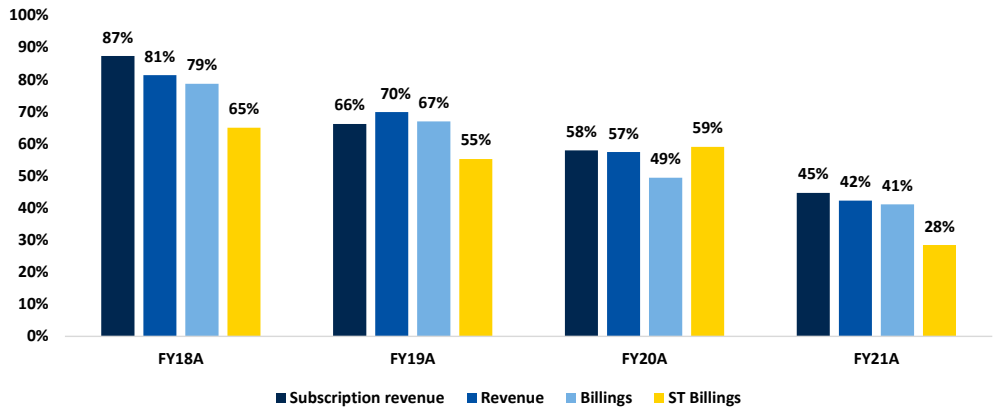
**Relevant commentary:** (CFO Janesh Jamnadas Moorjani Q3 2022 earnings call) *"On RPO specifically, again, just like deferred revenue, you know that the monthly cloud business does not have any RPO either. And so, again, RPO just becomes less meaningful for us. But the other important piece in a consumption model is you realize is that if a customer is ramping their usage faster and burning down their backlog, that's actually a good thing. And that's a big chunk of what we saw. So that will allow us to go back to those customers sooner than expected in the future to increase their commitments based on their higher usage, and that's obviously not reflected in the RPO today."*

(CFO Janesh Jamnadas Moorjani Q3 2022 earnings call) *"And then with respect to ARR, look, there isn't a perfect measure out there that addresses all the aspects of the business model differences and timing. We think revenue is probably the best of the measures at this point in time. And again, even in a consumption model, keep in mind that it's harder to use ARR as a revenue predictor, just given the variability of consumption patterns and the ramp in consumption that we're seeing, particularly when customers first join."*

(Billings 10-K definition and explanation as a key metric) *"We define calculated billings as total revenue plus the increase in total deferred revenue as presented on or derived from our consolidated statements of cash flows less the (increase) decrease in total unbilled accounts receivable in a given period. Calculated billings exclude deferred revenue and unbilled accounts receivable acquired through acquisitions in the period of the acquisition."*

Exhibit 30 - ESTC's Leading Metrics Y/Y Growth Rates FY18A-FY21A

Exhibit 30 shows Y/Y growth of ESTC's key leading metrics since FY18.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## EverCommerce (EVCM) – covered by Matt Hedberg

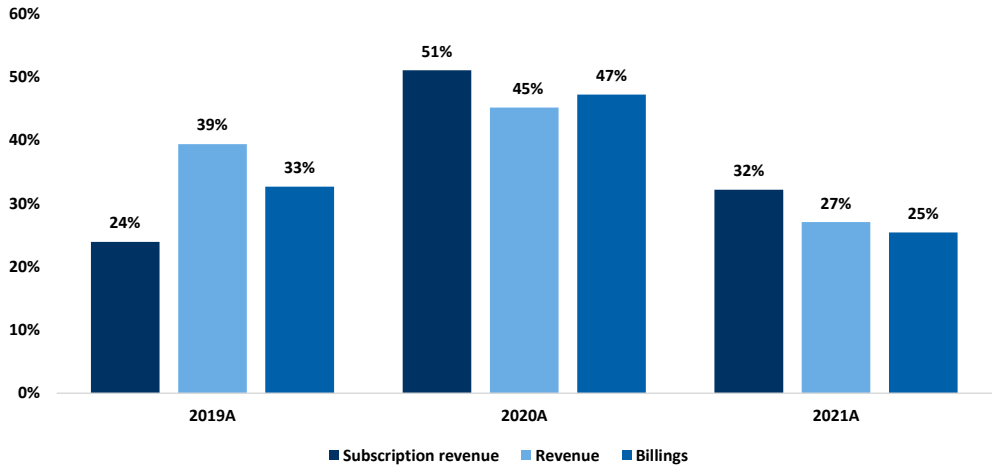
**Key metric(s):** N/A

**Leading metric practice/relevance:** Management noted that the key growth metric they focus on is revenue. That said, we also look at billings, though the company does not focus on it as a key metric. Additionally, the company doesn't focus on ARR, cRPO, RPO, or deferred revenue as key performance indicators. We'd note that the company expects to recognize 50% of its RPO as revenue within the next year and the remainder thereafter.

**Relevant commentary: (10-K)** *"Pro Forma Revenue Growth Rate is a key performance measure that our management uses to assess our consolidated operating performance over time. Management also uses this metric for planning and forecasting purposes."*

Exhibit 31 shows Y/Y growth of EVCM's key leading metrics since 2019.

Exhibit 31 - EVCM Leading Metrics Y/Y Growth Rates 2019A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## F5 (FFIV) – covered by Matt Hedberg

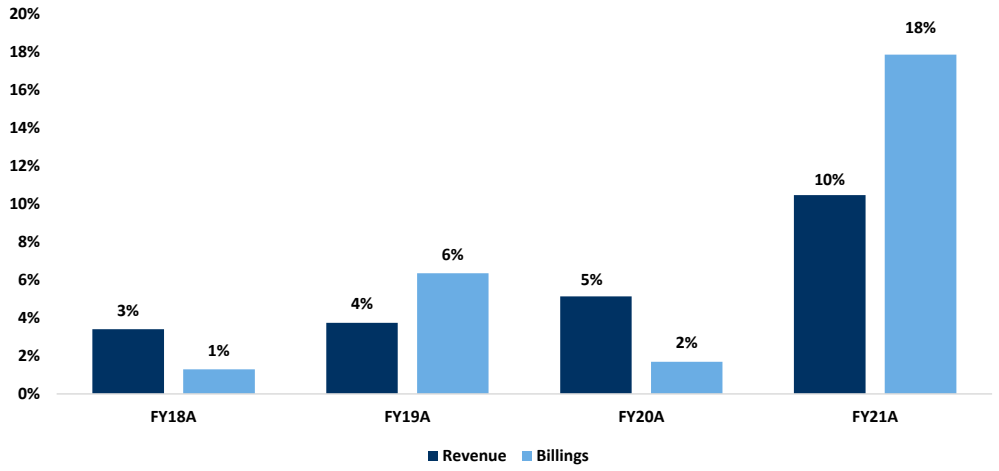
**Key metric(s):** Billings

**Leading metric practice/relevance:** Management largely looks to revenue as its key performance indicator, which was noted in its 10-K. Additionally, management guides to revenue, gross margin, operating expenses, and EPS, which reaffirms this point. That said, we can also look at billings, though it is not considered a key metric. The company also does not focus on ARR, cRPO, RPO, and deferred revenue as key performance indicators. We'd note that the company expects to recognize 65% of its RPO as revenue within the next year and the remainder thereafter.

**Relevant commentary: (10-K)** *"Our management team monitors and analyzes a number of key performance indicators in order to manage our business and evaluate our financial and operating performance on a consolidated basis. Those indicators include revenue, cost of revenues and gross margins, operating expenses, liquidity and cash flows, balance sheet."*

Exhibit 32 - FFIV Leading Metrics Y/Y Growth Rates FY18A-FY21A

Exhibit 32 shows Y/Y growth of FFIV's key leading metrics since FY18A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com

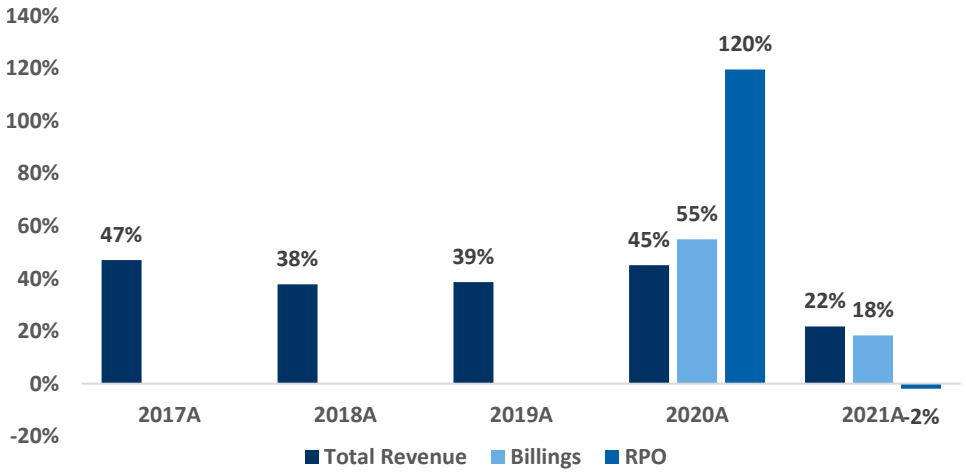


## Fastly, Inc. (FSLY) – covered by Rishi Jaluria

**Leading metric practice/relevance:** Fastly is a consumption-based model with customers invoiced monthly in arrears, therefore billings/RPO are not useful leading indicators.

Exhibit 33 shows Y/Y growth of FSLY's revenue, billings, and RPO since FY18A.

Exhibit 33 - FSLY Leading Metrics Y/Y Growth Rates, 2017A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Fortinet (FTNT) – covered by Dan Bergstrom

**Key metric(s):** Billings, ST Billings, cRPO, RPO

**Leading metric practice/relevance:** With revenue comprised of a mix of product, subscription and services, the company looks to billings as a leading metric for growth with billings reflecting the standard definition of revenue plus/minus the change in deferred revenue. Billings has provided visibility into a significant portion of future revenue at an average of 98% of next-year revenue while short-term billings has provided visibility into an average of 89% of next year revenue. cRPO contributes a moderate amount to revenue in any given year with cRPO coverage trends at ~50% over the last several years. The company guides to billings, which are reported with quarterly results; cRPO and RPO are reported in 10-Qs/10-Ks.

**Relevant commentary:** (CFO Keith Jensen on the Q4/21 earnings call) *"Bookings represent the value of all orders received from customers. Backlog represents the value of all orders received but not fulfilled. When an order is fulfilled, we recognize both billings and product revenue."*

(CFO Keith Jensen on the Q4/21 earnings call) *"For us, a booking is a distributor sends an order to us, and they would like to have shipment. So that counts as a booking. And if we ship it, then it becomes a billing and it becomes product revenue. And if we don't, it becomes backlog."*

Billings 10-K definition and explanation as a key metric: "*We define billings as revenue recognized in accordance with generally accepted accounting principles in the United States ("GAAP") plus the change in deferred revenue from the beginning to the end of the period, less any deferred revenue balances acquired from business combination(s) and an adjustment due to adoption of Accounting Standards Update ("ASU") 2021-08 during the period. We consider billings to be a useful metric for management and investors because billings drive current and future revenue, which is an important indicator of the health and viability of our business. There are a number of limitations related to the use of billings instead of GAAP revenue. First, billings include amounts that have not yet been recognized as revenue and are impacted by the term of FortiGuard security and FortiCare support agreements. Second, we may calculate billings in a manner that is different from peer companies that report similar financial measures.*"

Deferred revenue definition and explanation as a key metric: "*Our deferred revenue consists of amounts that have been invoiced but that have not yet been recognized as revenue. The majority of our deferred revenue balance consists of the unrecognized portion of service revenue from FortiGuard and other security subscription and FortiCare technical support service contracts, which is recognized as revenue ratably over the contractual service period. We monitor our deferred revenue balance, deferred revenue growth and the mix of short-term and long-term deferred revenue because deferred revenue represents a significant portion of free cash flow and of revenue to be recognized in future periods*."

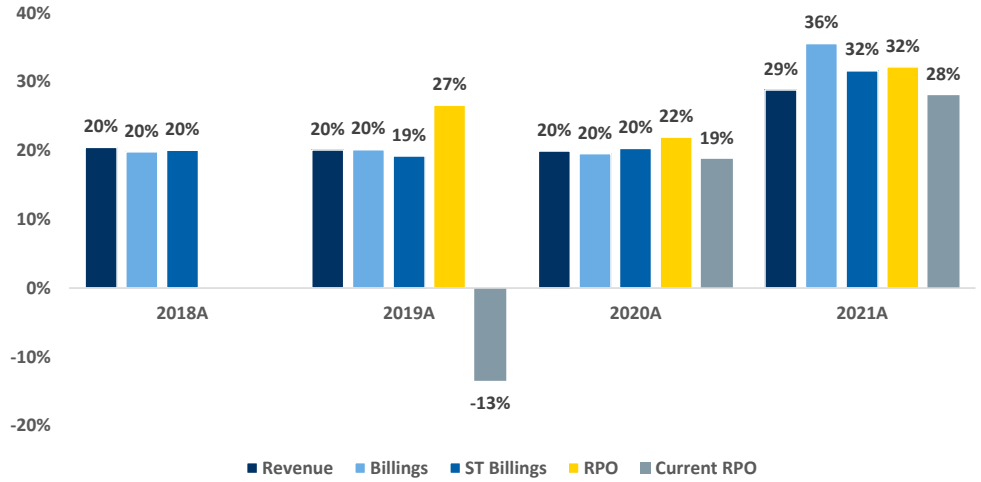
Performance obligation definition and explanation as a key metric: "*Our sales contracts typically contain multiple deliverables, such as hardware, software license, security subscription, technical support services and other services, which are generally capable of being distinct and accounted for as separate performance obligations. Our hardware and software licenses have significant standalone functionalities and capabilities. Accordingly, the hardware and software licenses are distinct from the security subscription and technical support services, as a customer can benefit from the product without the services and the services are separately identifiable within a contract. We allocate a transaction price to each performance obligation based on relative standalone selling price*."

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 34 shows Y/Y growth of FTNT's key leading metrics since 2018.

Exhibit 34 - FTNT Leading Metric Y/Y Growth Rates 2018A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Gitlab (GTLB) – covered by Matt Hedberg

**Key metric(s):** ST Billings, RPO

**Leading metric practice/relevance:** We believe that both revenue and RPO are better indicators than billings. This sentiment has been reiterated by management as it does not consider billings as a meaningful indicator of its business, which further supports our thesis that there are many limitations associated with billings. That said, management does think ST billings could be a useful metric. Additionally, management defines and provides insight into ARR as it uses it to break down its customer base and achieve the dollar-based net retention rate or DBNRR, which we also think provides good visibility into the stickiness of Gitlab's customer base. Overall, we think that revenue is a good indicator because a majority of its revenue is subscription-based along with RPO, but do not think billings is a great metric.

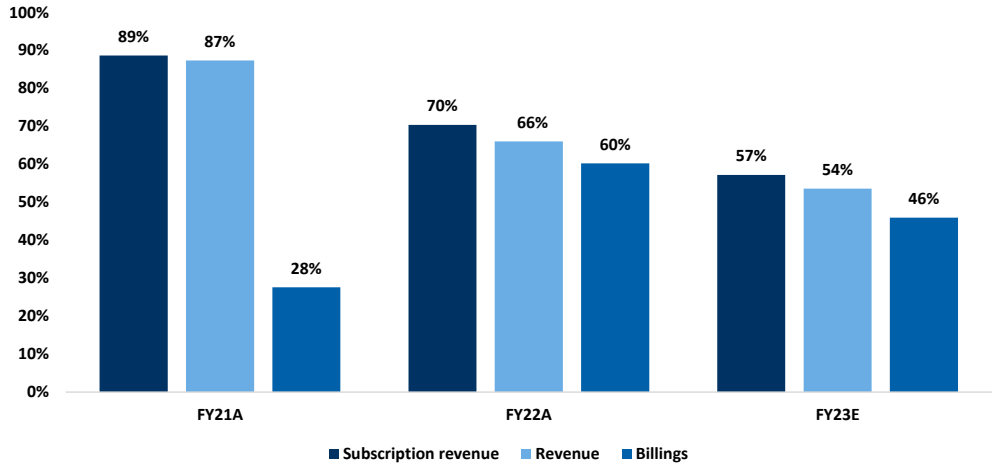
**Relevant commentary:** (CFO Brian G. Robins Q3 2022 earnings call) "*Billings is a little choppy, and so we tried to steer people away from billings when we're on the road*."

(CFO Brian G. Robins Q4 2022 earnings call) *"As many of you know, we do not believe calculated billings to be a good indicator for our business, given that prior period comparisons can be impacted by a number of factors, most notably our history of large prepaid multiyear deals."*

(ARR 10-K definition and explanation as a key metric) "*We calculate ARR by taking the monthly recurring revenue, or MRR, and multiplying it by 12. MRR for each month is calculated by aggregating, for all customers during that month, monthly revenue from committed contractual amounts of subscriptions, including our self-managed and SaaS offerings but excluding professional services."*

Exhibit 35 - GTLB's Leading Metrics Y/Y Growth Rates FY21A-FY23E

Exhibit 35 shows Y/Y growth of GTLB's key leading metrics since FY21A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Guidewire Software, Inc. (GWRE) – covered by Rishi Jaluria

**Key metric(s):** ARR, Cloud ARR, Fully Ramped ARR

**Leading metric practice/relevance:** Guidewire's financial model and visibility are tricky given the cloud transition. Guidewire reports and explicitly guides to ARR, which the company views as the best leading indicator. It provides incremental annual disclosures on cloud ARR, InsuranceSuite cloud ARR, and fully ramped ARR. Fully ramped ARR, in our view, is the best leading indicator for Guidewire since this calculation normalizes for large cloud contracts with multi-year ARR ramps (note we can also back into Fully Ramped Cloud ARR). In terms of the pace of the migration, management points investors to InsuranceSuite Cloud ARR (since it is the core platform and strips out cloud-only add-ons and InsuranceNow). We would also add that the focus on ARR means management relies less on unnatural contract structures for on-premise term licenses (to dampen the impacts of ASC 606). Last, we do not believe billings is currently useful given the cloud transition.

**Relevant commentary:** (CFO Jeff Cooper during FY21 Analyst Day) "*I wanted to take a quick step back to discuss why we are focused on ARR as opposed to our reported revenue. Over the last couple of years, we have focused 100% of internal Guidewire on executing to our ARR targets. We also think about expense ratios as compared with ARR. This is critical to drive the right business outcomes for Guidewire. The adoption of ASC 606, which materially changed the revenue patterns for term license revenue, combined with the increased cloud sales and the resulting ratable revenue recognition means we have a variety of revenue recognition patterns. While the revenue patterns may vary wildly, the cash collection cycle is very consistent. Generally our customers pay us annually upfront.*

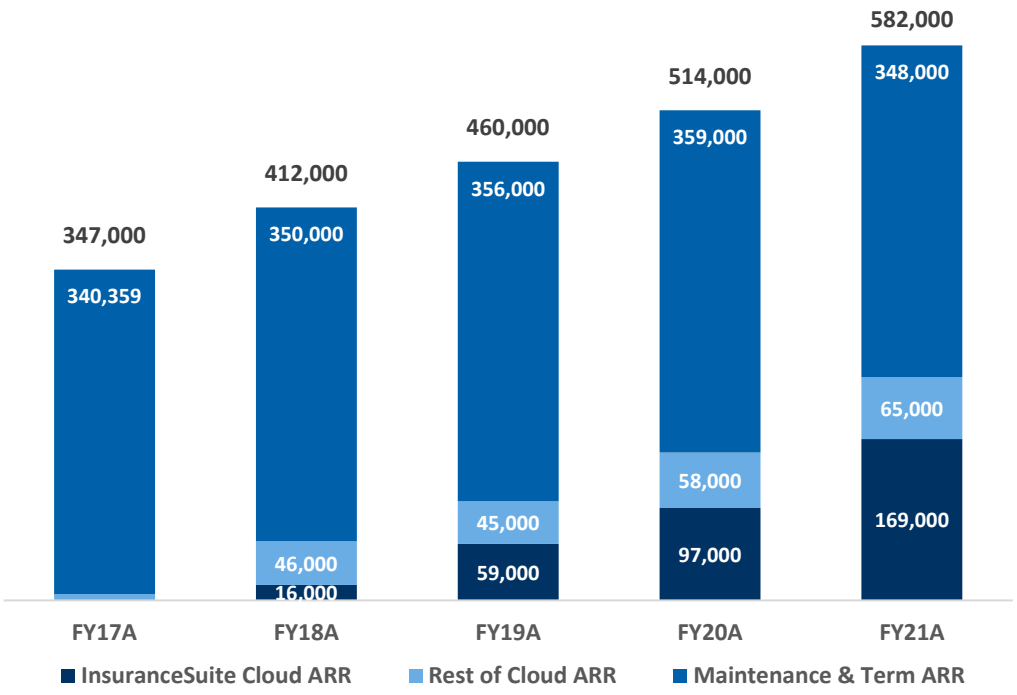
*Our standard term license contracts are two-year initial terms followed by annual renewals. However, in some instances, we will see deal durations that are longer than our standard terms and it is possible to see shorter deal durations as well. I will also note that a big opportunity in front of us continues to be the opportunity to migrate existing term license customers to our cloud products. This creates additional income statement complexity, because in the migration, the customer continues to use their on-premise software for a transition period. As a result, when we sell the cloud deal, some of the software revenue will be recognized to license and some will be allocated to subscription, which can be a bit counterintuitive.*

*While we're in the early stages of the cloud core system demand curve, we are seeing strong momentum with our cloud products. This is the view of our ARR by total cloud ARR and InsuranceSuite Cloud ARR… Another metric we have discussed during this transition is fully ramped ARR. Given the strategic and long-term nature of our customer relationships, it's common for large cloud projects to have a multi-year ARR ramp embedded into the customer agreement. As a result, our reported ARR from a new cloud customer is often significantly larger in year five than in year one… As we add new cloud customers and layer on new cohorts, we expect this to create a tailwind to future ARR growth and this will help with overall visibility into ARR.*"

This report is intended solely for cbass@marketstreetpartners.com

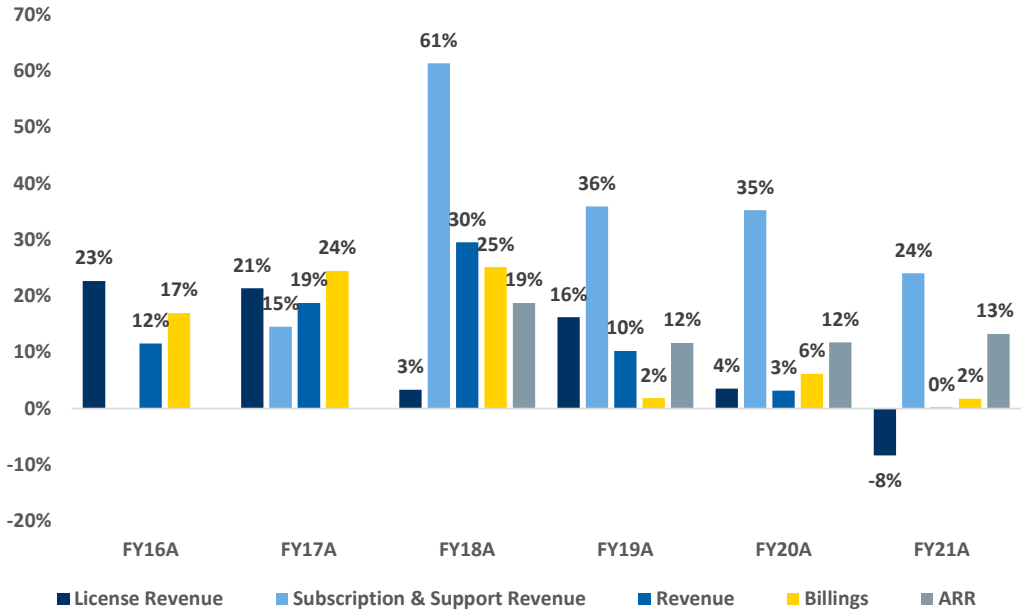


Exhibit 36 - GWRE ARR ($ in Thousands), FY17A-FY21A




Source: Company reports, RBC Capital Markets

Exhibit 37 - GWRE Leading Metrics Y/Y Growth Rates, FY16A-FY21A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## HubSpot Inc. (HUBS) – covered by Rishi Jaluria

**Key metric(s):** Billings, ARR

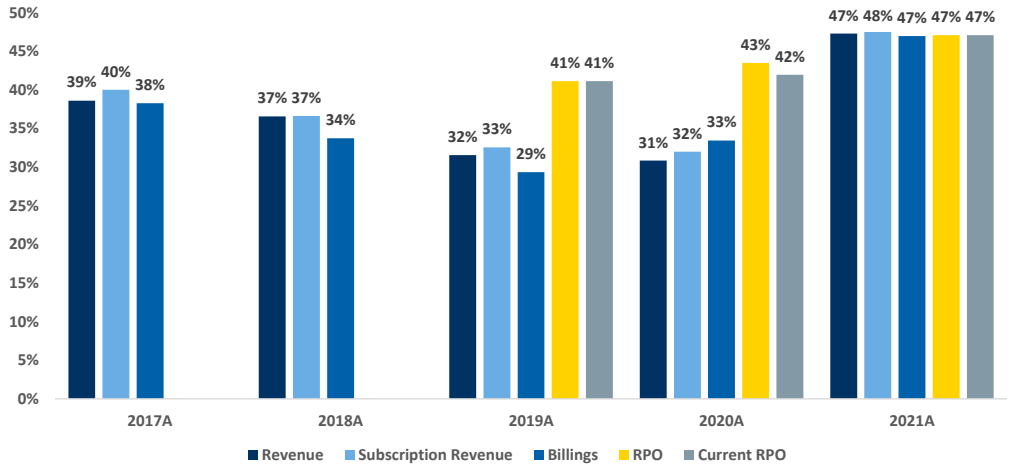
**Leading metric practice/relevance:** In our view, HubSpot's best leading metric is ARR since contracts are mostly one year or less in duration, though we believe most investors still pay attention to billings since ARR is only disclosed semi-annually (we do estimate ARR on a quarterly basis). While we would like to see regular ARR disclosures, management does provide helpful commentary on potential billings impacts from currency, co-terming, and duration. HubSpot's cRPO coverage is low (based on RPO expected to be recognized as revenue in the next 24 months), which to us reinforces the short contract durations. That said, we look to cRPO/RPO as indicators of HubSpot's progress becoming stickier and moving upmarket (longer, larger contracts should boost RPO).

**Relevant commentary:** (10-K commentary on billings) "*Most of our customers' subscriptions are 1 year or less in duration. Subs are billed in advance on various schedules. Because mix of billing terms for orders can vary from period to period, the annualized value of the orders we enter into with our customers will not be completely reflected in D/R at any single point in time. Accordingly, we do not believe that change in D/R is an accurate indicator of future revenue.*"

(CFO Kate Bueker on the 4Q21 earnings call) "*Constant currency revenue growth in Q4 exceeded constant currency billings growth, primarily as a result of the strong installed base selling mix in the quarter and modestly lower overall billing duration.*"

(Analyst Day slides definition of ARR) "*We define "ARR" as the annual value of our customer subscription contracts as of the specified point in time excluding any commissions owed to our partners. For each Hub, this is the sum of customer ARR for the Starter, Basic, Professional and Enterprise subscriptions, plus applicable Contacts (Marketing Only) or Add-Ons (e.g. Reporting or Ads). For multi-product customers, ARR would be distributed across based on the value of each SKU/Hub for which they pay. ARR can differ from Revenue due to several factors. ARR is converted into USD at fixed rates held consistent over time and may vary from those used for Revenue or D/R. ARR would exclude any impact for Bad Debt and Partner Commissions (as noted above) and would also differ from Revenue due to timing of revenue recognition.*"

Exhibit 38 - HUBS Leading Metrics Y/Y Growth Rates, 2017A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Informatica (INFA) – covered by Matt Hedberg

**Key metric(s):** Cloud ARR, Subscription ARR, ARR

**Leading metric practice/relevance:** We believe that ARR is a key leading metric for Informatica and provides good visibility into the health of the company. Breaking it down further, we like to look at Subscription ARR along with Cloud ARR. Additionally, management also guides to Total ARR and Subscription ARR along with revenue, operating income, and unlevered free cash flow. Overall, we think that these metrics are good indicators for the business and like the breakdown of its ARR, both in its results and in guidance, which we think helps investors better predict its growth trajectory. We'd note that the company expects to recognize 65% of its RPO as revenue within the next year and the remainder thereafter.

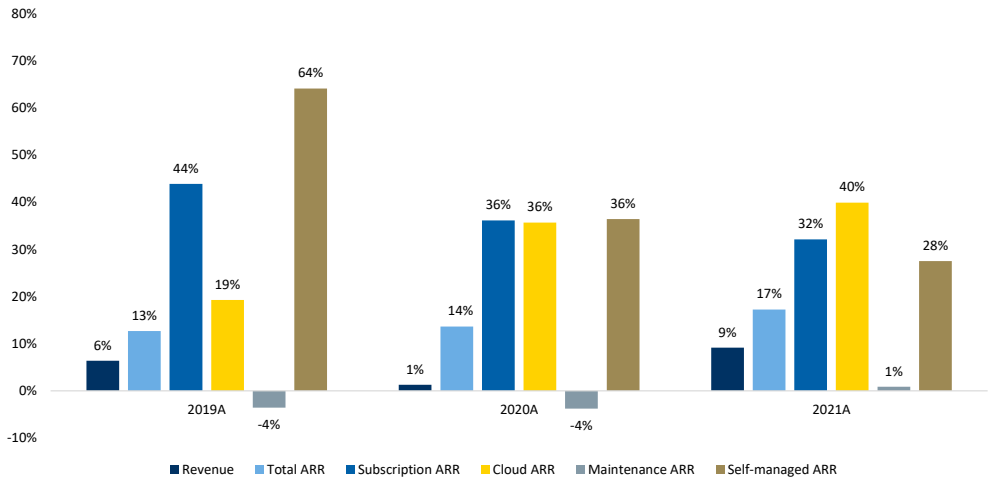
**Relevant commentary:** (CFO Eric F. Brown Q3 2021 earnings call) *"We provide guidance and disclosure on ARR metrics to provide an enhanced view of our business as ARR closely tracks the recurring cash flow characteristics of our business."*

(10-K)*"We believe ARR is an important metric for understanding our business since it tracks the annualized cash value collected over a 12-month period for all our recurring contracts, irrespective of whether it is a maintenance contract on a perpetual license, a ratable cloud contract, or an on-premises term-based subscription license."*

(ARR 10-K definition and explanation as a key metric) *"Annual Recurring Revenue (ARR) represents the expected annual billing amounts from all active maintenance and subscription agreements. ARR is calculated based on the contract Monthly Recurring Revenue (MRR) multiplied by 12. MRR is calculated based on the accounting adjusted total contract value divided by the number of months of the agreement based on the start and end dates of each contracted line item. The aggregate ARR calculated at the end of each reported period represents the value of all contracts that are active as of the end of the period, including those contracts that have expired but are still under negotiation for renewal."*

Exhibit 39 - INFA's Leading Metrics Y/Y Growth Rates 2019A-2021A

Exhibit 39 shows Y/Y growth of INFA's key leading metrics since 2019.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Jamf (JAMF) – covered by Matt Hedberg

**Key metric(s):** ARR

**Leading metric practice/relevance:** We like that Jamf's leading metric is ARR and believe it is the strongest metric as management notes that it mitigates fluctuations from seasonality, sales mix for subscriptions of SaaS and licenses, and contract terms. That said, management does not guide to ARR, and we feel that ARR guidance would help provide a more comprehensive picture of the business as well as management's confidence in its outlook. We'd also noted that in 2020 the company had 80% of RPO, which includes deferred revenue, to be recognized as revenue in the next 12 months and the remainder thereafter.

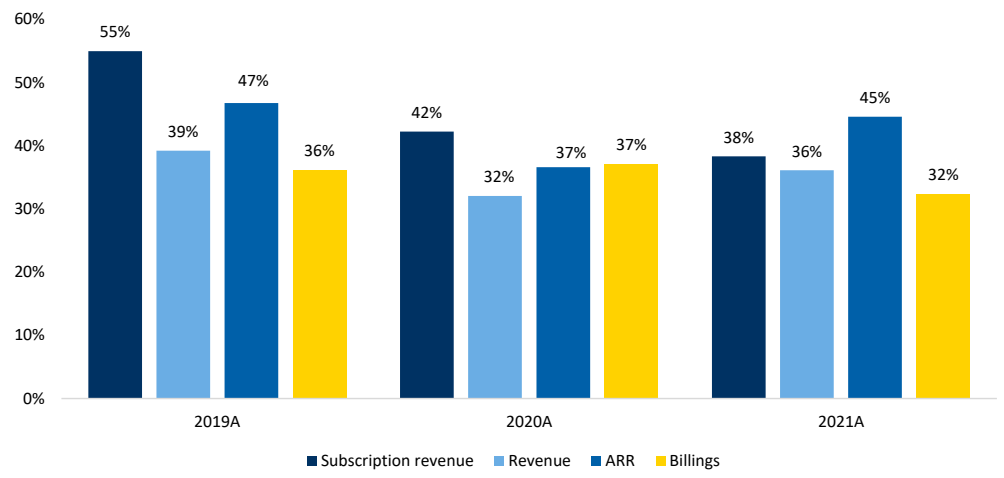
**Relevant commentary:** (Dean J. Hager CEO Q3 2021 earnings call) *"Going forward, we will not break out ARR growth by product or acquired company."*

(CFO Jill Putman Q2 2021 earnings call) *"The three primary drivers of our ARR growth are our consistently high device expansion rates, our strong new logo acquisition and the upselling and cross-selling of products into our installed base. We expect to continue benefiting from these trends going forward."*

(10-K definition and explanation as a key metric) *"Annual Recurring Revenue ("ARR") represents the annualized value of all subscription and support and maintenance contracts as of the end of the period."*

Exhibit 40 - JAMF's Leading Metrics Y/Y Growth Rates 2019A-2021A

Exhibit 40 shows Y/Y growth of JAMF's key leading metrics since 2019.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



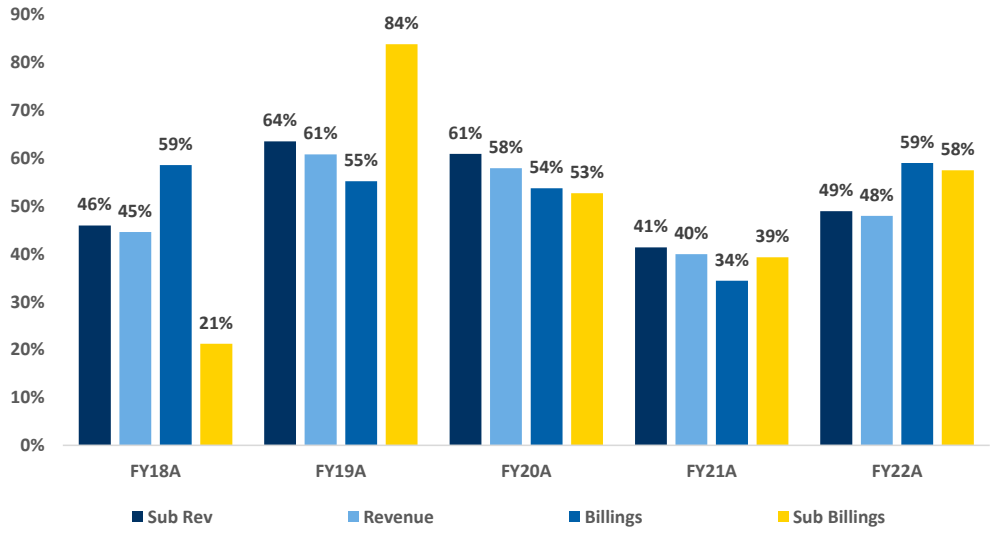
## MongoDB, Inc. (MDB) – covered by Rishi Jaluria

**Key metric(s):** Billings

**Leading metric practice/relevance:** Management infrequently discusses billings as it has become less relevant over time with Atlas mix-shift (since Atlas is consumption-based and billed in arrears, the majority doesn't flow into deferred revenue). We would note that alongside this, recognized deferred revenue used to account for ~60% of quarterly revenue, but now that number is 40%. Having said that, investors still do pay attention to billings, especially since MongoDB's RPO excludes contracts that are 12 months or less. We believe MDB could benefit from disclosing an ARR figure that normalizes EA for ASC 606 and captures Atlas based on the last month of the quarter (including pre-billed or pre-committed).

**Relevant commentary:** (COO/CFO Michael Gordon on F3Q20 earnings call) "*On your billings-related question… I think from the very beginning… we've highlighted how that's not a metric that we use internally or one that we focus on. I, obviously, understand that it's a popular one that investors and other folks look at. We tend to focus on the revenue… I think the other piece that's particularly challenging about billings, which we've talked about is, we're looking at sort of the revenue plus the change in deferred; historically, the change in deferred has really been the place where people have chosen to -- or people have a chance to outperform in recurring revenue subscription businesses, and I think Atlas sort of changes that. And so looking at deferred in isolation, in particular, we've heavily discouraged people from billings. We also are going to be, as I mentioned in the prepared remarks, continuing to focus on driving ease of adoption and ease of use of adopting Atlas, and that will continue to most likely involve a higher percentage of Atlas-related customers paying monthly for invoicing in arrears rather than signing big, large commitments upfront and paying for those, which would generate incremental deferred revenue. So I think while it's not a metric that we've ever focused on, I think you're correct it's of decreasing value or insight relative to understanding the business. We have looked at other metrics to try to discern or determine if there are other metrics that are useful. Unfortunately, we haven't found anything that actually provides an incrementally helpful view of the world, but we'll continue to look.*"

Exhibit 41 - MDB Y/Y Growth Rates, FY18A-FY22A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Microsoft Corporation (MSFT) – covered by Rishi Jaluria

**Leading metric practice/relevance:** Microsoft does disclose commercial bookings growth and commercial RPO in quarterly earnings releases, though regularly cautions investors of the volatility in these metrics given dynamics in the expiry base, contract terms, mix of contracts (e.g. term, subscription, consumption), and contract duration (note these are not "current" metrics i.e. Microsoft does not specify for commercial RPO that are expected to be recognized in the next 12 months). We do not believe investors pay too close attention to these metrics, though they do tell us the general health of the Microsoft Cloud (formerly Commercial Cloud) business.

This report is intended solely for cbass@marketstreetpartners.com



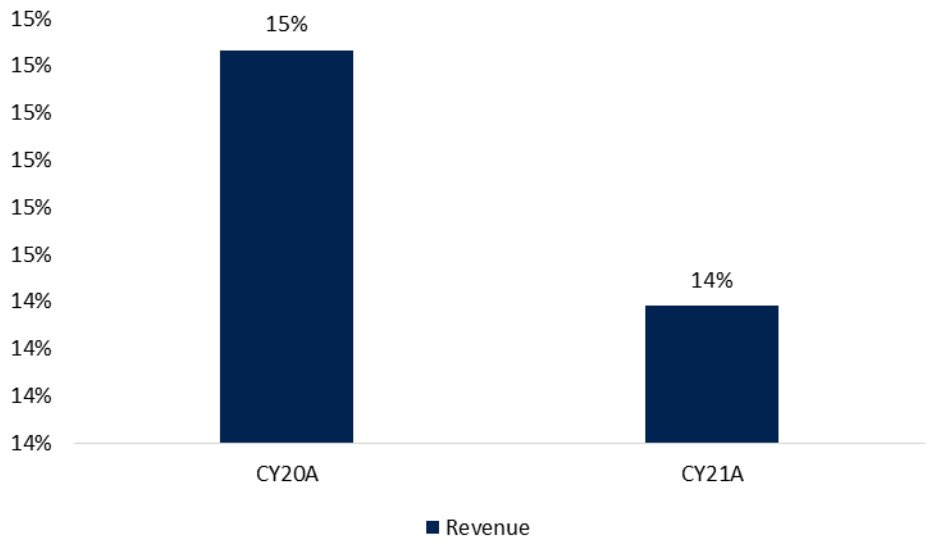
## N-Able (NABL) – covered by Matt Hedberg

**Key metric(s):** N/A

**Leading metric practice/relevance:** We believe revenue is the best metric for N-able given the company doesn't have much in the form of deferred revenue. The company derives ~97% of the revenue from subscriptions (as of CY/21), with substantially all of subscription revenue recognized ratably over the term of the subscription contract. Subscriptions are predominantly sold as SaaS deployments, while they also offer subscriptions with self-managed deployments. Management guides to revenue but doesn't focus on ARR, cRPO, RPO, billings or deferred revenue due to its business model.

Exhibit 42 shows Y/Y growth of NABL's key leading metrics since 2020.

Exhibit 42 - NABL Leading Metric Y/Y Growth Rates 2020-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



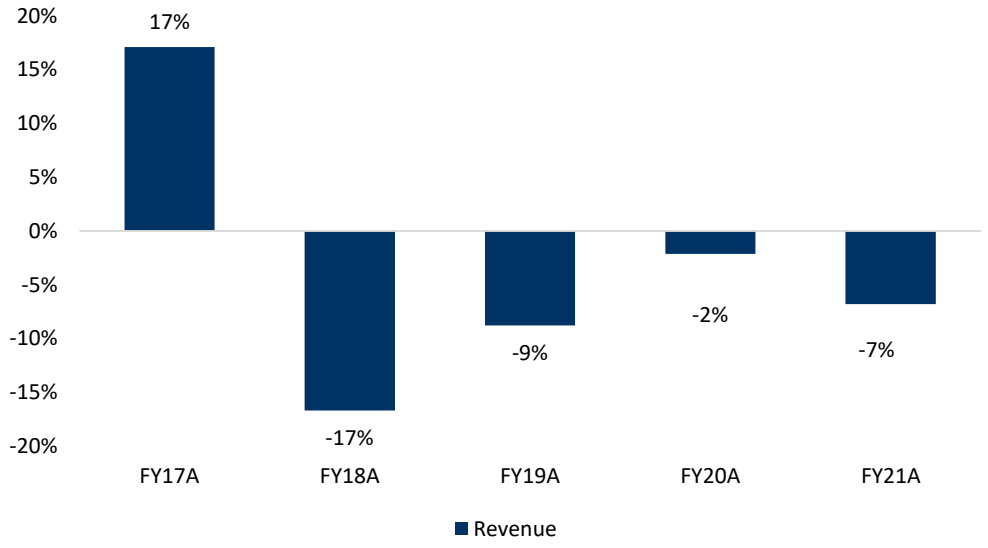
## NetScout (NTCT) – covered by Matt Hedberg

**Key metric(s):** N/A

**Leading metric practice/relevance:** Management guides to revenue but doesn't focus on ARR, cRPO, RPO, billings or deferred revenue due to its business model. NetScout generates 45% of revenue from product, which consists of sales of both hardware products and licensing of software products. Given the business model is largely on-premise (non-recurring), we believe there are no applicable key leading metrics. The company discloses product backlog in 10-Q/Ks, but we do not see it as a meaningful metric for indication of future sales levels.

Exhibit 43 shows Y/Y growth of NTCT's key leading metrics since FY17.

Exhibit 43 - NTCT Leading Metric Y/Y Growth Rates FY17A-FY21A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



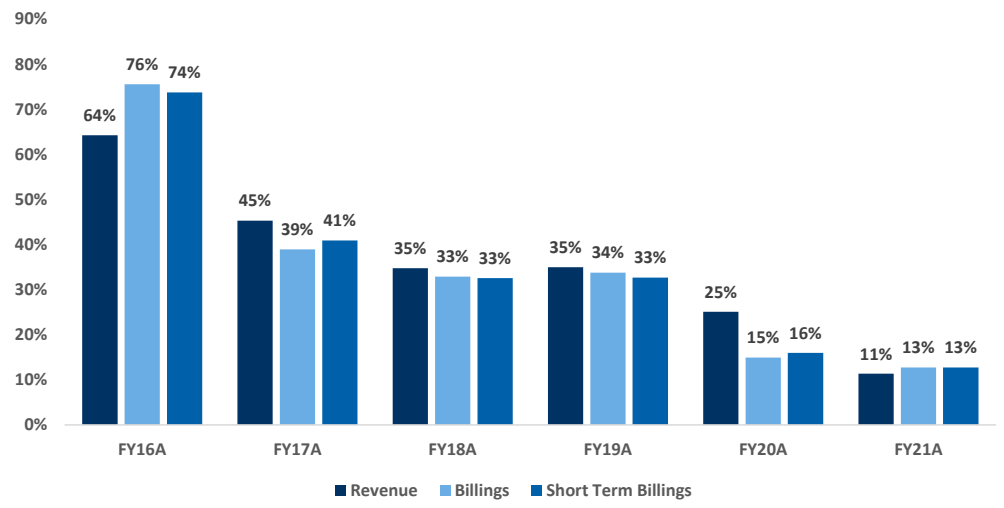
## New Relic, Inc. (NEWR) – covered by Rishi Jaluria

**Leading metric practice/relevance:** Management does not view ARR, billings, or RPO as useful leading indicators following New Relic's shift to consumption-based pricing, given the sales motion is now more focused on driving consumption and not upfront commitments (resulting in lower upfront commitments) and when variable consideration is recognized as revenue, it actually accelerates the drawdown of deferred revenue. Management contends data ingest volume / usage is a better leading indicator, though we believe investors would appreciate more consistent and precise disclosures on data ingest volume / usage.

**Relevant commentary:** (CFO Mark Sachleben) "*In terms of the impact of our transformation on the financial model, I want to highlight a couple things. The first is a reminder that we are no longer disclosing ARR or any ARR-related metrics as those are subscription-based metrics whereas we are now a consumption-based company. The second is that the move to a consumption model have changed the dynamics of deferred revenue and RPO. And those metrics are less useful in understanding the momentum of the business as we continue our transition. Specifically, these changes stem from a number of factors including, first, the aforementioned comp plan change means reps are driving consumption, not commitments. This means no early contract renewals or true-ups and less of a focus on large upgrades at renewal time or upon initial customer acquisition. Second, when we recognize variable consideration, we actually accelerate the drawdown of deferred revenue. Furthermore, incremental consumption in excess of commitment is billed in arrears and thus generally doesn't hit deferred revenue. And finally, some of our largest customers for one reason or another are opting for lower upfront commits in favor of a pay-as-you-consume structure*."

(F3Q22 10-Q filing commentary): "*In addition, the metrics we use to gauge the status and success of our consumption pricing model may continue to evolve as significant trends emerge. For example, beginning with the fiscal quarter ended June 30, 2021, we retired annual recurring revenue ("ARR") and all of our traditional subscription-based key operating metrics that rely upon ARR. In place of ARR and ARR-derived metrics, we are providing metrics that we believe provide better insight into our business now that we are entering into contracts that rely primarily upon consumption-based revenue.*"

Exhibit 44 - NEWR Y/Y Growth Rates, FY16A-FY21A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com

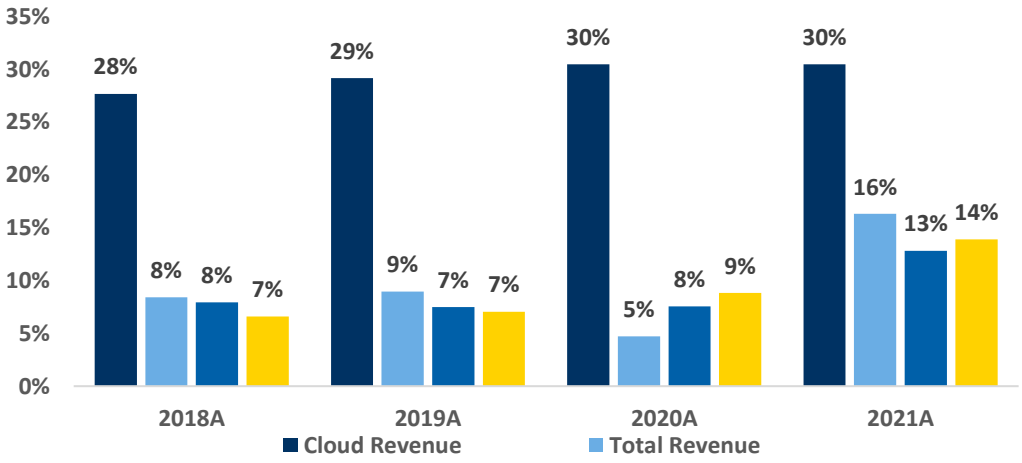


## NICE Ltd. (NICE) – covered by Rishi Jaluria

**Leading metric practice/relevance:** We do not believe billings is a key metric for NICE. NICE's cloud business is mostly comprised of inContact, which is billed in arrears and therefore doesn't have the large deferred revenue component. NICE does disclose RPO, although it uses an impartial definition, excluding contracts of less than one year in length. Management does not comment on these forward-looking metrics but will selectively comment on product-specific bookings. We do believe NICE should explore disclosing ARR as a leading indicator, especially since we believe a good portion of CXone is pre-committed (even if it's billed in arrears).

Exhibit 45 shows Y/Y growth of NICE's cloud revenue, total revenue, and billings since 2018.

Exhibit 45 - NICE Leading Metrics Y/Y Growth Rates, 2018A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## NortonLifeLock (NLOK) – covered by Matt Hedberg

**Key metric(s):** Billings, Bookings

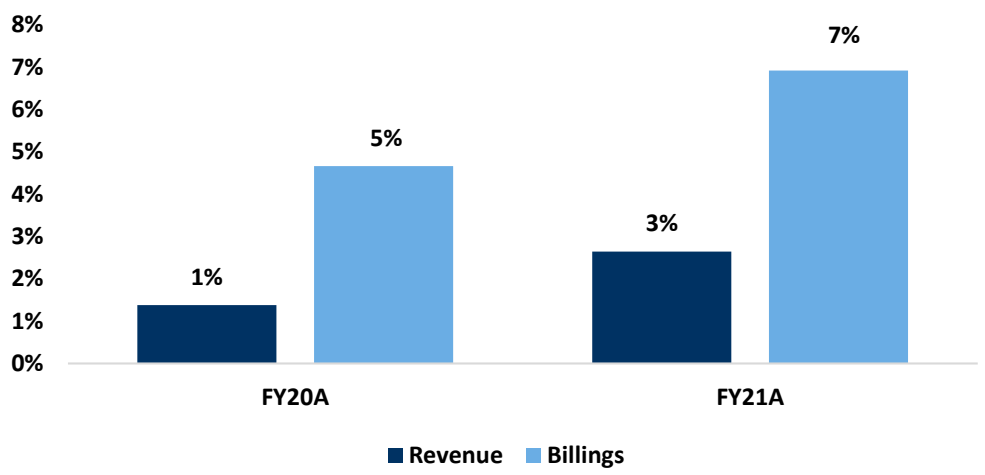
**Leading metric practice/relevance:** NortonLifeLock provides bookings as well as reported billings every quarter, although it doesn't guide to them. We note that both billings and bookings are impacted by differences in the duration of contracts, given the company sells subscriptions on both monthly and annual bases. Additionally, bookings/billings do not provide any insight into how the quarter was broken out between new business and renewals. As a result, we believe it is useful to evaluate bookings/billings in conjunction with other metrics including revenue, new customer additions and retention/renewal rates. We look at cc bookings growth, though we note there is limited history on the metric since it started disclosing in Q1/22. Management expects revenue and bookings to trend about the same way on a multi-quarter basis (not every quarter).

**Relevant commentary:** (Q3/22 earnings deck definition and explanation as a key metric) *"Bookings are defined as customer orders received that are expected to generate net revenues in the future. We present the operational metric of bookings because it reflects customers' demand for our products and services and to assist readers in analyzing our performance in future periods."*

*"We define reported billings as total revenue plus the change in adjusted contract liabilities. The change in contract liabilities excludes the change related to discontinued operations that does not amortize to revenue from continuing operations. There are several limitations related to the use of reported billings versus revenue calculated in accordance with GAAP. First, reported billings include amounts that have not yet been recognized as revenue. Second, our calculation of reported billings may be different from other companies in our industry, some of which may not use reported billings, may calculate reported billings differently, may have different reported billing frequencies, or may use other financial measures to evaluate their performance, all of which could reduce the usefulness of reported billings as a comparative measure. We compensate for these limitations by providing specific information regarding GAAP revenue and evaluating reported billings together with revenue calculated in accordance with GAAP."*

Exhibit 46 - NLOK Leading Metric Y/Y Growth Rates FY20A-FY21A

Exhibit 46 shows Y/Y growth of NLOK's key leading metrics since FY20.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Nutanix (NTNX) – covered by Matt Hedberg

**Key metric(s):** ARR, ACV Billings, Run-Rate ACV, Subscription Billings

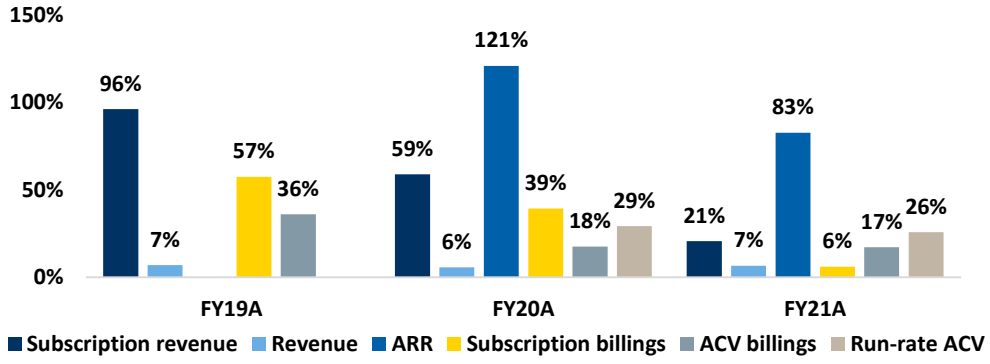
**Leading metric practice/relevance:** We believe ARR is the primary leading metric for Nutanix's subscription business. The company shifted focus from ACV to ARR, with ARR first disclosed in Q4/21. While management will disclose both ACV billings and ARR in the near term, we expect it to shift entirely to ARR over time. We see this as a positive given ARR is more of an industry standard and should help make the bridge to revenue easier to understand. While ACV billings excludes services and hardware, it includes non-portable software (LOD) transactions in addition to subscriptions, while ARR is pure subscription. Management expects average contract duration to continue to decline, expected to stabilize at 2.8-3 years beyond FY/23, after which revenue growth should accelerate and growth for ARR, ACV billings and revenue should start converging. While management guides to revenue and ACV billings, we'd like to see it guide to ARR ultimately.

**Relevant commentary:** (2021 Investor Day deck) *"Run rate ACV was established in lieu of ARR before term-based software business had significant traction. RR ACV includes all LOD (Life of Device) transactions (assumed 5-year term), as well as all subscription transactions. Complexities associated with LOD transactions can alter RR ACV and the associated growth rates. ARR eliminates all LOD transactions. This creates a much more predictable metric while showing the true growth of the subscription business. Growth rates for ARR, ACV billings and revenue converge over time. Going forward, we will morph to focus on ARR in lieu of RR ACV."*

(Q2/22 earnings deck and 10-Q commentary): *"ACV Billings, for any given period, is defined as the sum of the ACV for all contracts billed during the given period. ACV Billings is the sum of New ACV Billings and Renewals ACV Billings. Annual Contract Value, or ACV, is defined as the total annualized value of a contract, excluding amounts related to professional services and hardware. The total annualized value for a contract is calculated by dividing the total value of the contract by the number of years in the term of such contract, using, where applicable, an assumed term of five years for contracts that do not have a specified term. Annual Recurring Revenue, or ARR, for any given period, is defined as the sum of ACV for all non-life-of-device contracts in effect as of the end of a specific period. For the purposes of this calculation, we assume that the contract term begins on the date a contract is booked, unless the terms of such contract prevent us from fulfilling our obligations until a later period, and irrespective of the periods in which we would recognize revenue for such contract."*

Exhibit 47 - NTNX Leading Metric Y/Y Growth Rates FY19A-FY21A

Exhibit 47 shows Y/Y growth of NTNX's key leading metrics since FY19.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Okta (OKTA) – covered by Matt Hedberg

**Key metric(s):** cRPO, cRPO Bookings, RPO, Billings

**Leading metric practice/relevance:** With 96% of revenue generated from subscriptions, we believe cRPO and RPO are good leading indicators for the overall health of the business. Looking at the historical cRPO coverage trend, it has remained between 65-71% of FY+1 revenue, which we believe provides helpful insight into the future revenue trend. Management believes cRPO is a strong forward-looking metric and is more insightful than billings, or ARR. While it currently guides to billings, we wouldn't be surprised to see it guide to cRPO at some point in the future. We like the move to cRPO as we believe it is a cleaner metric than billings, which remains unaffected by invoicing timing as well as duration challenges. We also note than less than 50% of revenue in any given year comes from deferred revenue.

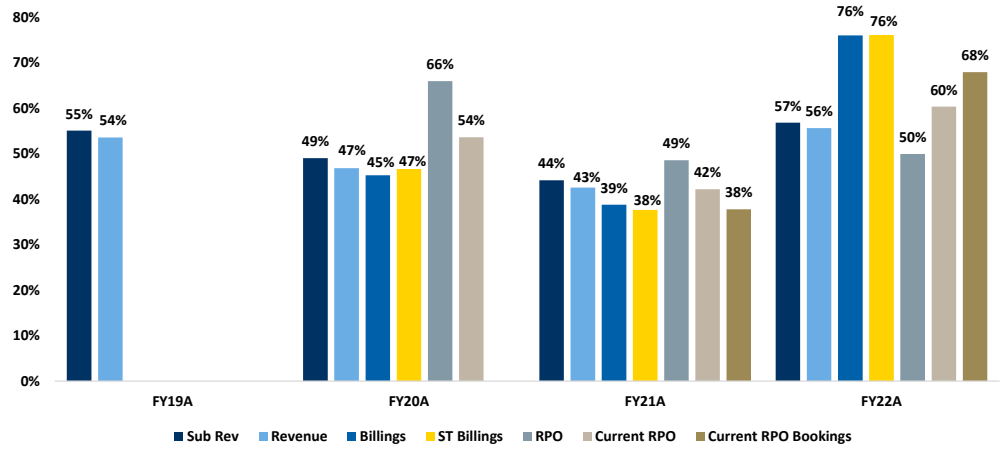
**Relevant commentary:** (CFO Brett Tighe on Q3/22 earnings call) *"We continue to believe that current RPO is a better metric than calculated billings to measure quarterly performance as it provides a more predictable and consistent view of the business."*

(10-K definition and explanation as a key metric) *"RPO represent all future, non-cancelable, contracted revenue under our subscription contracts with customers that has not yet been recognized, inclusive of deferred revenue that has been invoiced and non-cancelable amounts that will be invoiced and recognized as revenue in future periods. Current RPO represents the portion of RPO expected to be recognized during the next 12 months. RPO fluctuates due to a number of factors, including the timing, duration and dollar amount of customer contracts.*

*Calculated Billings represent our total revenue plus the change in deferred revenue, net of acquired deferred revenue, and less the change in unbilled receivables, net of acquired unbilled receivables, in the period. Calculated Billings in any particular period reflect sales to new customers plus subscription renewals and upsells to existing customers, and represent amounts invoiced for subscription, support and professional services. We typically invoice customers in advance in annual installments for subscriptions to our platform."*

Exhibit 48 shows Y/Y growth of OKTA's key leading metrics since FY19.

Exhibit 48 - OKTA Leading Metric Y/Y Growth Rates FY19A-FY22A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



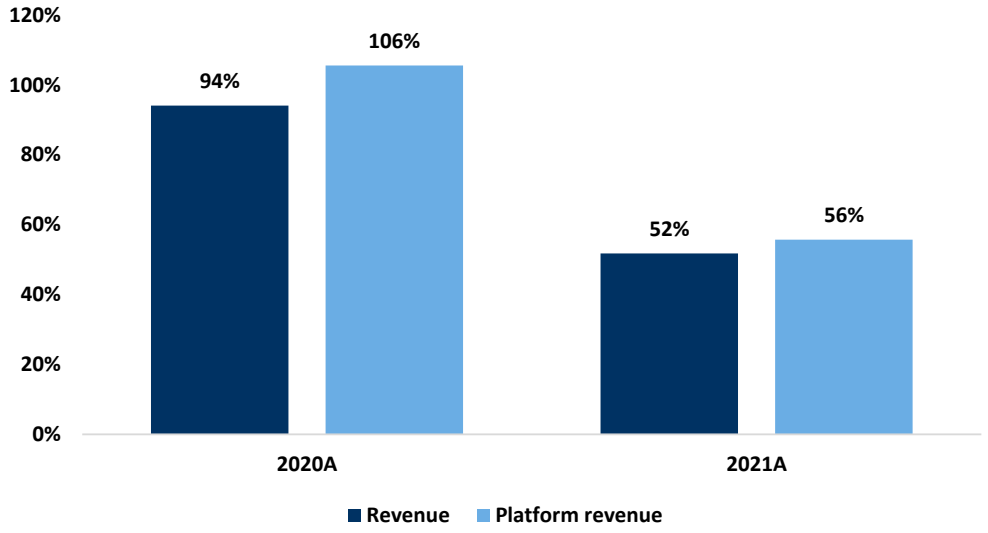
## Olo (OLO) – covered by Matt Hedberg

**Key metric(s):** N/A

**Leading metric practice/relevance:** Olo is a transactional SaaS model, which means revenue is generated through the sale of subscriptions and transaction-based revenue streams. In CY/21, 46% of the platform revenue was subscription revenue while 54% was transactional revenue, as transactional revenue continues to trend up. Additionally, Olo bills monthly in arrears, which means deferred revenue, billings and RPO are less relevant for the company. As a result, we focus on revenue.

Exhibit 49 shows Y/Y growth of OLO's key leading metrics since 2020.

Exhibit 49 - OLO Leading Metric Y/Y Growth Rates 2020A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## PagerDuty (PD) – covered by Matt Hedberg

**Key metric(s):** ARR, TTM billings, Billings

**Leading metric practice/relevance:** We believe ARR should be the primary growth metric for the business. PagerDuty discloses and guides to billings and TTM billings on a quarterly basis while ARR was disclosed for the first time in Q4/22. Management expects to provide ARR on an annual basis for now. Management believes ARR provides better value as a long-term metric, as its bookings are not linear from quarter to quarter, which can make quarterly ARR more variable. Management believes looking at annual ARR in conjunction with quarterly billings trend should be useful. As such, we would like to see a more frequent quarterly ARR disclosure at some point and a move away from billings. We believe ARR is a cleaner growth metric than billings, as it is unaffected by co-terming and duration. Additionally, we note only ~44% and 41% of total revenue in FY/22 and FY/21, respectively, was from deferred revenue.

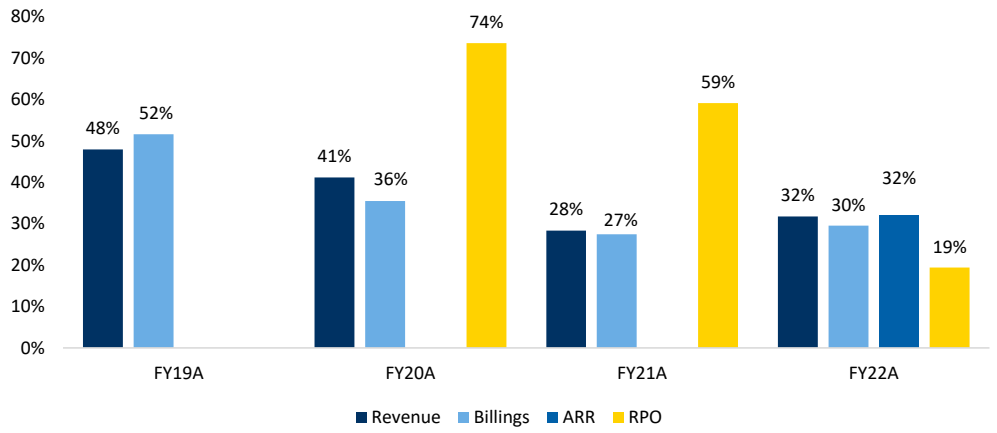
**Relevant commentary:** (CFO Howard Wilson on Q4/22 earnings call) *"I would like to introduce the second annual metric, annual recurring revenue, or ARR. Given the increasing scale of PagerDuty and inherent fluctuation in quarterly billings, we wanted to provide additional color on the fiscal year…. So our plan is to provide it on an annual basis, just because we feel that there's – some of the movements by quarter effectively could end up disclosing proprietary movement from a company perspective. So we're thinking about this on an annual basis. But obviously, I think when people look at that, triangulate that with the billings information that we'll be sharing, they're going to have a much better view. And we'll continue to provide guidance around our quarterly billings and trailing 12-month billings to give you that more complete picture*."

(CFO Howard Wilson on Q2/22 earnings call) *"As a reminder, we focus on trailing 12 months billings to eliminate some of the noise associated with the variable timing of renewals and co-terms."*

(10-K definition and explanation as a key metric) "*We define ARR as the annualized recurring value of all active contracts at the end of a reporting period."*

Exhibit 50 - PD Leading Metric Y/Y Growth Rates FY19A-FY22A

Exhibit 50 shows Y/Y growth of PD's key leading metrics since FY19.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



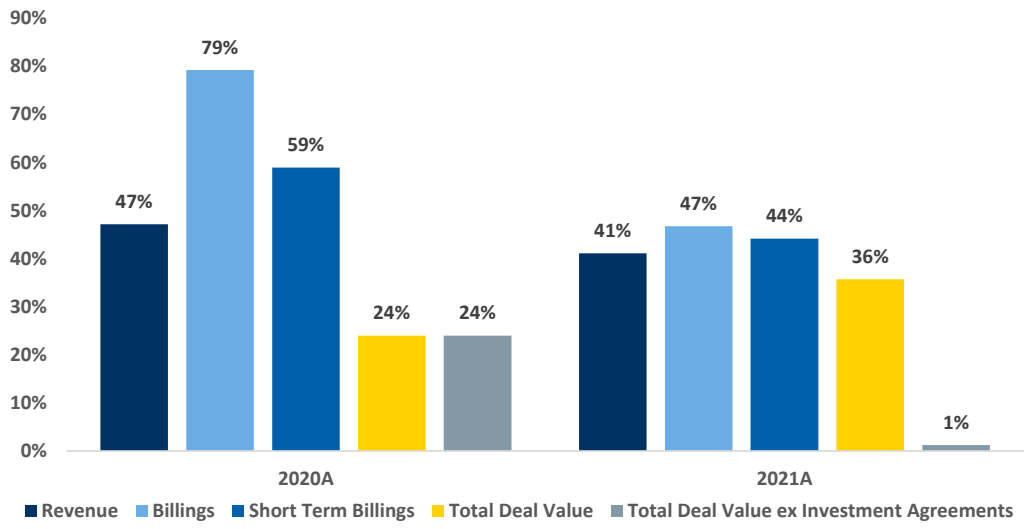
## Palantir Technologies Inc. (PLTR) – covered by Rishi Jaluria

**Key metric(s):** Total Remaining Deal Value

**Leading metric practice/relevance:** We have written extensively on the topic of Palantir's leading indicators. Management points to Total Remaining Deal Value as the primary leading indicator for the business, though we prefer to remove contribution from investment agreements (primarily related to SPACs) given we don't believe many of these deals are sustainable. Because Total Remaining Deal Value is a non-GAAP metric, we believe it is less subject to write-downs than deferred revenue or RPO would be. We also note Palantir's RPO excludes contracts with initial duration of 12 months or less, making the metric less relevant.

**Relevant commentary:** (10-K definition of Total Remaining Deal Value) "*Total remaining deal value is the total remaining value of contracts that have been awarded by our government and commercial customers and includes existing contractual obligations and unexercised contract options available to those customers. Total remaining deal value presumes the exercise of all contract options and no termination of contracts; however, the majority of our contracts are subject to termination provisions, including for convenience, and there can be no guarantee that contracts are not terminated or that contract options will be exercised. Also included within total remaining deal value is remaining contract value from commercial contracts entered into in connection with our strategic investments, many of which are subject to termination, including for convenience in the event the proposed business combination is not completed.*"

Exhibit 51 - PLTR Leading Metrics Y/Y Growth Rates, 2020A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Palo Alto Networks (PANW) – covered by Matt Hedberg

**Key metric(s):** NGS ARR, NGS Billings, cRPO, RPO

**Leading metric practice/relevance:** We believe NGS ARR is the primary growth metric for the health of recurring subscription cloud business. Management first started disclosing NGS ARR in Q3/20 and also provides annual guidance for the metric. In addition to that the company provides NGS billings on a quarterly basis, though we feel NGS ARR is a better, cleaner metric. Palo Alto Networks generates ~74% of the revenue from subscription and support, while 26% of the revenue comprises of product (sales of appliances), which is recognized on an upfront basis. As product revenue continues to decline in the mix, we believe cRPO and RPO can be increasingly helpful, in addition to billings, in evaluating future revenue potential and momentum in the business.

**Relevant commentary:** (CFO Dipak Golechha on Q2/22 earnings call) *"Going forward, we encourage investors to focus on our NGS ARR metric as we view this measure as being more indicative of the underlying drivers of this business. We will not be updating NGS billings in the future."*

(CFO Dipak Golechha at 2021 Analyst Day) *"RPO is a measure of future revenue that is contractually committed. This metric includes both revenue that is on our balance sheet in the form of deferred revenue as well as customer commitments that have not yet been invoiced. You can see the clear relationship between prior-year ending RPO and our current year revenue. The revenue coverage in our current RPO has steadily grown from 49% entering fiscal year 2020 to 58% entering this year. This gives us a greater degree of predictability entering the year, enabling us not only to forecast with more confidence, but also to invest with more confidence. We expect revenue coming from our current RPO will continue to increase as we proceed through our transformation journey over time."*

(10-K definition and explanation as a key metric) "*We define billings as total revenue plus the change in total deferred revenue, net of acquired deferred revenue, during the period. We consider billings to be a key metric used by management to manage our business. We believe billings provides investors with an important indicator of the health and visibility of our business because it includes subscription and support revenue, which is recognized ratably over the contractual service period, and product revenue, which is recognized at the time of shipment, provided that all other conditions for revenue recognition have been met. We consider billings to be a useful metric for management and investors, particularly if we continue to experience increased sales of subscriptions and strong renewal rates for subscription and support offerings, and as we monitor our near-term cash flows."*

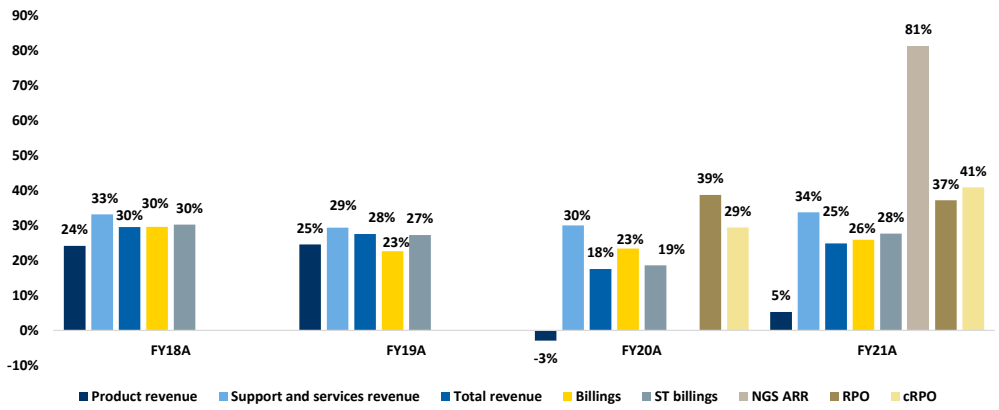
(Q2/22 earnings deck definition and explanation as a key metric) *"ARR = Annualized Recurring Revenue. Next-Gen Security ARR is annualized allocated revenue of all active contracts as of the final day of the reporting period for Prisma and Cortex offerings inclusive of the VM-Series and related services, and certain cloud-delivered security services."*

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 52 shows Y/Y growth of PANW's key leading metrics since FY18.

Exhibit 52 - PANW Leading Metric Y/Y Growth Rates FY18A-FY21A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Pegasystems Inc. (PEGA) – covered by Rishi Jaluria

**Key metric(s):** ACV, RPO

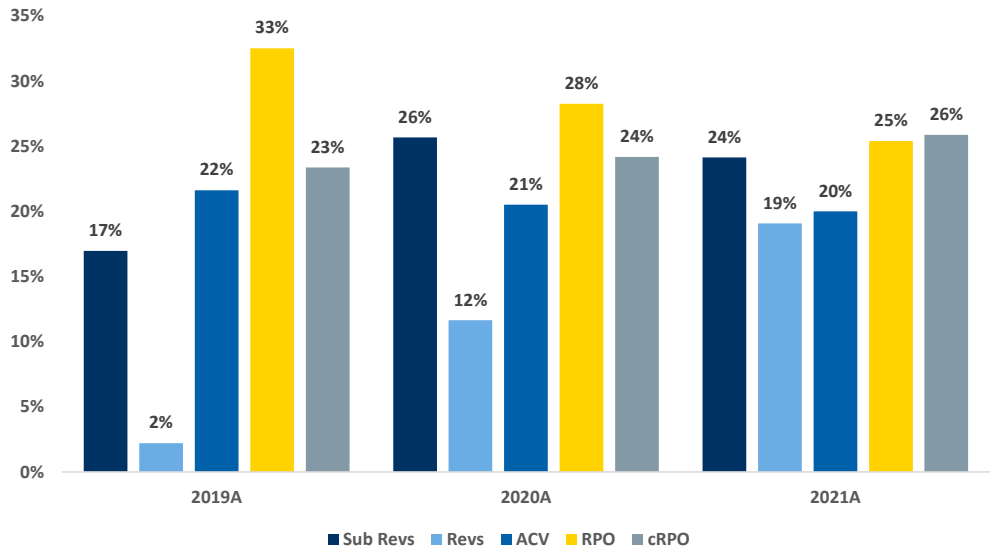
**Leading metric practice/relevance:** Pega is arguably the most transparent company in software when it comes to ACV and RPO disclosures. Pega's ACV (annual contract value) metric is the most important metric both to management and investors, and the company discloses ACV separated by cloud, maintenance, and term (term ACV normalizes for ASC 606-related noise). Management regularly highlights ACV as the most critical metric for investors while Pega is going through its cloud transition (Pega is in year 4 of a ~5-year transition) and has noted that the metric will still be the best leading indicator for a post-transition Pega. Similarly, Pega also provides a detailed breakdown of RPO, separated by source (perpetual license, term license, maintenance, cloud, services) and duration (<1 year, 1-2 years, 2-3 years, 3+ years). In fact, we note Pega is the only software company we are aware of to provide this level of detail, and we note Pega was one of the first software companies to highlight RPO for investors.

**Relevant commentary:** (CFO Ken Stillwell at Investor Conference) *"So, ACV – think of ACV as ARR. They're interchangeable for us. They're really – its annual billings, right? So, we use the annual contract value because of the fact that we sell Pega Cloud and Client Cloud. So, we've just chosen that term. We could just as easily use ARR. So that is the most important measure. RPO is another measure and revenue will just kind of – now we're getting to the point where the revenue is kind of really kind of trailing the ACV in a much more, at least on an annual basis, it's much more connected."*

(10-K Key Metrics commentary on ACV) *"ACV represents the annualized value of our active contracts as of the measurement date. The contract's total value is divided by its duration in years to calculate ACV for term license and Pega Cloud contracts. Maintenance revenue for the quarter then ended is multiplied by four to calculate ACV for maintenance. ACV is a performance measure that we believe provides useful information to our management and investors, particularly during our subscription transition."*

Exhibit 53 - PEGA Leading Metrics Y/Y Growth Rates, 2019A-2021A

Exhibit 53 shows Y/Y growth of PEGA's key leading metrics since 2019.



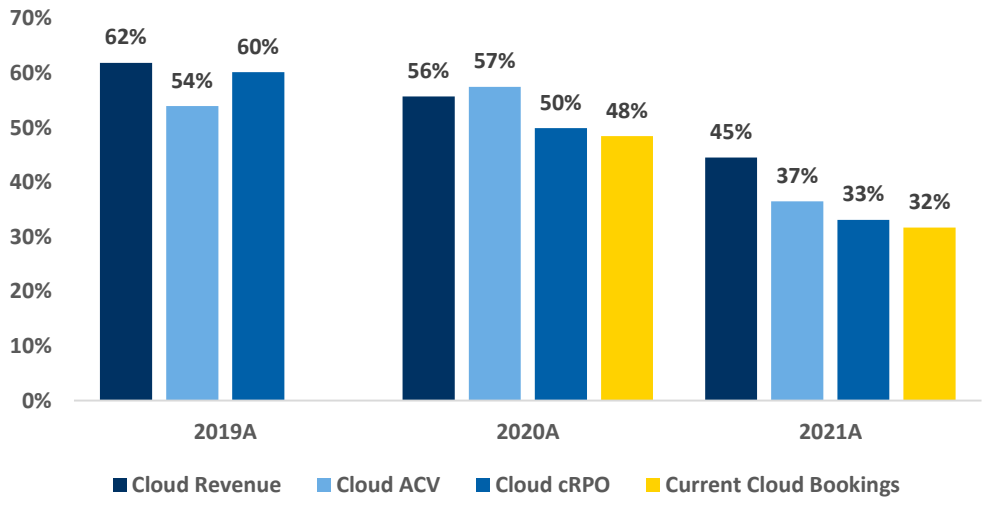

Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 54 shows Y/Y growth of PEGA's leading cloud metrics since 2018.

Exhibit 54 - PEGA Leading Cloud Metrics Y/Y Growth Rates, 2019A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



Importantly, we can derive additional metrics based on the disclosures, which helps us "sanity check" the ACV numbers (since RPO is a GAAP metric). For example, as seen in our new RPO worksheet (Exhibit 55), we can calculate a current cloud bookings numbers (cloud revenue plus the changed in cloud RPO <1 year), which, on a NTM basis, closely tracks with cloud ACV.

## Exhibit 55 - PEGA RPO and ACV Breakdown

| | | | 2020A | | | | | 2021A | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2018A | 2019A | Q1A Mar | Q2A Jun | Q3A Sep | Q4A Dec | 2020A | Q1A Mar | Q2A Jun | Q3A Sep | Q4A Dec | 2021A |
| **Remaining Performance Obligations** | | | | | | | | | | | | |
| **Perpetual License** | **18,669** | **4,484** | **6,163** | **9,820** | **10,408** | **11,909** | **11,909** | **10,278** | **6,941** | **22,971** | **17,709** | **17,709** |
| *YoY Growth* | | *-76%* | *-54%* | *-8%* | *6%* | *166%* | *166%* | *67%* | *-29%* | *121%* | *49%* | *49%* |
| 1 Year or Less | 14,665 | 2,305 | 3,995 | 8,120 | 8,708 | 11,514 | 11,514 | 9,649 | 6,707 | 15,686 | 10,952 | 10,952 |
| 1-2 Years | 2,343 | 2,179 | 2,168 | 1,700 | 1,700 | 395 | 395 | 629 | 234 | 1,064 | 4,505 | 4,505 |
| 2-3 Years | 1,661 | - | - | - | - | - | - | - | - | 4,094 | 2,252 | 2,252 |
| Greater Than 3 Years | - | - | - | - | - | - | - | - | - | 2,127 | - | - |
| **Term License** | **84,380** | **116,833** | **43,189** | **66,843** | **63,185** | **118,814** | **118,814** | **58,331** | **63,018** | **66,802** | **172,155** | **172,155** |
| *YoY Growth* | | *38%* | *-13%* | *53%* | *108%* | *2%* | *2%* | *35%* | *-6%* | *6%* | *45%* | *45%* |
| 1 Year or Less | 72,378 | 97,826 | 30,962 | 53,550 | 50,788 | 105,920 | 105,920 | 41,025 | 46,146 | 49,265 | 153,467 | 153,467 |
| *YoY Growth* | | *35%* | *-30%* | *41%* | *96%* | *8%* | *8%* | *33%* | *-14%* | *-3%* | *45%* | *45%* |
| 1-2 Years | 10,355 | 12,014 | 5,088 | 6,187 | 5,341 | 7,962 | 7,962 | 9,874 | 15,708 | 16,872 | 14,968 | 14,968 |
| *YoY Growth* | | *16%* | *19%* | *32%* | *41%* | *-34%* | *-34%* | *94%* | *154%* | *216%* | *88%* | *88%* |
| 2-3 Years | 1,414 | 3,132 | 6,504 | 6,460 | 7,052 | 4,928 | 4,928 | 7,055 | 909 | 420 | 1,955 | 1,955 |
| *YoY Growth* | | *121%* | *760%* | *908%* | *1093%* | *57%* | *57%* | *8%* | *-86%* | *-94%* | *-60%* | *-60%* |
| Greater Than 3 Years | 233 | 3,861 | 635 | 646 | 4 | 4 | 4 | 377 | 255 | 245 | 1,765 | 1,765 |
| *YoY Growth* | | *1557%* | *370%* | *249%* | *-95%* | *-100%* | *-100%* | *-41%* | *-61%* | *6025%* | *44025%* | *44025%* |
| **Maintenance** | **207,914** | **268,294** | **269,723** | **257,959** | **239,148** | **329,915** | **329,915** | **322,637** | **336,449** | **327,696** | **367,008** | **367,008** |
| *YoY Growth* | | *29%* | *33%* | *33%* | *26%* | *23%* | *23%* | *20%* | *30%* | *37%* | *11%* | *11%* |
| 1 Year or Less | 192,274 | 206,882 | 205,083 | 186,618 | 170,643 | 227,803 | 227,803 | 220,100 | 214,645 | 196,667 | 234,917 | 234,917 |
| *YoY Growth* | | *8%* | *9%* | *8%* | *8%* | *10%* | *10%* | *7%* | *15%* | *15%* | *3%* | *3%* |
| *Coverage (% of NTM Maintenance)* | *69%* | *70%* | *69%* | *61%* | *54%* | *71%* | *71%* | *67%* | *64%* | *57%* | *40%* | *40%* |
| 1-2 Years | 10,436 | 30,291 | 34,633 | 40,153 | 40,631 | 54,509 | 54,509 | 52,366 | 59,164 | 59,360 | 65,502 | 65,502 |
| *YoY Growth* | | *190%* | *270%* | *220%* | *119%* | *80%* | *80%* | *51%* | *47%* | *46%* | *20%* | *20%* |
| 2-3 Years | 3,644 | 17,844 | 19,411 | 20,671 | 18,277 | 28,320 | 28,320 | 33,337 | 36,076 | 37,734 | 38,432 | 38,432 |
| *YoY Growth* | | *390%* | *337%* | *256%* | *120%* | *59%* | *59%* | *72%* | *75%* | *106%* | *36%* | *36%* |
| Greater Than 3 Years | 1,560 | 13,277 | 10,596 | 10,517 | 9,597 | 19,283 | 19,283 | 16,834 | 26,564 | 33,935 | 28,157 | 28,157 |
| *YoY Growth* | | *751%* | *428%* | *274%* | *94%* | *45%* | *45%* | *59%* | *153%* | *254%* | *46%* | *46%* |
| **Cloud** | **298,817** | **422,170** | **414,474** | **457,391** | **506,807** | **590,137** | **590,137** | **564,316** | **602,735** | **571,461** | **731,563** | **731,563** |
| *YoY Growth* | | *41%* | *18%* | *26%* | *39%* | *40%* | *40%* | *36%* | *32%* | *13%* | *24%* | *24%* |
| 1 Year or Less | 103,354 | 165,571 | 174,277 | 191,187 | 211,661 | 248,223 | 248,223 | 252,104 | 281,793 | 284,359 | 330,426 | 330,426 |
| *YoY Growth* | | *60%* | *51%* | *54%* | *58%* | *50%* | *50%* | *45%* | *47%* | *34%* | *33%* | *33%* |
| *Coverage (% of NTM Cloud)* | *77%* | *79%* | *75%* | *74%* | *75%* | *82%* | *82%* | *78%* | *81%* | *48%* | *53%* | *53%* |
| 1-2 Years | 80,214 | 128,109 | 125,473 | 140,860 | 157,500 | 193,064 | 193,064 | 187,456 | 194,841 | 177,214 | 220,231 | 220,231 |
| *YoY Growth* | | *60%* | *37%* | *43%* | *50%* | *51%* | *51%* | *49%* | *38%* | *13%* | *14%* | *14%* |
| 2-3 Years | 61,906 | 84,788 | 81,187 | 88,273 | 93,283 | 104,542 | 104,542 | 91,861 | 88,855 | 79,775 | 124,969 | 124,969 |
| *YoY Growth* | | *37%* | *14%* | *16%* | *28%* | *23%* | *23%* | *13%* | *1%* | *-14%* | *20%* | *20%* |
| Greater Than 3 Years | 53,343 | 43,702 | 33,537 | 37,071 | 44,363 | 44,308 | 44,308 | 32,895 | 37,246 | 30,113 | 55,937 | 55,937 |
| *YoY Growth* | | *-18%* | *-54%* | *-41%* | *-14%* | *1%* | *1%* | *-2%* | *0%* | *-32%* | *26%* | *26%* |
| **Consulting and Training** | **21,193** | **24,362** | **20,277** | **25,166** | **18,442** | **21,612** | **21,612** | **24,248** | **22,263** | **45,096** | **56,459** | **56,459** |
| *YoY Growth* | | *15%* | *34%* | *44%* | *24%* | *-11%* | *-11%* | *20%* | *-12%* | *145%* | *161%* | *161%* |
| 1 Year or Less | 17,235 | 20,798 | 18,945 | 21,923 | 14,977 | 19,226 | 19,226 | 21,068 | 17,863 | 31,673 | 41,411 | 41,411 |
| 1-2 Years | 2,810 | 1,439 | 1,215 | 1,986 | 2,042 | 346 | 346 | 914 | 2,675 | 6,561 | 8,917 | 8,917 |
| 2-3 Years | 940 | 132 | 107 | 631 | 770 | 851 | 851 | 1,756 | 762 | 5,165 | 5,512 | 5,512 |
| Greater Than 3 Years | 208 | 1,993 | 10 | 626 | 653 | 1,189 | 1,189 | 510 | 963 | 1,697 | 619 | 619 |
| **Total** | **630,973** | **836,143** | **753,826** | **817,179** | **837,990** | **1,072,387** | **1,072,387** | **979,810** | **1,031,406** | **1,034,026** | **1,344,894** | **1,344,894** |
| *YoY Growth* | | *33%* | *19%* | *30%* | *38%* | *28%* | *28%* | *30%* | *26%* | *23%* | *25%* | *25%* |
| 1 Year or Less | 399,906 | 493,382 | 433,262 | 461,398 | 456,777 | 612,686 | 612,686 | 543,946 | 567,154 | 577,650 | 771,173 | 771,173 |
| *YoY Growth* | | *23%* | *17%* | *28%* | *35%* | *24%* | *24%* | *26%* | *23%* | *26%* | *26%* | *26%* |
| 1-2 Years | 106,158 | 174,032 | 168,577 | 190,886 | 207,214 | 256,276 | 256,276 | 251,239 | 272,622 | 261,071 | 314,123 | 314,123 |
| *YoY Growth* | | *64%* | *57%* | *62%* | *60%* | *47%* | *47%* | *49%* | *43%* | *26%* | *23%* | *23%* |
| 2-3 Years | 69,565 | 105,896 | 107,209 | 116,035 | 119,382 | 138,641 | 138,641 | 134,009 | 126,602 | 127,188 | 173,120 | 173,120 |
| *YoY Growth* | | *52%* | *35%* | *38%* | *42%* | *31%* | *31%* | *25%* | *9%* | *7%* | *25%* | *25%* |
| Greater Than 3 Years | 55,344 | 62,833 | 44,778 | 48,860 | 54,617 | 64,784 | 64,784 | 50,616 | 65,028 | 68,117 | 86,478 | 86,478 |
| *YoY Growth* | | *14%* | *-40%* | *-26%* | *-4%* | *3%* | *3%* | *13%* | *33%* | *25%* | *33%* | *33%* |
| **Total (Subscription)** | **591,111** | **807,297** | **727,386** | **782,193** | **809,140** | **1,038,866** | **1,038,866** | **945,284** | **1,002,202** | **965,959** | **1,270,726** | **1,270,726** |
| *YoY Growth* | | *37%* | *20%* | *30%* | *39%* | *29%* | *29%* | *30%* | *28%* | *19%* | *22%* | *22%* |
| 1 Year or Less | 368,006 | 470,279 | 410,322 | 431,355 | 433,092 | 581,946 | 581,946 | 513,229 | 542,584 | 530,291 | 718,810 | 718,810 |
| *YoY Growth* | | *28%* | *18%* | *29%* | *36%* | *24%* | *24%* | *25%* | *26%* | *22%* | *24%* | *24%* |
| 1-2 Years | 101,005 | 170,414 | 165,194 | 187,200 | 203,472 | 255,535 | 255,535 | 249,696 | 269,713 | 253,446 | 300,701 | 300,701 |
| *YoY Growth* | | *69%* | *57%* | *61%* | *60%* | *50%* | *50%* | *51%* | *44%* | *25%* | *18%* | *18%* |
| 2-3 Years | 66,964 | 105,764 | 107,102 | 115,404 | 118,612 | 137,790 | 137,790 | 132,253 | 125,840 | 117,929 | 165,356 | 165,356 |
| *YoY Growth* | | *58%* | *40%* | *40%* | *45%* | *30%* | *30%* | *23%* | *9%* | *-1%* | *20%* | *20%* |
| Greater Than 3 Years | 55,136 | 60,840 | 44,768 | 48,234 | 53,964 | 63,595 | 63,595 | 50,106 | 64,065 | 64,293 | 85,859 | 85,859 |
| *YoY Growth* | | *10%* | *-40%* | *-27%* | *-5%* | *5%* | *5%* | *12%* | *33%* | *19%* | *35%* | *35%* |
| **Bookings Build** | | | | | | | | | | | | |
| Total Bookings | | 1,116,553 | 183,274 | 290,728 | 246,762 | 532,997 | 1,253,761 | 220,922 | 377,298 | 258,888 | 627,052 | 1,484,160 |
| *YoY Growth* | | | *-14%* | *44%* | *25%* | *6%* | *12%* | *21%* | *30%* | *5%* | *18%* | *18%* |
| *Sequential Growth* | | | *-64%* | *59%* | *-15%* | *116%* | | *-59%* | *71%* | *-31%* | *142%* | |
| **Current Bookings** | | **1,004,859** | **205,471** | **255,511** | **221,330** | **454,509** | **1,136,821** | **244,759** | **348,910** | **266,764** | **509,707** | **1,370,140** |
| *YoY Growth* | | | *12%* | *31%* | *13%* | *5%* | *13%* | *19%* | *37%* | *21%* | *12%* | *21%* |
| *Sequential Growth* | | | *-52%* | *24%* | *-13%* | *105%* | | *-46%* | *43%* | *-24%* | *91%* | |
| Cloud Bookings | | 257,099 | 35,770 | 91,755 | 104,192 | 144,518 | 376,235 | 42,037 | 111,712 | 47,095 | 241,548 | 442,392 |
| *YoY Growth* | | | *-55%* | *116%* | *186%* | *48%* | *46%* | *18%* | *22%* | *-55%* | *67%* | *18%* |
| *Sequential Growth* | | | *-63%* | *157%* | *14%* | *39%* | | *-71%* | *166%* | *-58%* | *413%* | |
| **Current Cloud Bookings** | | **195,963** | **52,172** | **65,748** | **75,250** | **97,750** | **290,920** | **71,739** | **102,982** | **80,935** | **127,513** | **383,169** |
| *YoY Growth* | | | *31%* | *63%* | *68%* | *38%* | *48%* | *38%* | *57%* | *8%* | *30%* | *32%* |
| *Sequential Growth* | | | *-26%* | *26%* | *14%* | *30%* | | *-27%* | *44%* | *-21%* | *58%* | |
| *TTM (% of Cloud ACV)* | | *116%* | *114%* | *111%* | *114%* | *109%* | *109%* | *110%* | *113%* | *110%* | *105%* | *105%* |
| Subscription Bookings | | 829,945 | 127,507 | 220,133 | 192,325 | 462,933 | 1,002,898 | 161,346 | 313,289 | 160,735 | 553,961 | 1,189,331 |
| *YoY Growth* | | | *-19%* | *80%* | *39%* | *12%* | *21%* | *27%* | *42%* | *-16%* | *20%* | *19%* |
| *Sequential Growth* | | | *-69%* | *73%* | *-13%* | *141%* | | *-65%* | *94%* | *-49%* | *245%* | |
| **Current Subscription Bookings** | | **716,032** | **147,461** | **186,359** | **167,115** | **382,061** | **882,996** | **186,211** | **285,726** | **184,685** | **437,713** | **1,094,335** |
| *YoY Growth* | | | *20%* | *63%* | *22%* | *12%* | *23%* | *26%* | *53%* | *11%* | *15%* | *24%* |
| *Sequential Growth* | | | *-57%* | *26%* | *-10%* | *129%* | | *-51%* | *53%* | *-35%* | *137%* | |
| *TTM (% of Total ACV)* | | *103%* | *104%* | *110%* | *108%* | *106%* | *106%* | *108%* | *114%* | *183%* | *109%* | *109%* |
| **ACV** | | | | | | | | | | | | |
| Cloud ACV | 109,973 | 169,329 | 182,165 | 210,741 | 232,176 | 266,642 | 266,642 | 282,389 | 306,919 | 320,653 | 363,970 | 363,970 |
| *YoY Growth* | | *54%* | *42%* | *55%* | *57%* | *57%* | *57%* | *55%* | *46%* | *38%* | *37%* | *37%* |
| *Sequential Growth* | | | *8%* | *16%* | *10%* | *15%* | | *6%* | *9%* | *4%* | *14%* | |
| *% of ACV* | *19%* | *24%* | *26%* | *29%* | *30%* | *32%* | *32%* | *33%* | *34%* | *34%* | *36%* | *36%* |
| Maintenance ACV | 269,708 | 292,696 | 294,780 | 288,888 | 298,680 | 304,488 | 304,488 | 302,244 | 315,128 | 332,752 | 330,904 | 330,904 |
| *YoY Growth* | *8%* | *9%* | *9%* | *4%* | *6%* | *4%* | *4%* | *3%* | *9%* | *11%* | *9%* | *9%* |
| *Sequential Growth* | | | *1%* | *-2%* | *3%* | *2%* | | *-1%* | *4%* | *6%* | *-1%* | |
| *% of ACV* | *47%* | *42%* | *41%* | *39%* | *38%* | *36%* | *36%* | *35%* | *35%* | *35%* | *33%* | *33%* |
| Term ACV | 190,349 | 231,267 | 234,361 | 238,268 | 246,000 | 264,346 | 264,346 | 267,931 | 277,388 | 294,320 | 307,750 | 307,750 |
| *YoY Growth* | | *21%* | *23%* | *20%* | *18%* | *14%* | *14%* | *14%* | *16%* | *20%* | *16%* | *16%* |
| *Sequential Growth* | | | *1%* | *2%* | *3%* | *7%* | | *1%* | *4%* | *6%* | *5%* | |
| *% of ACV* | *33%* | *33%* | *33%* | *32%* | *32%* | *32%* | *32%* | *31%* | *31%* | *31%* | *31%* | *31%* |
| **Total ACV** | **570,030** | **693,292** | **711,306** | **737,897** | **776,856** | **835,476** | **835,476** | **852,564** | **899,435** | **947,725** | **1,002,624** | **1,002,624** |
| *YoY Growth* | *28%* | *22%* | *20%* | *20%* | *22%* | *21%* | *21%* | *20%* | *22%* | *22%* | *20%* | *20%* |
| *YoY Growth (Constant Currency)* | | *22%* | *21%* | *21%* | *21%* | *19%* | *19%* | *17%* | *18%* | *20%* | *21%* | *21%* |
| *Sequential Growth* | | | *3%* | *4%* | *5%* | *8%* | | *2%* | *5%* | *5%* | *6%* | |

Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Ping Identity (PING) – covered by Matt Hedberg

**Key metric(s):** ARR

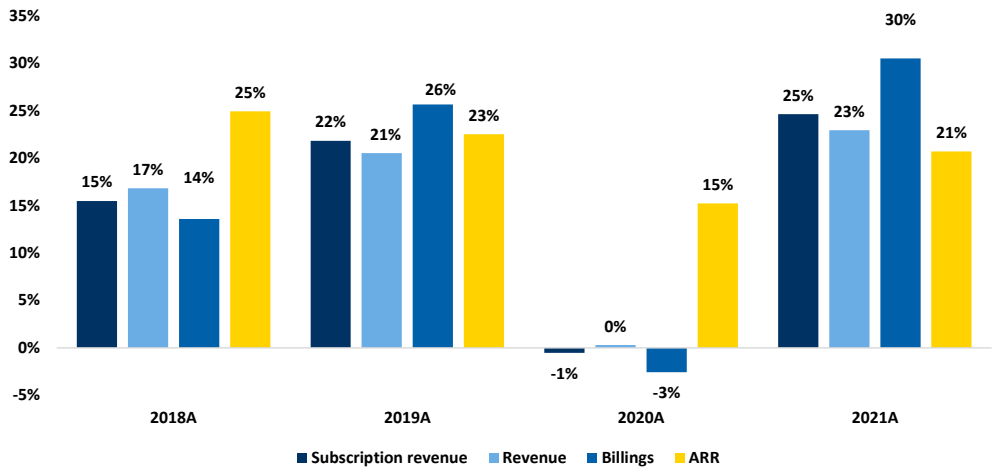
**Leading metric practice/relevance:** ARR is the key growth metric for Ping. As Ping continues to transition to more of a SaaS-based model, it sees ARR as a smoother and cleaner metric to help think through future growth assumptions. As evident in the following exhibit, Ping's revenue and billings growth are more varied than ARR due to timing of revenue recognition. Management noted that SaaS ARR is 25%+ of total ARR in the latest quarter and as that number grows revenue growth rates and ARR growth rates will converge. With the stability of the metric and management's focus on it, including guiding towards it, we see ARR as the best metric to track the growth of the company.

**Relevant commentary:** (CFO Raj Dani at Q4 2021 earnings call) *"Over time, we expect a difference between our trailing 12-month revenue and ARR will shrink with growth rates roughly converging. As we have said for several quarters, these revenue fluctuations support our ongoing belief that annual recurring revenue or ARR is the best metric for measuring Ping's growth trajectory."*

(CFO Raj Dani at Q4 2021 earnings call) *"Subscription term-based license revenue grew by 25% year-over-year in Q3. Given the impact that deployment mix and contract duration have on GAAP revenue, we continue to believe that ARR is the key growth metric of a subscription business."*

Exhibit 56 shows Y/Y growth of PING's key leading metrics since 2018.

Exhibit 56 - PING Leading Metric Y/Y Growth Rates 2018-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## PowerSchool (PWSC) – covered by Matt Hedberg

**Key metric(s):** ARR

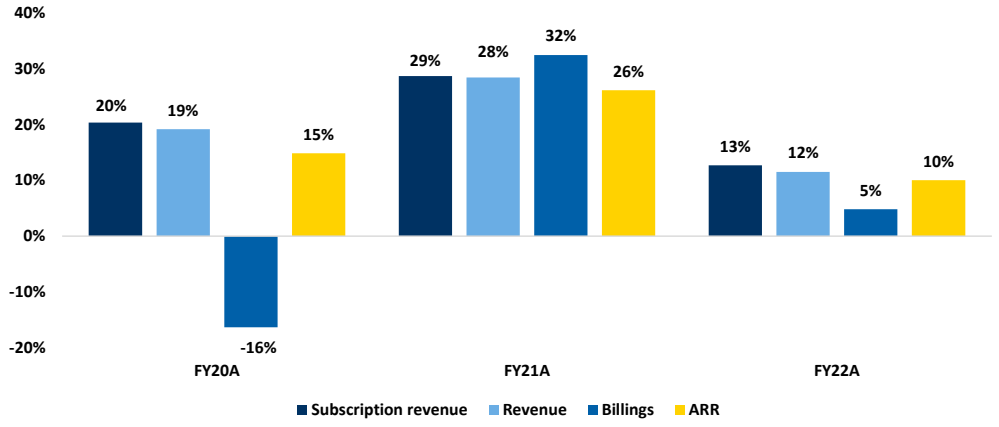
**Leading metric practice/relevance:** Due to the seasonality of customer spend for PowerSchool, ARR is the best metric to show how smooth and linear its growth actually is. This is because almost 60% renewals hit in Q3, coinciding with new school-year budgets. This seasonality can effect revenue and billings more so than ARR and without more knowledge of the customer base could confuse anyone analyzing PowerSchool. While management does not guide to ARR, it continues to use it as a key metric in filings and in investor presentations. We continue to see ARR as a good metric to track PowerSchool's growth trajectory without the noise of seasonality.

**Relevant commentary:** (CEO Hardeep Gulati at 2021 RBC Technology Conference) *"Now there is a seasonality difference between how a typical software company, which has quarter-over-quarter versus our] rare seasonality of Q2 versus Q3, where most of the renewals, almost 60% of renewals hit in Q3 because that's where the school budget year opens after the back-to-school."*

(2021 Investor Day Slides) *"ARR represents the annualized value of all recurring contracts as of the end of the period. ARR mitigates fluctuations due to seasonality, contract term, one-time discounts given to help customers meet their budgetary and cash flow needs and the sales mix for recurring and non-recurring revenue. ARR does not have any standardized meaning and is therefore unlikely to be comparable to similarly titled measures presented by other companies. ARR should be viewed independently of revenue and deferred revenue and is not intended to be combined with or to replace either of those items. ARR is not a forecast, and the active contracts at the end of a reporting period used in calculating ARR may or may not be extended or renewed by our customers."*

Exhibit 57 - PWSC Leading Metric Y/Y Growth Rates FY20A-FY22A

Exhibit 57 shows Y/Y growth of PWSC's key leading metrics since FY20.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## PTC (PTC) – covered by Matt Hedberg

**Key metric(s):** ARR

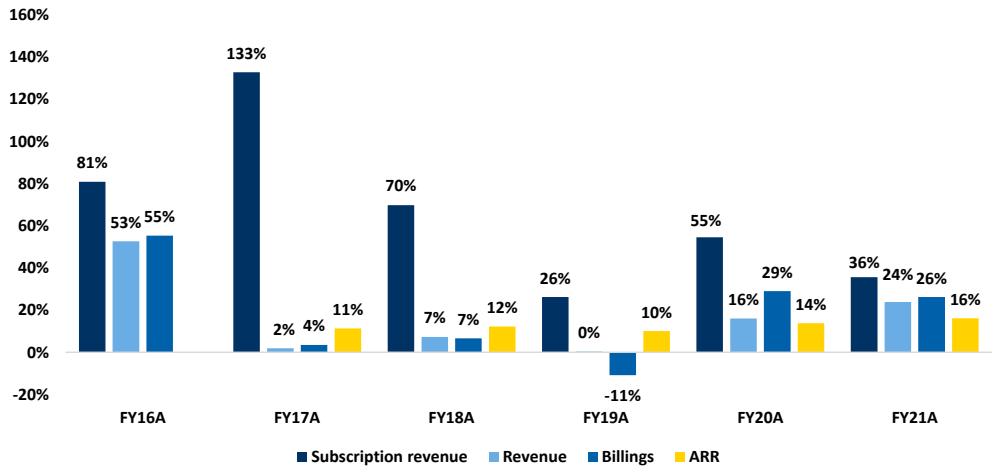
**Leading metric practice/relevance:** As PTC continues in its SaaS journey, it has chosen ARR as the best metric to show its growth. As seen in the exhibit that follows, revenue and billings growth have been highly variable over the years while ARR has shown stable growth. Due to the varying product offerings, contract lengths and recognition policies, PTC does not think revenue is the best metric to use. Management guides to ARR and continues to point to the metric as its main focus when it comes to measuring top-line growth.

**Relevant commentary:** (CFO Kristian Talvitie at 2021 Investor Day) *"We believe that ARR is the best indicator of PTC's top-line performance, and free cash flow is the best indicator of PTC's bottom-line performance. […] So, ultimately, the mix of contract types, changes in term lengths, movement from ratable to upfront revenue recognition and vice versa make revenue a very complicated metric, which impacts the utility of the income statement as a tool to cleanly understand the company's financial performance. This is why we focus on ARR and free cash flow."*

(CFO Kristian Talvitie at 2021 Investor Day) *"ARR, or annual run rate, represents the annualized value of our portfolio of active subscription software, cloud, SaaS, and support contracts at the end of the reporting period. We believe ARR is a valuable operating metric to measure the health of a subscription business because it captures the expected subscription and support cash generation from our customers. As depicted in the chart and for our internal planning purposes, we use ARR plus trailing-12-month perpetual license and professional services revenue as a close proxy for the cash we expect to collect in any given year."*

Exhibit 58 - PTC Leading Metric Y/Y Growth Rates FY16A-FY21A

Exhibit 58 shows Y/Y growth of PTC's key leading metrics since FY16.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



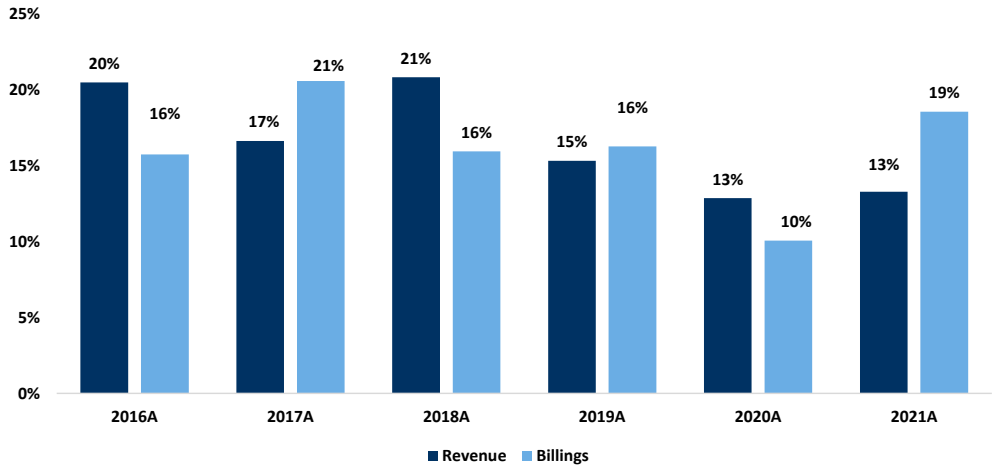
## Qualys (QLYS) – covered by Matt Hedberg

**Key metric(s):** Billings

**Leading metric practice/relevance:** Qualys does not offer further metrics beyond revenue, billings, and deferred revenue. Although Qualys does sell subscription services, management is most focused on total revenue growth. There is some focus from management on billings growth, but this is still secondary to revenue growth.

Exhibit 59 shows Y/Y growth of QLYS's key leading metrics since 2016.

Exhibit 59 - QLYS Leading Metric Y/Y Growth Rates 2016A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Rapid7 (RPD) – covered by Matt Hedberg

**Key metric(s):** ARR

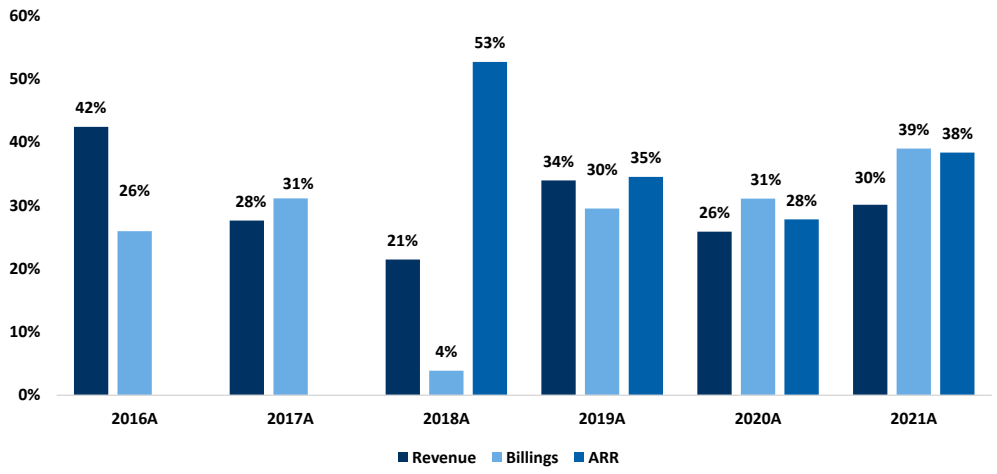
**Leading metric practice/relevance:** ARR is the key metric management has noted it uses and wants investors to use to evaluate Rapid7's growth. Management guides to full-year ARR and talks at length about ARR in its investor days and conference calls. Rapid7 is focused on ARR per customer, which drives its land-and-expand sales model. As of Q4 2021, about half of its 38% of ARR growth was from landing new customers and the other was from upselling and cross-selling to current customers. YoY ARR growth for Rapid7 has stayed around or above 30% since disclosing ARR in 2017 with acceleration closer to 40% happening in the latest fiscal year.

**Relevant commentary:** (CFO Jeff Kalowski at 2021 Investor Day) *"So, investors sometimes ask, what are the right metrics to track our progress? Many software companies look at a variety of metrics like billings growth, deferred revenue growth and the net retention rate as their key metrics. However, we don't feel that these metrics are the best indicator of our growth for our business. This is in part because many other software companies do not disclose ARR. The key growth metric that most directly translates to revenue growth and reflects the health of our business is ARR. And ARR growth is comprised of two core drivers, number of customers and ARR per customer, which incorporates the cross-sell and adoption of our platform products, driving more wallet share per customer. These will be the two core drivers of growth in our business over time, and these are the metrics we manage the business to. We've disclosed this metric publicly since 2017. This is what we encourage investors to track our progress by."*

(2021 Investor Day Slides) *"Annualized Recurring Revenue (ARR) is defined as the annual value of all recurring revenue related to contracts in place at the end of the period. ARR should be viewed independently of revenue and deferred revenue as ARR is an operating metric and is not intended to be combined with or replace these items. ARR is not a forecast of future revenue and can be impacted by contract start and end dates and renewal rates, and does not include revenue reported as perpetual license or professional services revenue in our consolidated statement of operations."*

Exhibit 60 - RPD Leading Metric Y/Y Growth Rates 2016A-2021A

Exhibit 60 shows Y/Y growth of RPD's key leading metrics since 2016.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Smartsheet, Inc. (SMAR) – covered by Rishi Jaluria

**Key metric(s):** Billings

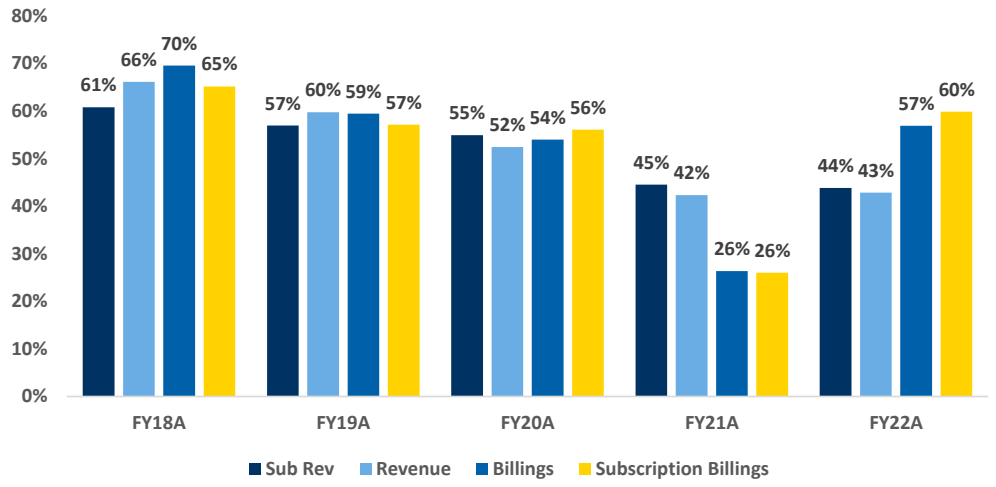
**Leading metric practice/relevance:** SMAR regularly discusses billings on earnings calls, provides a breakdown of contract duration (monthly, annual, and other), and explicitly guides to full-year billings (during COVID through FY22, the company also guided to quarterly billings). We find billings to be a good leading indicator, as the COVID slowdown showed up first in billings, then revenue, and the same happened with the reacceleration coming out of COVID.

**Relevant commentary:** (10-K commentary on billings) "*We define calculated billings as total revenue plus the change in deferred revenue in the period. Because we recognize subscription revenue ratably over the subscription term, calculated billings can be used to measure our subscription sales activity for a particular period, to compare subscription sales activity across particular periods, and as an indicator of future subscription revenue… Because we generate most of our revenue from customers who are invoiced on an annual basis, and because we have a wide range of customers, from those who pay us less than $200 per year to those who pay us more than $3.0 million per year, we experience seasonality and variability that is tied to typical enterprise buying patterns and contract renewal dates of our largest customers. We expect that our billings trends will continue to vary in future periods based on the timing and size of new and renewal bookings, changes to the economic environment, and other factors.*"

(CFO Pete Godbole on F4Q22 earnings call) "*Turning to billings, fourth quarter billings came in strong at $224.3 million, representing year-over-year growth of 48%. Approximately 93% of our subscription billings were annual, with 4% monthly. Quarterly and semiannual represented approximately 3% of the total. Multi-year billings represented less than 1% of total billings.*

*Starting FY23, we will be reverting to our pre-COVID approach by providing billings and free cash flow guidance on a FY basis only. Given the growing seasonality in our business, we expect our quarterly billings cadence to be more weighted towards the back half of the year, with the lowest % of billings on an absolute and growth basis occurring in the F1Q and then building towards the back of the year. The factors that contribute to this growing seasonality are the expansion of additional territories, internal promotions to quota-carrying and management roles, the frontloaded hiring of new sales reps, and an in-person sales kickoff in March.*"

Exhibit 61 - SMAR Leading Metrics Y/Y Growth Rates, FY18A-FY22A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Salesforce, Inc. (CRM) – covered by Rishi Jaluria

**Key metric(s):** Current RPO, billings

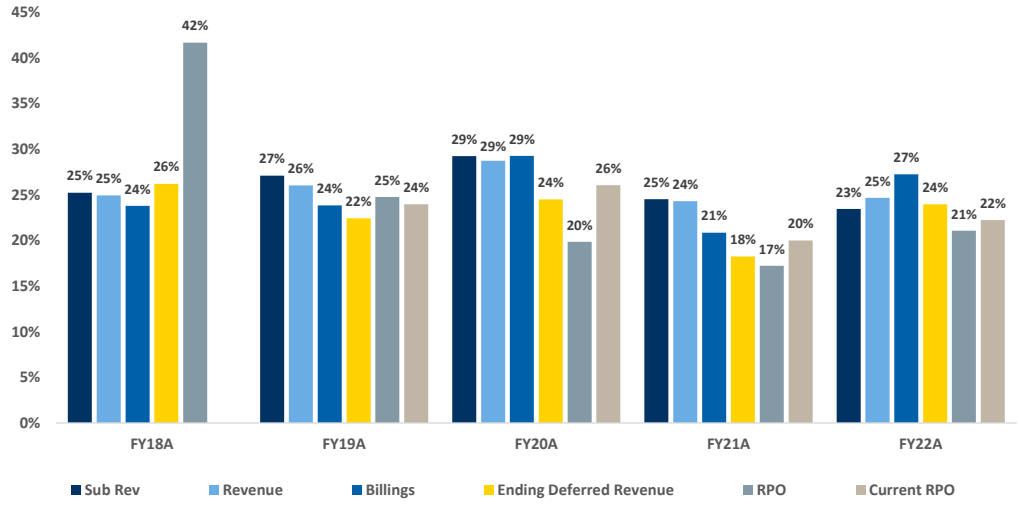
**Leading metric practice/relevance:** Salesforce changed to guiding to quarterly cRPO in FY20, versus deferred revenue (and implicitly billings) previously. We believe both remain important. Salesforce provides useful commentary around cRPO, which is based on revenue expected to be recognized in the next 12 months and includes contracts under 12 months, which we view favorably.

**Relevant commentary:** (Former President & CFO Mark Hawkins on F1Q21 earnings call) "*We don't really – billings is not a metric that we focus a lot on. We think CRPO helps us a bit more in terms of actually managing the business, and that's a metric that we track carefully.*"

(Former President & CFO Mark Hawkins on F3Q19 earnings call) *"We've always said that unearned revenue can be lumpy due to invoice timing, renewal timing, duration changes, et cetera... First, the FX environment has changed significantly over last year, and we now anticipate a year-over-year FX headwind… Secondly, as you may recall, in Q4 of last year, we had an extremely strong renewal quarter, including some of the largest renewals in history. This drove outsized growth in our billed and unbilled deferred revenue in Q4 of last year. And thirdly and finally, as we discussed at the most recent Investor Day, the timing of invoices and renewals can impact the UR balance in any given quarter. Third quarter UR came in ahead of our guidance and is a result of this dynamic which has a direct impact on the UR balance in successive quarters and the related sequential changes. As a reminder, this is the last quarter we'll provide unearned revenue guidance. That said, when we report our fourth quarter results, you will have 2 full years of data on the current remaining performance obligation, which we think is a more complete metric because it's contract-based versus invoice-based."*

Exhibit 62 - CRM's Leading Metrics Y/Y Growth, F17A-FY22A

Exhibit 62 shows Y/Y growth of CRM's key leading metrics since FY17A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Samsara (IOT) – covered by Matt Hedberg

**Key metric(s):** Revenue, Billings, ARR

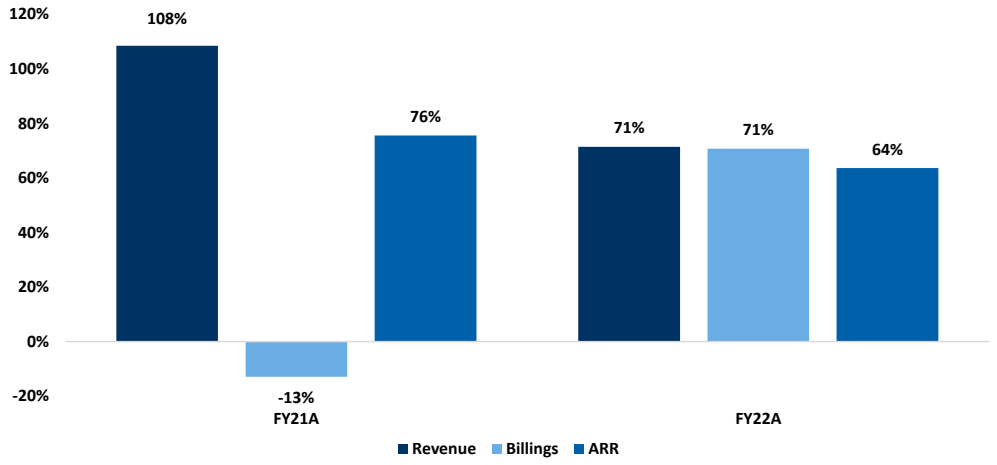
**Leading metric practice/relevance:** Samara has only reported one quarter as a public company but ARR is a large focus of its growth story. Samsara is almost 100% a subscription business so revenue and ARR growth should track fairly close going forward but there is an increased focus on ARR as the top-line metric to follow. The lack of historical data makes it harder to examine trends in ARR, but for the two years we have we can see it is a less variable metric than revenue and billings growth. Moving forward as Samara builds more history as a public company we see ARR as the most important metric to examine its growth story.

**Relevant commentary:** (CEO Sanjit Biswas in Q4 2022 earnings press release) *"We achieved significant milestones in the fourth quarter, including becoming a publicly listed company. We ended the year with over half a billion in ARR, while growing 64% year-over-year, which capped off a very successful year for the company."*

(Q4 2022 earnings call slides) *"We define ARR as the annualized value of subscription contracts that have commenced revenue recognition as of the measurement date."*

Exhibit 63 - IOT Leading Metric Y/Y Growth Rates FY21A-FY22A

Exhibit 63 shows Y/Y growth of IOT's key leading metrics since FY21.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## ServiceNow (NOW) – covered by Matt Hedberg

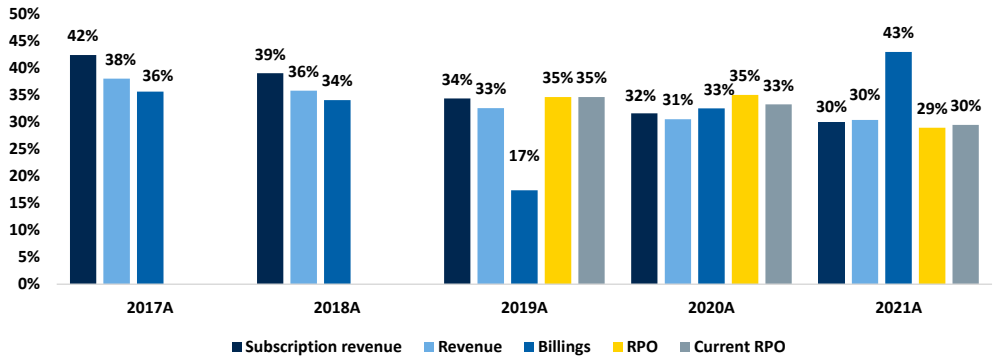
**Key metric(s):** cRPO

**Leading metric practice/relevance:** cRPO is a focus metric for management (and is guided to on both a quarterly and annual basis). We like cRPO, which helps normalize duration in billings, and view it as the best leading indicator over billings (noisy quarter-to-quarter) and RPO (which provides limited visibility over NTM).

**Relevant commentary:** (CFO Gina Mastantuono at 2021 Financial Analyst Day) *"I want to take a quick moment for a sidebar here. We believe cRPO provides better visibility and is a more consistent indicator of business performance, normalizing the timing and duration noise that impacts quarterly billings... subscription revenue growth [] tracks very closely to cRPO []. On the contrary, subscription billings [], has wide fluctuations and often requires adjustments for timing-related noise such as can customer co-terming which can change the timing of renewals from one quarter to another. In addition, invoice durations both short-term and multi-year can cause fluctuations in growth rates. These changes are purely timing-driven by customer requests and have nothing to do with the underlying health of our business. A perfect example of that is the 80 million of billings that we saw pulled forward into Q4 2020 from 2021 due to early customer payments of their second or third year invoice. This resulted in us recognizing billings from these customers twice in 2020 both in Q1 and in Q4. We believe these are one-time in nature and the result of customers having excess cash at the end of the year given the incremental cost savings enterprises saw from COVID. Normalizing for that timing, you can see on the right 2021 subscription billings guidance cause for 28% year over year or about 27% in constant currency… So you can see, there's often a lot more behind the reported billings number and understandably it can become confusing. Looking at billings from a four quarter rolling basis will help normalize for these large quarterly swings as this metric is much more correlated to cRPO and revenue that's why you saw us introduce cRPO guidance this year and why you'll see us focus more of our commentary on cRPO going forward."*

Exhibit 64 - NOW Leading Metric Y/Y Growth Rates 2017A-2021A

Exhibit 64 shows Y/Y growth of NOW's key leading metrics since 2017.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Snowflake Inc. (SNOW) – Not Covered

**Key metric(s):** RPO

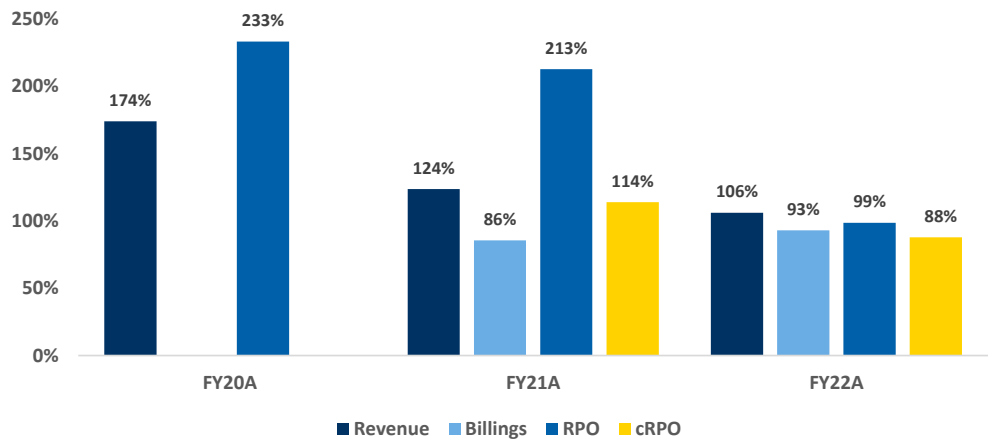
**Leading metric practice/relevance:** Snowflake is unique among consumption-based software peers, in our view, in that the company uses very direct and consistent messaging for investors on its revenue model and leading indicators. To this point, Snowflake has two slides in its quarterly earnings deck that, in simple language, clearly state that the company is a consumption-based model (93% of total revenue), the nature of its contract terms (e.g. primarily billed in advance with some on-demand in arrears), that billings is not a relevant metric (and why), and why investors should focus on revenue and RPO. RPO notably excludes some on-demand arrangements (primarily onboarding; represent a low-single-digit percent of revenue) and "certain time and material contracts, billed in arrears", which we believe is primarily professional services based. Overall, we take the view RPO will become a critical forward-looking metric for investors with the caveat that Snowflake's current state of hyper revenue growth (up 106% Y/Y in FY22) and the current macro climate make the model difficult to predict at this time.

**Relevant commentary:** (CFO Michael Scarpelli at Investor Conference) *"Yeah. Well, RPO isn't the most important thing. The most important thing is revenue. How our customers are consuming, because when they consume they have to buy more. And when they come back to buy more, that's all about timing. And when we started disclosing RPO going public and we had these massive growths, I kept reminding people that don't get too excited about that RPO growth if you go back on all the calls, because last year was the first time we've really ever sold multi-year contracts. And so, yes, we were growing over 200% in total RPO, but it wasn't a true compare."*

(10-K Key Metrics commentary on RPO) "*Remaining performance obligations (RPO) represent the amount of contracted future revenue that has not yet been recognized, including (i) deferred revenue, and (ii) non-cancelable contracted amounts that will be invoiced and recognized as revenue in future periods. RPO excludes performance obligations from on-demand arrangements and certain time and materials contracts that are billed in arrears. Portions of RPO that are not yet invoiced and are denominated in foreign currencies are revalued into USD each period based on the applicable period-end exchange rates.*"

Exhibit 65 - SNOW Leading Metrics Y/Y Growth Rates, FY20A-FY22A

Exhibit 65 shows Y/Y growth of SNOW's leading metrics since FY20A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## SolarWinds (SWI) – covered by Matt Hedberg

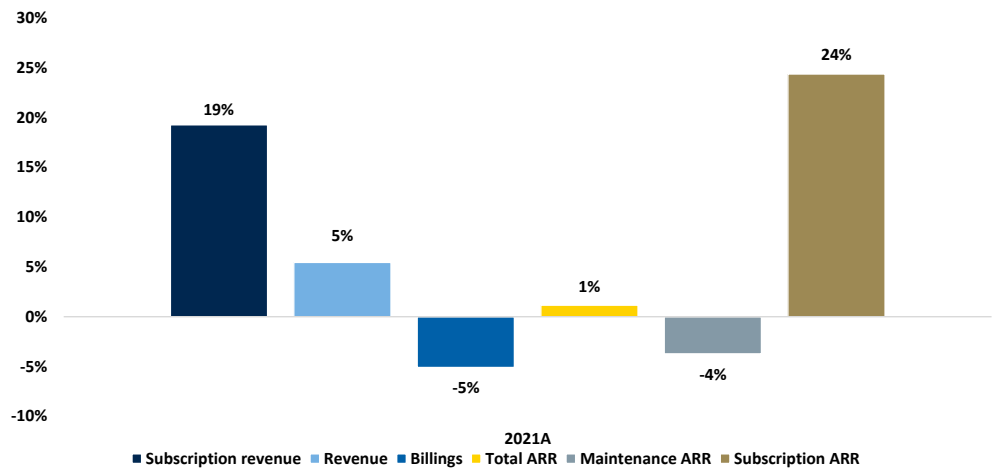
**Key metric(s):** Subscription ARR, ARR

**Leading metric practice/relevance:** Following the N-able spin-off, management focuses on ARR and subscription ARR as the best growth metrics to monitor growth. Its FY/25 target is for +$1 billion in total ARR with subscription ARR growing +30% compared to current ARR of $630 million and subscription ARR growth of +24%. As SolarWinds pivots more to subscription revenue, ARR growth will come into more focus for the company versus total revenue growth. With the spin-off still fresh, there is not much historical data to study, but moving forward we will continue to watch ARR as an indicator of SolarWinds' future growth.

**Relevant commentary:** (CFO James Barton Kalsu at 2021 RBC Technology Conference) *"One of the things Sudhakar [CEO] talked earlier about, you know what our current ARR, we – that's going to be a focus for us is what our ARR is, and how our ARR grows over time. And a big chunk of that ARR today is maintenance revenue. And what we are focused, one of our big focuses over the next two or three years is going to be trying to get that maintenance base to shift to subscription revenue. That'll happen more as more and more of the functionality becomes available. So we're talking like the back half of 2022, and then end of 2023. But the fact that we have such a big customer base to leverage from, we think that there's upside in that ARR just from our current customer base."*

Exhibit 66 - SWI Leading Metric Y/Y Growth Rates 2021A

Exhibit 66 shows Y/Y growth of SWI's key leading metrics since 2021 (post N-able spin-off).




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Splunk (SPLK) – covered by Matt Hedberg

**Key metric(s):** Cloud ARR, ARR, cRPO, RPO

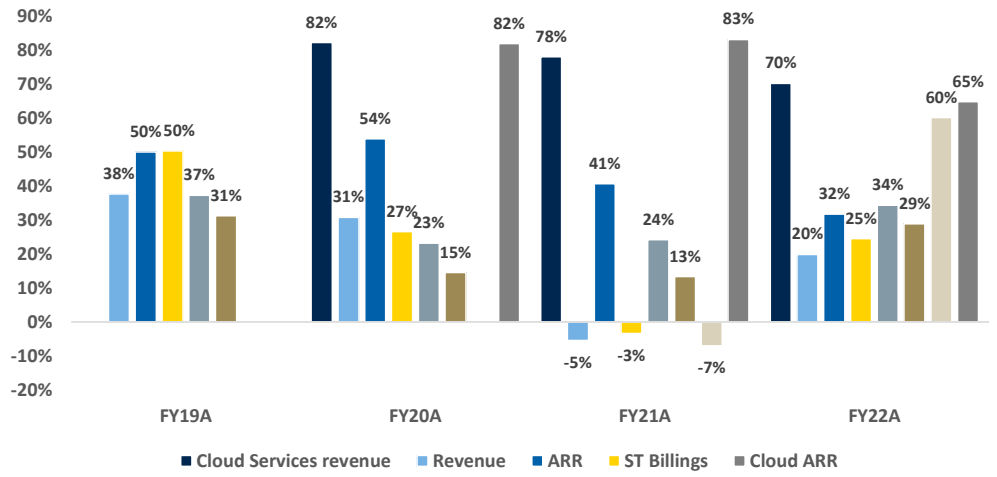
**Leading metric practice/relevance:** Through Splunk's model transition, ARR and Cloud ARR are a better leading indicator than billings or revenue growth, which are both backward looking in comparison. Given the accounting changes from ASC 606 as well the shift to annual billings, the company has aligned its reporting to provide visibility to this transition, which is impacted by the purchasing method of its customers, currently offering license, term license and cloud, which are all recognized as revenue on different schedules. In addition to ARR, cRPO, which is expected to be recognized in the next 12 months, helps to normalize for these purchasing models, while RPO in total helps to gain a better understanding of the larger multi-year commitments.

**Relevant commentary:** (Former CEO Douglas Merritt on F2Q22 earnings call) "*We were really excited about the continued progress of workload-based pricing. 80-plus percent of that new cloud ARR was workload, 25% now of all cloud ARR's workload-based, so I think that program – when you talk to customers that have moved to cloud and have abandoned infrastructure-based or – sorry, ingest-based pricing and moved to infrastructure, if you're aligned with everything I'm hearing, you'll hear a lot of enthusiasm from customers. It was a stronger term quarter than we thought, and that really comes back to the craziness of the transformation and why ARR is such an important metric, and we are thrilled that we got some big TCG term deals in this quarter.*"

(10-K definition and explanation as a key metric) "*Cloud ARR represents the annualized revenue run-rate of active cloud services contracts at the end of a reporting period. Total ARR represents the annualized revenue run-rate of active cloud services, term license and maintenance contracts at the end of a reporting period. Each contract is annualized by dividing the contract value by the number of days in the contract term and then multiplying by 365. ARR should be viewed independently of revenue, and does not represent our revenue under U.S. GAAP on an annualized basis, as it is an operating metric that can be impacted by contract start and end dates and renewal rates. ARR is not intended to be a replacement for forecasts of revenue.*"

Exhibit 67 - SPLK Leading Metrics Y/Y Growth Rates FY17A-FY22A

Exhibit 67 shows Y/Y growth of SPLK's key leading metrics since FY17.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Sumo Logic (SUMO) – covered by Matt Hedberg

**Key metric(s):** ARR, cRPO

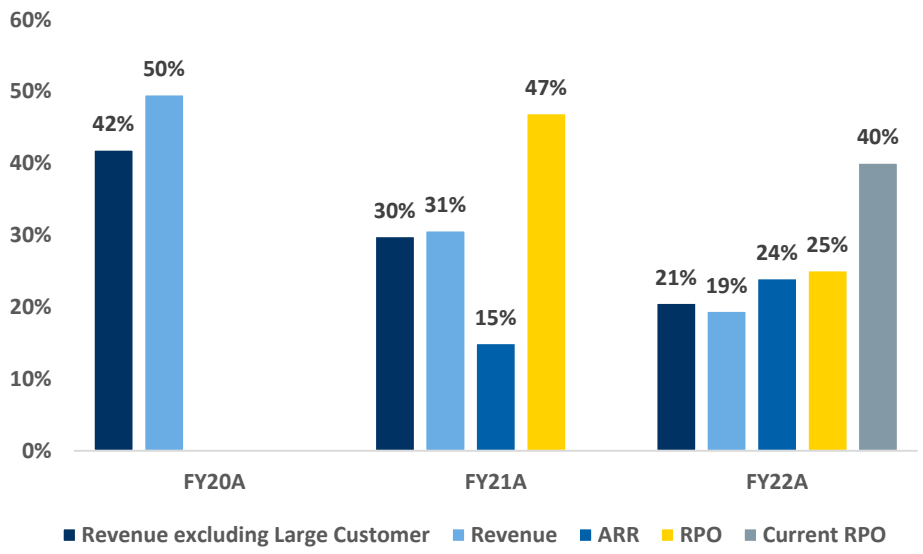
**Leading metric practice/relevance:** Traditionally Sumo Logic has predominately been judged by revenue due to its unique consumption model where customers can adjust their contracts to match consumption. The company recently disclosed ARR, which is a better forward-looking metric, as well as cRPO, which gives greater visibility to future results.

**Relevant commentary:** (CFO Stewart Grierson on F4Q22 earnings call) "*I'd like to introduce a new metric that we will be providing on a quarterly basis which is ARR. We believe the ARR is the best leading indicator of growth in a SaaS business and will provide greater transparency to investors as it reflects our current business and growth trajectory. We ended the year with ARR of $259 million, representing 24% year-over-year growth. This is a significant acceleration from the prior year ARR growth which was 15%.*"

(10-K definition and explanation as a key metric) "*We define ARR as the annualized recurring revenue run-rate from all customers that are under contract with us at the end of the period or with which we are negotiating a renewal contract. Given our historical experience of customer renewals, if we are in active discussions for a renewal, we continue to include customers with expired contracts in our ARR until the customer either renews its contract or negotiations terminate without renewal. For certain customers whose revenue may fluctuate from month to month based upon their specific contractual arrangements, we calculate ARR using the annualized monthly recurring revenue, or MRR, run-rate (MRR multiplied by 12). This enables us to calculate our anticipated recurring revenue for all customers based on our packaging and licensing models, which we believe provides a more accurate view of our anticipated recurring revenue.*"

Exhibit 68 - SUMO Leading Metrics Y/Y Growth Rates FY20A-FY22A

Exhibit 68 shows Y/Y growth of SUMO's key leading metrics since FY20.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## TeamViewer AG (TMV) – covered by Sherri Malek

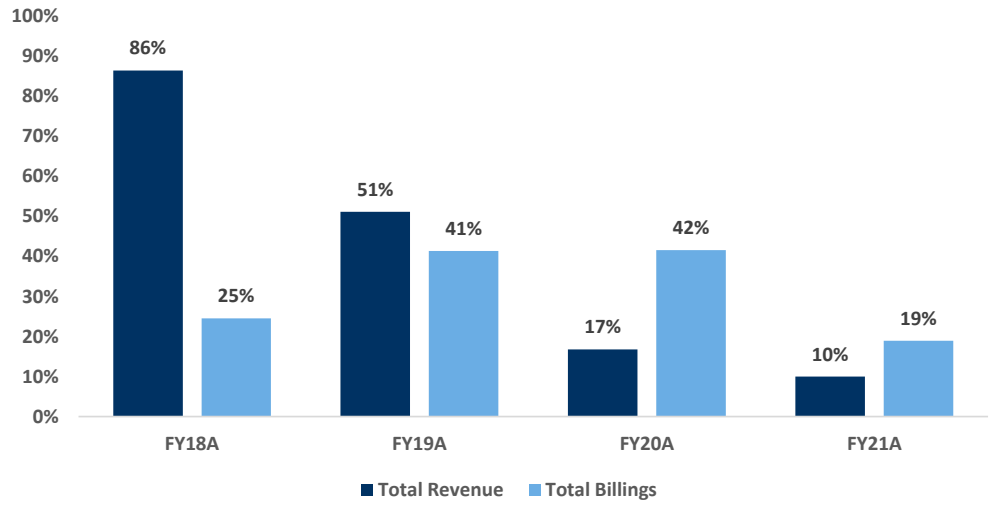
**Key metric(s):** Billings

**Leading metric practice/relevance:** TeamViewer sees billings, the value of goods and services invoiced to customers in a given period, as one of its primary performance indicators. This is because they are derived directly from customer contracts and are not affected by the timing of revenue recognition. In the view of management, billings are a better reflection of the TeamViewer Group's underlying business development than revenue because they are not impacted by the transition from the perpetual license to the subscription license. We believe investors also mainly focus on billings in order to exclude the impact from changes in deferred revenue recognition.

**Relevant commentary:** (CFO Stefan Markus Gaiser on 3Q2) *"The revenue growth rate is below billings growth due to the base effect from the recognized perpetual license revenue last year. As you all know, we discontinued perpetual license sales in 2017, but the recognition of those license sales still continued for a few years. And therefore, during 2020, we still recognize close to $40 million in the first 9 months of perpetual license revenue. And obviously, we don't have that anymore in 2021, and that means it depresses our overall reporting revenue growth. But however, this effect will completely disappear in 2022."*

(From IPO prospectus) *"TeamViewer believes billings more closely reflect its cash flow than revenue as billings do not include changes in deferred revenue."*

Exhibit 69 - TMV Leading Metrics Y/Y Growth Rates, FY18A-FY21A




Source: Company reports, RBC Capital Markets.

This report is intended solely for cbass@marketstreetpartners.com



## Teradata (TDC) – covered by Matt Hedberg

**Key metric(s):** Cloud ARR, ARR

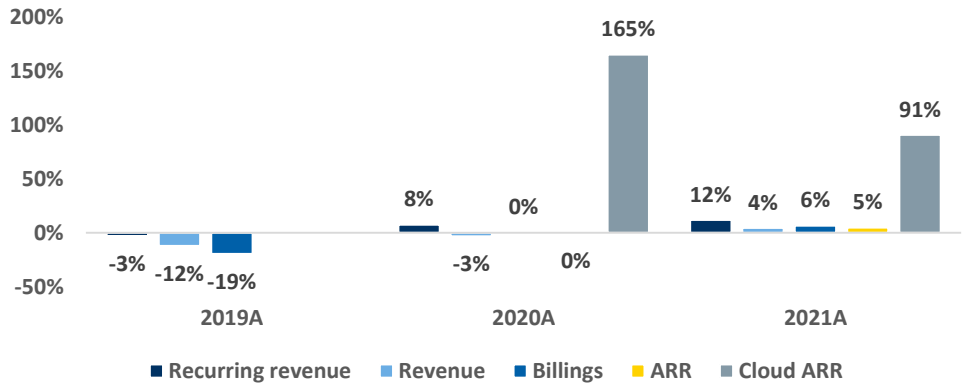
**Leading metric practice/relevance:** As Teradata transitions towards recurring revenue as a primary top-line metric, ARR and more specifically cloud ARR have become the best forward-looking metrics for the business. The growing mix of cloud ARR is important since it does not include underlying storage representing the revenue generated from Teradata software running on top of public could environments as a service.

**Relevant commentary:** (CFO Claire Bramley on 2021 Analyst Day) "*Cloud ARR growth is clearly a primary focus for us. We believe that key decision-makers at the global 10,000 enterprises will want a connected multi-cloud platform for data warehousing and analytics from start to scale. And we are an essential part of that new environment because we give customers choice, choice with a modern data fabric which enables access and analysis of data no matter where it resides, choice from our open platform connecting to multiple public clouds, choice from our integration with Native Object Stores driving more data flowing to Vantage, and finally, choice from the ongoing development and investment in our partner ecosystem, all of which enable more opportunity for the new Teradata.*"

(10-K definition and explanation as a key metric) "*Annual value at a point in time of all contracts related to public cloud implementations of Teradata Vantage and does not include ARR related to private or managed cloud implementations.*"

Exhibit 70 - TDC Leading Metrics Y/Y Growth Rates 2017A-2021A

Exhibit 70 shows Y/Y growth of TDC's key leading metrics since 2017.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com

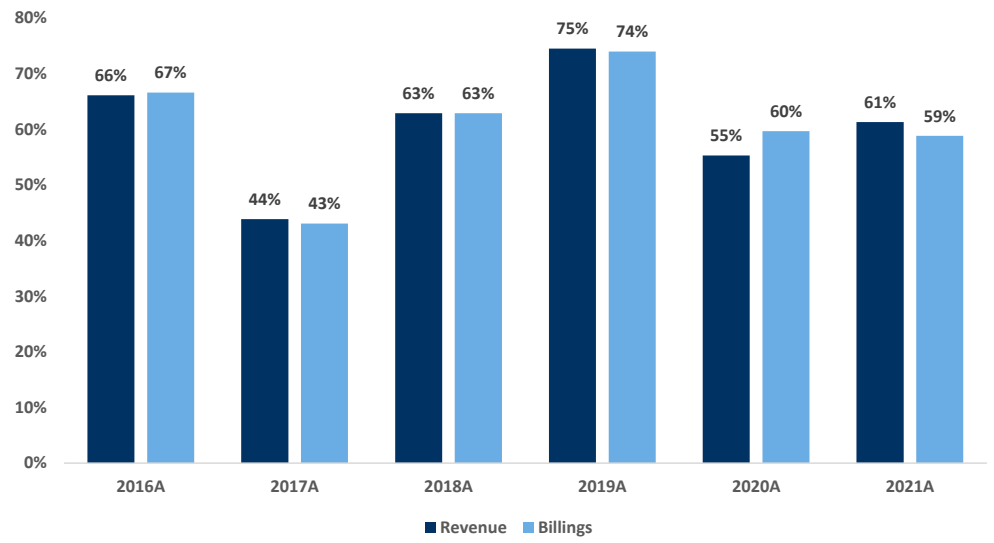


## Twilio, Inc. (TWLO) – covered by Rishi Jaluria

**Leading metric practice/relevance:** Twilio is a consumption-based model with customers invoiced monthly in arrears, therefore billings/RPO are not useful leading indicators.

**Relevant commentary:** (COO Khozema Shipchandler during an investor conference in December 2021) *"I think the only difficulty that we've created in any of this is, is that we've massively over-delivered relative to that first year's 30% goal. Nevertheless, we look at it and say, okay, we're now off of a much higher base. Has anything changed? And the answer flatly is no, that we continue to see our ability to generate on a just Twilio basis, Twilio Classic, whatever you want to call it, pre-Segment basis, our ability to deliver a 30% organic growth over a long-term format… I'm not putting out new guidance today, obviously, but we have three years left in that story that reconstructed last year, and we have very, very high confidence in our ability to deliver those numbers. With Segment in the bag, obviously, that's sort of the cherry on top and allows us to generate perhaps even additional growth beyond that. But we feel really, really strong about that 30%... Now, in terms of the data that I look at that ultimately underpins that, we have great data in terms of customer usage, customer behavior. We have an FP&A team that's obviously constantly running scenarios around sales capacity, customer use cases. What if there was something idiosyncratic that were to happen in terms of attrition, which is never something that we've really experienced on our platform with customers, what if expansion rates were to come down markedly, what about tough compares, all this sort of thing, like what are all of the various factors?... Barring some extreme event and I would suggest that we just went through one with the pandemic, so it had to be even more extreme than that. We just don't see a scenario in which we're not able to deliver on 30% organic growth over a multiple-year timeframe. And, well, I'll just leave it there. We have very strong conviction in that forecast. We did not put a four-year commitment out there lightly, and we definitely believe that we can deliver it."*

Exhibit 71 - TWLO Y/Y Growth Rates, 2016A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## UiPath (PATH) – covered by Matt Hedberg

**Key metric(s):** ARR

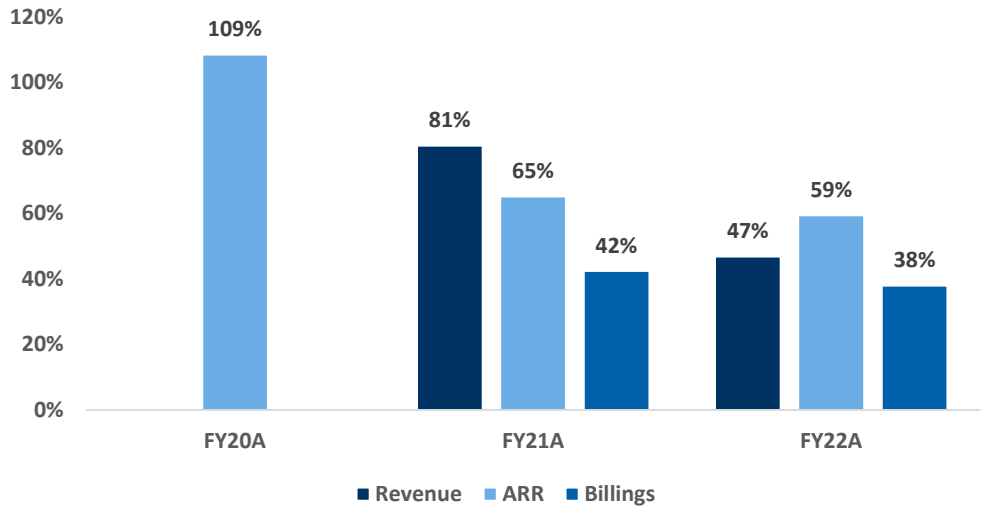
**Leading metric practice/relevance:** The primary growth metric for UiPath is ARR. Within ARR, there is a growing focus on the percentage that is made up of cloud revenue including hybrid and SaaS deployments, which crossed $140 million in ARR in the fourth quarter equating to 15% of total ARR. The flow-through from a top-line perspective ARR is the best leading indicator for revenue growth, which in turn is more of a backward-looking metric in terms of the health of the overall business.

**Relevant commentary:** (CFO Ashim Gupta FQ122) "*I would like to emphasize the financial strengths of our business model. We have a highly recurring subscription-based business primarily contracted annually and build in advance. Our technology is incredibly sticky resulting in best-in-class dollar-based gross retention and net retention rates. We offer our customers flexible deployment models on prem, hybrid, cloud and SaaS, the mix of which can create lumpy revenue recognition particularly from license revenue under ASC 606. As a result, we remind you that we run and manage our business on ARR which is most representative of the underlying performance of our business.* "

(10-K definition and explanation as a key metric) "*We define ARR as annualized invoiced amounts per solution SKU from subscription licenses and maintenance obligations assuming no increases or reductions in their subscriptions. ARR does not include the costs we may incur to obtain such subscription licenses or provide such maintenance, and does not reflect any actual or anticipated reductions in invoiced value due to contract non-renewals or service cancellations other than for specific bad debt or disputed amounts.*"

Exhibit 72 - PATH Leading Metrics Y/Y Growth Rates FY20A-FY22A

Exhibit 72 shows Y/Y growth of PATH's key leading metrics since FY20.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Varonis (VRNS) – covered by Matt Hedberg

**Key metric(s):** ARR

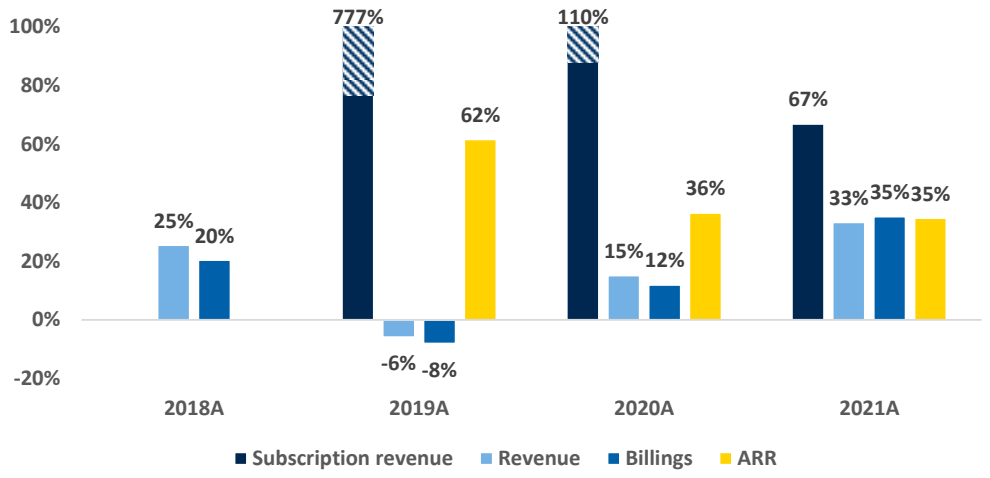
**Leading metric practice/relevance:** Post the company's subscription transition, the primary forward-looking metric for Varonis is ARR with subscription revenue growth also meaningful, in our opinion. The transition to subscription has unlocked the full-power of Varonis's platform enabling customers to see the compounding value of multiple solutions. This has shown up in net expansion rates, which along with new customer additions are the primary drivers for ARR growth. The maturity of the transition was highlighted by ARR now being given as an annual guidance metric with expectations that revenue growth and ARR growth begin to converge in CY/22.

**Relevant commentary:** (CFO Guy Melamed FQ421) "*I want to note that as our SaaS offering continues to increase in the years ahead, ARR will become even more important as a leading indicator for our business, given the ratable revenue recognition of SaaS. Additionally, some color on our 2022 guidance.*"

(10-K definition and explanation as a key metric) "*Annual recurring revenues is a key performance indicator defined as the annualized value of active term-based subscription license contracts and maintenance contracts related to perpetual licenses in effect at the end of that period. Subscription license contracts and maintenance for perpetual license contracts are annualized by dividing the total contract value by the number of days in the term and multiplying the result by 365.*"

Exhibit 73 - VRNS Leading Metrics Y/Y Growth Rates 2017A-2021A

Exhibit 73 shows Y/Y growth of VRNS's key leading metrics since CY17.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



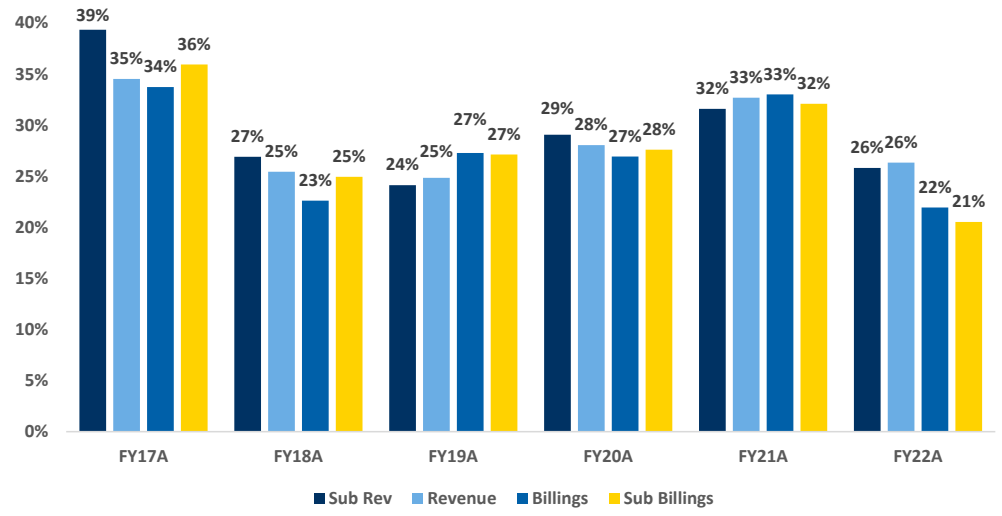
## Veeva Systems Inc. (VEEV) – covered by Rishi Jaluria

**Key metric(s):** Billings

**Leading metric practice/relevance:** Veeva has a strong billings practice, in our view. The company explicitly guides to quarterly and annual billings, while providing commentary of the puts and takes. Investors view billings as a key metric. We note Veeva has unusual billings seasonality (due to co-terminus billings) and the metric can be noisy on a quarterly basis, but we find it to be a good leading indicator on an annual basis. Management doesn't view RPO as a useful leading indicator since ~80% of deals are one year and several factors cause variability in RPO, more so than billings, and the company excludes professional services from RPO.

**Relevant commentary:** (10-K commentary on billings) "*Many of our customers, incl. many of our large customers, are billed on a quarterly basis and therefore a substantial portion of the value of contracts billed on a quarterly basis will not be reflected in our deferred revenue at the end of any given quarter. Also, particularly with respect to our Commercial Solutions orders, because the term of orders for additional end users or apps is commonly <1 year, the annualized value of such orders may not be completely reflected in deferred revenue at any single point in time. We have also agreed from time to time, and may agree in the future, to allow customers to change the renewal dates of their orders to, for ex., align more closely with a customer's annual budget process or to align with the renewal dates of other orders placed by other entities within the same corporate control group, or to change payment terms from annual to quarterly, or vice versa. Such changes typically result in an order of less than 1 year to align all orders to the desired renewal date and, thus, may result in a lesser increase to deferred revenue than if the adjustment had not occurred. Additionally, changes in renewal dates may change the quarter in which deferred revenue associated with a particular order is booked. Accordingly, we do not believe changes on a quarterly basis in deferred revenue, unbilled A/R, or calculated billings, a metric commonly cited by financial analysts, are accurate indicators of the underlying momentum of our business or future revenues. We believe that our subs. revenue guidance and calculated billings guidance for the full FY are the best indicators of the momentum of our business or future revenues.*"

Exhibit 74 - VEEV Leading Metrics Y/Y Growth Rates, 2020A-2021A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## VMware (VMW) – covered by Matt Hedberg

**Key metric(s):** Subscription and SaaS ARR

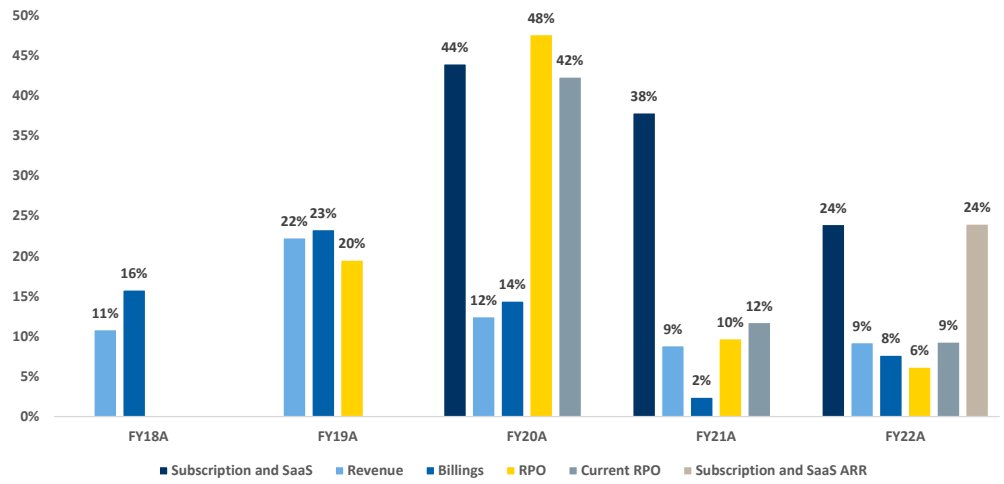
**Leading metric practice/relevance:** Following the spin-off from Dell, VMware continues to focus on being Switzerland to multi-cloud environments with a growing set of SaaS and subscription solutions to align with its enterprise customers purchasing preferences. Throughout FY/23, the company will continue to make all major product offerings available through these purchasing methods including all SDDC offerings to go along with Tanzu, cloud-management, security and end-user SaaS offerings to continue and accelerate this business model transformation. SaaS ARR continues to be the primary leading metric for the success of this transition.

**Relevant commentary:** (CFO Zane Rowe FQ422) "*Our large installed base provides an opportunity to convert recurring software maintenance streams to subscription and SaaS, delivering incremental value to customers by driving continuous innovation as well as enabling additional upsell and cross-sell opportunities….Q4 FY 2022 performance marked continued progress towards VMware's strategy of becoming a multi-cloud subscription and SaaS company given our mission critical portfolio and deep customer relationships.*"

(10-K definition and explanation as a key metric) "*Annual recurring revenue ("ARR") represents the annualized value of our committed customer subscription and SaaS contracts as of the end of the reporting period, assuming any contract that expires during the next 12 months is renewed on its existing terms, except that, for consumption-based subscription and SaaS offerings, ARR represents the annualized quarterly revenue based on revenue recognized for the current reporting period. ARR is an operating measure we use to assess the strength of our subscription and SaaS offerings. ARR is a performance metric and should be viewed independently of, and not as a substitute for or combined with, revenue and unearned revenue.*"

Exhibit 75 - VMW Leading Metrics Y/Y Growth Rates FY18A-FY22A

Exhibit 75 shows Y/Y growth of VMW's key leading metrics since FY18A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Workday, Inc. (WDAY) – Not Covered

**Key metric(s):** Subscription Revenue Backlog, Billings

**Leading metric practice/relevance:** Workday reports on and guides to 24-month Subscription Revenue Backlog (i.e. subscription RPO) and additionally provides total Subscription Revenue Backlog. Management does not view a 12-month metric as practical since Workday's contracts are typically 3+ years and notes a 24-month metric does mostly eliminate the duration factor. While we still would prefer a 12-month metric, we acknowledge Workday's Subscription Backlog coverage ratio (as a percentage of N24M subscription revenue) is notably high at 66% if we look back to FY20 (FY20 Subscription Backlog represented 66% of FY21 and FY22 revenue) and was consistently high in FY19/FY18. We believe investors still pay attention to billings too.

**Relevant commentary:** (10-K Subscription Revenue Backlog Definition) "*Our subscription revenue backlog, which is also referred to as remaining performance obligations for subscription contracts, represents contracted subscription services revenues that have not yet been recognized and includes billed and unbilled amounts. Subscription revenue backlog may fluctuate from period to period due to a number of factors, including the timing of renewals and overall renewal rates, new business growth, average contract duration, and seasonality.*"

(Former CFO Robynne Sisco on WDAY's FY20 Analyst Day) "*Subscription revenue backlog… is a metric that we've been reporting now for a few years with the new revenue recognition standard. All of us, I know you included, us included, are still trying to wrap our heads around what's the best way to use these numbers. Our overall subscription revenue growth that we reported on our last earnings call, 27%. When you look at the 0 to 24, 28% that takes out a lot of the duration dynamics, one of the questions that we get a lot is… are we going to disclose this 0-12 month backlog, and we still firmly believe that given the duration of our contracts that the 0 to 24 is the most meaningful metric for us to be disclosing. So, we don't plan to change that. But I will say, we went back and we looked at the last couple of years, 0-12 and 0-24 month growth, fairly consistent, right? It varies by a few % points with the 0-12 generally being higher because you get even less of a duration impact. And so, you can use the 0-24 month growth as a pretty strong proxy for 0-12. Another thing that we've seen with this is that we've seen the analysts and the investors do with this subscription revenue backlog number is to calculate a bookings number from that and then to take those bookings numbers and to calculate growth rates and then to make conclusion – draw conclusions on our net new ACV growth based on that bookings growth. And I just want to caution you about doing that. We went back and looked at 4 separate quarters pre-Adaptive acquisition that it wasn't a factor in this analysis, and we looked at the difference between the bookings growth rates that all of you are calculating and our actual net new ACV growth rates, and we saw wild fluctuations between those two being off in both directions as much as 45%. And the dynamic that's driving that is the renewal base, right? And our renewal base, the timing of that can move around and it can fluctuate. And so, if you look at the renewals as a percent of total bookings in any given quarter, that can wildly fluctuate as well, which is why subscription bookings growth is actually not a good reflection of our net new ACV growth. If you look at the TTM bookings and you look at the growth on that, it's a better metric, still not perfect, but it's better one to use.*"

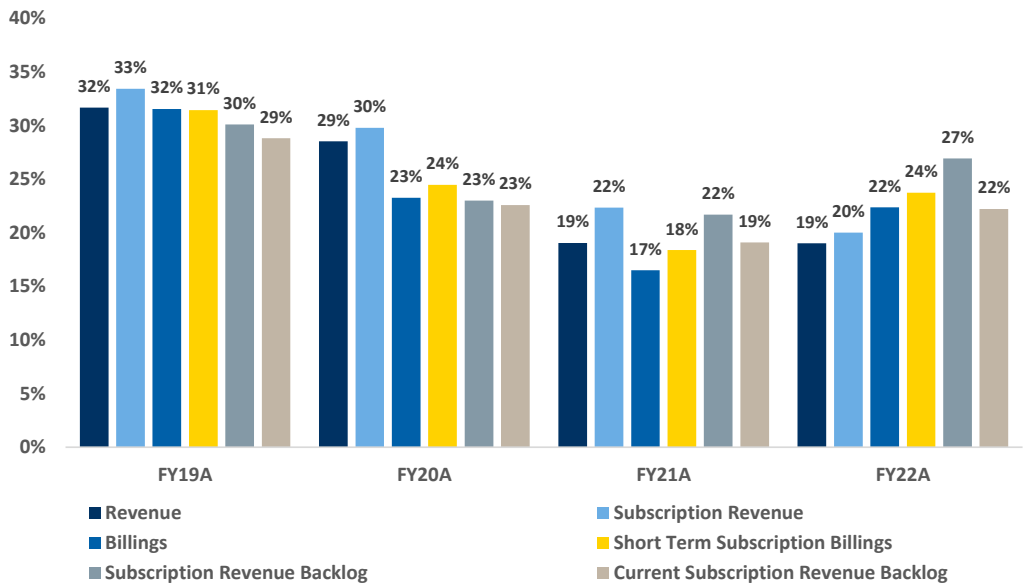
(Former CFO Robynne Sisco on F4Q19 earnings call) "*As we enter FY20, I want to reiterate that we don't run the business to maximize either unearned revenue or calculated billings. As we discussed throughout this past fiscal year, our focus continues to be on new customer acquisition and long-term customer economics, which means that we will likely continue to see variability in unearned revenue throughout the year. We believe that our subscription revenue backlog is a much better gauge of the health of our business over the long term. And even though backlog will have some variability quarter-to-quarter based on contract duration and*

This report is intended solely for cbass@marketstreetpartners.com



*the timing of renewals, including early renewals associated with add-on business, unlike billings and unearned, it filters out the noise caused by varying invoicing terms."*

Exhibit 76 - WDAY Leading Metrics Y/Y Growth Rates FY19A-FY22A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Xometry (XMTR) – covered by Matt Hedberg

**Key metric(s):** N/A

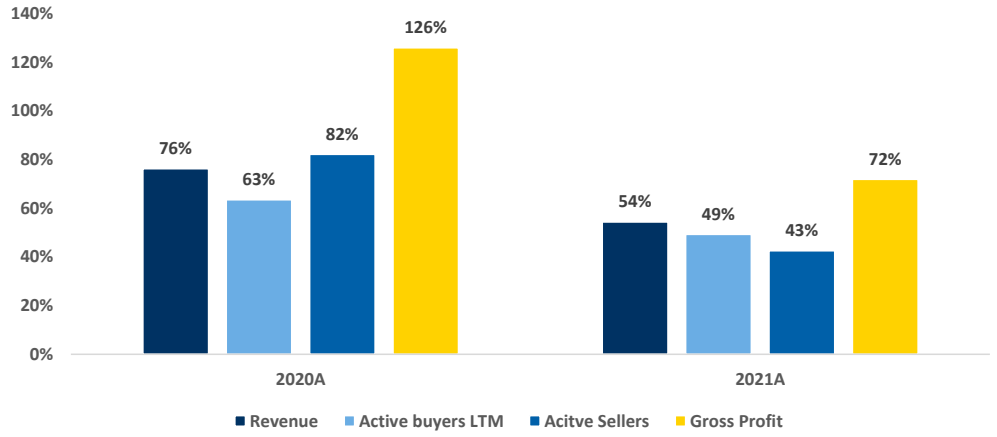
**Leading metric practice/relevance:** Xometry has a unique business model with its software enabling a large marketplace of buyers and sellers to enter into manufacturing transactions. The number of buyers and sellers on the platform are good leading indicators for future revenue growth due to the high-retention rate of the service and percentage of revenue, which comes from existing customers. Additionally, the company generates revenue from seller services, which come at a higher gross margin than the core business. Due to a lower gross margin than software peers, investors focus on gross profit as a primary growth metric.

**Relevant commentary:** (CFO James Rallo FQ421) "*Q4 active buyers increased 49% year-over-year to 28,130. In Q4, the percentage of revenue from existing accounts was 95%, underscoring the efficiency and transparency of our business model that lead to increasing account stickiness and spend over time. We believe the repeat purchase activity from existing accounts reflects the underlying strength of our business and provides us with substantial revenue visibility and predictability. Once an account joins our platform, we aim to expand the relationship and increase engagement and spending activities from the account over time.*"

(10-K definition and explanation as a key metric) "*We define Active Buyers as the number of buyers who have made at least one purchase on our marketplace during the last twelve months. An increase or decrease in the number of Active Buyers is a key indicator of our ability to attract, retain and engage buyers on our platform.*"

Exhibit 77 - XMTR Leading Metrics Y/Y Growth Rates 2020A-2021A

Exhibit 77 shows Y/Y growth of XMTR's key leading metrics since 2020.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Ziff Davis, Inc. (ZD) – covered by Rishi Jaluria

**Leading metric practice/relevance:** We do not view billings as a useful indicator for ZD as the company has a large advertising-based business (with relatively low average spend per advertiser) and primarily monthly-based subscriptions. Additionally, ZD has yet to provide historical financials following the consensus spin-off, making analysis difficult.

This report is intended solely for cbass@marketstreetpartners.com



## Zendesk, Inc. (ZEN) – Not Covered

**Key metric(s):** RPO, billings

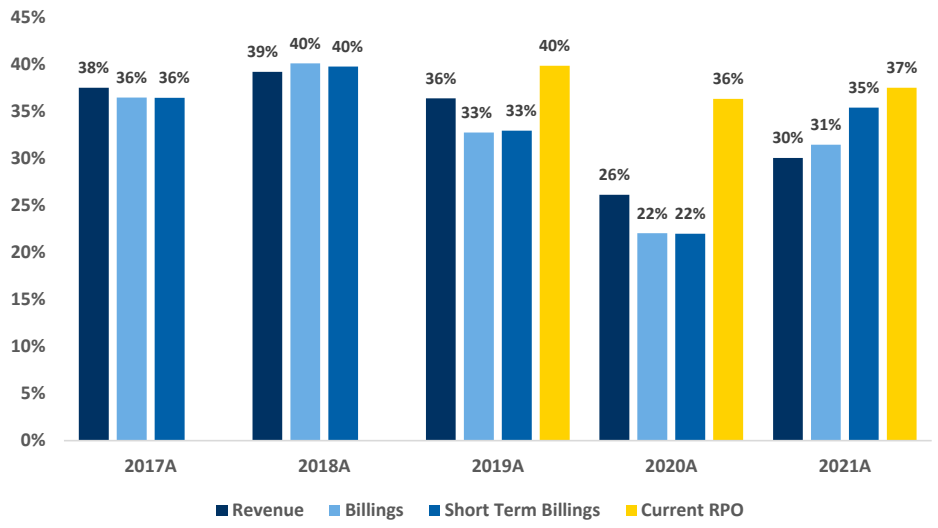
**Leading metric practice/relevance:** Zendesk focuses on RPO and cRPO, and provides quarterly commentary on RPO during earnings calls. Zendesk's RPO figures include contracts of 12 months or less, and cRPO is based on revenue expected to be recognized in the next 12 months, both of which we view favorably. Management views RPO as the best measure of success in the company's signing of longer, more strategic contracts with customers, though we acknowledge RPO gets a boost as mix shifts to quarterly/annual billings cycles from monthly, thus it is not necessarily the best leading indicator for revenue growth, in particular. Despite duration-led volatility, we believe investors still pay attention to billings alongside RPO/cRPO.

**Relevant commentary:** (Former CFO Elena Gomez on 1Q21 earnings all) "*I would just say RPO continues to be a great proxy for us. In fact, I would encourage you to look at that because our billings, as you know, many of our customers are monthly and that billings number can move in any given quarter. But RPO is a good proxy for us in particular. But just remember it doesn't always capture all of our customers so you have to look at a couple of different data points. But we definitely encourage you guys to look at RPO, especially as our proxy for enterprise*."

(Former CFO Elena Gomez on 4Q20 earnings call) "*We have been focused on signing multi-year deals. And so, you're not going to see that in the current billings in revenue unless they do annual billing, which we're definitely moving more and more as we can to annual billing. But more importantly, we're getting longer term commitments, which will show up in RPO but may not show up in the current billing cycle or revenue*."

Exhibit 78 - ZEN Leading Metric Y/Y Growth Rates, 2017A-2021A

Exhibit 78 shows Y/Y growth of ZEN's key leading metrics since 2017.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## ZoomInfo Technologies, Inc. (ZI) – covered by Rishi Jaluria

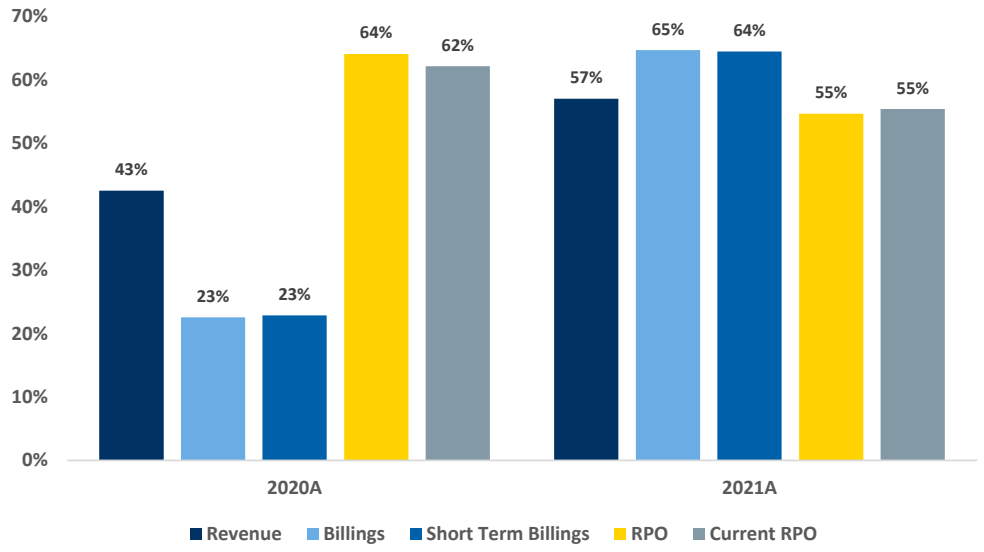
**Key metric(s):** RPO, billings

**Leading metric practice/relevance:** As highlighted below, management views sequential revenue growth as the best indicator of forward momentum in the business, though management still provides quarterly commentary on RPO, and sometimes billings, which we still see as relevant despite their volatility. ZoomInfo implements best practices, in our view, with RPO including contracts that are 12 months or less and by providing a cRPO figure based on revenue expected to be recognized in the next 12 months.

**Relevant commentary:** (CFO on 4Q21 earnings call) "*With respect to liabilities and future performance obligations, unearned revenue at the end of the quarter was $364 million and remaining performance obligations, or RPO, were $864 million, of which $672 million are expected to be delivered in the next 12 months. We believe calculated billings and RPO are imprecise metrics to assess in-period activity and forward momentum. As a result, we focus on days adjusted sequential revenue growth. We delivered 13% days adjusted sequential revenue growth in the fourth quarter, up from 12% in the third quarter.*"

Exhibit 79 - ZI Leading Metric Y/Y Growth Rates, 2020A-2021A

Exhibit 79 shows Y/Y growth of ZI's key leading metrics since 2020.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Zoom Video Communications, Inc. (ZM) – covered by Rishi Jaluria

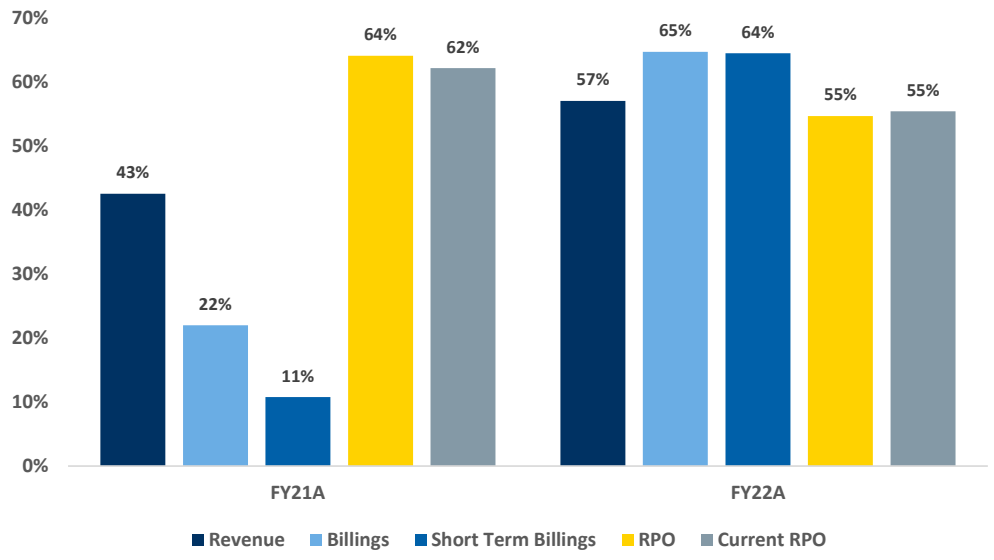
**Key metric(s):** Billings

**Leading metric practice/relevance:** Zoom does not view billings as a reliable forward-looking metric considering the large mix of monthly contracts (which grew materially as a percent of total business in the pandemic). We believe that investors do view billings as important and watch the metric closely but generally agree that its usefulness has diminished considering Zoom's unique positioning as *the* COVID-19 beneficiary. We do believe Zoom could benefit from disclosing ARR, especially with the ongoing consumer churn.

**Relevant commentary:** (CFO Kelly Steckelberg on F4Q22 earnings call) *"So remember that, unfortunately, billings is not a good forward-looking metric. And it's due to the fact that we have this 50-50 split in our business. And the Enterprise business is – the billings associated with that are what you would expect in a normal SaaS business. And they're multiyear – annual to multiyear. In the Online business, it is not. But there are many of those customers that are still buying and pay on a monthly basis. So it really doesn't make sense for a metric. And that's why you're going to continue to see volatility in that metric. And because of that, I think what we've tried to give you was at least some color around deferred revenue. So you can understand that because also the seasonality and the linearity that we have in our renewals and our billings. Unfortunately, the best that I have to give you is our revenue guidance, and that's it. We're trying to give you metrics that better reflect now how we think about that."*

Exhibit 80 shows Y/Y growth of ZM's key leading metrics since FY21A.

Exhibit 80 - ZM Leading Metrics Y/Y Growth Rates, FY21A-FY22A




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Zscaler (ZS) – covered by Matt Hedberg

**Key metric(s):** Billings, ST Billings, cRPO, RPO

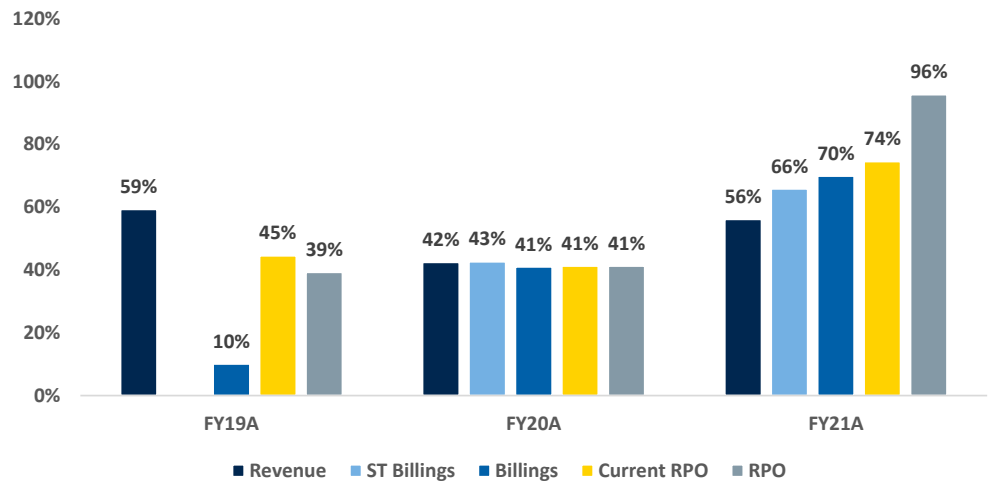
**Leading metric practice/relevance:** Zscaler's business model lends itself well to traditional metrics due the cloud-based delivery model. Outside of revenue and billings, forward-looking metrics such as ST billings, RPO and cRPO all remain in focus supported by a strong +125% DBNRR. While some companies view billings as more of a backward-looking metric based on their revenue model, Zscaler continues to feel that it is the best indicator for the overall business, while RPO, cRPO and ST Billings are all useful metrics. This is in part due to RPO and cRPO's sensitivity to large deals and the timing of renewals.

**Relevant commentary:** (CFO Remo Canessa FQ222) "*If you take a look at our RPO growth year over year, it was about 90% and CRPO growth was 79%. What we've always called out is when the RPO growth rates and CRPO growth rates are growing triple digit, we brought back investors that said billings is the best way to look at our business. And when you take a look at RPO and CRPO, they're more sensitive to the timing of large deals, the timing of renewals, contract durations, and other specific terms. And it's for these reasons that we want – we feel that people should be looking at billings versus RPO or CRPO, especially when we're in the range of 10 to 14 months. And both in the quarter, our duration was right in the middle of the range of 10 to 14 months. One thing that I'd also call out that investors probably look at is short-term billings growth. Our short-term billings growth was 61%. So I would look at everything as an investor. But from a Zscaler perspective, it's really billings.*"

(10-K definition and explanation as a key metric) "*Calculated billings is a non-GAAP financial measure that we believe is a key metric to measure our periodic performance. Calculated billings represents our total revenue plus the change in deferred revenue in a period. Calculated billings in any particular period aims to reflect amounts invoiced for subscriptions to access our cloud platform, together with related support services for our new and existing customers. We typically invoice our customers annually in advance, and to a lesser extent quarterly in advance, monthly in advance or multi-year in advance.*"

Exhibit 81 - ZS Leading Metrics Y/Y Growth Rates FY19A-FY21A

Exhibit 81 shows Y/Y growth of ZS's key leading metrics since FY19A.




Source: Company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Corporate Access Recap

**CWAN:** We recently hosted investor meetings with CWAN's CEO Sandeep Sahai, CFO Jim Cox and IR Joon Park, and we came away more positive on the acceleration in the business and long runway for growth in the years ahead. We like the improved efficiency in onboarding new clients, broad-based demand environment and strong traction internationally/new end markets, underappreciated durability of the underlying AUM, and focused investment plans. For further reading, please see our note titled: "Encouraging Takeaways from vNDR with Management."

**NICE:** We recently hosted investor meetings with VP of Investor Relations Marty Cohen and Chief Financial Officer Beth Gaspich. Our key takeaways were: 1) NICE's Digital self-service/AI solutions are addressing a large, growing trend with an attractive competitive moat; 2) we see multiple trends in the contact center labor market (demand for WFH, hiring challenges) benefiting NICE and furthering CCaaS adoption; 3) underpenetrated TAM, with Digital being a notable expansion opportunity; and 4) we continue to view NICE's competitive positioning and leadership in CCaaS as sustainable. For further reading, please see our note titled: "On the (virtual) road with NICE; Digital, Labor Market, Competition, and the Cloud Opportunity."

**ZD:** We hosted ESG-focused virtual investor meetings with ZD CEO Vivek Shah, CFO Bret Richter, VP Sustainability and Responsibility Darrah Feldman, and VP Corporate Finance Alan Steier. Our key highlights were: 1) progress on ZD's board refresh should be a positive for investors; 2) DE&I-based exec bonuses show real commitment; 3) ZD's environmental transparency is ahead of peers; and 4) improvements on ESG rating-agency-friendly formatting should drive continued improvement in ESG scores. For further reading, please see our note titled: "Encouraging Takeaways from vNDR with Management."

**VEEV:** We hosted Veeva's CFO Brent Bowman and EVP of Strategy Paul Shawah for a group investor call. We came away with increased confidence in the reacceleration of revenue / billings growth to 20%+, as large R&D deals begin to close (recently announced large R&D win with Boehringer Ingelheim) and new hires become more productive. Other key highlights include: 1) large opportunities in Data Cloud, and OLS with MedTech CRM; 2) recapping billings guidance dynamics with 2H reacceleration; and 3) no impact to date from geopolitical tensions. For further reading, please see our note titled: "Veeva Investor Group Call: Large R&D Deals and Positive Momentum in Data Cloud."

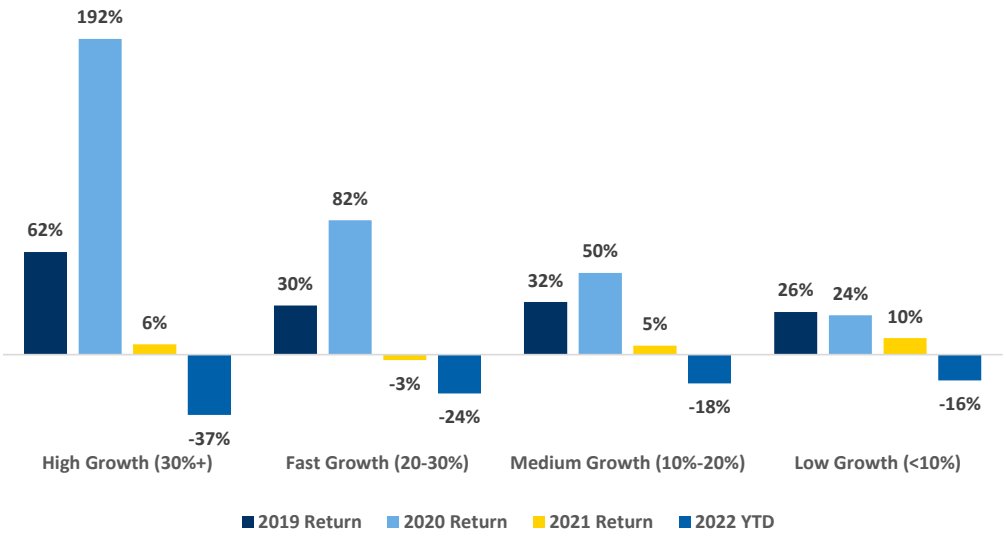
This report is intended solely for cbass@marketstreetpartners.com



## Returns by Style

High Growth (30%+) and Fast Growth (20%-30%) software stocks outperformed Medium Growth (10%-20%) and Low Growth (<10%) software stocks in 2019 and 2020. However, in late 2021 and YTD in 2022 (based on prices at close on April 24, 2022), this trend was upended as rising rates became center stage and impacted software broadly, but especially High Growth software. In 2021, High Growth returned up 6%, Fast Growth returned down 3%, Medium Growth returned up 5%, and Low Growth returned up 10%. YTD in 2022, High Growth is down 37%, Fast Growth is down 24%, Medium Growth is down 18%, and Low Growth is down 16%. That said, from 2019 to present, High Growth returned 319% (42% CAGR), Fast Growth 174% (18% CAGR), Medium Growth returned 172% (18% CAGR), and Low Growth returned 145% (12% CAGR).

Exhibit 82 - Stock Returns for High, Medium and Low Growth Software Companies, 2019-2021




Source: FactSet, RBC Capital Markets

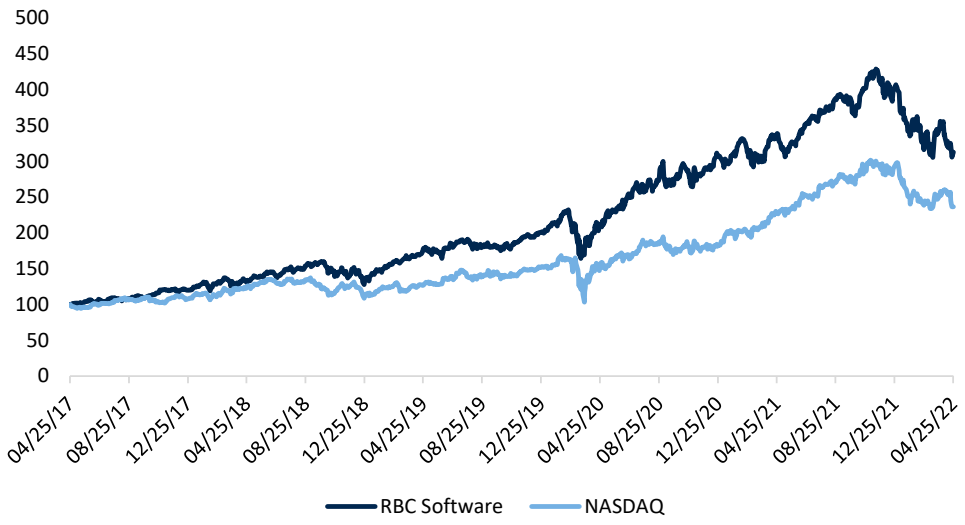
This report is intended solely for cbass@marketstreetpartners.com



# Valuation

Our analysis of the software sector through this report is based on our proprietary RBC Software Index ("The Index"). The index is an equal-weighted index of 164 publicly traded software companies. Exhibit 83 shows the performance of the Index since 2016. Exhibits 84 and 85 show the components of the Index, separated by Application and Infrastructure software.

Exhibit 83 - RBC Software Index (end of 2016 = 100)




Source: FactSet, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Exhibit 84 - Application Software Comps

***APPLICATIONS***

| | | 04/22/2022 | CY Revenue Growth | | | | CY EV/Revenue | | | | CY EV/EBITDA | | | | CY EV/FCF | | | | CY PE | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Company Name** | **Ticker** | **Price** | **2020** | **2021E** | **2022E** | **2023E** | **2020** | **2021E** | **2022E** | **2023E** | **2020** | **2021E** | **2022E** | **2023E** | **2020** | **2021E** | **2022E** | **2023E** | **2020** | **2021E** | **2022E** | **2023E** |
| 2U, Inc. | TWOU | $10.55 | 21% | 22% | 13% | 13% | 1.9X | 1.5X | 1.4X | 1.2X | 90.1X | 21.7X | 18.5X | 12.1X | NM | NM | NM | NM | NM | NM | NM | NM |
| Adobe Incorporated | ADBE | $408.67 | 15% | 23% | 13% | 15% | 15.2X | 12.4X | 11.0X | 9.6X | 31.2X | 24.8X | 22.4X | 19.4X | 37.0X | 28.4X | 26.6X | 23.0X | 40.5X | 32.7X | 30.0X | 25.4X |
| Alkami Technology Inc | ALKT | $11.86 | 51% | 36% | 25% | 25% | 8.3X | 6.1X | 4.9X | 3.9X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Altair Engineering Inc. Class A | ALTR | $55.51 | 0% | 13% | 8% | 9% | 10.0X | 8.8X | 8.1X | 7.4X | 81.7X | 54.9X | 45.8X | 38.4X | 174.6X | 87.0X | 145.2X | 48.0X | 179.1X | 81.6X | 70.4X | 60.1X |
| Alteryx, Inc. Class A | AYX | $67.87 | 19% | 8% | 34% | 19% | 10.3X | 9.5X | 7.1X | 6.0X | 64.9X | 403.0X | NM | 96.0X | 103.6X | 167.1X | NM | 171.4X | 73.0X | NM | NM | NM |
| Amplitude Inc Class A | AMPL | $18.82 | 50% | 63% | 38% | 34% | 20.5X | 12.6X | 9.1X | 6.8X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Anaplan, Inc. | PLAN | $65.15 | 29% | 32% | 26% | 24% | 23.0X | 17.4X | 13.8X | 11.1X | NM | NM | NM | 324.4X | NM | NM | NM | NM | NM | NM | NM | NM |
| ANSYS, Inc. | ANSS | $276.52 | 11% | 14% | 8% | 10% | 14.5X | 12.7X | 11.8X | 10.7X | 32.3X | 29.1X | 26.0X | 22.7X | 47.9X | 46.6X | 42.8X | 36.0X | 41.3X | 37.5X | 34.7X | 31.4X |
| AppFolio Inc Class A | APPF | $108.88 | 21% | 16% | 26% | 20% | 13.6X | 11.7X | 9.3X | 7.7X | 85.3X | 125.6X | 84.6X | 66.8X | NM | NM | 222.8X | 135.4X | 24.5X | 4199.5X | NM | 593.9X |
| AppLovin Corp. Class A | APP | $41.21 | 46% | 92% | 32% | 23% | 11.8X | 6.1X | 4.6X | 3.8X | 49.6X | 23.6X | 16.4X | 11.8X | NM | 47.5X | 17.4X | 14.8X | NM | 588.7X | 95.3X | 33.5X |
| Asana, Inc. Class A | ASAN | $29.09 | 59% | 67% | 40% | 33% | 26.0X | 15.6X | 11.2X | 8.4X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Aspen Technology, Inc. | AZPN | $156.91 | 13% | -2% | 14% | 7% | 15.7X | 16.0X | 14.0X | 13.0X | 27.3X | 31.0X | 26.6X | 25.2X | 42.0X | 37.1X | 37.2X | 29.7X | 31.7X | 34.5X | 30.2X | 29.5X |
| Atlassian Corp. Plc Class A | TEAM | $245.29 | 27% | 35% | 22% | 32% | 35.6X | 26.4X | 21.6X | 16.4X | 137.7X | 95.5X | 101.2X | 61.4X | 134.6X | 82.0X | 92.9X | 82.8X | 209.6X | 146.0X | 170.8X | 126.9X |
| Autodesk, Inc. | ADSK | $187.31 | 16% | 16% | 15% | 16% | 11.2X | 9.7X | 8.4X | 7.2X | 35.7X | 30.7X | 21.4X | 17.4X | 31.5X | 28.7X | 19.9X | 23.2X | 46.2X | 36.9X | 27.8X | 22.5X |
| Avalara Inc | AVLR | $84.41 | 31% | 40% | 23% | 22% | 14.3X | 10.2X | 8.4X | 6.9X | NM | 364.1X | NM | NM | 207.7X | NM | NM | 206.0X | 844.1X | NM | NM | 1651.5X |
| Benefitfocus, Inc. | BNFT | $11.47 | -9% | -2% | -3% | NM | 1.7X | 1.8X | 1.8X | NM | 10.5X | 9.4X | 10.0X | 9.4X | 29.8X | 19.4X | 16.5X | 15.4X | NM | NM | NM | NM |
| BigCommerce Holdings Inc | BIGC | $17.32 | 36% | 44% | 27% | 21% | 8.4X | 5.8X | 4.6X | 3.8X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Bill.com Holdings, Inc. | BILL | $172.53 | 37% | 124% | 72% | 32% | 107.1X | 47.9X | 27.9X | 21.0X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Blackbaud, Inc. | BLKB | $54.45 | 1% | 2% | 16% | 6% | 4.2X | 4.2X | 3.6X | 3.4X | 15.9X | 15.9X | 14.8X | 13.3X | 50.6X | 23.8X | 25.5X | 19.0X | 18.5X | 18.0X | 20.0X | 18.1X |
| BlackLine, Inc. | BL | $65.63 | 22% | 21% | 23% | 21% | 11.3X | 9.3X | 7.6X | 6.3X | 67.8X | 66.3X | 127.8X | 75.3X | 111.0X | 68.8X | 422.8X | 130.6X | 86.4X | 113.2X | 793.3X | 168.8X |
| Blend Labs, Inc. Class A | BLND | $4.49 | 92% | 27% | -37% | 11% | 2.5X | 1.9X | 3.1X | 2.8X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Cadence Design Systems, Inc. | CDNS | $148.07 | 15% | 11% | 13% | 9% | 15.5X | 13.9X | 12.3X | 11.3X | 41.5X | 34.6X | 29.5X | 27.9X | 51.1X | 31.2X | 35.9X | 33.3X | 53.1X | 44.7X | 39.0X | 35.1X |
| Ceridian HCM Holding, Inc. | CDAY | $56.72 | 2% | 22% | 18% | 17% | 11.4X | 9.4X | 7.9X | 6.8X | 60.3X | 59.0X | 51.0X | 40.1X | NM | NM | 265.5X | 247.6X | 115.8X | 218.2X | 154.6X | 96.5X |
| Chegg, Inc. | CHGG | $26.74 | 57% | 20% | 9% | 14% | 5.8X | 4.9X | 4.5X | 3.9X | 18.2X | 14.2X | 14.1X | 11.8X | 33.1X | 21.8X | 25.1X | 20.8X | 20.4X | 20.7X | 23.1X | 19.0X |
| Clearwater Analytics Holdings, Inc. Class A | CWAN | $17.63 | 21% | 24% | 20% | 20% | 21.1X | 17.0X | 14.2X | 11.8X | 75.2X | 59.0X | 52.7X | 42.5X | NM | NM | 128.6X | 101.2X | NM | NM | 82.3X | 69.1X |
| CoStar Group, Inc. | CSGP | $61.20 | 19% | 17% | 11% | 15% | 12.9X | 11.0X | 9.9X | 8.6X | 38.9X | 33.1X | 36.2X | 29.7X | 50.3X | 50.4X | 51.2X | 42.4X | 61.9X | 53.9X | 61.1X | 49.9X |
| Coupa Software, Inc. | COUP | $89.48 | 39% | 34% | 16% | 21% | 14.6X | 10.9X | 9.4X | 7.8X | 114.8X | 76.6X | 157.6X | 83.2X | 86.2X | 50.9X | 61.5X | 46.2X | 117.7X | 107.8X | 451.7X | 169.3X |
| Coursera Inc | COUR | $21.05 | 59% | 41% | 30% | 25% | 9.3X | 6.6X | 5.1X | 4.0X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| CS Disco, Inc. | LAW | $30.30 | 41% | 67% | 31% | 32% | 23.1X | 13.8X | 10.6X | 8.0X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| DocuSign, Inc. | DOCU | $86.33 | 49% | 45% | 17% | 16% | 12.4X | 8.6X | 7.3X | 6.3X | 77.5X | 38.1X | 36.2X | 34.1X | 84.0X | 40.5X | 38.1X | 34.4X | 98.1X | 43.8X | 43.5X | 37.2X |
| Domo, Inc. Class B | DOMO | $41.86 | 21% | 23% | 25% | 20% | 7.6X | 6.2X | 5.0X | 4.1X | NM | NM | NM | NM | NM | NM | NM | 279.7X | NM | NM | NM | NM |
| Duck Creek Technologies, Inc. | DCT | $15.79 | 24% | 23% | 13% | 15% | 8.0X | 6.5X | 5.7X | 5.0X | 126.8X | 84.6X | 87.5X | 57.5X | 269.6X | NM | 302.1X | 92.8X | NM | 157.9X | 170.7X | 103.1X |
| DoubleVerify Holdings, Inc. | DV | $22.08 | 33% | 36% | 30% | 27% | 14.7X | 10.8X | 8.3X | 6.5X | 48.9X | 32.6X | 27.6X | 20.4X | 325.6X | 49.0X | 39.6X | 30.6X | NM | 74.5X | 44.5X | 32.2X |
| Enfusion, Inc. Class A | ENFN | $12.59 | 35% | 40% | 33% | 30% | 17.5X | 12.5X | 9.4X | 7.2X | 166.8X | 60.9X | 42.1X | 26.3X | NM | NM | 75.4X | 36.2X | NM | NM | 57.7X | 39.7X |
| Envestnet, Inc. | ENV | $80.91 | 11% | 19% | 14% | 13% | 5.5X | 4.6X | 4.0X | 3.6X | 22.5X | 20.9X | 20.2X | 16.7X | NM | 40.5X | 37.4X | 26.2X | 31.5X | 33.4X | 34.7X | 27.6X |
| eGain Corporation | EGAN | $10.52 | 9% | 12% | 16% | NM | 3.9X | 3.5X | 3.0X | NM | 27.0X | 28.6X | 42.3X | NM | NM | NM | NM | NM | 36.3X | 33.9X | NM | NM |
| EverCommerce, Inc. | EVCM | $12.96 | 40% | 45% | 27% | 17% | 9.0X | 6.2X | 4.9X | 4.2X | 38.6X | 28.4X | 24.7X | 19.6X | 68.3X | 116.2X | 39.0X | 26.5X | NM | NM | 39.8X | 29.6X |
| Freshworks, Inc. Class A | FRSH | $17.78 | 45% | 49% | 32% | 32% | 21.5X | 14.5X | 10.9X | 8.3X | NM | NM | NM | NM | 228.2X | NM | NM | 489.0X | NM | NM | NM | NM |
| Guidewire Software, Inc. | GWRE | $89.73 | 9% | 0% | 8% | 12% | 9.6X | 9.6X | 8.8X | 7.9X | 65.2X | NM | NM | 412.1X | 78.8X | NM | 110.6X | 79.4X | 74.8X | NM | NM | 1131.1X |
| HubSpot, Inc. | HUBS | $396.28 | 31% | 47% | 33% | 28% | 21.6X | 14.7X | 11.0X | 8.7X | 169.9X | 118.7X | 89.9X | 63.0X | 241.2X | 93.8X | 80.9X | 64.6X | 289.3X | 217.7X | 166.2X | 116.4X |
| Instructure Holdings, Inc. | INST | $17.63 | 17% | 36% | 12% | 11% | 9.6X | 7.1X | 6.3X | 5.7X | NM | 19.8X | 17.7X | 15.4X | NM | 28.8X | 19.5X | 15.8X | NM | 21.5X | 18.0X | 16.2X |
| Intuit Inc. | INTU | $443.29 | 8% | 48% | 16% | 14% | 17.5X | 11.8X | 10.2X | 8.9X | 46.1X | 30.6X | 26.2X | 21.6X | 56.1X | 41.0X | 34.1X | 27.7X | 56.0X | 39.9X | 35.8X | 30.2X |
| Kaltura, Inc. | KLTR | $1.49 | 24% | 37% | 7% | 17% | 0.8X | 0.6X | 0.5X | 0.4X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Mimecast Limited | MIME | $79.82 | 19% | 19% | 15% | NM | 11.6X | 9.7X | 8.4X | NM | 47.3X | 33.8X | 31.0X | 24.0X | 72.8X | 44.6X | 38.5X | 30.1X | 78.3X | 57.8X | 52.0X | 45.6X |
| Model N, Inc. | MODN | $26.02 | 14% | 22% | 9% | 14% | 6.0X | 4.9X | 4.5X | 3.9X | 40.9X | 38.2X | 38.4X | 27.6X | 52.6X | 110.8X | 29.3X | 30.8X | 50.0X | 49.1X | 53.6X | 40.0X |
| Momentive Global Inc. | MNTV | $17.40 | 22% | 18% | 12% | 14% | 7.0X | 5.9X | 5.3X | 4.7X | 57.8X | 63.2X | 56.8X | 29.9X | 57.7X | 54.2X | 46.6X | 32.0X | NM | NM | NM | NM |
| monday.com Ltd. | MNDY | $137.63 | 106% | 91% | 54% | 41% | 37.7X | 19.7X | 12.8X | 9.1X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Microsoft Corporation | MSFT | $274.03 | 14% | 21% | 14% | 12% | 13.1X | 10.9X | 9.5X | 8.4X | 27.6X | 21.2X | 18.7X | 15.8X | 40.3X | 33.3X | 29.3X | 26.1X | 40.8X | 30.9X | 27.8X | 23.7X |
| nCino Inc | NCNO | $36.45 | 48% | 34% | 46% | 27% | 20.3X | 15.2X | 10.4X | 8.2X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | 2226.7X |
| Olo, Inc. Class A | OLO | $10.21 | 94% | 52% | 30% | 28% | 13.9X | 9.1X | 7.0X | 5.5X | 81.4X | 61.0X | 128.1X | 71.2X | 70.0X | NM | 80.7X | 77.8X | NM | 92.8X | 255.3X | 102.1X |
| ON24, Inc. | ONTF | $13.37 | 76% | 30% | -1% | 12% | 2.4X | 1.8X | 1.8X | 1.6X | 15.0X | 64.1X | NM | NM | 10.2X | 221.2X | NM | NM | NM | 360.0X | NM | NM |
| Paychex, Inc. | PAYX | $131.35 | -1% | 9% | 9% | 7% | 11.9X | 10.9X | 10.0X | 9.4X | 28.8X | 24.8X | 22.7X | 21.0X | NM | 38.3X | 29.5X | 29.0X | 44.7X | 37.7X | 33.8X | 31.1X |
| Paycom Software, Inc. | PAYC | $295.97 | 14% | 25% | 25% | 23% | 19.8X | 15.8X | 12.7X | 10.3X | 50.3X | 39.7X | 31.7X | 25.2X | 134.2X | 83.7X | 63.3X | 51.6X | 84.8X | 66.2X | 54.4X | 43.2X |
| Paycor HCM, Inc. | PYCR | $24.76 | 7% | 15% | 16% | 17% | 12.8X | 11.2X | 9.7X | 8.3X | 51.6X | 51.5X | 41.8X | 28.5X | NM | NM | 207.2X | 232.0X | NM | 141.4X | 119.2X | 85.7X |
| Paylocity Holding Corp. | PCTY | $191.67 | 13% | 25% | 25% | 22% | 18.6X | 14.8X | 11.8X | 9.7X | 65.8X | 55.1X | 44.8X | 35.6X | 192.7X | 116.6X | 80.8X | 61.9X | 99.8X | 74.6X | 73.2X | 57.7X |
| PowerSchool Holdings, Inc. Class A | PWSC | $14.40 | 19% | 28% | 12% | 10% | 8.3X | 6.5X | 5.8X | 5.2X | 26.6X | 22.4X | 19.8X | 17.2X | 62.4X | 35.0X | 26.3X | 22.4X | NM | 19.6X | 19.9X | 18.1X |
| Procore Technologies Inc | PCOR | $51.14 | 38% | 29% | 29% | 24% | 17.4X | 13.5X | 10.5X | 8.4X | 321.9X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| PROS Holdings, Inc. | PRO | $29.12 | 1% | 0% | 7% | 7% | 5.7X | 5.8X | 5.4X | 5.0X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| PTC Inc. | PTC | $99.12 | 18% | 20% | 8% | 14% | 8.5X | 7.1X | 6.6X | 5.8X | 24.5X | 18.5X | 16.7X | 14.7X | 40.5X | 35.4X | 30.1X | 24.4X | 33.5X | 25.0X | 21.8X | 18.2X |
| Q2 Holdings, Inc. | QTWO | $52.87 | 28% | 23% | 16% | 19% | 8.0X | 6.5X | 5.6X | 4.7X | 145.7X | 85.5X | 76.0X | 51.0X | NM | 329.0X | 278.4X | 111.6X | 755.3X | 165.2X | 148.7X | 83.3X |
| Qualtrics International, Inc. Class A | XM | $21.78 | 29% | 41% | 33% | 23% | 15.5X | 11.0X | 8.3X | 6.7X | NM | 185.6X | 192.6X | 98.5X | NM | NM | NM | 270.4X | NM | NM | 1322.0X | 301.9X |
| Salesforce, Inc. | CRM | $171.43 | 24% | 25% | 21% | 18% | 8.2X | 6.6X | 5.5X | 4.6X | 26.4X | 21.3X | 17.7X | 15.8X | 43.5X | 33.1X | 26.5X | 22.2X | 34.8X | 35.7X | 36.8X | 29.7X |
| SAP SE Sponsored ADR | SAP | $103.85 | 4% | 1% | 1% | 6% | 4.2X | 4.2X | 4.1X | 3.9X | 12.1X | 12.2X | 12.2X | 11.6X | 19.3X | 22.9X | 26.8X | 19.9X | 16.3X | 13.2X | 18.6X | 16.4X |
| SEMrush Holdings, Inc. Class A | SEMR | $10.10 | 35% | 51% | 30% | 26% | 9.7X | 6.4X | 4.9X | 3.9X | NM | 413.2X | NM | NM | 464.2X | 59.5X | NM | 77.0X | NM | NM | NM | NM |
| Shopify, Inc. Class A | SHOP | $459.97 | 86% | 57% | 31% | 32% | 17.7X | 11.2X | 8.6X | 6.5X | 105.8X | 67.2X | 117.6X | 67.9X | 149.3X | 105.6X | 178.8X | 261.0X | 116.4X | 71.6X | 170.8X | 113.5X |
| Similarweb Ltd. | SMWB | $13.13 | 32% | 47% | 41% | 32% | 10.3X | 7.0X | 5.0X | 3.8X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Smartsheet, Inc. Class A | SMAR | $49.05 | 42% | 43% | 37% | 32% | 16.7X | 11.7X | 8.5X | 6.5X | NM | NM | NM | NM | NM | NM | NM | 322.3X | NM | NM | NM | NM |
| SPS Commerce, Inc. | SPSC | $123.47 | 12% | 23% | 15% | 15% | 14.0X | 11.3X | 9.8X | 8.5X | 50.2X | 40.8X | 34.5X | 28.7X | 60.5X | 46.8X | 41.8X | 35.6X | 80.2X | 67.8X | 60.8X | 50.8X |
| SS&C Technologies Holdings, Inc. | SSNC | $68.69 | 0% | 8% | 6% | 6% | 4.6X | 4.3X | 4.0X | 3.8X | 11.7X | 10.5X | 9.9X | 9.2X | NM | 15.6X | 14.5X | 13.6X | 16.0X | 13.7X | 13.0X | 12.2X |
| Sprinklr, Inc. Class A | CXM | $13.03 | 19% | 27% | 24% | 21% | 8.2X | 6.5X | 5.2X | 4.3X | 143.5X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Sprout Social Inc Class A | SPT | $62.31 | 29% | 41% | 33% | 31% | 25.2X | 17.8X | 13.4X | 10.2X | NM | NM | NM | NM | NM | 241.3X | 345.6X | 176.0X | NM | NM | NM | NM |
| Synopsys, Inc. | SNPS | $285.00 | 13% | 18% | 7% | 15% | 11.5X | 9.8X | 9.2X | 8.0X | 33.7X | 28.5X | 27.0X | 20.6X | 42.9X | 32.3X | 26.3X | 28.0X | 47.0X | 36.9X | 37.3X | 29.4X |
| Toast, Inc. Class A | TOST | $18.80 | 24% | 107% | 40% | 32% | 13.6X | 6.6X | 4.7X | 3.6X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Tyler Technologies, Inc. | TYL | $385.40 | 3% | 43% | 16% | 10% | 15.6X | 11.0X | 9.4X | 8.6X | 53.6X | 40.2X | 36.2X | 32.5X | 54.8X | 55.8X | 45.7X | 40.9X | 69.8X | 54.9X | 51.4X | 46.7X |
| Unity Software, Inc. | U | $75.70 | 43% | 44% | 35% | 29% | 32.7X | 22.7X | 16.9X | 13.1X | NM | NM | NM | 229.7X | NM | NM | NM | NM | NM | NM | NM | 1517.2X |
| Veeva Systems Inc Class A | VEEV | $183.97 | 26% | 26% | 17% | 17% | 18.8X | 14.8X | 12.7X | 10.9X | 45.5X | 35.4X | 32.4X | 27.5X | 50.8X | 36.6X | 35.0X | 28.7X | 62.6X | 49.5X | 45.8X | 39.5X |
| Vertex, Inc. Class A | VERX | $14.49 | 16% | 14% | 13% | 13% | 5.8X | 5.1X | 4.5X | 4.0X | 27.8X | 27.9X | 29.7X | 24.7X | 44.0X | 46.8X | 45.2X | 40.4X | 32.0X | 46.7X | 54.6X | 43.6X |
| WalkMe Ltd. | WKME | $14.80 | 41% | 30% | 30% | 30% | 6.9X | 5.3X | 4.1X | 3.1X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Workday, Inc. Class A | WDAY | $206.13 | 19% | 19% | 21% | 20% | 12.5X | 10.5X | 8.7X | 7.3X | 47.7X | 35.8X | 35.4X | 27.9X | 53.6X | 40.3X | 44.9X | 33.2X | 71.8X | 51.8X | 57.3X | 45.5X |
| Workiva Inc. Class A | WK | $103.92 | 18% | 26% | 20% | 19% | 17.0X | 13.5X | 11.2X | 9.4X | 488.6X | 241.6X | NM | NM | 191.4X | 129.0X | NM | 225.5X | 1039.2X | 280.9X | NM | NM |
| Yext, Inc. | YEXT | $5.82 | 19% | 10% | 4% | 9% | 1.6X | 1.4X | 1.4X | 1.3X | NM | NM | NM | NM | NM | 244.5X | 355.8X | 77.3X | NM | NM | NM | NM |
| Zendesk, Inc. | ZEN | $120.85 | 26% | 30% | 27% | 25% | 16.7X | 12.8X | 10.2X | 8.1X | 172.7X | 183.9X | 101.2X | 69.7X | NM | 122.2X | 97.8X | 66.3X | 232.4X | 191.8X | 151.3X | 101.3X |
| Zeta Global Holdings Corp. Class A | ZETA | $10.55 | 20% | 25% | 19% | 20% | 9.7X | 7.8X | 6.6X | 5.5X | 90.4X | 56.6X | 43.8X | 32.1X | 347.5X | 183.8X | 134.8X | 57.1X | NM | NM | 52.7X | 33.3X |
| Zoom Video Communications, Inc. Class A | ZM | $100.17 | 326% | 55% | 11% | 13% | 9.7X | 6.3X | 5.7X | 5.0X | 24.4X | 14.5X | 16.5X | 14.4X | 18.5X | 17.0X | 17.5X | 14.8X | 30.1X | 19.7X | 28.2X | 25.0X |
| ZoomInfo Technologies Inc | ZI | $49.79 | 43% | 56% | 38% | 32% | 44.9X | 28.7X | 20.8X | 15.7X | 91.2X | 66.8X | 50.2X | 36.7X | 140.4X | 77.8X | 56.6X | 40.1X | 146.3X | 85.8X | 69.7X | 51.2X |
| Zuora, Inc. Class A | ZUO | $12.71 | 11% | 14% | 16% | 15% | 5.4X | 4.8X | 4.1X | 3.6X | 332.3X | NM | 149.8X | 40.7X | NM | 161.9X | 121.0X | 36.0X | NM | NM | NM | 211.8X |
| **Average Application** | | | **31%** | **32%** | **21%** | **20%** | **14.2X** | **10.2X** | **8.2X** | **6.9X** | **76.8X** | **68.3X** | **48.3X** | **48.6X** | **105.5X** | **76.5X** | **88.7X** | **82.0X** | **130.3X** | **173.4X** | **114.5X** | **182.3X** |
| **Median Application** | | | **22%** | **26%** | **18%** | **19%** | **11.9X** | **9.6X** | **8.3X** | **6.7X** | **49.9X** | **36.9X** | **33.4X** | **27.9X** | **60.5X** | **47.2X** | **43.9X** | **38.2X** | **62.2X** | **51.8X** | **52.3X** | **43.4X** |
| **Average Software** | | | **28%** | **29%** | **21%** | **19%** | **15.5X** | **11.0X** | **8.6X** | **7.2X** | **75.7X** | **74.0X** | **63.2X** | **49.1X** | **94.9X** | **78.5X** | **83.5X** | **75.0X** | **147.2X** | **157.2X** | **150.7X** | **192.5X** |
| **Median Software** | | | **21%** | **25%** | **18%** | **19%** | **11.5X** | **9.3X** | **7.6X** | **6.4X** | **47.7X** | **39.9X** | **34.6X** | **28.5X** | **57.7X** | **47.4X** | **44.7X** | **40.1X** | **56.0X** | **53.6X** | **53.1X** | **46.4X** |

Source: FactSet consensus for all estimates, company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



## Exhibit 85 - Infrastructure Software Comps

***INFRASTRUCTURE***

| | | 04/22/2022 | CY Revenue Growth | | | | CY EV/Revenue | | | | CY EV/EBITDA | | | | CY EV/FCF | | | | CY PE | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Company Name** | **Ticker** | **Price** | **2020** | **2021E** | **2022E** | **2023E** | **2020** | **2021E** | **2022E** | **2023E** | **2020** | **2021E** | **2022E** | **2023E** | **2020** | **2021E** | **2022E** | **2023E** | **2020** | **2021E** | **2022E** | **2023E** |
| 8x8, Inc. | EGHT | $10.25 | 22% | 18% | 26% | 17% | 2.9X | 2.5X | 2.0X | 1.7X | NM | 84.7X | 42.0X | 27.7X | NM | 347.0X | 75.5X | NM | NM | 256.3X | 154.9X | 49.6X |
| Akamai Technologies, Inc. | AKAM | $117.14 | 11% | 8% | 8% | 9% | 7.1X | 6.6X | 6.1X | 5.6X | 16.3X | 14.6X | 14.1X | 12.9X | 47.8X | 26.4X | 25.4X | 23.0X | 22.4X | 20.4X | 19.8X | 17.9X |
| Appian Corporation Class A | APPN | $49.60 | 15% | 21% | 21% | 18% | 11.8X | 9.7X | 8.1X | 6.8X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Bandwidth Inc. Class A | BAND | $25.60 | 48% | 43% | 12% | 17% | 2.4X | 1.7X | 1.5X | 1.3X | 31.5X | 16.1X | 21.4X | 14.7X | NM | 107.8X | NM | 112.0X | 47.4X | 26.7X | 394.5X | 52.8X |
| Bentley Systems, Incorporated Class B | BSY | $41.81 | 9% | 20% | 17% | 10% | 16.8X | 14.0X | 12.0X | 10.9X | 50.7X | 41.6X | 36.1X | 31.9X | 55.6X | 44.0X | 44.0X | 37.5X | NM | 50.4X | 50.9X | 46.3X |
| Box, Inc. Class A | BOX | $30.98 | 11% | 13% | 14% | 12% | 7.5X | 6.6X | 5.8X | 5.2X | 29.9X | 23.6X | 19.9X | 17.2X | 48.1X | 34.0X | 24.7X | 19.3X | 44.3X | 36.4X | 27.4X | 21.8X |
| C3.ai, Inc. Class A | AI | $17.85 | 24% | 35% | 34% | 30% | 7.3X | 5.4X | 4.0X | 3.1X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Check Point Software Technologies Ltd. | CHKP | $137.27 | 4% | 5% | 6% | 5% | 8.0X | 7.6X | 7.2X | 6.8X | 15.6X | 15.4X | 15.8X | 15.0X | 14.5X | 14.1X | 13.9X | 13.0X | 20.2X | 19.5X | 18.9X | 17.2X |
| Citrix Systems, Inc. | CTXS | $101.50 | 8% | -1% | 3% | 4% | 5.0X | 5.0X | 4.9X | 4.7X | 13.9X | 13.8X | 13.9X | 14.5X | 18.1X | 27.6X | 25.1X | 18.6X | 16.6X | 19.0X | 19.3X | 17.2X |
| Cloudflare Inc Class A | NET | $95.22 | 50% | 52% | 42% | 35% | 74.6X | 49.0X | 34.6X | 25.6X | NM | NM | 322.6X | 183.8X | NM | NM | NM | NM | NM | NM | 3023.4X | 1043.2X |
| CommVault Systems, Inc. | CVLT | $62.51 | 1% | 9% | 7% | 7% | 4.0X | 3.7X | 3.4X | 3.2X | 20.5X | 16.9X | 15.2X | 13.2X | 32.9X | 18.7X | 15.0X | 14.2X | 34.2X | 26.5X | 23.5X | 20.5X |
| Confluent Inc Class A | CFLT | $34.06 | 7% | 7% | 41% | 36% | 45.2X | 27.6X | 19.6X | 14.4X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Couchbase, Inc. | BASE | $17.03 | 25% | 20% | 19% | 22% | 6.2X | 5.2X | 4.4X | 3.6X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| CrowdStrike Holdings, Inc. Class A | CRWD | $205.62 | 82% | 66% | 48% | 34% | 55.6X | 33.5X | 22.6X | 16.8X | 460.6X | 200.1X | 128.0X | 86.4X | 169.2X | 109.0X | 76.2X | 55.1X | 790.8X | 302.4X | 185.7X | 126.8X |
| CyberArk Software Ltd. | CYBR | $160.37 | 7% | 8% | 17% | 21% | 15.1X | 13.9X | 11.9X | 9.8X | 69.1X | 220.1X | NM | NM | 70.5X | 106.4X | NM | 312.9X | 78.2X | 501.2X | NM | NM |
| Datadog Inc Class A | DDOG | $118.40 | 66% | 70% | 49% | 36% | 71.2X | 41.7X | 28.1X | 20.6X | NM | 270.5X | 208.6X | 122.3X | NM | 178.0X | 167.0X | 109.4X | 538.2X | 246.7X | 234.1X | 148.0X |
| Datto Holding Corp. | MSP | $34.69 | 13% | 19% | 17% | 18% | 11.2X | 9.4X | 8.0X | 6.8X | 38.6X | 33.1X | 34.6X | 27.1X | 85.1X | 74.3X | 77.0X | 57.0X | NM | 53.4X | 61.1X | 49.2X |
| Dropbox, Inc. Class A | DBX | $21.94 | 15% | 13% | 8% | 8% | 4.4X | 3.9X | 3.6X | 3.4X | 15.1X | 10.7X | 10.2X | 9.3X | 17.3X | 12.0X | 11.0X | 9.8X | 23.6X | 14.3X | 13.5X | 11.8X |
| DigitalOcean Holdings, Inc. | DOCN | $42.90 | 25% | 35% | 32% | 30% | 15.6X | 11.6X | 8.8X | 6.7X | 51.9X | 36.5X | 26.6X | 19.7X | NM | 190.0X | 96.6X | 57.8X | NM | 182.1X | 64.1X | 38.2X |
| Dynatrace, Inc. | DT | $39.90 | 29% | 33% | 24% | 24% | 17.7X | 13.3X | 10.7X | 8.6X | 58.1X | 48.7X | 47.3X | 36.6X | 61.9X | 51.8X | 44.5X | 29.0X | 67.6X | 59.6X | 56.9X | 46.4X |
| Elastic NV | ESTC | $79.50 | 44% | 44% | 27% | 26% | 14.0X | 9.7X | 7.7X | 6.1X | NM | 127.6X | 285.5X | 102.5X | NM | NM | NM | 150.7X | NM | NM | NM | 1157.4X |
| Everbridge, Inc. | EVBG | $45.74 | 35% | 36% | 16% | 16% | 7.6X | 5.6X | 4.8X | 4.1X | 255.2X | 183.0X | 61.2X | 33.6X | NM | NM | 79.6X | 29.5X | NM | 217.8X | 208.2X | 67.9X |
| Mandiant, Inc. | MNDT | $22.06 | 6% | -35% | -9% | 25% | 5.3X | 8.1X | 8.8X | 7.1X | 29.2X | 146.9X | NM | 101.4X | 71.1X | NM | NM | 182.8X | 73.5X | NM | NM | NM |
| F5, Inc. | FFIV | $200.46 | 6% | 11% | 5% | 10% | 5.0X | 4.5X | 4.3X | 3.9X | 14.3X | 12.6X | 12.5X | 11.0X | 19.8X | 21.4X | 16.8X | 14.5X | 21.3X | 18.0X | 18.5X | 15.1X |
| Fastly, Inc. Class A | FSLY | $17.02 | 45% | 22% | 15% | 16% | 9.1X | 7.5X | 6.5X | 5.6X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Five9, Inc. | FIVN | $102.85 | 33% | 40% | 24% | 25% | 18.2X | 13.0X | 10.5X | 8.4X | 92.5X | 72.2X | 67.2X | 42.6X | 214.1X | NM | 225.8X | 161.2X | 103.9X | 88.7X | 91.8X | 61.7X |
| ForgeRock Inc Class A | FORG | $19.96 | 22% | 39% | 21% | 26% | 12.8X | 9.2X | 7.6X | 6.1X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Fortinet, Inc. | FTNT | $304.29 | 20% | 29% | 29% | 20% | 19.0X | 14.7X | 11.4X | 9.5X | 64.5X | 51.3X | 41.5X | 35.0X | 52.3X | 40.3X | 31.6X | 23.7X | 90.6X | 76.5X | 61.6X | 50.9X |
| Gitlab, Inc. Class A | GTLB | $48.16 | 87% | 66% | 54% | 40% | 53.4X | 32.2X | 20.9X | 14.9X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| HashiCorp, Inc. Class A | HCP | $45.14 | 75% | 51% | 31% | 30% | 37.5X | 24.8X | 18.9X | 14.6X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Informatica, Inc. Class A | INFA | $21.25 | 1% | 9% | 11% | 13% | 6.4X | 5.9X | 5.3X | 4.7X | 22.7X | 22.6X | 23.3X | 18.8X | 55.4X | 39.1X | 28.4X | 28.5X | NM | NM | 30.7X | 22.5X |
| Jamf Holding Corp | JAMF | $30.30 | 32% | 36% | 28% | 23% | 15.4X | 11.3X | 8.8X | 7.2X | 126.0X | 167.1X | 103.6X | 123.0X | 82.5X | 74.8X | 61.8X | 43.3X | 198.2X | 178.2X | 219.0X | 102.3X |
| JFrog Ltd. | FROG | $22.67 | 44% | 37% | 32% | 28% | 13.7X | 10.0X | 7.6X | 5.9X | NM | 208.1X | 208.9X | 74.3X | 79.8X | 87.7X | 65.1X | 74.2X | NM | 755.7X | NM | 295.9X |
| KnowBe4, Inc. Class A | KNBE | $23.33 | 45% | 41% | 34% | 27% | 23.7X | 16.8X | 12.6X | 9.9X | 262.1X | 98.3X | 84.1X | 58.1X | 112.8X | 58.0X | 74.7X | 47.9X | NM | 179.1X | 178.9X | 114.8X |
| Limelight Networks, Inc. | LLNW | $4.85 | 15% | -5% | 12% | 9% | 3.4X | 3.6X | 3.2X | 2.9X | 31.9X | 61.3X | 29.9X | 19.3X | NM | NM | NM | 83.1X | NM | NM | NM | 60.6X |
| LivePerson, Inc. | LPSN | $23.51 | 26% | 28% | 18% | 18% | 5.0X | 3.9X | 3.3X | 2.8X | 48.6X | 63.3X | NM | 45.5X | NM | NM | NM | 292.8X | NM | NM | NM | NM |
| MicroStrategy Incorporated Class A | MSTR | $409.08 | -1% | 6% | 3% | 8% | 14.5X | 13.7X | 13.2X | 12.3X | NM | 66.5X | 65.9X | 60.3X | NM | NM | NM | NM | NM | NM | 83.3X | 78.7X |
| MongoDB, Inc. Class A | MDB | $371.04 | 40% | 48% | 35% | 32% | 49.2X | 33.2X | 24.6X | 18.6X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | 3147.0X |
| New Relic, Inc. | NEWR | $64.54 | 14% | 15% | 16% | 19% | 6.5X | 5.7X | 4.9X | 4.1X | 51.4X | 134.9X | 59.4X | 37.0X | 63.7X | NM | 112.9X | 99.4X | 806.8X | NM | NM | 131.3X |
| NetScout Systems, Inc. | NTCT | $32.30 | -6% | 4% | -4% | 7% | 2.7X | 2.6X | 2.7X | 2.5X | 11.4X | 10.0X | 12.1X | 9.4X | 10.8X | 10.2X | 14.2X | 11.4X | 18.9X | 15.8X | 20.1X | 15.5X |
| NICE Ltd Sponsored ADR | NICE | $203.05 | 5% | 16% | 12% | 12% | 8.2X | 7.0X | 6.3X | 5.6X | 24.0X | 20.3X | 20.1X | 23.9X | 31.5X | 27.8X | 24.2X | 24.0X | 35.4X | 31.1X | 28.2X | 24.8X |
| NortonLifeLock Inc. | NLOK | $25.52 | -32% | 11% | 6% | 8% | 7.2X | 6.5X | 6.1X | 5.7X | 12.8X | 12.3X | 11.6X | 11.0X | NM | 18.8X | 15.7X | 13.9X | 19.5X | 15.1X | 13.9X | 12.6X |
| Nutanix, Inc. Class A | NTNX | $25.01 | 5% | 17% | 16% | 18% | 4.7X | 4.0X | 3.5X | 3.0X | NM | NM | NM | 55.5X | NM | NM | 341.5X | 69.9X | NM | NM | NM | NM |
| Okta, Inc. Class A | OKTA | $136.33 | 43% | 56% | 37% | 33% | 26.8X | 17.3X | 12.6X | 9.5X | NM | NM | NM | NM | 197.8X | 255.1X | 350.3X | 140.9X | 1363.3X | NM | NM | NM |
| Ooma Inc | OOMA | $13.40 | 11% | 14% | 9% | 11% | 2.0X | 1.7X | 1.6X | 1.4X | 23.8X | 21.4X | 26.9X | 17.5X | 241.3X | 142.1X | 71.9X | 43.2X | 27.3X | 26.8X | 36.5X | 25.6X |
| Open Text Corporation | OTEX | $40.88 | 12% | 4% | 6% | NM | 4.2X | 4.0X | 3.8X | NM | 10.9X | 10.9X | 10.5X | 9.9X | 15.8X | 16.1X | 13.4X | NM | 12.6X | 12.5X | 12.4X | NM |
| Oracle Corporation | ORCL | $76.21 | 0% | 5% | 4% | 6% | 7.0X | 6.6X | 6.4X | 6.0X | 13.8X | 12.8X | 12.3X | 11.2X | 22.6X | 38.3X | 23.6X | 21.4X | 18.3X | 15.4X | 15.6X | 13.9X |
| PagerDuty, Inc. | PD | $29.43 | 28% | 32% | 29% | 25% | 12.4X | 9.4X | 7.3X | 5.9X | NM | NM | NM | 174.3X | NM | NM | 362.4X | 107.7X | NM | NM | NM | 1471.5X |
| Palantir Technologies Inc. Class A | PLTR | $11.96 | 47% | 41% | 29% | 28% | 23.0X | 16.3X | 12.6X | 9.9X | NM | 112.0X | 44.4X | 32.1X | NM | 59.3X | 66.3X | 41.9X | NM | 85.4X | 62.8X | 44.9X |
| Palo Alto Networks, Inc. | PANW | $571.58 | 21% | 28% | 24% | 21% | 18.9X | 14.7X | 11.9X | 9.8X | 74.4X | 65.0X | 49.4X | 41.3X | 61.0X | 47.3X | 36.3X | 30.2X | 98.2X | 89.9X | 70.8X | 58.3X |
| Pegasystems Inc. | PEGA | $71.94 | 12% | 19% | 21% | 21% | 6.1X | 5.2X | 4.3X | 3.5X | NM | 75.0X | 50.5X | 32.0X | NM | 267.7X | 62.2X | 31.8X | NM | 553.4X | 87.3X | 51.8X |
| Ping Identity Holding Corp. | PING | $27.54 | 0% | 23% | 12% | 19% | 10.5X | 8.6X | 7.7X | 6.4X | 208.5X | 115.7X | 489.7X | 106.2X | 274.9X | 117.0X | NM | 142.6X | 81.0X | 306.0X | NM | 238.0X |
| Progress Software Corporation | PRGS | $48.42 | 6% | 22% | 10% | 2% | 5.8X | 4.7X | 4.3X | 4.2X | 14.2X | 11.3X | 12.1X | 9.4X | 18.6X | 14.7X | NM | NM | 15.7X | 12.5X | 11.9X | 11.2X |
| Qualys, Inc. | QLYS | $139.33 | 13% | 13% | 18% | 18% | 14.7X | 13.0X | 11.1X | 9.4X | 31.5X | 27.9X | 28.9X | 25.5X | 35.4X | 30.3X | 32.3X | 26.2X | 48.5X | 43.1X | 47.9X | 41.0X |
| Rapid7 Inc. | RPD | $105.23 | 26% | 30% | 28% | 22% | 21.2X | 16.3X | 12.7X | 10.4X | NM | 366.3X | 166.0X | 129.4X | NM | 248.6X | 191.8X | 124.4X | NM | NM | 998.7X | 224.6X |
| RingCentral, Inc. Class A | RNG | $91.45 | 31% | 35% | 26% | 24% | 8.5X | 6.3X | 5.0X | 4.0X | 62.2X | 45.8X | 34.8X | 28.8X | NM | 117.5X | 75.3X | 42.8X | 93.3X | 68.2X | 53.4X | 41.2X |
| SailPoint Technologies Holdings, Inc. | SAIL | $64.27 | 27% | 20% | 18% | 21% | 19.7X | 16.4X | 13.9X | 11.5X | 138.7X | 340.5X | NM | NM | 133.4X | NM | NM | NM | 160.7X | 714.1X | NM | NM |
| Samsara, Inc. Class A | IOT | $13.14 | NM | 71% | 34% | 30% | 25.7X | 15.0X | 11.1X | 8.6X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| SecureWorks Corp. Class A | SCWX | $11.62 | 1% | -5% | -10% | NM | 1.5X | 1.5X | 1.7X | NM | 24.6X | 43.9X | NM | NM | 14.2X | 93.9X | NM | NM | 55.3X | 116.2X | NM | NM |
| SentinelOne, Inc. Class A | S | $32.68 | 100% | 120% | 80% | 61% | 92.3X | 42.0X | 23.3X | 14.5X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| ServiceNow, Inc. | NOW | $471.40 | 31% | 30% | 26% | 25% | 21.2X | 16.3X | 12.9X | 10.4X | 66.1X | 49.2X | 40.8X | 32.0X | 68.0X | 52.3X | 41.7X | 32.9X | 101.2X | 79.2X | 64.3X | 50.7X |
| Snowflake, Inc. Class A | SNOW | $173.80 | 124% | 106% | 66% | 56% | 100.9X | 49.0X | 29.6X | 19.0X | NM | NM | NM | 258.6X | NM | NM | 187.4X | 115.3X | NM | NM | 1131.7X | 398.3X |
| SolarWinds Corp. | SWI | $12.33 | 9% | -13% | -17% | 5% | 3.2X | 3.7X | 4.4X | 4.2X | 6.6X | 8.7X | 10.7X | 9.9X | 8.8X | 19.0X | 14.1X | 12.1X | 11.8X | 9.1X | 11.7X | 10.6X |
| Splunk Inc. | SPLK | $123.58 | -5% | 20% | 22% | 23% | 10.2X | 8.5X | 6.9X | 5.7X | NM | NM | 258.0X | 66.0X | NM | 185.7X | 55.7X | 36.8X | NM | NM | NM | 166.6X |
| Sumo Logic, Inc. | SUMO | $9.66 | 31% | 20% | 20% | 19% | 4.9X | 4.1X | 3.4X | 2.9X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Telos Corporation | TLS | $8.19 | 13% | 35% | -3% | 24% | 2.7X | 2.0X | 2.0X | 1.7X | 84.8X | 20.9X | 20.7X | 13.3X | NM | NM | 104.2X | 21.3X | NM | 32.8X | 40.0X | 23.2X |
| Tenable Holdings, Inc. | TENB | $57.00 | 24% | 23% | 23% | 20% | 15.4X | 12.5X | 10.1X | 8.5X | 194.8X | 111.7X | 121.9X | 74.4X | 153.9X | 72.9X | 66.1X | 50.1X | 335.3X | 167.6X | 309.5X | 146.1X |
| Teradata Corporation | TDC | $43.74 | -3% | 4% | 1% | 0% | 2.6X | 2.5X | 2.4X | 2.3X | NM | 8.7X | 10.2X | 12.5X | 21.9X | 10.9X | 11.8X | 10.4X | 33.1X | 18.0X | 23.0X | 19.6X |
| Tufin Software Technologies Ltd | TUFN | $12.90 | -2% | 10% | 13% | 0% | 5.1X | 4.6X | 4.1X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Twilio, Inc. Class A | TWLO | $122.78 | 51% | 61% | 36% | 29% | 11.0X | 6.8X | 5.0X | 3.9X | 105.0X | 74.6X | 82.9X | 47.9X | NM | NM | NM | NM | 533.8X | NM | NM | 733.0X |
| UiPath Inc Class A | PATH | $17.95 | 81% | 47% | 21% | 0% | 14.0X | 9.5X | 7.8X | 6.1X | NM | 106.4X | 416.3X | 93.5X | 240.9X | NM | NM | 162.4X | NM | 224.4X | NM | 288.3X |
| Varonis Systems, Inc. | VRNS | $44.54 | 15% | 33% | 25% | 0% | 17.2X | 12.9X | 10.3X | 8.5X | NM | 139.5X | 122.3X | 72.4X | NM | NM | NM | 142.4X | NM | 371.2X | 302.1X | 141.3X |
| Verint Systems Inc. | VRNT | $55.75 | -4% | -32% | 7% | 0% | 3.4X | 4.9X | 4.6X | 4.2X | 13.8X | 18.6X | 17.0X | 15.5X | 21.3X | 28.3X | 20.7X | 17.8X | 24.2X | 24.5X | 22.2X | 19.6X |
| Vonage Holdings Corp. | VG | $20.10 | 5% | 13% | 13% | 0% | 5.0X | 4.4X | 3.9X | 3.4X | 36.7X | 31.6X | 27.1X | 22.2X | 124.8X | NM | NM | NM | 111.7X | 154.6X | 117.4X | 53.5X |
| VMware, Inc. Class A | VMW | $108.18 | 15% | 9% | 7% | 0% | 4.9X | 4.4X | 4.1X | 3.9X | 12.2X | 11.7X | 11.7X | 10.8X | 14.0X | 14.4X | 15.1X | 13.8X | 15.0X | 14.9X | 15.5X | 14.0X |
| Xometry, Inc. Class A | XMTR | $36.79 | 76% | 54% | 82% | 0% | 12.3X | 8.0X | 4.4X | 3.2X | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM | NM |
| Ziff Davis, Inc. | ZD | $92.55 | 9% | 6% | -4% | 0% | 3.1X | 2.9X | 3.0X | 2.8X | 7.5X | 7.0X | 8.5X | 7.5X | 11.5X | 11.8X | 14.1X | 13.2X | 12.9X | 10.9X | 13.9X | 11.7X |
| Zscaler, Inc. | ZS | $206.47 | 49% | 60% | 43% | 0% | 59.6X | 37.2X | 26.1X | 19.8X | 436.0X | 291.4X | 191.4X | 123.4X | 391.4X | 156.0X | 111.5X | 81.1X | 573.5X | 368.7X | 293.8X | 175.4X |
| **Average Infrastructure** | | | **25%** | **27%** | **21%** | **18%** | **17.0X** | **11.8X** | **9.1X** | **7.4X** | **74.1X** | **79.9X** | **78.3X** | **49.5X** | **83.0X** | **80.8X** | **77.4X** | **66.9X** | **165.3X** | **141.0X** | **188.4X** | **202.5X** |
| **Median Infrastructure** | | | **15%** | **22%** | **18%** | **19%** | **10.4X** | **8.0X** | **7.1X** | **6.0X** | **31.9X** | **47.3X** | **34.8X** | **31.9X** | **55.5X** | **49.6X** | **50.1X** | **41.9X** | **48.5X** | **59.6X** | **55.2X** | **50.7X** |
| **Average Software** | | | **28%** | **29%** | **21%** | **19%** | **15.5X** | **11.0X** | **8.6X** | **7.2X** | **75.7X** | **74.0X** | **63.2X** | **49.1X** | **94.9X** | **78.5X** | **83.5X** | **75.0X** | **147.2X** | **157.2X** | **150.7X** | **192.5X** |
| **Median Software** | | | **21%** | **25%** | **18%** | **19%** | **11.5X** | **9.3X** | **7.6X** | **6.4X** | **47.7X** | **39.9X** | **34.6X** | **28.5X** | **57.7X** | **47.4X** | **44.7X** | **40.1X** | **56.0X** | **53.6X** | **53.1X** | **46.4X** |

Source: FactSet consensus for all estimates, company reports, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com

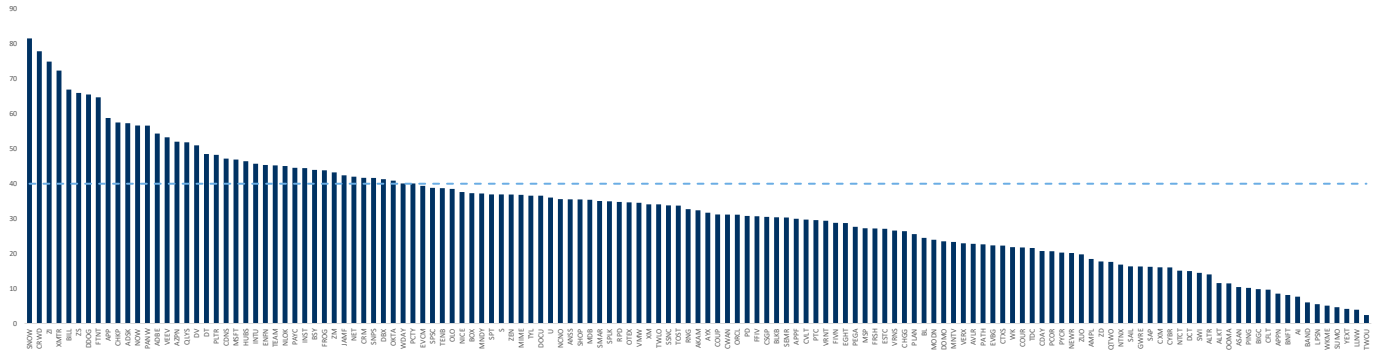


# Rule of 40

We view Rule of 40 as an important metric for investors, though perhaps more of a guideline than a rule. The Rule attempts to take a holistic and balanced view of growth and profitability, with the idea that organic revenue growth plus a profitability margin should exceed 40%. We choose to use FCF margin as the measure of profitability, as we see it as a better metric than EBITDA despite its greater volatility from year to year, and we also use multi-year averages, which we believe reduced the problem of outlier years for both components of the measure.

Exhibit 86 shows the Rule of 40 for CY22 for every company in The Index that we have consensus estimates for that show a positive score.

Exhibit 86 - Rule of 40




Source: FactSet, RBC Capital Markets

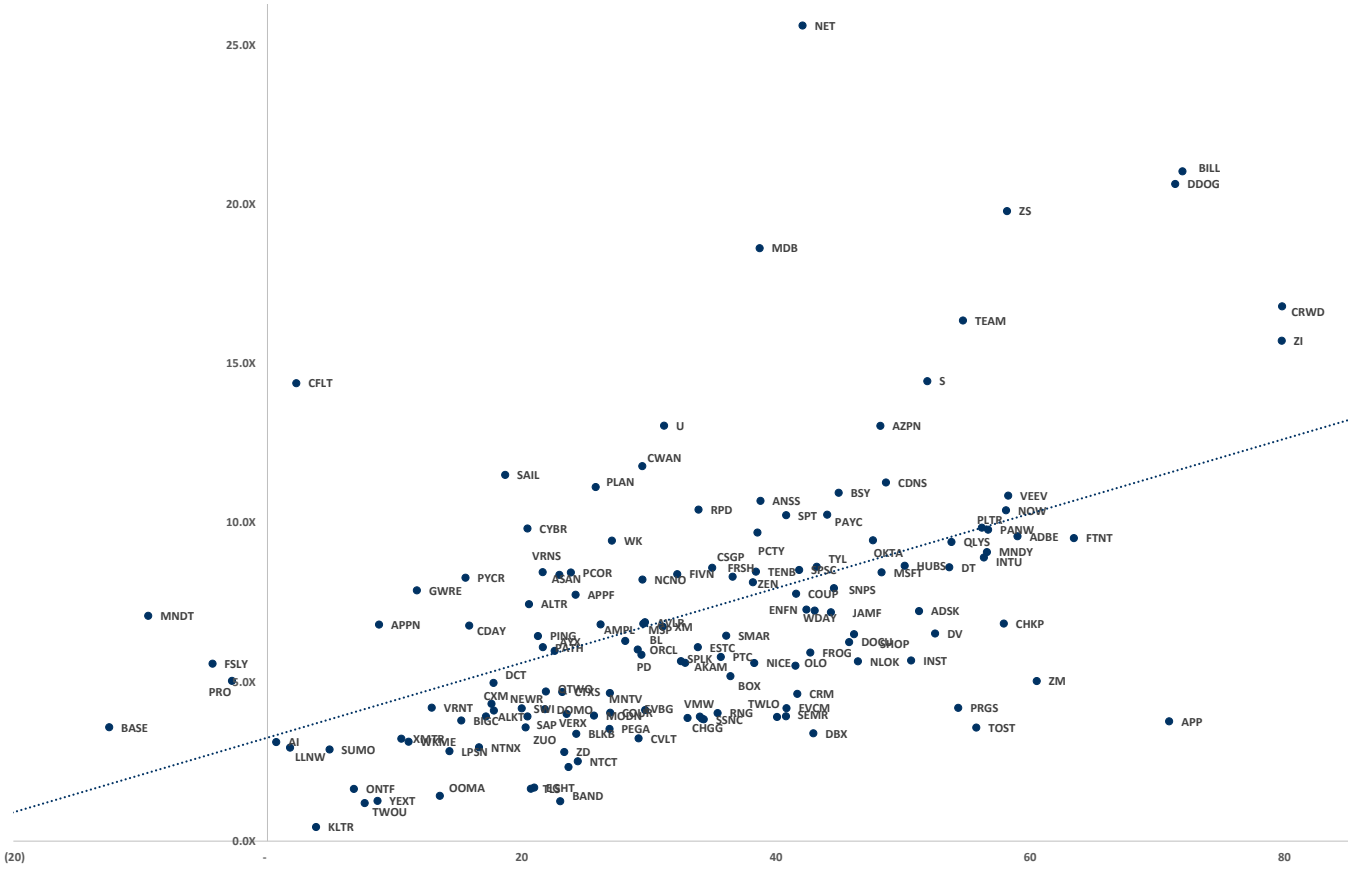
**We view the Rule of 40 as useful for at least three reasons:**

1) It is a simple but effective heuristic for identifying companies that are underperforming or are executing at a high level. Software companies tend to converge around the 40% level, so a company with a score of 20% almost certainly has significant room for improvement, while a company with a score of 60% is almost certainly best-in-class.
2) It allows a normalized method for comparing companies at different points in their lifecycles. A company growing 20% with 20% FCF margins is likely a better company than one growing 30% with FCF burn, or one that earns 25% margins and is growing less than 5%.
3) There is a reasonably strong correlation between a company's Rule of 40 score and its valuation. Across the public software universe, there is a 54% correlation between the two. Exhibit 87 shows this correlation.

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 87 - Rule of 40 vs. EV/CY22E Revenue




Source: FactSet, RBC Capital Markets

Caveats to keep in mind:

- Organic growth is not always easy to calculate, let alone disclosed. We can estimate it fairly accurately within our active coverage list, but not across all 100+ software companies. Acquisitions tend to be margin dilutive near term, partially mitigating this problem in the combined score, and we can adjust for larger deals, but the numbers will never be perfectly clean.
- Out-year consensus estimates may underestimate a company's ongoing growth and earnings power, especially given most management teams' guidance philosophies. This is one reason why we use multi-year averages. There is a lag between when a company makes an investment and its impact on revenue (and vice versa). If a company is underinvesting now, particularly in long-term areas like R&D and pipeline generation, it can boost margins short term with only minimal revenue growth effects. Again, this is a reason we use multi-year averages.
- There is not a 1:1 trade-off between growth and margin from investors' perspectives, based on our experience, as investors value growth above margin in the current environment. For example, a company growing 40% with breakeven margins would garner a higher multiple than one growing 10% with 30% FCF margins. This is partly because a company can sacrifice margin now and grow instead before focusing on margins, thus growing the terminal FCF (in dollar terms) even if the long-term margin is the same.

This report is intended solely for cbass@marketstreetpartners.com

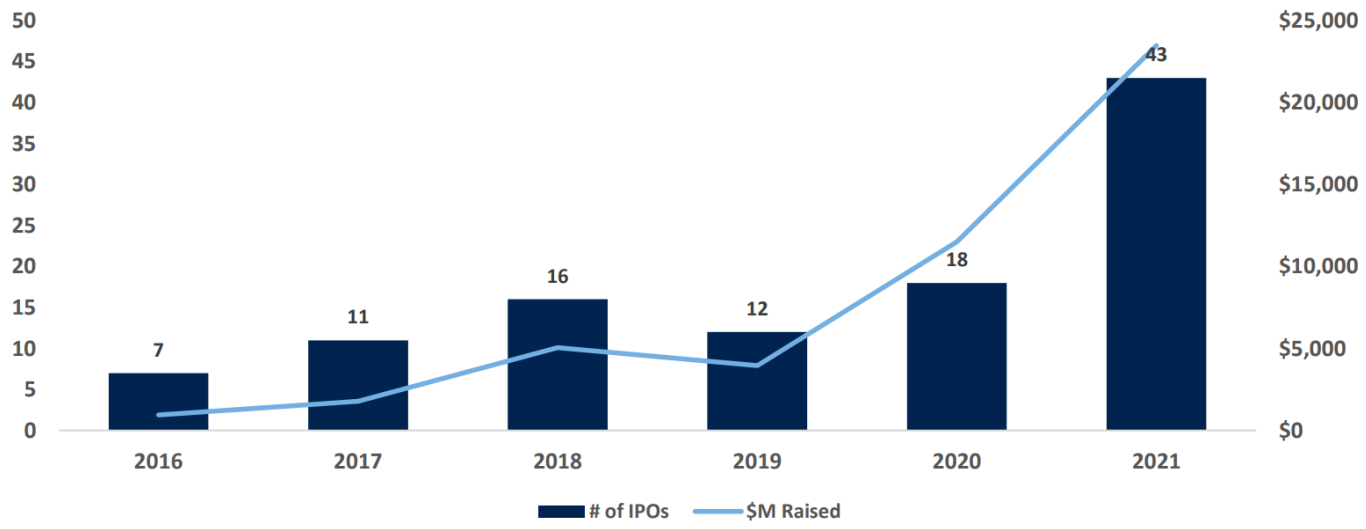


# IPOs

While 2021 was a record year for software IPOs (eclipsing even 2020), 2022 has been quiet with no software public offerings that we've tracked YTD. Going forward, we do see some puts and takes, but overall we expect to see a return of software IPOs in 2022 (although more than likely below prior years). On the positive side, higher multiples and greater appetite from investors for unprofitable companies (we note that Confluent had negative (38%) non-GAAP operating margins in its last fiscal year) should help drive more companies to public markets. On the other hand, the price rationalization among some high-growth or digital-transformation beneficiaries does point to a less frothy or a (relatively) more price-sensitive market, and we do expect the "rush for the gates" on public offerings to subside somewhat in 2022.

Exhibit 88 shows the number of software IPOs since 2016 and amounts raised. We note that SPACs are not included in this analysis.

Exhibit 88 - IPOs 2016-2022, Count and Funding Raised ($, Millions)



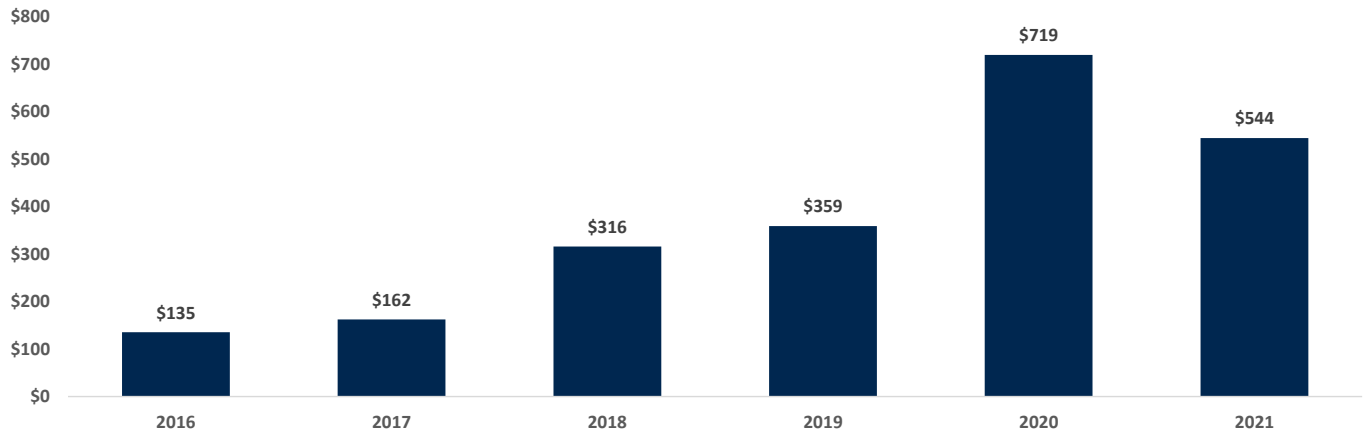

Source: FactSet, RBC Capital Markets

Exhibit 89 shows the average money raised per IPO (for purposes of money raised per IPO, we exclude WORK, ASAN, PLTR, and AMPL as these direct listings raised no capital).

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 89 - IPOs 2016-2021, $ in Millions




Source: FactSet, RBC Capital Markets

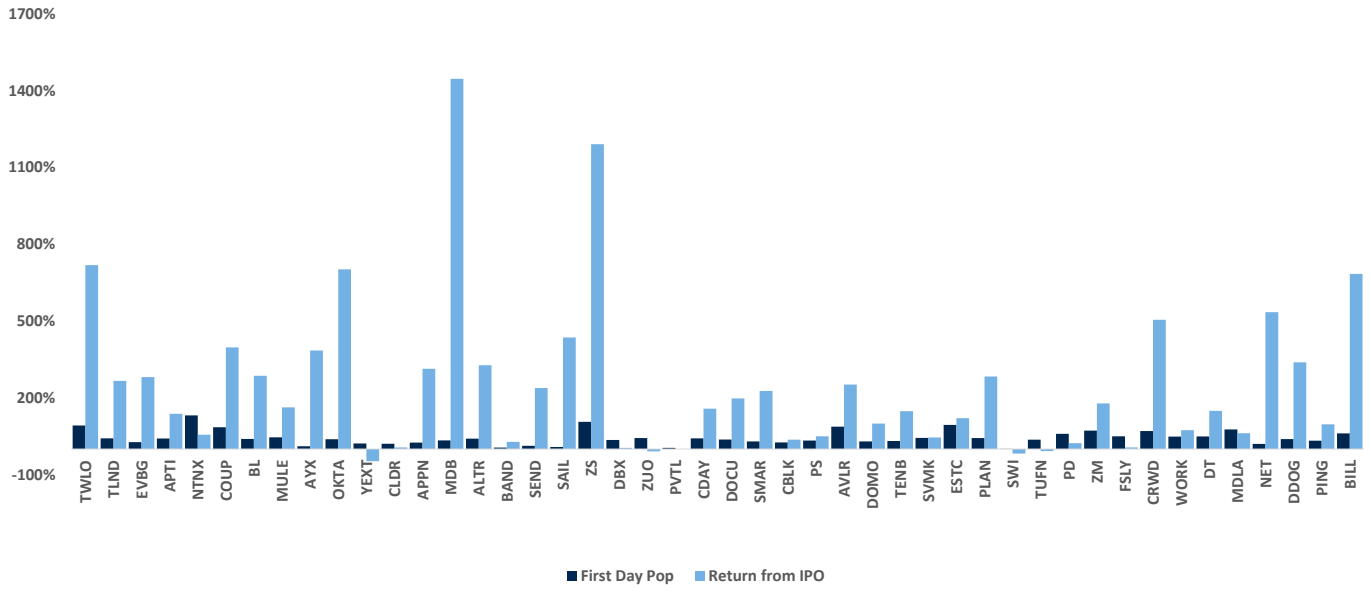
Exhibits 90 and 91 show the performance of the 107 software IPOs we've tracked since 2016. More than half (51%) are actually down from their IPO (or reference) price. This underperformance is most notable among the 61 software IPOs from 2020 and 2021 with only 11 names in those years actually up from their reference price (including ZI, JAMF, SNOW, U, PLTR, ASAN, and MSP). That said, we continue to see strong performance from IPOs in prior years, a theoretical investor buying equal weights of all 2016 software IPOs at the reference price would have an aggregate return of 306%, 363% for 2017 IPOs, 174% for 2018, and 220% for 2019.

Of the 46 software IPOs we tracked in 2016-2019 (Exhibit 90), 41 are trading above their IPO price with seven names having more than 5x'd since IPO (TWLO, OKTA, MDB, ZS, CRWD, NET, BILL). The aggregate return for a theoretical investor buying into all 2016-2019 software IPOs at the IPO price would be 251%.

This report is intended solely for cbass@marketstreetpartners.com



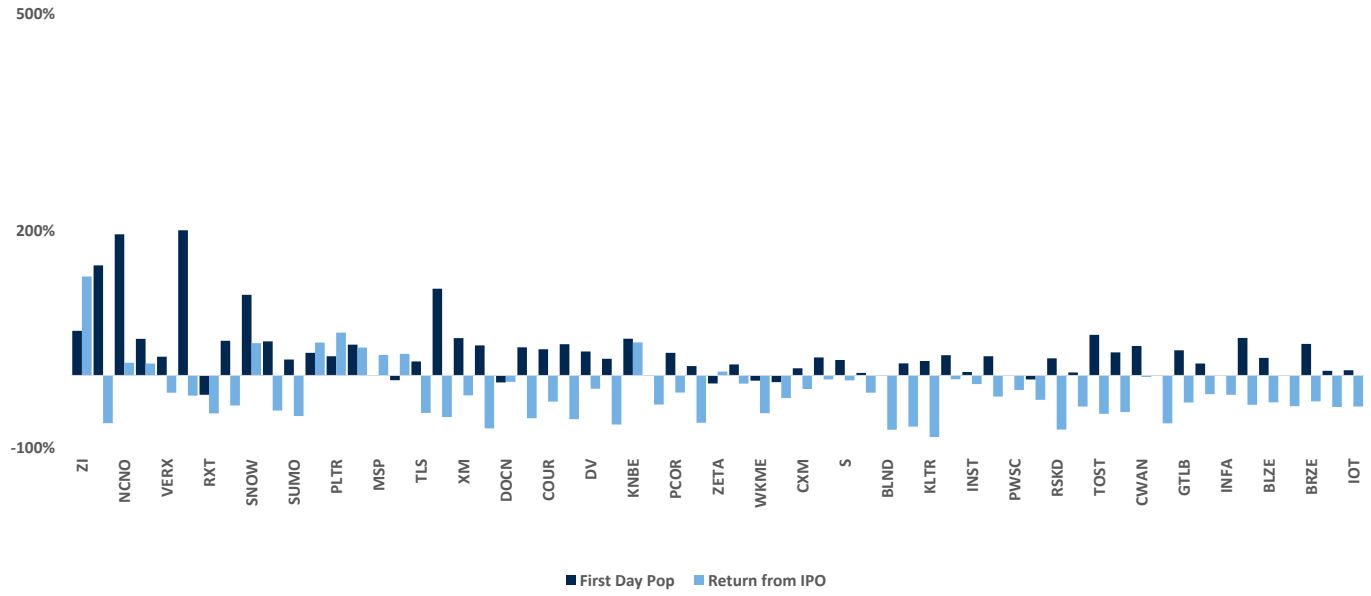
Exhibit 90 - IPOs 2016-2019, Stock Returns




Source: FactSet, RBC Capital Markets

Of the 61 software IPOs we tracked in 2020-2021 (Exhibit 91), only 11 are trading above their IPO price. With recent market pullbacks, a theoretical investor buying into these names at the IPO price would have lost 25% in aggregate. Still, a number of names in this group continue to exhibit strong returns from IPOs such as ZI (up 137% from IPO), PLTR (59%), U (46%), and SNOW (45%).

Exhibit 91 - IPOs 2020-2021, Stock Returns




Source: FactSet, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com

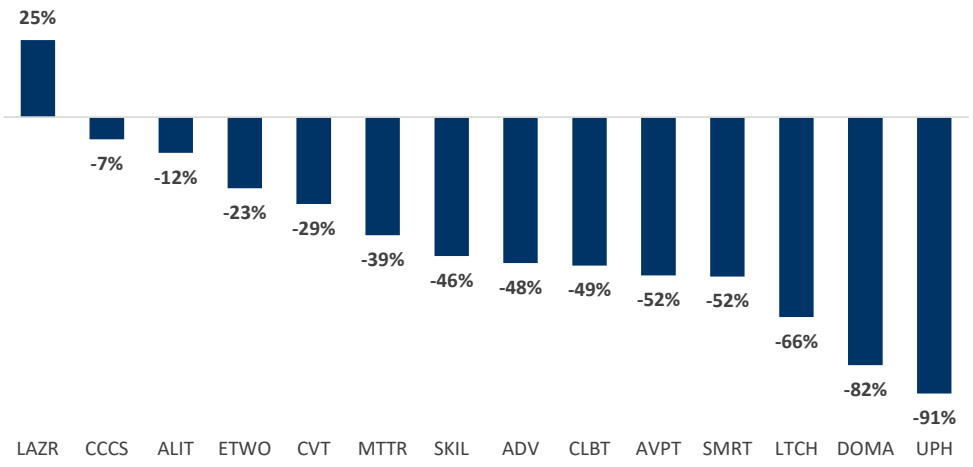


## SPACs

Our SPAC tracker below shows returns for public software companies that went public through a SPAC versus their initial reference price of $10/share (which is effectively equivalent to an IPO price). SPACs became popular for a number of reasons, including the ability to provide long-term forecasts, but we note enterprise software SPAC traction has been somewhat limited and overall SPAC issuance has slowed down in 2022. The companies below all peg themselves as software companies; however, we would note most of these companies are not included in the RBC Software Index for various reasons, especially if a majority of revenue is not software.

Exhibit 92 shows SPAC returns versus $10/share (which is effectively the equivalent to an IPO price or reference price).

Exhibit 92 - SPAC Returns from Initial Reference Price of $10/Share




Source: FactSet, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



# M&A

## Coming Off Strong M&A Deal Volume in 2021…

In 2021, software saw an acceleration in the quantity and dollar value of M&A deals, with both high-profile strategic acquisitions (Intuit-Mailchimp, Okta-Auth0, Ericsson-Vonage, Oracle-Cerner, Microsoft-Nuance) and private equity takeouts (Investor Group-McAfee, Thoma Bravo-Proofpoint, Thoma Bravo-Medallia, Permira Advisors-Mimecast, Clearlake-Cornerstone OnDemand). This trend seems to be holding steady, if not accelerating into 2022.

## … The Software Drawdown Has Only Accelerated M&A Activity in 2022

We count 20 software M&A deals announced YTD, with 14 in 1Q22 and 6 in 2Q22-to-date. On a dollar basis, there has been $123.8B in deal value announced YTD, with $106.7B in 1Q22 (up from $44.8B in 1Q21) and $17.1B in 2Q22-to-date (vs. $100.5B in all of 2Q21).

On the strategic side, we highlight Microsoft's pending $68.7B acquisition of Activision (represents 64% of the total deal value in 1Q22), Kaseya Ltd.'s announced $5.6B acquisition of Datto, and Google's proposed $5.4B acquisition of Mandiant. We expect the strategic acquisition market to remain strong, as large software vendors with healthy cash balances likely look to take advantage of the drawdown in public market valuations, similar to the actions already by Microsoft (ATVI) and Google (MNDT). That said, strategic deals inherently carry more risk as shareholders don't always agree with the Board's buying decisions and stock deals are sensitive to market conditions, as we saw another deal get broken up with Zendesk-Momentum, following last year's Zoom-Five9 deal collapsing.

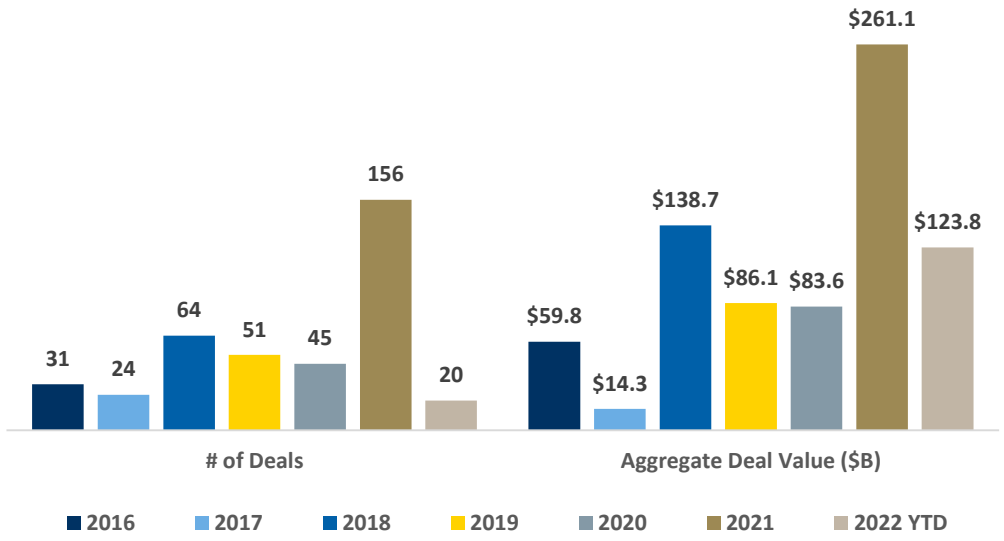
For private equity deals, we point out Vista Equity's announced $16.5B buy of Citrix, Thoma Bravo's proposed takeouts of both Anaplan for $10.7B and Sailpoint for $6.1B, and KKR's purchase of Barracuda Networks, a private security software vendor. We also note Zendesk's $4.1B acquisition of Momentive (formerly SurveyMonkey) fell through, and now Bloomberg reported Zendesk is thought to be exploring a potential sale. We continue to see momentum in what we have dubbed the new private equity playbook in which financial acquirers seek out unprofitable growth or stalled growth companies (e.g., recently Anaplan and Sailpoint, in addition to SolarWinds, Cloudera, Cornerstone OnDemand, and Medallia) as opposed to the standard model of buying declining business and optimizing cash flow, though the recent takeout of Citrix is more in line with these historical norms.

Exhibit 93 shows the number of deals and aggregate deal value in each year since 2016. Note we selectively exclude certain acquisitions that we do not consider to be software transactions, such as Teledoc's acquisition of Livongo and Square's acquisition of AfterPay. Additionally, we exclude SPACs from this analysis.

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 93 - M&A 2016-2022, Count and Aggregate Value ($B)




Source: FactSet, RBC Capital Markets

## The Most Viable M&A Targets in Our Coverage Universe

*Disclaimer: In the section below, we discuss the most viable M&A targets in our coverage universe. All potential M&A content discussing BASE, BOX, COUP, DBX, DOCU, FSLY, NABL, NEWR, NLOK, NTNX, QLYS, SMAR, SPLK, SUMO, SWI, TDC, and ZM is based on our own opinions/research, and we have no knowledge of the mentioned companies discussing such transactions.*

**Following our industry note on M&A and the announced acquisitions of Sailpoint and Datto, we are refreshing our most likely takeout candidates and rationale for each across both strategic and private equity.**

- **BASE (covered by Matt Hedberg; 3.6x EV/CY23E revenue; $640M EV).** We think the company is making the correct pivot to a fully hosted cloud database and believe a larger strategic vendor could deploy even more capital than Couchbase can as an independent vendor to target a large and growing opportunity similar to MDB.
- **BOX (covered by Rishi Jaluria; 5.2x EV/CY23E revenue; $5.8B EV).** First, we believe BOX's valuation now makes the company a less-attractive takeout candidate. While BOX has guided to an attractive financial profile (implies revenue growth acceleration and margin expansion), we do believe any misstep would increase BOX's attractiveness as a private equity takeout. We could see a private equity sponsor step in to find more low-hanging fruit in the model and drive accelerated margin expansion (in light of BOX's relatively low sales efficiency).
- **COUP (covered by Rishi Jaluria; 7.8x EV/CY23E revenue; $7.9B EV).** COUP's valuation certainly is not cheap, but given the premium multiple paid for PLAN, which is also in the back-office space (11.5x EV/CY23E revenue and 14.3x EV/NTM revenue) and the drastic pullback from 52-week high (down 69% from 52-week high), we could see private equity get interested, especially following disappointing F4Q22 earnings. Additionally, we do think a large back-office vendor could take a look, such as SAP or WDAY, though we see this as less likely.

This report is intended solely for cbass@marketstreetpartners.com



- **DBX (covered by Rishi Jaluria; 3.4x EV/CY23E revenue; $8.9B EV).** As we have opined before, we increasingly wonder if DBX would be better off as part of a larger technology platform, as content management could be more valuable being integrated and cross-sold with other collaboration solutions. We note that competitors Microsoft and Google have had success selling their content management solutions as part of larger suites (Office 365/Teams and Google Workspace, respectively), while Dropbox doesn't have this muscle, nor does it have a full-fledged enterprise salesforce. We see ADBE, CRM, or even ZM as the most likely strategic buyers.
- **DOCU (covered by Rishi Jaluria; 6.2x EV/CY23E revenue; $18.0B EV).** DOCU is attempting a move to become a broader platform with "Agreement Cloud", but is still early in the journey (non-esignature is <10% of ARR). With DOCU's recent sales execution issues and the COVID boost dissipating, we see opportunity for DOCU to become part of a larger platform, given its market-leadership position, wide array of integrations, and profitability. The best fit, in our view, is CRM, but MSFT could make sense too (though, we point out MSFT has a robust partnership with ADBE).
- **FSLY (covered by Rishi Jaluria; 5.5x EV/CY23E revenue; $2.6B EV).** FSLY has been facing issues with customer concentration and a comedown post-COVID. We still believe edge computing is a real opportunity and view FSLY as good technology that could be attractive to an acquirer. To us, CSCO could make sense, as well as GOOGL in order to bring edge cloud together with GOOGL's GCP central cloud (which we view as a distant #3 in the cloud wars).
- **NABL (covered by Matt Hedberg; 4.8x EV/CY23E revenue; $2.1B EV).** Similar to MSP, we think NABL is an interesting PE acquisition candidate as it too has a large MSP and SME market opportunity that remains fragmented and underserved from a technology perspective. Overall, we feel this market could consolidate over time.
- **NEWR (covered by Rishi Jaluria; 4.0x EV/CY23E revenue; $4.1B EV).** NEWR has undergone a number of transitions in the last few years, including the shift to a consumption model. The success of these transitions has been mixed and meanwhile, NEWR has lost share to competitors, most notably DDOG. We could make the case for NEWR to be bought by private equity (where the company can invest more aggressively out of the public eye) or a strategic acquirer that wants to layer observability on top of a broader DevOps platform.
- **NLOK (covered by Matt Hedberg; 3.0x EV/CY23E revenue; $17.9B EV).** Following the acquisition of McAfee, we think NLOK too could be consolidated given its growth and strong margin profile that could be well suited for PE ownership that wants a dominant position in consumer security.
- **NTNX (covered by Matt Hedberg; 5.7x EV/CY23E revenue; $6.2B EV).** We believe Nutanix represents an interesting strategic asset in hybrid-cloud infrastructure as it continues to pivot to a software and increasingly SaaS model. As large IT vendors need a view into both public and on-premise spending, Nutanix could help solve that given its large scale, salesforce and market opportunity.
- **QLYS (covered by Matt Hedberg; 9.4x EV/CY23E revenue; $5.3B EV).** Given its SaaS portfolio and high margins, we think Qualys represents an interesting acquisition candidate for a vendor looking to get deeper into vulnerability management (VM) and the overall cloud security market. This could make even more sense if the VM market consolidates up to the platform level. It's also worth noting we believe other SMID cap security names (such as PING, VRNS, RPD) could represent consolidation opportunities.
- **SMAR (covered by Rishi Jaluria; 6.4x EV/CY23E revenue; $6.4B EV).** We believe the collaborative work management space is crowded and, over time, could benefit from consolidation with adjacent collaboration solutions. In addition, we would note there have been a number of acquisitions within the space, including CTXS-Wrike, ADBE-Workfront, and TEAM-Trello. We could see larger software companies become serious about collaboration

This report is intended solely for cbass@marketstreetpartners.com



buying SMAR, including CRM and Seattle neighbor, MSFT (SMAR could replace MSFT's legacy solution, Project, though we acknowledge MSFT has been able to organically build Teams and could look to continue to build rather than buy in this space).

- **SPLK (covered by Matt Hedberg; 5.7x EV/CY23E revenue; $22.8B EV).** We think Splunk is too good a business/asset to trade where it is and believe the work it's done from a product and go-to-market perspective, along with a new CEO, will position the company as a good investment in 2022 if it can execute. While we have no knowledge of a potential sale and management hasn't commented on this previously, if the company is unable to execute its strategy, it might make sense for the board to explore a sale.
- **SUMO (covered by Matt Hedberg; 2.3x EV/CY23E revenue; $801M EV).** We believe Sumo is in a good position to accelerate growth as a cloud-only security and observability vendor. Given intense competition around it, we think the company could benefit from additional scale and could be an interesting asset to a bigger platform vendor that could spend even more than Sumo can as an independent company.
- **SWI (covered by Matt Hedberg; 4.2x EV/CY23E revenue; $3.2B EV).** Given the company's strong margin profile but still recovering from the prior cyber-attack, we think SWI could be an interesting PE candidate for a firm that can leverage a growing SaaS portfolio, unique low-touch sales model and high margins.
- **TDC (covered by Matt Hedberg; 2.3x EV/CY23E revenue; $4.7B EV).** We think the company has made good progress on its cloud transition and, as such, it could represent an interesting acquisition candidate for a large vendor or PE shop that wants to get deeper into cloud analytics.
- **ZM (covered by Rishi Jaluria; 5.0x EV/CY23E revenue; $25.8B EV).** Once one of the most expensive stocks in software, ZM has had a major pullback (off 80% from the highs) as a result of softer demand post-COVID, particularly at the lower end of the market, and intensifying competition with MSFT. Having said that, we still believe ZM has the best video conferencing technology out there and the potential to become a larger platform, while also bringing an attractive financial profile (noting ZM trades at 16.7x EV/CY23E FCF). While CEO and founder Eric Yuan may be hesitant to sell, given the history with CSCO's acquisition of Webex, the right acquirer may make sense. We believe CRM (bringing together Slack and ZM) and ORCL could make sense for ZM (we note ZM uses OCI and that ORCL founder and CTO Larry Ellison was very vocal about supporting ZM early in the pandemic when there were controversies around security and privacy).

This report is intended solely for cbass@marketstreetpartners.com



# Top Picks

## Rishi Jaluria's top picks (in alphabetical order)

**HubSpot (HUBS): Outperform, $925 PT:** We believe HUBS' pullback (down 54% from 52-week highs on Nov 17) reflects investor concerns around HUBS' EU exposure and SMB's sensitivity to a potential recession, though we contend HUBS has become increasingly mission-critical and has moved more upmarket, making its business model more resilient. We highlight three key points: 1) successful evolution from a point marketing solution into a multi-product front office platform, unlocking potential to consolidate SMB spending; 2) continued traction upmarket in midmarket, with improving net and gross retention rates and double-digit ASRPC growth; and 3) solid financial profile, with sustainable high growth (following strong 45% cc revenue growth in 2021 that accelerated 15 points, we see potential for HUBS to achieve 40% cc in 2022) and expanding margins (9% / 14% EBIT / FCF margins in 2021). HUBS trades at 8.7x EV/CY23E consensus revenue, relatively in-line with the High Growth peer group, which we see as attractive given HUBS strong execution track-record, potential to consolidate SMB spend, and above-average profitability.

**MongoDB (MDB): Outperform, $505 PT:** In our view, MongoDB provides mission-critical software in an uncertain macroeconomic environment, and we see potential for sustained Atlas hyper-growth (82% Atlas revenue growth in 2021) to drive meaningful potential upside to total revenue growth (consensus calls for 35% in 2022 vs. 48% in 2021). We highlight four key points: 1) MDB competes in a large and growing database market and is uniquely positioned to address the market's fastest growing categories (NoSQL and DBaaS); 2) powerful secular drivers including big data, modern application development, and the mission-critical nature of databases; 3) MDB has become the clear NoSQL leader with significant developer mindshare, an effective licensing model, and technical leadership backed by an impressive product development velocity; and 4) impressive growth at scale driven by Atlas ($1B+ revenue run-rate growing 56% Y/Y) coupled with a clear path to profitability. MDB trades at 18.6x EV/CY23E consensus revenue, towards the high-end of High Growth peers, though we believe MongoDB can continue to report upside to numbers and drive shares higher.

**Twilio (TWLO): Outperform, $400 PT:** We expect a 2H rebound for Twilio, driven by: 1) reopening tailwinds (return of business travel); 2) gradually easier compares (plus a boost from mid-term elections); 3) accelerating Segment revenue growth; and 4) a cleaner overall model (Segment / Zipwhip become organic in 1Q / 3Q and 10DLC A2P fees become better understood). We remain bullish on Twilio's potential to become a much larger software platform in a post-pandemic world where technology increasingly has become table-stakes and communication sits at the center of all digital customer experiences. As such, we see CPaaS as a large and fast-growing market with Twilio being the clear leader. Additionally, we believe Twilio, in many ways, parallels AWS with the ability to move "up the stack" with Application Services. Twilio trades at just 3.9x EV/CY23E consensus revenue, a discount to the High Growth peer group, and even on an EV/GP basis is still a notable discount, which we see as attractive given Twilio's strong financial profile (30%+ sustainable organic growth and improving operating margins), competitive positioning, and very large TAM.

**Veeva (VEEV): Outperform, $300 PT:** We believe investor concerns around pharma sales rep reductions is overblown, as Veeva remains the gold standard in vertical software and in our view can achieve a unique financial profile of 20% revenue growth and 40% FCF margins. We see multiple upside growth drivers in FY23, in both: 1) Commercial Solutions, with traction on CRM add-ons, especially in data and analytics (Crossix, Link, OpenData, DataCloud); and 2) R&D Solutions, with larger deals, broad-based demand across the product portfolio, and a strong lineup of early stage products (CDMS, Safety, Quality, Regulatory, etc.). Long-term, we like Veeva for four primary reasons: 1) Life Sciences remains an attractive end market; 2) Veeva

This report is intended solely for cbass@marketstreetpartners.com



has a sustainable competitive moat with its deep domain expertise and customer relationships; 3) Veeva has an impressive flywheel of product innovation, which continues to open up new market opportunities (e.g., new MedTech CRM built on Vault); and 4) Veeva has a unique go-to-market motion centered around reference selling, product-led growth, and disciplined investments, leading to a best-in-class financial model. Veeva trades at 28.9x EV/CY23E consensus FCF, which is attractive to us relative to vertical software and medium growth peers given its high-quality financial profile.

## Matt Hedberg's top picks

**CrowdStrike (CRWD), Outperform $275 PT:** We believe COVID has fundamentally changed the security landscape which should enable customers to continue to compound, wallet share gains and building cash flow as we believe security spending remains a top priority in 2022. As such, we believe the model likely shows durable, high growth and continued upside to estimates while the install base can grow 10x over time, with continued cross/up-sell. We believe several potential catalysts exist including: 1) monetizing the untapped workload protection market that takes on a whole new level of importance in 2022 as both security and digital transformations continue to accelerate, which is the largest catalyst in our mind; 2) continued share gains from legacy AV/endpoint vendors; 3) rapid adoption of newer modules such as Spotlight for vulnerability management, Discovery for IT hygiene and DLP/FIM for data governance – all of which helps offset the risk of discounting with EPP/EDR; 4) momentum with GSIs; and 5) profitable growth as FCF margins likely show minimal leverage in CY/22, but remain elevated and differentiated given growth.

**Palo Alto (PANW), Outperform $710 PT:** With our expectations that security spending remains a top priority in 2022 and vendor consolidation could increase in focus, we believe Palo Alto is in a good position to deliver consistent revenue growth/upside with expanding margins. With the product platform "90% there", the focus is more on scaling sales productivity than adding new products, which importantly should position margins to move higher as we believe the rate of M&A is likely to slow. We believe several catalysts exist including: 1) a firewall product cycle that gives the company strong viability despite supply chain concerns; 2) Prisma Cloud that benefits from an increased focus on cloud workload protection; 3) Prisma Access that benefits from increased focus on zero-trust security; 4) Cortex that benefits from an increased focus on security analytics and XDR; and 5) margin expansion as we believe the product platform is largely built out which reduces the risk of material M&A.

**ServiceNow (NOW), Outperform $660 PT:** We believe investors will increasingly look to quality growth ideas that also have above-average margins, something ServiceNow represents, in our view. Following 2021 that saw accelerating new ACV, we believe upside to 2022 subscription revenue and cRPO remains likely. As an organic growth story, ServiceNow is targeting 2024 subscription revenue of $10B+ growing to $15B+ in 2026, with 2024 operating margins of 26.5% and 33% FCF margins. We believe the "playbook" to get to $10B and $15B is largely the same one as the company has used so far; organic innovation supplemented with tuck-in acquisitions. We believe several catalysts exist including: 1) Customer/Employee Workflows and Creator Workflows grow in the mix; 2) bullish on the opportunity for observability (following the Lightstep acquisition) and RPA cross-sell within the base; 3) increased customer focus on automation in a tight labor market; 4) accelerating new ACV in 2021 likely points to strong subscription growth in 2022; 5) additional sales leverage from GIS momentum; and 6) expanding federal opportunities.

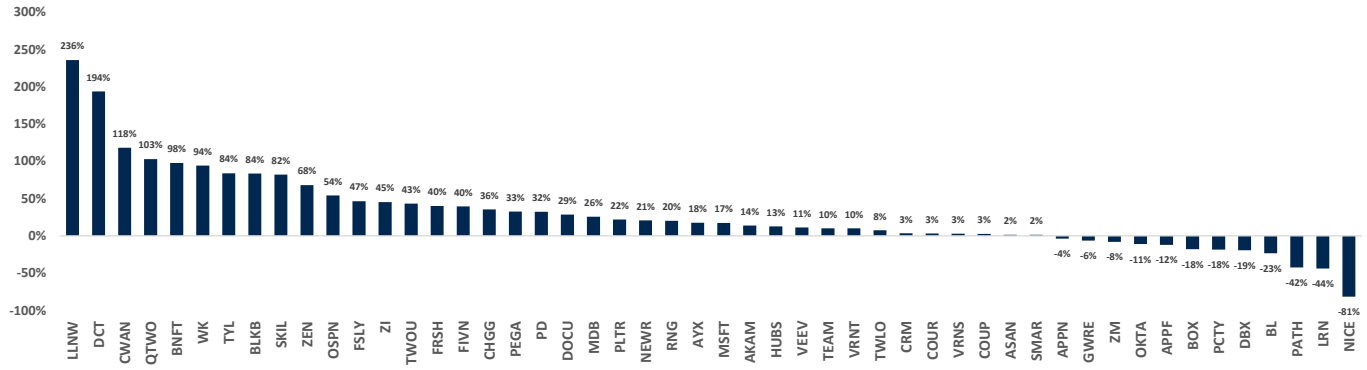
This report is intended solely for cbass@marketstreetpartners.com



## LinkedIn/Glassdoor Trackers

Exhibit 94 shows the sequential change in job postings (on LinkedIn) for 50 software companies between 1Q22 and 4Q21. While we wouldn't read too significantly into a sequential change, especially when considering the current environment, we believe that job openings can be a leading indicator of business momentum over time.

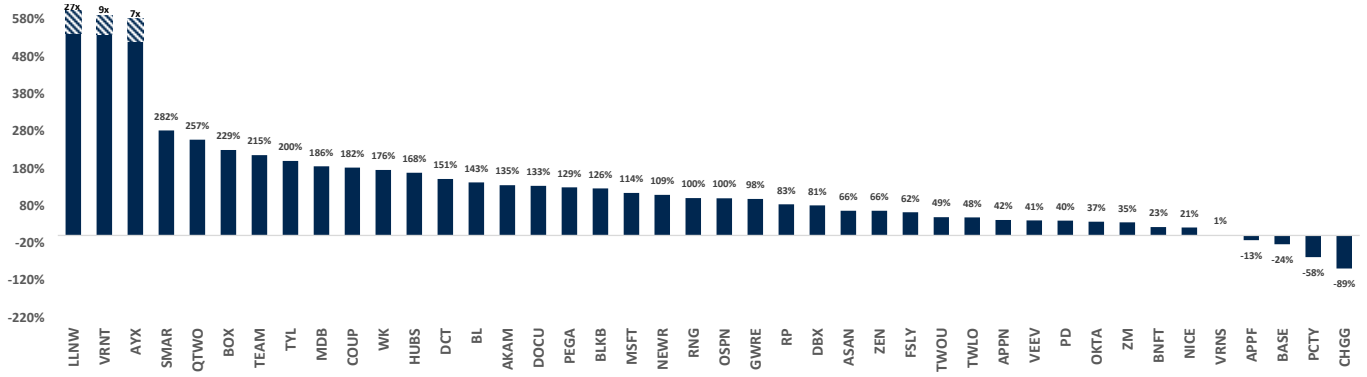
Exhibit 94 - Sequential Change in Job Postings (on LinkedIn), 1Q22 vs. 4Q21




Source: LinkedIn, RBC Capital Markets

Exhibit 95 shows the year-over-year change in job postings (on LinkedIn) for 42 software companies between 1Q22 and 4Q21. Many of the names we tracked have materially increased hiring initiatives relative to this time last year as 22 companies have 2x+ the number of job postings Y/Y and 3 companies have at least 3x the number of job postings. Y/Y job posting growth continues normalize from 3Q21 when we saw job postings 2x+ Y/Y for 34 companies (15 had at least 3x'd job postings Y/Y in 3Q21). We note that this dataset only includes companies we have data for in this time period.

Exhibit 95 - Annual Change in Job Postings (on LinkedIn), 1Q22 vs. 1Q21



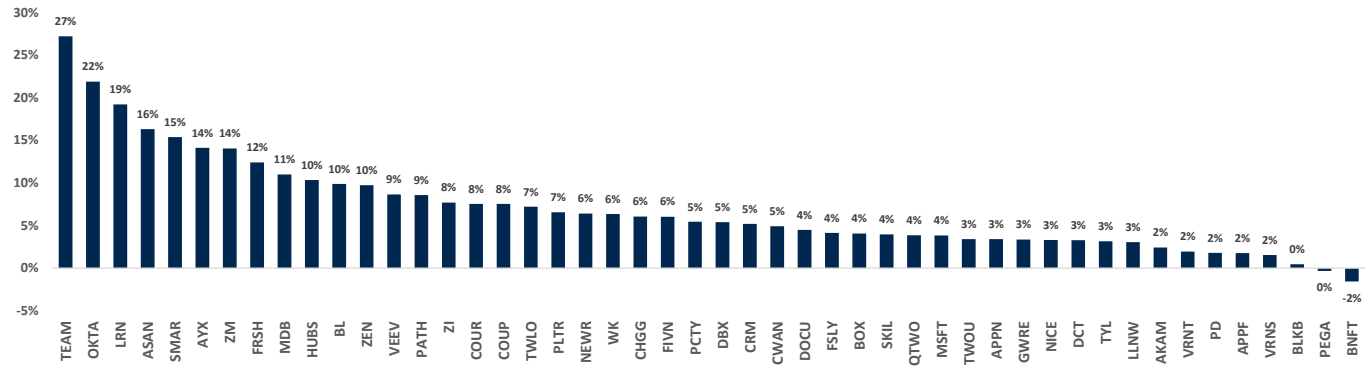

Source: LinkedIn, RBC Capital Markets

Exhibit 96 shows the sequential change in employee count (according to LinkedIn) for 50 software companies between 1Q22 and 4Q21.

This report is intended solely for cbass@marketstreetpartners.com



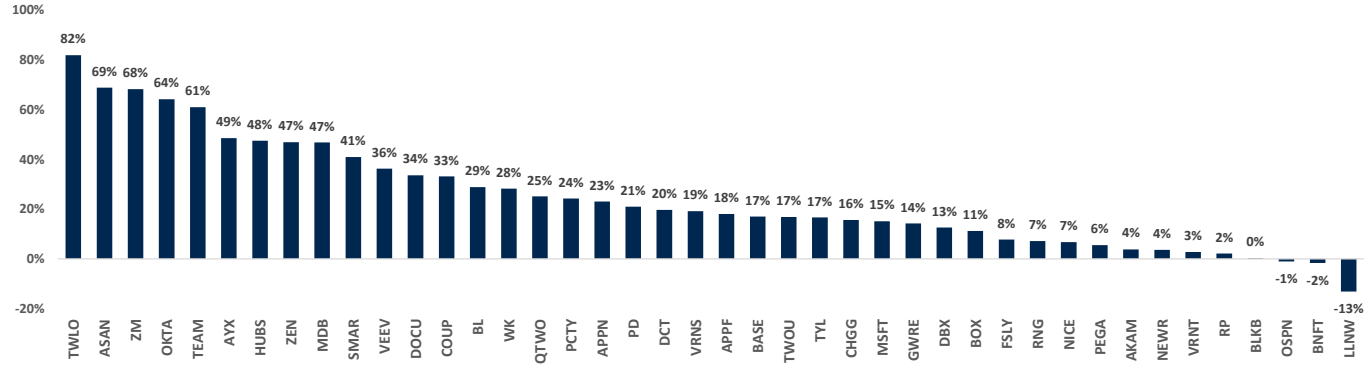
Exhibit 96 - Sequential Change in Employee Count (on LinkedIn), 1Q22 vs. 4Q21




Source: LinkedIn, RBC Capital Markets

Exhibit 97 shows the annual change in employee count (according to LinkedIn) for 42 software companies between 1Q22 and 1Q21.

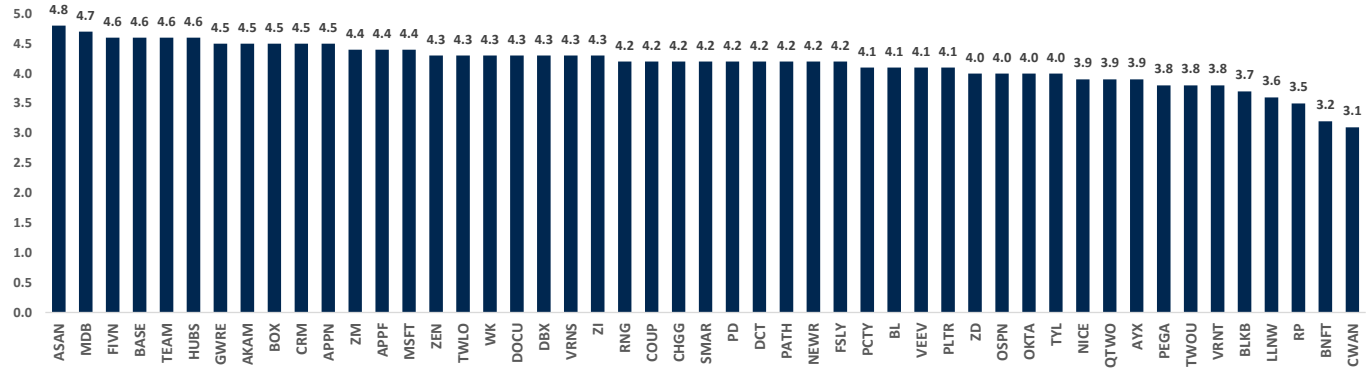
Exhibit 97 - Annual Change in Employee Count (on LinkedIn), 1Q22 vs. 1Q21




Source: LinkedIn, RBC Capital Markets

Exhibit 98 shows the Glassdoor ratings for 49 software companies.

Exhibit 98 - Glassdoor Ratings (out of 5), 1Q22



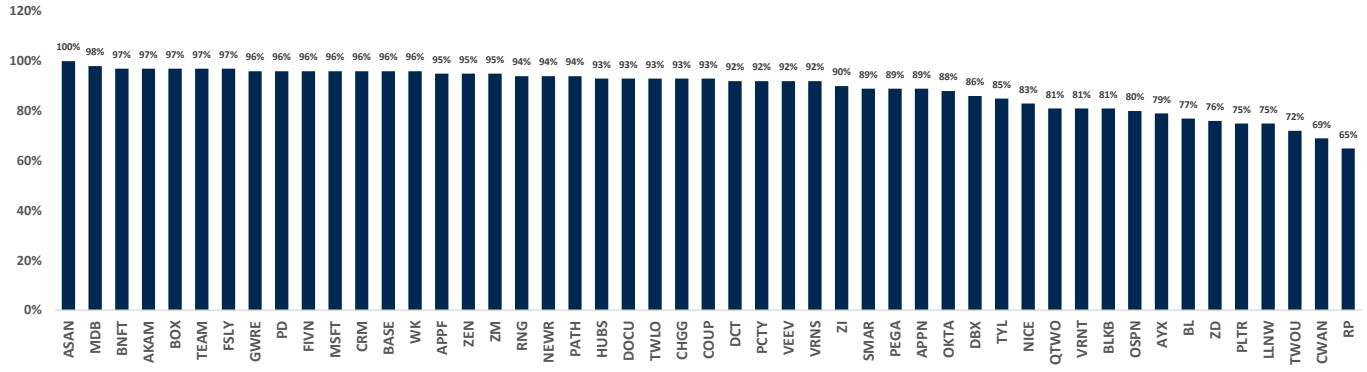

Source: Glassdoor, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



Exhibit 99 shows the Glassdoor CEO ratings for 49 software companies.

Exhibit 99 - Glassdoor CEO Ratings, 2Q21




Source: Glassdoor, RBC Capital Markets

This report is intended solely for cbass@marketstreetpartners.com



# Contributing Authors

**RBC Capital Markets, LLC**

| | | |
|---|---|---|
| Rishi Jaluria (Analyst) | (415) 633-8798 | rishi.jaluria@rbccm.com |
| Matthew Hedberg (Analyst) | (612) 313-1293 | matthew.hedberg@rbccm.com |
| Richard Poland (Senior Associate) | (347) 920-6958 | richard.poland@rbccm.com |
| Philip Rigby (Senior Associate) | (971) 400-3331 | philip.rigby@rbccm.com |
| Dan Bergstrom (Analyst) | (612) 313-1254 | dan.bergstrom@rbccm.com |
| Matthew Swanson (Analyst) | (612) 313-1237 | matthew.swanson@rbccm.com |
| Anushtha Mittal (Associate) | (347) 637-0003 | anushtha.mittal@rbccm.com |
| Simran Biswal (Associate) | (646) 618-6894 | simran.biswal@rbccm.com |

**RBC Europe Limited**

| | | |
|---|---|---|
| Sherri Malek (Analyst) | +44 0 20 7653 4510 | sherri.malek@rbccm.com |

This report is intended solely for cbass@marketstreetpartners.com



# Required disclosures

## Non-U.S. analyst disclosure

One or more research analysts involved in the preparation of this report (i) may not be registered/qualified as research analysts with the NYSE and/or FINRA and (ii) may not be associated persons of the RBC Capital Markets, LLC and therefore may not be subject to FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

## Conflicts disclosures

This product constitutes a compendium report (covers six or more subject companies). As such, RBC Capital Markets chooses to provide specific disclosures for the subject companies by reference. To access conflict of interest and other disclosures for the subject companies, clients should refer to https://www.rbccm.com/GLDisclosure/PublicWeb/DisclosureLookup.aspx?entityId=1. These disclosures are also available by sending a written request to RBC Capital Markets Research Publishing, P.O. Box 50, 200 Bay Street, Royal Bank Plaza, 29th Floor, South Tower, Toronto, Ontario M5J 2W7 or an email to rbcinsight@rbccm.com.

The analyst(s) responsible for preparing this research report received compensation that is based upon various factors, including total revenues of the member companies of RBC Capital Markets and its affiliates, a portion of which are or have been generated by investment banking activities of the member companies of RBC Capital Markets and its affiliates.

With regard to the MAR investment recommendation requirements in relation to relevant securities, a member company of Royal Bank of Canada, together with its affiliates, may have a net long or short financial interest in excess of 0.5% of the total issued share capital of the entities mentioned in the investment recommendation. Information relating to this is available upon request from your RBC investment advisor or institutional salesperson.

## Distribution of ratings

For the purpose of ratings distributions, regulatory rules require member firms to assign ratings to one of three rating categories - Buy, Hold/Neutral, or Sell - regardless of a firm's own rating categories. Although RBC Capital Markets' ratings of Outperform (O), Sector Perform (SP), and Underperform (U) most closely correspond to Buy, Hold/Neutral and Sell, respectively, the meanings are not the same because our ratings are determined on a relative basis.

**Distribution of ratings**
**RBC Capital Markets, Equity Research**
**As of 31-Mar-2022**

| | | | Investment Banking Serv./Past 12 Mos. | |
|---|---|---|---|---|
| Rating | Count | Percent | Count | Percent |
| BUY [Outperform] | 841 | 57.68 | 330 | 39.24 |
| HOLD [Sector Perform] | 569 | 39.03 | 172 | 30.23 |
| SELL [Underperform] | 48 | 3.29 | 3 | 6.25 |

## Conflicts policy

RBC Capital Markets Policy for Managing Conflicts of Interest in Relation to Investment Research is available from us on request. To access our current policy, clients should refer to
https://www.rbccm.com/global/file-414164.pdf
or send a request to RBC Capital Markets Research Publishing, P.O. Box 50, 200 Bay Street, Royal Bank Plaza, 29th Floor, South Tower, Toronto, Ontario M5J 2W7. We reserve the right to amend or supplement this policy at any time.

## Dissemination of research and short-term trade ideas

RBC Capital Markets endeavors to make all reasonable efforts to provide research content simultaneously to all eligible clients, having regard to local time zones in overseas jurisdictions. RBC Capital Markets provides eligible clients with access to Equity Research Reports and to SPARC on the Firm's proprietary INSIGHT website, via email and via third-party vendors. SPARC contains market color and commentary regarding subject companies on which the Firm currently provides equity research coverage. Research Analysts may, from time to time, include short-term trade ideas in Research Reports and / or in SPARC. A short-term

This report is intended solely for cbass@marketstreetpartners.com



trade idea offers a short-term view on how a security may trade, based on market and trading events, and the resulting trading opportunity that may be available. A short-term trade idea may differ from the price targets and recommendations in our published Research Reports reflecting the Research Analyst's views of the longer-term (one year) prospects of the subject company, as a result of the differing time horizons, methodologies and/or other factors. Thus, it is possible that a subject company's common equity that is considered a long-term 'Sector Perform' or even an 'Underperform' might present a short-term buying opportunity as a result of temporary selling pressure in the market; conversely, a subject company's common equity rated a long-term 'Outperform' could be considered susceptible to a short-term downward price correction. Short-term trade ideas are not ratings, nor are they part of any ratings system, and the firm generally does not intend, nor undertakes any obligation, to maintain or update short-term trade ideas. Short-term trade ideas may not be suitable for all investors and have not been tailored to individual investor circumstances and objectives, and investors should make their own independent decisions regarding any securities or strategies discussed herein. Please contact your investment advisor or institutional salesperson for more information regarding RBC Capital Markets' research.

For a list of all recommendations on the company that were disseminated during the prior 12-month period, please click on the following link: https://rbcnew.bluematrix.com/sellside/MAR.action

The 12 month history of SPARCs can be viewed at https://www.rbcinsightresearch.com.

## Analyst certification

All of the views expressed in this report accurately reflect the personal views of the responsible analyst(s) about any and all of the subject securities or issuers. No part of the compensation of the responsible analyst(s) named herein is, or will be, directly or indirectly, related to the specific recommendations or views expressed by the responsible analyst(s) in this report.

## Third-party-disclaimers

The Global Industry Classification Standard ("GICS") was developed by and is the exclusive property and a service mark of MSCI Inc. ("MSCI") and Standard & Poor's Financial Services LLC ("S&P") and is licensed for use by RBC. Neither MSCI, S&P, nor any other party involved in making or compiling the GICS or any GICS classifications makes any express or implied warranties or representations with respect to such standard or classification (or the results to be obtained by the use thereof), and all such parties hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability and fitness for a particular purpose with respect to any of such standard or classification. Without limiting any of the foregoing, in no event shall MSCI, S&P, any of their affiliates or any third party involved in making or compiling the GICS or any GICS classifications have any liability for any direct, indirect, special, punitive, consequential or any other damages (including lost profits) even if notified of the possibility of such damages.

RBC Capital Markets disclaims all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any statements made to the media or via social media that are in turn quoted in this report, or otherwise reproduced graphically for informational purposes.

References herein to "LIBOR", "LIBO Rate", "L" or other LIBOR abbreviations means the London interbank offered rate as administered by ICE Benchmark Administration (or any other person that takes over the administration of such rate).

## Disclaimer

RBC Capital Markets is the business name used by certain branches and subsidiaries of the Royal Bank of Canada, including RBC Dominion Securities Inc., RBC Capital Markets, LLC, RBC Europe Limited, RBC Capital Markets (Europe) GmbH, Royal Bank of Canada, Hong Kong Branch and Royal Bank of Canada, Sydney Branch. The information contained in this report has been compiled by RBC Capital Markets from sources believed to be reliable, but no representation or warranty, express or implied, is made by Royal Bank of Canada, RBC Capital Markets, its affiliates or any other person as to its accuracy, completeness or correctness. All opinions and estimates contained in this report constitute RBC Capital Markets' judgement as of the date of this report, are subject to change without notice and are provided in good faith but without legal responsibility. Nothing in this report constitutes legal, accounting or tax advice or individually tailored investment advice. This material is prepared for general circulation to clients and has been prepared without regard to the individual financial circumstances and objectives of persons who receive it. The investments or services contained in this report may not be suitable for you and it is recommended that you consult an independent investment advisor if you are in doubt about the suitability of such investments or services. This report is not an offer to sell or a solicitation of an offer to buy any securities. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. RBC Capital Markets research analyst compensation is based in part on the overall profitability of RBC Capital Markets, which includes profits attributable to investment banking revenues. Every province in Canada, state in the U.S., and most countries throughout the world have their own laws regulating the types of securities and other investment products which may be offered to their residents, as well as the process for doing so. As a result, the securities discussed in this report may not be eligible for sale in some jurisdictions. RBC Capital Markets may be restricted from publishing research reports, from time to time, due to regulatory restrictions and/ or internal compliance policies. If this is the case, the latest published research reports available to clients may not reflect recent material changes in the applicable industry and/or applicable subject companies. RBC Capital Markets research reports are current only as of the date set forth on the research reports. This report is not, and under no circumstances should be construed as, a solicitation to act as securities broker or dealer in any jurisdiction by any person or company that is not legally permitted to carry on the business of a securities broker or dealer in that jurisdiction. To the full extent permitted by law neither RBC Capital Markets nor any of its affiliates, nor any other person, accepts any liability whatsoever for any direct, indirect or consequential loss arising from, or in connection with, any use of this report or the information contained herein. No matter contained in this document may be reproduced or copied by any means without the prior written consent of RBC Capital Markets in each instance.

Additional information is available on request.

**To U.S. Residents:**

This publication has been approved by RBC Capital Markets, LLC (member FINRA, NYSE, SIPC), which is a U.S. registered broker-dealer and which accepts responsibility for this report and its dissemination in the United States. Any U.S. recipient of this report that is not a registered broker-dealer or a bank acting in

This report is intended solely for cbass@marketstreetpartners.com



a broker or dealer capacity and that wishes further information regarding, or to effect any transaction in, any of the securities discussed in this report, should contact and place orders with RBC Capital Markets, LLC.

**To Canadian Residents:**

This publication has been approved by RBC Dominion Securities Inc.(member IIROC). Any Canadian recipient of this report that is not a Designated Institution in Ontario, an Accredited Investor in British Columbia or Alberta or a Sophisticated Purchaser in Quebec (or similar permitted purchaser in any other province) and that wishes further information regarding, or to effect any transaction in, any of the securities discussed in this report should contact and place orders with RBC Dominion Securities Inc., which, without in any way limiting the foregoing, accepts responsibility for this report and its dissemination in Canada.

**To U.K. Residents:**

This publication has been approved by RBC Europe Limited ('RBCEL') which is authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority ('FCA') and the Prudential Regulation Authority, in connection with its distribution in the United Kingdom. This material is not for general distribution in the United Kingdom to retail clients, as defined under the rules of the FCA. RBCEL accepts responsibility for this report and its dissemination in the United Kingdom.

**To EEA Residents:**

This material is distributed in the EU by either RBCEL on an authorised cross-border basis, or by RBC Capital Markets (Europe) GmbH (RBC EG) which is authorised and regulated in Germany by the Bundesanstalt für Finanzdienstleistungsaufsicht (German Federal Financial Supervisory Authority) (BaFin).

**To Persons Receiving This Advice in Australia:**

This material has been distributed in Australia by Royal Bank of Canada, Sydney Branch (ABN 86 076 940 880, AFSL No. 246521). This material has been prepared for general circulation and does not take into account the objectives, financial situation or needs of any recipient. Accordingly, any recipient should, before acting on this material, consider the appropriateness of this material having regard to their objectives, financial situation and needs. If this material relates to the acquisition or possible acquisition of a particular financial product, a recipient in Australia should obtain any relevant disclosure document prepared in respect of that product and consider that document before making any decision about whether to acquire the product. This research report is not for retail investors as defined in section 761G of the Corporations Act.

**To Hong Kong Residents:**

This publication is distributed in Hong Kong by Royal Bank of Canada, Hong Kong Branch, which is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission (SFC) in Hong Kong, RBC Investment Services (Asia) Limited and RBC Global Asset Management (Asia) Limited, both entities are regulated by the SFC. This material is not for general distribution in Hong Kong to persons who are not professional investors (as defined in the Securities and Futures Ordinance of Hong Kong (Cap. 571) and any rules made thereunder.

**To Singapore Residents:**

This publication is distributed in Singapore by the Royal Bank of Canada, Singapore Branch, a registered entity licensed by the Monetary Authority of Singapore. This material has been prepared for general circulation and does not take into account the objectives, financial situation, or needs of any recipient. You are advised to seek independent advice from a financial adviser before purchasing any product. If you do not obtain independent advice, you should consider whether the product is suitable for you. Past performance is not indicative of future performance. If you have any questions related to this publication, please contact the Royal Bank of Canada, Singapore Branch. Royal Bank of Canada, Singapore Branch accepts responsibility for this report and its dissemination in Singapore.

**To Japanese Residents:**

Unless otherwise exempted by Japanese law, this publication is distributed in Japan by or through RBC Capital Markets (Japan) Ltd. which is a Financial Instruments Firm registered with the Kanto Local Financial Bureau (Registered number 203) and a member of the Japan Securities Dealers Association (JSDA) and the Financial Futures Association of Japan (FFAJ).

**® Registered trademark of Royal Bank of Canada. RBC Capital Markets is a trademark of Royal Bank of Canada. Used under license.**
**Copyright © RBC Capital Markets, LLC 2022 - Member SIPC**
**Copyright © RBC Dominion Securities Inc. 2022 - Member Canadian Investor Protection Fund**
**Copyright © RBC Europe Limited 2022**
**Copyright © Royal Bank of Canada 2022**
**All rights reserved**